CHAPTER 1

Numbers

INTRODUCTION

In Hindu Arabic System, we use ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 called *digits* to represent any number. This is the *decimal system* where we use the numbers 0 to 9. 0 is called *insignificant digit* whereas 1, 2, 3, 4, 5, 6, 7, 8, 9 are called *significant digits*.

A group of figures, denoting a number is called a *numeral*. For a given numeral, we start from extreme right as Unit's place, Ten's place, Hundred's place and so on.

Illustration 1: We represent the number 309872546 as shown below:

Ten Crore 10^8	Crores 10^7	Ten Lakhs (million) 10^6	Lakhs 10^5	Ten Thousand 10^4	Thousand 10^3	Hundred 10^2	Ten's 10^1	Units 10^0
3	0	9	8	7	2	5	4	6

We read it as

'Thirty crores, ninety-eight lakhs, seventy-two thousands five hundred and forty-six.'

In this numeral:

The place value of 6 is $6 \times 1 = 6$

The place value of 4 is $4 \times 10 = 40$

The place value of 5 is $5 \times 100 = 500$

The place value of 2 is $2 \times 1000 = 2000$ and so on.

The face value of a digit in a number is the value itself wherever it may be.

Thus, the face value of 7 in the above numeral is 7. The face value of 6 in the above numeral is 6 and so on.

NUMBER SYSTEM

Natural Numbers

Counting numbers 1, 2, 3, 4, 5, ... are known as *natural numbers*.

The set of all natural numbers can be represented by

$$N = \{1, 2, 3, 4, 5, ...\}$$

Whole Numbers

If we include 0 among the natural numbers, then the numbers 0, 1, 2, 3, 4, 5, ... are called *whole numbers*.

The set of whole numbers can be represented by

$$W = \{0, 1, 2, 3, 4, 5, ...\}$$

Clearly, every natural number is a whole number but 0 is a whole number which is not a natural number.

Integers

All counting numbers and their negatives including zero are known as *integers*.

The set of integers can be represented by

$$Z \text{ or } I = \{..., -4, -3, -2, -1, 0, 1, 2, 3, 4, ...\}$$

Positive Integers

The set $I^+ = \{1, 2, 3, 4, ...\}$ is the set of all *positive integers*. Clearly, positive integers and natural numbers are synonyms.

Negative Integers

The set $I^- = \{-1, -2, -3, ...\}$ is the set of all *negative integers*. 0 is neither positive nor negative.

Non-negative Integers

The set $\{0, 1, 2, 3, ...\}$ is the set of all *non-negative integers*.

Rational Numbers

The numbers of the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$, are known as *rational numbers*, e.g., $\frac{4}{7}$, $\frac{3}{2}$, $-\frac{5}{8}$, $\frac{0}{1}$, $-\frac{2}{3}$, etc.

The set of all rational numbers is denoted by Q.

That is, $Q = \{x : x = \frac{p}{q};\ p, q \in I, q \neq 0\}$

Since every natural number 'a' can be written as $\frac{a}{1}$, every natural number is a rational number. Since 0 can be written as $\frac{0}{1}$ and every non-zero integer 'a' can be written as $\frac{a}{1}$, every integer is a rational number.

Every rational number has a peculiar characteristic that when expressed in decimal form is expressible either in terminating decimals or in non-terminating repeating decimals.

For example, $\frac{1}{5} = 0.2$, $\frac{1}{3} = 0.333...$, $\frac{22}{7} = 3.1428714287$, $\frac{8}{44} = 0.181818\ ...$, etc.

The recurring decimals have been given a short notation as

$$0.333... = 0.\overline{3}$$
$$4.1555... = 4.0\overline{5}$$
$$0.323232... = 0.\overline{32}$$

Irrational Numbers

Those numbers which when expressed in decimal form are neither terminating nor repeating decimals are known as *irrational numbers*, e.g., $\sqrt{2}, \sqrt{3}, \sqrt{5}, \pi$, etc.

Note that the exact value of π is not $\frac{22}{7} \cdot \frac{22}{7}$ is rational while π is irrational number. $\frac{22}{7}$ is approximate value of π. Similarly, 3.14 is not an exact value of it.

Real Numbers

The rational and irrational numbers combined together to form *real numbers*, e.g., $\frac{13}{21}, \frac{2}{5}, -\frac{3}{7}, \sqrt{3}, 4+\sqrt{2}$, etc. are real numbers.

The set of all real numbers is denoted by R.

Note that the sum, difference or product of a rational and irrational number is irrational, e.g., $3+\sqrt{2}$, $4-\sqrt{3}$, $\frac{2}{5}-\sqrt{5}$, $4\sqrt{3}$, $-7\sqrt{5}$ are all irrational.

Even Numbers

All those numbers which are exactly divisible by 2 are called *even numbers*, e.g., 2, 6, 8, 10, etc., are even numbers.

Odd Numbers

All those numbers which are not exactly divisible by 2 are called *odd numbers*, e.g., 1, 3, 5, 7, etc., are odd numbers.

Prime Numbers

A natural number other than 1, is a *prime number* if it is divisible by 1 and itself only.

For example, each of the numbers 2, 3, 5, 7, etc., are prime numbers.

Composite Numbers

Natural numbers greater than 1 which are not prime, are known as composite numbers.

For example, each of the numbers 4, 6, 8, 9, 12, etc., are composite numbers.

Notes

1. The number 1 is neither a prime number nor a composite number.
2. 2 is the only even number which is prime.
3. Prime numbers up to 100 are:
   2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, i.e., 25 prime numbers between 1 and 100.
4. Two numbers which have only 1 as the common factor are called *co-primes* or *relatively prime* to each other, e.g., 3 and 5 are co-primes.

Note that the numbers which are relatively prime need not necessarily be prime numbers, e.g., 16 and 17 are relatively prime, although 16 is not a prime number.

BASIC FORMULAE

ADDITION AND SUBTRACTION

Addition and Subtraction is best illustrated with the help of following example:

Illustration 1: 54321 – (9876 + 8967 + 7689) = ?

Step 1 Add Ist column:

$6 + 7 + 9 = 22$

To obtain 1 at unit's place add 9 to make 31. In the answer, write 9 at unit's place and carry over 3.

$$\begin{array}{r} 54321 \\ 9876 \\ 8967 \\ 7689 \\ \hline 27789 \end{array}$$

Step 2 Add 2nd column:

$3 + 7 + 6 + 8 = 24$

To obtain 2 at ten's place, add 8 to make 32. In the answer, write 8 at ten's place and carry over 3.

Step 3 Add 3rd column:

$3 + 8 + 9 + 6 = 26$

To obtain 3 at hundred's place, add 7 to make 33. In the answer, write 7 at hundred's place and carry over 3.

Step 4 Add 4th column:

$3 + 9 + 8 + 7 = 27$

To obtain 4 at thousand's place, add 7 to make 34. In the answer, write 7 at thousand's place and carry over 3.

Step 5 5th column:

To obtain 5 at ten-thousand's place add 2 to it to make 5. In the answer, write 2 at the ten-thousand's place.

$\therefore$ 54321 – (9876 + 8967 + 7689)

= 27789.

MULTIPLICATION

01 Multiplication of a given number by 9, 99, 999, etc., that is by $10^n - 1$

Method: Put as many zeros to the right of the multiplicant as there are nines in the multiplier and from the result subtract the multiplicant and get the answer.

Illustration 2: Multiply:

(a) 3893 by 99 (b) 4327 by 999

(c) 5863 by 9999

Solution: (a) $3893 \times 99 = 389300 - 3893 = 385407$

(b) $4327 \times 999 = 4327000 - 4327 = 4322673$

(c) $5863 \times 9999 = 58630000 - 5863 = 58624137$

02 Multiplication of a given number by 11, 101, 1001, etc., that is, by $10^n + 1$.

Method: Place n zeros to the right of the multiplicant and then add the multiplicant to the number so obtained.

Illustration 3: Multiply:

(a) 4782×11 (b) 9836×101

(c) 6538×1001

Solution: (a) $4782 \times 11 = 47820 + 4782 = 52602$

(b) $9836 \times 101 = 983600 + 9836 = 993436$

(c) $6538 \times 1001 = 6538000 + 6538 = 6544538$

03 Multiplication of a given number by 15, 25, 35, etc.

Method: Double the multiplier and then multiply the multiplicant by this new number and finally divide the product by 2.

Illustrations 4: Multiply:

(a) 7054×15 (b) 3897×25

(c) 4563×35

Solution: (a) $7054 \times 15 = \frac{1}{2}(7054 \times 30)$

$= \frac{1}{2}(211620) = 105810$

(b) $3897 \times 25 = \frac{1}{2}(3897 \times 50) = \frac{1}{2}(194850)$

$= 97425$

(c) $4536 \times 35 = \frac{1}{2}(4563 \times 70) = \frac{1}{2}(319410)$

$= 159705$

04 Multiplication of a given number by 5, 25, 125, 625, etc., that is, by a number which is some power of 5.

Method: Place as many zeros to the right of the multiplicant as is the power of 5 in the multiplier, then divide the number so obtained by 2 raised to the same power as is the power of 5.

Illustration 5: Multiply:

(a) 3982×5 (b) 4739×25

(c) 7894×125 (d) 4863×625

Solution: (a) $3982 \times 2 = \dfrac{39820}{2} = 19910$

(b) $4739 \times 25 = \dfrac{473900}{2^2} = \dfrac{473900}{4} = 118475$

(c) $7894 \times 125 = \dfrac{7894000}{2^3} = \dfrac{7894000}{8}$
$= 986750$

(d) $4863 \times 625 = \dfrac{48630000}{2^4} = \dfrac{48630000}{16}$
$= 3039375$

Distributive Laws

For any three numbers a, b, c, we have

(a) $a \times b + a \times c = a \times (b + c)$

(b) $a \times b - a \times c = a \times (b - c)$

Illustration 6: $438 \times 637 + 438 \times 367 = ?$

Solution: $438 \times 637 + 438 \times 367 = 438 \times (637 + 367)$
$= 430 \times 1000$
$= 438000$

Illustration 7: $674 \times 832 - 674 \times 632 = ?$

Solution: $674 \times 832 - 674 \times 632$
$= 674 \times (832 - 632)$
$= 674 \times 200 = 134800$

SQUARES

01 To square any number ending with 5.

Method: $(A5)^2 = A(A + 1)/25$

Illustration 8:

(a) $(25)^2 = 2(2 + 1)/25 = 6/25 = 625$

(b) $(45)^2 = 4(4 + 1)/25 = 20/25 = 2025$

(c) $(85)^2 = 8(8 + 1)/25 = 72/25 = 7225$

02 To square a number in which every digit is one.

Method: Count the number of digits in the given number and start writing numbers in ascending order from one to this number and then in descending order up to one.

Illustration 9:

(a) $11^2 = 121$

(b) $111^2 = 12321$

(c) $1111^2 = 1234321$

(d) $222^2 = 2^2(111)^2 = 4(12321) = 49284$

(e) $3333^2 = 3^2(1111)^2 = 9(1234321) = 11108889$

03 To square a number which is nearer to $10x$.

Method: Use the formula:

$$x^2 = (x^2 - y^2) + y^2 = (x + y)(x - y) + y^2$$

Illustration 10:

(a) $(97)^2 = (97 + 3)(97 - 3) + 3^2$
$= 9400 + 9 = 9409$

(b) $(102)^2 = (102 - 2)(102 + 2) + 2^2$
$= 10400 + 4 = 10404$

(c) $(994)^2 = (994 + 6)(994 - 6) + 6^2$
$= 988000 + 36 = 988036$

(d) $(1005)^2 = (1005 - 5)(1005 + 5) + 5^2$
$= 1010000 + 25 = 1010025$

DIVISION

Division is repeated subtraction.

For example, when we divide 63289 by 43, it means 43 can be repeatedly subtracted 1471 times from 63289 and the remainder 36 is left.

```
                1471 ← Quotient
Divisor → 43 ) 63289 ← Dividend
               43
               202
               172
                308
                301
                 79
                 43
                 36   ← Remainder
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Dividend = (Divisor × Quotient) + Remainder

or, $\text{Divisor} = \dfrac{\text{Dividend} - \text{Remainder}}{\text{Quotient}}$

Illustration 11: On dividing 7865321 by a certain number, the quotient is 33612 and the remainder is 113. Find the divisor.

Solution: $\text{Divisor} = \dfrac{\text{Dividend} - \text{Remainder}}{\text{Quotient}}$

$= \dfrac{7865321 - 113}{33612} = \dfrac{7865208}{33612} = 234$

Illustration 12: A number when divided by 315 leaves remainder 46 and the value of quotient is 7. Find the number.

Solution: Number = (Divisor × Quotient) + Remainder
$= (315 \times 7) + 46 = 2205 + 46 = 2251$

Illustration 13: Find the least number of 5 digits which is exactly divisible by 632.

Solution: The least number of 5 digits is 10000. Dividing this number by 632, the remainder is 520. So, the required number = 10000 + (632 + 520) = 10112.

$$\begin{array}{r} 15 \\ 632\overline{)\,10000} \\ \underline{632} \\ 3680 \\ \underline{3160} \\ 520 \end{array}$$

Illustration 14: Find the greatest number of 5 digits which is exactly divisible by 463.

Solution: The greatest number of 5 digits is 99999. Dividing this number by 463, the remainder is 454. So, the required number = 99999 – 454 = 99545.

$$\begin{array}{r} 215 \\ 463\overline{)\,99999} \\ \underline{926} \\ 739 \\ \underline{463} \\ 2769 \\ \underline{2315} \\ 454 \end{array}$$

Illustration 15: Find the number nearest to 13700 which is exactly divisible by 235.

Solution: On dividing the number 13700 by 235, the remainder is 70. Therefore, the nearest number to 13700, which is exactly divisible by 235 = 13700 – 70 = 13630.

$$\begin{array}{r} 58 \\ 235\overline{)\,13700} \\ \underline{1175} \\ 1950 \\ \underline{1880} \\ 70 \end{array}$$

TESTS OF DIVISIBILITY

1. **Divisibility by 2** A number is divisible by 2 if the unit's digit is zero or divisible by 2.
   For example, 4, 12, 30, 18, 102, etc., are all divisible by 2.
2. **Divisibility by 3** A number is divisible by 3 if the sum of digits in the number is divisible by 3.
   For example, the number 3792 is divisible by 3 since 3 + 7 + 9 + 2 = 21, which is divisible by 3.
3. **Divisibility by 4** A number is divisible by 4 if the number formed by the last two digits (ten's digit and unit's digit) is divisible by 4 or are both zero.
   For example, the number 2616 is divisible by 4 since 16 is divisible by 4.
4. **Divisibility by 5** A number is divisible by 5 if the unit's digit in the number is 0 or 5.
   For example, 13520, 7805, 640, 745, etc., are all divisible by 5.
5. **Divisibility by 6** A number is divisible by 6 if the number is even and sum of its digits is divisible by 3.
   For example, the number 4518 is divisible by 6 since it is even and sum of its digits 4 + 5 + 1 + 8 = 18 is divisible by 3.
6. **Divisibility by 7** The unit digit of the given number is doubled and then it is subtracted from the number obtained after omitting the unit digit. If the remainder is divisible by 7, then the given number is also divisible by 7.
   For example, consider the number 448. On doubling the unit digit 8 of 448 we get 16.
   Then, 44 – 16 = 28.
   Since 28 is divisible by 7, 448 is divisible by 7.
7. **Divisibility by 8** A number is divisible by 8, if the number formed by the last 3 digits is divisible by 8.
   For example, the number 41784 is divisible by 8 as the number formed by last three digits, i.e., 784 is divisible by 8.
8. **Divisibility by 9** A number is divisible by 9 if the sum of its digits is divisible by 9.
   For example, the number 19044 is divisible by 9 as the sum of its digits 1 + 9 + 0 + 4 + 4 = 18 is divisible by 9.
9. **Divisibility by 10** A number is divisible by 10, if it ends in zero.
   For example, the last digit of 580 is zero, therefore, 580 is divisible by 10.
10. **Divisibility by 11** A number is divisible by 11 if the difference of the sum of the digits at odd places and sum of the digits at even places is either zero or divisible by 11.
    For example, in the number 38797, the sum of the digits at odd places is 3 + 7 + 7 = 17 and the sum of the digits at even places is 8 + 9 = 17. The difference is 17 – 17 = 0, so the number is divisible by 11.
11. **Divisibility by 12** A number is divisible by 12 if it is divisible by 3 and 4.
12. **Divisibility by 18** An even number satisfying the divisibility test of 9 is divisible by 18.

13. **Divisibility by 25** A number is divisible by 25 if the number formed by the last two digits is divisible by 25 or the last two digits are zero.

For example, the number 13675 is divisible by 25 as the number formed by the last two digits is 75 which is divisible by 25.

14. **Divisibility by 88** A number is divisible by 88 if it is divisible by 11 and 8.
15. **Divisibility by 125** A number is divisible by 125 if the number formed by the last three digits is divisible by 125 or the last three digits are zero.

For example, the number 5250 is divisible by 125 as 250 is divisible by 125.

SHORT-CUT METHODS

01 **Test to find whether a given number is a prime**

Step 1 Select a least positive integer n such that n^2 > given number.

Step 2 Test the divisibility of given number by every prime number less than n.

Step 3 The given number is prime only if it is not divisible by any of these primes.

Illustration 16: Investigate, whether 571 is a prime number

Solution: Since $(23)^2 = 529 < 571$ and $(24)^2 = 576 > 571$

$\therefore \quad n = 24$

Prime numbers less than 24 are 2, 3, 5, 7, 11, 13, 17, 19, 23. Since 24 is divisible by 2, 571 is not a prime number.

Illustration 17: Investigate whether 923 is a prime number.

Solution: Since $(30)^2 = 900 < 923$ and $(31)^2 = 961 > 923$

$\therefore \quad n = 31$

Prime numbers less than 31 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. Since 923 is not divisible by any of these primes, therefore, 923 is a prime number.

02 The least number, which when divided by d_1, d_2 and d_3 leaves the remainders r_1, r_2 and r_3 respectively such that $(d_1 - r_1) = (d_2 - r_2) = (d_3 - r_3)$, is = (L.C.M. of d_1, d_2 and d_3) − $(d_1 - r_1)$ or $(d_2 - r_2)$ or $(d_3 - r_3)$.

Illustration 18: Find the least number which when divided by 9, 10 and 15 leaves the remainders 4, 5 and 10, respectively.

Solution: Here, $9 - 4 = 10 - 5 = 15 - 10 = 5$

Also, L.C.M. (9, 10, 15) = 90,

$\therefore$ the required least number = $90 - 5 = 85$.

03 A number on being divided by d_1 and d_2 successively leaves the remainders r_1 and r_2, respectively. If the number is divided by $d_1 \times d_2$, then the remainder is = $(d_1 \times r_2 + r_1)$.

Illustration 19: A number on being divided by 10 and 11 successively leaves the remainders 5 and 7, respectively. Find the remainder when the same number is divided by 110.

Solution: The required remainder

$$= d_1 \times r_2 + r_1 = 10 \times 7 + 5 = 75.$$

04 To find the number of numbers divisible by a certain integer.

The method is best illustrated with the help of following example.

Illustration 20: How many numbers up to 532 are divisible by 15?

Solution: We divide 532 by 15

$$\frac{532}{15} = 35 \times 15 + 7$$

The quotient obtained is the required number of numbers. Thus, there are 35 such numbers.

Illustration 21: How many numbers up to 300 are divisible by 5 and 7 together?

Solution: L.C.M. of 5 and 7 = 35

We divide 300 by 35

$$300 = \underline{8} \times 35 + 20$$

Thus, there are 8 such numbers.

05 Two numbers when divided by a certain divisor give remainders r_1 and r_2. When their sum is divided by the same divisor, the remainder is r_3. The divisor is given by $r_1 + r_2 - r_3$.

Illustration 22: Two numbers when divided by a certain divisor give remainders 473 and 298, respectively. When their sum is divided by the same divisor, the remainder is 236. Find the divisor.

Solution: The required divisor

$= 437 + 298 - 236 = 499.$

EXERCISE-I

1. $7372 \times 7372 + 7372 \times 628 = ?$

(a) 58976000 (b) 58967000
(c) 5897600 (d) None of these

2. $9999 + 8888 + 777 + ? = 19700$

(a) 36 (b) 16
(c) 64 (d) 26

3. $60\ ?\ 6 \times 111 = 666666$

(a) 0 (b) 2
(c) 1 (d) 6

4. $3149 \times 1\ ?\ 5 = 425115$

(a) 3 (b) 2
(c) 4 (d) 6

5. If the two digits of the age of Mr Manoj are reversed then the new age so obtained is the age of his wife. $\frac{1}{11}$ of the sum of their ages is equal to the difference between their ages. If Mr Manoj is older than his wife then find the difference between their ages.

(a) Cannot be determined
(b) 8 years
(c) 10 years
(d) 9 years
(e) 7 years

6. If in a long division sum, the dividend is 380606 and the successive remainders from the first to the last are 434, 125 and 413, then divisor is:

(a) 451 (b) 843
(c) 4215 (d) 3372

7. If $\frac{x}{y} = \frac{3}{4}$, then the value of $\left(\frac{6}{7} + \frac{y-x}{y+x}\right)$ equals:

(a) $\frac{5}{7}$ (b) $1\frac{1}{7}$
(c) 1 (d) 2

8. The largest natural number by which the product of three consecutive even natural numbers is always divisible, is:

(a) 16 (b) 24
(c) 48 (d) 96

9. Which number should replace both the '*'s in $\left(\frac{*}{21}\right) \times \left(\frac{*}{189}\right) = 1$?

(a) 21 (b) 63
(c) 3969 (c) 147

10. In a division sum, the divisor is 12 times the quotient and 5 times the remainder. If the remainder be 48, then the dividend is:

(a) 240 (b) 576
(c) 4800 (d) 4848

11. What least number must be subtracted from 1294 so that the remainder when divided by 9, 11, 13 will leave in each case the same remainder 6?

(a) 0 (b) 1
(c) 2 (d) 3

12. 24 is divided into two parts such that 7 times the first part added to 5 times the second part makes 146. The first part is:

(a) 11 (b) 13
(c) 16 (d) 17

13. $\frac{1}{4}$ of a number subtracted from $\frac{1}{3}$ of the same number gives 12. The number is:

(a) 144 (b) 120
(c) 72 (d) 63

14. $\frac{4}{3}$ of a certain number is 64. Half of that number is:

(a) 32 (b) 40
(c) 80 (d) 16

15. A fraction becomes 4 when 1 is added to both the numerator and denominator; and it becomes 7 when 1 is subtracted from both the numerator and denominator. The numerator of the given fraction is:

(a) 2 (b) 3
(c) 7 (d) 15

16. Three numbers are in the ratio 3:4:5. The sum of the largest and the smallest equals the sum of the third and is 52. The smallest number is:

(a) 20 (b) 27
(c) 39 (d) 52

17. The sum of three numbers is 68. If the ratio between first and second is 2:3 and that between second and third is 5:3, then the second number is:

(a) 30 (b) 20
(c) 58 (d) 48

18. If 1 is added to the denominator of a fraction, the fraction becomes $\frac{1}{2}$. If 1 is added to the numerator, the fraction becomes 1. The fraction is:

(a) $\frac{4}{7}$ (b) $\frac{5}{9}$

(c) $\frac{2}{3}$ (c) $\frac{10}{11}$

19. $\frac{4}{5}$ of a number exceeds its $\frac{2}{3}$ by 8. The number is:

(a) 30 (b) 60
(c) 90 (d) None of these

20. What is the sum of all prime numbers from 60 to 80?

(a) 361 (b) 341
(c) 351 (d) 349

21. The quotient arising from the division of 24446 by a certain divisor is 79 and the remainder is 35, what is the divisor?

(a) 39 (b) 309
(c) 390 (d) 3009

22. In a division sum, the quotient is 120, the divisor 456 and the remainder 333, find the dividend.

(a) 5533 (b) 50553
(c) 56053 (d) 55053

23. The quotient arising from a division of a number by 62 is 463 and the remainder is 60, what is the number?

(a) 28666 (b) 28766
(c) 28576 (d) 28676

24. A number when divided by 221 gives a remainder 43. What remainder will be obtained by dividing the same number by 17?

(a) 11 (b) 8
(c) 9 (d) 13

25. Which one of the following is the largest prime number of three digits?

(a) 997 (b) 999
(c) 991 (d) 993

26. When a certain number is multiplied by 7, the product consists entirely of fives; find the least value of such a number.

(a) 79365 (b) 78365
(c) 77365 (d) 79265

27. In a division sum, the divisor is 10 times the quotient and five times the remainder. What is the dividend, if the remainder is 46?

(a) 5636 (b) 5536
(c) 5336 (d) 5436

28. Which one of the following is the least number of four digits divisible by 71?

(a) 1006 (b) 1065
(c) 1094 (d) 1056

29. How many numbers up to 100 are divisible by 7?

(a) 14 (b) 107
(c) 93 (d) 100

30. How many numbers up to 500 are divisible by 23?

(a) 23 (b) 27
(c) 21 (d) 19

31. How many numbers up to 200 are divisible 2 and 3 both?

(a) 35 (b) 33
(c) 29 (d) 27

32. How many numbers between 100 and 300 are divisible by 11?

(a) 11 (b) 10
(c) 12 (d) 18

33. How many numbers between 150 and 500 are divisible by 2, 3 and 7 together?

(a) 9 (b) 8
(c) 10 (d) 11

34. The number of five figures to be added to a number of four fives to obtain the least number of six figures exactly divisible by 357 is:

(a) 94762 (b) 94802
(c) 94485 (d) None of these

35. The nearest figure to 58701 which is divisible by 567 is:

(a) 58968 (b) 58434
(c) 58401 (d) None of these

36. The digits indicated by * in 3422213 ** so that this number is divisible by 99 are:

(a) 1, 9 (b) 3, 7
(c) 4, 6 (d) 5, 5

37. The least value to be given to * so that the number 5 * 3457 is divisible by 11 is:
(a) 2 (b) 3
(c) 0 (d) 4

38. The nearest whole number to one million which is divisible by 537 is:
(a) 1000106 (b) 999894
(c) 1000437 (d) 999563

39. The smallest number between 400 and 500 which is divisible by 9 is:
(a) 414 (b) 405
(c) 423 (d) None of these

40. Which one of the following is the greatest number of five digits divisible by 231?
(a) 99792 (b) 99892
(c) 99692 (d) 99972

41. Find the number nearest to 16386 which is exactly divisible by 425.
(a) 16575 (b) 16375
(c) 16050 (d) 16450

42. Find the least number which must be subtracted from 9269 so that resulting number is exactly divisible by 73?
(a) 17 (b) 57
(c) 71 (d) 63

43. Find the least number which must be added to 15463 so that the resulting number is exactly divisible by 107?
(a) 52 (b) 71
(c) 55 (d) 19

44. What is the number just more than 5000 which is exactly divisible by 73?
(a) 5001 (b) 5009
(c) 5037 (d) 5027

45. The sum of two numbers is 100 and their difference is 37. The difference of their squares is:
(a) 37 (b) 100
(c) 63 (d) 3700

46. The number of times 79 be subtracted from 50000, so that the remainder be 43759; is:
(a) 69 (b) 79
(c) 59 (d) None of these

47. The ratio between two numbers is 3:4 and their sum is 420. The greater of the two numbers is:
(a) 175 (b) 200
(c) 240 (d) 315

48. The difference between the squares of two consecutive numbers is 35. The numbers are:
(a) 14, 15 (b) 15, 16
(c) 17, 18 (d) 18, 19

49. Three-fourths of one-fifth of a number is 60. The number is:
(a) 300 (b) 400
(c) 450 (d) 1200

50. The sum of squares of two numbers is 80 and the square of their difference is 36. The product of the two numbers is:
(a) 22 (b) 44
(c) 58 (d) 116

51. A number when divided by 357 gives a remainder 37. By dividing the same number by 17, the remainder would be:
(a) 3 (b) 4
(c) 2 (d) None of these

52. The product of two numbers is 120. The sum of their squares is 289. The sum of the two numbers is:
(a) 20 (b) 23
(c) 169 (d) None of these

53. Three numbers are in the ratio 4:5:6 and their average is 25. The largest number is:
(a) 42 (b) 36
(c) 30 (d) 32

54. A number exceeds 20% of itself by 40. The number is:
(a) 50 (b) 60
(c) 80 (d) 320

55. If 16% of 40% of a number is 8, the number is:
(a) 200 (b) 225
(c) 125 (d) 320

56. 4767 exactly divides *** 341, the missing digits are:
(a) 468 (b) 586
(c) 363 (d) None of these

57. A number when divided by a certain divisor left remainder 241, when twice the number was divided by the same divisor, the remainder was 112. Find the divisor.
(a) 370 (b) 365
(c) 380 (d) 456

58. Two numbers when divided by a certain divisor give remainders 43 and 37 respectively, when their sum is divided by the same divisor, the remainder is 13. Find the divisor.
(a) 71 (b) 67
(c) 57 (d) 77

59. Two numbers are such that the ratio between them is 3:5; but if each is increased by 10, the ratio between them becomes 5:7. The numbers are:
(a) 3, 5 (b) 7, 9
(c) 13, 22 (d) 15, 25

60. Divide 50 into two parts so that the sum of their reciprocals is 1/12.
(a) 20, 30 (b) 24, 26
(c) 28, 22 (d) 36, 14

61. The sum of seven numbers is 235. The average of the first three is 23 and that of last three is 42. The fourth number is:
(a) 40 (b) 126
(c) 69 (d) 195

62. The sum of squares of two numbers is 68 and the squares of their difference is 36. The product of the two numbers is:
(a) 16 (b) 32
(c) 58 (d) 104

63. What is the least value of K so that the number 6735K1 is divisible by 9?
(a) 5 (b) 7
(c) 4 (d) 3

64. For what value of K, the number 7236K2 is divisible by 8?
(a) 7 (b) 5
(c) 4 (d) 9

65. Find the least values of x and y so that the number 5x423y is divisible by 88.
(a) 8, 2 (b) 7, 3
(c) 9, 4 (d) 6, 5

66. 24 is divided into two parts such that 7 times the first part added to 5 times the second part makes 146. The first part is:
(a) 13 (b) 15
(c) 17 (d) 19

67. Sum of three numbers is 132. First number is twice the second and third number is one-third of the first. Find the second number.
(a) 18 (b) 36
(c) 20 (d) 16

68. What least number must be added to 7231 so that the resulting number is exactly divisible by 5 and 9 together?
(a) 20 (b) 18
(c) 14 (d) 16

69. Find a number nearest to 9231 which is exactly divisible by 3 as well as by 11.
(a) 9240 (b) 9340
(c) 9540 (d) 9440

70. Find a nearest number to 12199 which is exactly divisible by the product of the first four prime numbers.
(a) 12181 (b) 12179
(c) 11281 (d) 11279

71. The sum of squares of two numbers is 90 and the squares of their difference is 46. The product of the two numbers is:
(a) 22 (b) 24
(c) 26 (d) 28

72. If 40% of a number is 360, what will be 15% of that number?
(a) $20\frac{1}{4}$ (b) $20\frac{1}{2}$
(c) $22\frac{1}{4}$ (d) $22\frac{1}{2}$

73. The sum of the digits of a two-digit number is 8. If the digits are reversed the number is increased by 54. Find the number.
(a) 17 (b) 19
(c) 21 (d) 23

EXERCISE-2
(BASED ON MEMORY)

1. N is the largest two digit number, which when divided by 3, 4 and 6 leaves the remainder 1, 2 and 4 respectively, What is the remainder when N is divided by 5?
(a) 4 (b) 2
(c) 0 (d) 1
[SSC CGL Tier-II CBE, 2018]

2. A and B are positive integers. If $A + B + AB = 65$ then What is the difference between A and B (A, $B \leq 15$)?
(a) 3 (b) 4
(c) 5 (d) 6
[SSC CGL Tier-II CBE, 2018]

3. Which one is the largest among the fractions $\left(\frac{5}{113}\right),\left(\frac{7}{120}\right),\left(\frac{13}{145}\right)$ and $\left(\frac{17}{160}\right)$?

(a) $\frac{5}{113}$ (b) $\frac{7}{120}$

(c) $\frac{13}{145}$ (d) $\frac{17}{160}$

[SSC CGL Tier-I CBE, 2018]

4. On dividing a number by 38, we get 90 as quotient and 19 as remainder. What is the number?

(a) 3401 (b) 3382

(c) 3458 (d) 3439

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

5. (90 + 92 + 93 + ...+ 110) is equal to

(a) 4020 (b) 2010

(c) 6030 (d) 8040

[SSC CGL Tier-I CBE, 2017]

6. If the number 583-437 is completely divisible by 9, then the smallest whole number in the place of the blank digit will be

(a) 4 (b) 5

(c) 3 (d) 6

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

7. The least number of five digits exactly divisible by 88 is:

(a) 10032 (b) 10132

(c) 10088 (d) 10023

[SSC Matric Level MTS, 2017]

8. If 34N is divisible by 11, then what is the value of N?

(a) 1 (b) 3

(c) 4 (d) 9

[SSC CAPFs ASI and Delhi Police SI, 2017]

9. How many positive factors of 36 are there?

(a) 4 (b) 6

(c) 9 (d) 12

[SSC CAPFs ASI and Delhi Police SI, 2017]

10. A boy added all natural numbers from 1 to 12, however he added one number twice due to which the sum became 80. What is the number which he added twice?

(a) 3 (b) 2

(c) 7 (d) 8

[SSC CAPFs ASI and Delhi Police SI, 2017]

11. How many numbers are there between 1 and 200 which are divisible by 3 but not by 7?

(a) 38 (b) 45

(c) 57 (d) 66

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

12. What least number be subtracted from 3401, so that the sum is completely divisible by 11?

(a) 3 (b) 1

(c) 2 (d) 0

[SSC CGL Tier-I CBE, 2017]

13. What is the remainder when 2468 is divided by 37?

(a) 26 (b) 36

(c) 18 (d) 14

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

14. If 432 PI is completely divisible by 9, then what is the value of *P*?

(a) 7 (b) 8

(c) 7 (d) 2

[SSC Multi-Tasking Start, 2017]

15. For what value of M. 34 M is divisible by 12?

(a) 2 (b) 0

(c) 8 (d) 6

[SSC Delhi police Constable, 2017]

16. How many positive factors of 160 are there?

(a) 6 (b) 8

(c) 5 (d) 12

[SSC Multi-Tasking Staff, 2017]

17. Which fraction among $\frac{2}{3},\frac{4}{5}$ and $\frac{7}{11}$ is the largest?

(a) $\frac{2}{3}$ (b) $\frac{4}{5}$

(c) $\frac{7}{11}$ (d) All are equal

[SSC Multi-Tasking Staff, 2017]

18. Which fraction among $\frac{3}{7},\frac{5}{11}$ and $\frac{6}{13}$ is largest ?

(a) $\frac{3}{7}$ (b) $\frac{5}{11}$

(c) $\frac{6}{13}$ (d) All are equal

[SSC Multi-Tasking Staff, 2017]

19. What is the largest two digits prime number?

(a) 91 (b) 93
(c) 97 (d) 99

[SSC Delhi Police constable, 2017]

20. A mason can build a wall in 70 hours. After 7 hours he takes a break. What fraction of the wall is yet to be build?

(a) 0.9 (b) 0.8
(c) 0.5 (d) 0.75

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

21. Two baskets together have 640 oranges. If $\left(\frac{1}{5}\right)$th of the oranges in the first basket be taken to the second baskets became then, no. of oranges in both baskets equal. The number of oranges in the first basket is

(a) 800 (b) 600
(c) 400 (5) 300

[SSC CGL Tier-II (CBE), 2017]

22. What is the value of $\left(\frac{3}{5}+\frac{7}{9}\right)$?

(a) $\frac{62}{45}$ (b) $\frac{31}{28}$
(c) $\frac{5}{7}$ (d) $\frac{1}{7}$

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

23. The sum of a non-zero number and twice its reciprocal is $\frac{33}{4}$ Find the number.

(a) 9 (b) 10
(c) 11 (d) 8

[SSC CGL Tier-I CBE, 2017]

24. The reciprocal of the sum of the reciprocals of $\frac{6}{5}$ and $\frac{3}{7}$

(a) $\frac{57}{18}$ (b) $\frac{35}{57}$
(c) $\frac{18}{57}$ (d) $\frac{57}{35}$

[SSC CGL Tier-I CBE, 2017]

25. In a college, $\frac{1}{5}$th of the girls and $\frac{1}{8}$th of the boys took part in a social camp. The part of students in the college who took part in the camp is:

(a) $\frac{13}{80}$ (b) $\frac{2}{13}$
(c) $\frac{13}{40}$ (d) $\frac{4}{15}$

[SSC Matric Level MTS, 2017]

26. A fraction is greater than twice its reciprocal by $\frac{7}{15}$. What is the fraction?

(a) $\frac{3}{5}$ (b) $\frac{5}{3}$
(c) $\frac{3}{4}$ (d) $\frac{4}{3}$

[SSC CGL Tier-I CBE, 2017]

27. A fraction is greater than its reciprocal by $\frac{9}{20}$. What is the fraction?

(a) $\frac{5}{4}$ (b) $\frac{4}{5}$
(c) $\frac{3}{4}$ (d) $\frac{4}{3}$

[SSC CGL Tier-I CBE, 2017]

28. A number is greater than 58 times its reciprocal by $\frac{3}{4}$. What is the number?

(a) –8 (b) 12
(c) –12 (d) 8

[SSC CGL Tier-I CBE, 2017]

29. Natu and Buchku each have certain number of oranges. Natu says to Buchku. "If you give me 10 of your oranges. I will have twice the number of oranges left with you". Buchku replies, "If you give me 10 of your oranges, I will have the same number of oranges as left with you". What is the number of oranges with Natu and Buchku respectively?

(a) 50, 20 (b) 70, 50
(c) 20, 50 (d) 50, 70

[SSC CGL Tier-II (CBE), 2017]

30. The ten's digit of a two-digit number is greater than the unit's digit by 7. If we subtract 63 from the number. the new number obtained is a number formed by interchange of the digits. Find the number.

(a) 81 (b) 18
(c) 63 (d) 26

[SSC CGL Tier-I CBE, 2017]

31. The numerator of a fraction is multiple of two numbers. One of the numbers is greater than the other by 2. The greater number is smaller than the denominator $7 + c(c > -7)$ is a constant, then the ninimum value of the fraction is

(a) 5 (b) $\frac{1}{5}$

(c) –5 (d) $\frac{-1}{5}$

[SSC CGL Tier-II (CBE), 2016]

32. In a division sum, the divisor 'd' is 10 times the quotient 'q' and 5 times the remainder 'r'. If r = 46 the dividend will be

(a) 5042 (b) 5328
(c) 5336 (d) 4276

[SSC CGL Tier-II, 2016]

33. A number when divided by 44, gives 432 as quotient and 0 as remainder. What will be the remainder when dividing the same number by 31?

(a) 3 (b) 4
(c) 5 (d) 6

[SSC CPO ASI, 2016]

34. A number when divide by 729 gives a remainder if the same number is divided by 27?

(a) 4 (b) 2
(c) 0 (d) 1

[SSC CPO ASI, 2016]

35. What is the smallest 6-digit number that is completely divisible by 108?

(a) 100003 (b) 100004
(c) 100006 (d) 100008

[SSC CPO, 2016]

36. When a number is divided by 56, the remainder will be 29. If the same number is divided by 8 then the remainder will be

(a) 6 (b) 7
(c) 5 (d) 3

[SSC CGL Tier-I (CBE), 2016]

37. A number when divided by the sum of 555 and 445 gives two times their difference as quotient and 30 as the remainder. The number is

(a) 220030 (b) 22030
(c) 1220 (d) 1250

[SSC CGL Tier-II (CBE), 2016]

38. The sum of three numbers is 2 the 1st number is $\frac{1}{2}$ times the 2nd number and the 3rd number is $\frac{1}{4}$ times the 2nd number. The 2nd number is

(a) $\frac{7}{6}$ (b) $\frac{8}{7}$

(c) $\frac{9}{8}$ (d) $\frac{10}{9}$

[SSC CGL Tier-II CBE, 2016]

39. If $\frac{1}{2}$ is added to a number and the sum is multiplied by 3, the result is 21. Then the number is:-

(a) 6.5 (b) 5.5
(c) 4.5 (d) –6.5

[SSC CGL Tier-I (CBE), 2016]

40. The value of x in the following equation is:

$0.3 + 0.6 + 0.7 + 0.8 = x$

(a) 5.3 (b) $2\frac{3}{10}$

(c) $2\frac{2}{5}$ (d) 2.35

[SSC CAPFs (CPO) SI and ASI. Delhi, 2016]

41. The least number which when divided by 6, 9,12, 15, 18 leaves the same remainder 2 in each case is:

(a) 178 (b) 182
(c) 176 (d) 180

[SSC, 2015]

42. If 3/4 of a number is 7 more than 1/6 of the number, then 5/3 of the number is:

(a) 15 (b) 18
(c) 12 (d) 20

[SSC, 2015]

43. What is the arithmetic mean of first 20 odd natural numbers?

(a) 17 (b) 19
(c) 22 (d) 20

[SSC, 2015]

44. What is the Arithmetic mean of the first 'n' natural numbers?

(a) $\frac{n+1}{2}$ (b) $\frac{n^2(n+1)}{2}$

(c) 2(n + 1) (d) $\frac{n(n+1)}{2}$

[SSC, 2015]

45. There is a number consisting of two digits, the digit in the units place is twice that in the tens place and if 2 be subtracted from the sum of the digits, the difference is equal to $1/6^{th}$ of the number. The number is

(a) 23 (b) 24
(c) 25 (d) 26

[SSC, 2015]

46. Let x be the smallest number, which when added to 2000 makes the resulting number divisible by 12, 16, 18 and 21. The sum of the digits of x is

(a) 7 (b) 4
(c) 6 (d) 5

[SSC, 2015]

47. The unit digit in the product $(2467)^{153} \times (341)^{72}$ is

(a) 3 (b) 9
(c) 7 (d) 1

[SSC, 2015]

48. Let x be the least number, which when divided by 5, 6, 7 and 8 leaves a remainder 3 in each case but when divided by 9 leaves no remainder. The sum of digits of x is

(a) 22 (b) 18
(c) 21 (d) 24

[SSC, 2015]

49. A number when divided by 361 gives a remainder 47. If the same number is divided by 19, the remainder obtained is

(a) 8 (b) 9
(c) 1 (d) 3

[SSC, 2015]

50. The greatest number among 3^{50}, 4^{40}, 5^{30} and 6^{20} is

(a) 6^{20} (b) 3^{50}
(c) 4^{40} (d) 5^{30}

[SSC, 2015]

51. A number exceeds its two fifth by 75. The number is

(a) 100 (b) 112
(c) 150 (d) 125

[SSC, 2015]

52. If the sum of two numbers, one of which is $\frac{2}{5}$ times the other is 50, then the numbers are:

(a) $\frac{250}{7}, \frac{100}{7}$ (b) $\frac{115}{7}, \frac{235}{7}$
(c) $\frac{150}{7}, \frac{200}{7}$ (d) $\frac{240}{7}, \frac{110}{7}$

[SSC, 2015]

53. If $x[-2\{-4(-a)\}] + 5\ [-2\{-2(-a)\}] = 4a$, then x =

(a) –2 (b) –3
(c) –4 (d) –5

[SSC, 2014]

54. In a Mathematics examination the numbers scored by 5 candidates are 5 successive odd integers. If their total marks is 185, the highest score is

(a) 39 (b) 43
(c) 41 (d) 37

[SSC, 2014]

55. Arranging the following in descending order, we get $\sqrt[3]{4}, \sqrt{2}, \sqrt[6]{3}, \sqrt[4]{5}$

(a) $\sqrt[3]{4} > \sqrt[4]{5} > \sqrt{2} > \sqrt[6]{3}$
(b) $\sqrt[3]{4} > \sqrt[4]{5} > \sqrt{2} > \sqrt[6]{3}$
(c) $\sqrt{2} > \sqrt[6]{3} > \sqrt[3]{4} > \sqrt[4]{5}$
(d) $\sqrt[6]{3} > \sqrt[4]{5} > \sqrt[3]{4} > \sqrt{2}$

[SSC, 2014]

56. For any integral value of n, $3^{2n} + 9n + 5$ when divided by 3 will leave the remainder

(a) 1 (b) 2
(c) 0 (d) 5

[SSC, 2014]

57. The least prime number is

(a) 3 (b) 2
(c) 0 (d) 1

[SSC, 2014]

58. Find the least number which when divided by 12, 18, 36 and 45 leaves the remainder 8, 14, 32 and 41 respectively.

(a) 178 (b) 186
(c) 176 (d) 180

[SSC, 2014]

59. The value of 204×197 is

(a) 39812 (b) 40218
(c) 40188 (d) 40212

[SSC, 2014]

60. A piece of cloth measured with a metre stick, one cm short, is 100 metres long. Reckoning the metre stick as being right, the actual length of the cloth (in cm) is

(a) 8,000 (b) 6,100
(c) 3,900 (d) 9,900

[SSC, 2014]

61. The value of (1001)3 is

(a) 100300301 (b) 103003001
(c) 1003003001 (d) 100303001

[SSC, 2014]

62. When 335 is added to 5*A*7, the result is 8*B*2. 8*B*2 is divisible by 3. What is the largest possible value of *A*?

(a) 8 (b) 2
(c) 1 (d) 4

[SSC, 2013]

63. Which one of the following numbers is divisible by 25?

(a) 303310 (b) 373355
(c) 303375 (d) 22040

[SSC, 2013]

64. The units digit in $3 \times 38 \times 537 \times 1256$ is:

(a) 4 (b) 2
(c) 6 (d) 8

[SSC, 2013]

65. A number is of two digits. The position of digits is interchanged and the new number is added to the original number. The resultant number will always be divisible by:

(a) 8 (b) 9
(c) 10 (d) 11

[UPPCS, 2012]

66. The last digit, that is, the digit in the unit's place of the number $[(57)^{25} - 1]$ is:

(a) 6 (b) 8
(c) 0 (d) 5

[SSC Assistant Grade III, 2012]

67. A number N is a positive three-digit number. If x is in its hundred's place and y is in its unit's place, then the number $N - 100x - y$ is always divisible by:

(a) 8 (b) 9
(c) 10 (d) 11

[SSC Assistant Grade III, 2012]

68. n is a whole number which when divided by 4 gives the remainder 3. The remainder when $2n$ is divided by 4 is:

(a) 1 (b) 2
(c) 3 (d) 0

[SSC, 2012]

69. If m and n are positive integers and $(m - n)$ is an even number, then $(m^2 - n^2)$ will be always divisible by:

(a) 4 (b) 6
(c) 8 (d) 12

[SSC, 2012]

70. Rachita enters a shop to buy ice-creams, cookies and pastries. She has to buy at least 9 units of each. She buys more cookies than ice-creams and more pastries than cookies. She picks up a total of 32 items. How many cookies does she buy?

(a) Either 12 or 13 (b) Either 11 or 12
(c) Either 10 or 11 (d) Either 9 or 11
(e) Either 9 or 10

[IBPS PO/MT, 2012]

71. The product of three consecutive even number is 4032. The product of the first and the third number is 252. What is five times the second number?

(a) 80 (b) 100
(c) 60 (d) 70
(e) 90

[IBPS PO/MT, 2012]

72. When 2^{23} is divided by 10, the remainder will be:

(a) 2 (b) 3
(c) 4 (d) 8

[SSC (GL), 2011]

73. Find the unit digit in the product $(4387)^{245} \times (621)^{72}$.

(a) 1 (b) 2
(c) 5 (d) 7

[SSC (GL), 2011]

74. If a and b are odd numbers, then which of the following is even?

(a) $a + b + ab$ (b) $a + b - 1$
(c) $a + b + 1$ (d) $a + b + 2ab$

[SSC (GL), 2011]

75. $2^{16} - 1$ is divisible by:

(a) 11 (b) 13
(c) 17 (d) 19

[SSC (GL), 2011]

76. The sum of two numbers is 24 and their product is 143. The sum of their squares is:

(a) 296 (b) 295
(c) 290 (d) 228

[SSC (GL), 2011]

77. The unit digit in the sum $(124)^{372} + (124)^{373}$ is:

(a) 5 (b) 4
(c) 2 (d) 0

[SSC (GL), 2011]

78. If the sum of two numbers be multiplied by each number separately, the products so obtained are 247 and 114. The sum of the numbers is:

(a) 19 (b) 20
(c) 21 (d) 23

[SSC (GL), 2011]

79. Find a number, one-seventh of which exceeds its eleventh part by 100.

(a) 1925 (b) 1825
(c) 1540 (d) 1340

[SSC (GL), 2011]

80. $\sqrt{6+\sqrt{6+\sqrt{6+\cdots=?}}}$

(a) 2.3 (b) 3
(c) 6 (d) 6.3

[SSC (GL), 2011]

81. In a division sum, the divisor is 4 times the quotient and twice the remainder. If a and b are respectively the divisor and the dividend, then:

(a) $\frac{4b-a^2}{a}=3$ (b) $\frac{4b-2a}{a^2}=2$

(c) $(a+1)^2=4b$ (d) $\frac{a(a+2)}{b}=4$

[SSC, 2011]

82. If 738*A*6*A* is divisible by 11, then the value of A is:

(a) 6 (b) 3
(c) 9 (d) 1

[SSC, 2011]

83. The product of two numbers is 1575 and their quotient is $\frac{9}{7}$. Then the sum of the numbers is:

(a) 74 (b) 78
(c) 80 (d) 90

[SSC, 2011]

84. When $(67^{67}+67)$ is divided by 68, the remainder is:

(a) 1 (b) 63
(c) 66 (d) 67

[SSC, 2011]

85. The least positive integer that should be subtracted from 3011 × 3012 so that the difference is a perfect square, is:

(a) 3009 (b) 3010
(c) 3011 (d) 3012

[SSC, 2011]

86. The sum of nine consecutive odd numbers of Set A is 621. What is the sum of a different set of six consecutive even numbers whose lowest number is 15 more than the lowest number of Set A?

(a) 498 (b) 468
(c) 478 (d) 488
(e) None of these

[IOB PO, 2011]

87. The sum of the squares of two consecutive even numbers is 6500. Which is the smaller number?

(a) 54 (b) 52
(c) 48 (d) 56

[Punjab National Bank PO, 2010]

88. The sum of five consecutive even numbers of set-A is 220. What is the sum of a different set of five consecutive numbers whose second lowest number in 37 less than double of the lowest number of set-A?

(a) 223 (b) 225
(c) 235 (d) None of these

[CBI PO, 2010]

89. A number, when divided by 296, gives 75 as the remainder. If the same number is divided by 37 then the remainder will be:

(a) 1 (b) 2
(c) 19 (d) 31

[SSC, 2010]

90. The sum and product of two numbers are 12 and 35 respectively. The sum of their reciprocals will be:

(a) $\frac{1}{3}$ (b) $\frac{1}{5}$

(c) $\frac{12}{35}$ (d) $\frac{35}{12}$

[SSC, 2010]

91. The greatest number among $\sqrt{5}, \sqrt[3]{4}, \sqrt[5]{2}, \sqrt[7]{3}$ is:

(a) $\sqrt[3]{4}$ (b) $\sqrt[7]{3}$

(c) $\sqrt{5}$ (d) $\sqrt[5]{2}$

[SSC, 2010]

92. Two numbers are such that their difference, their sum and their product are in the ratio of 1:7:24. The product of the numbers is:

(a) 24 (b) 36
(c) 48 (d) 60

[SSC, 2010]

CHAPTER 2

H.C.F. and L.C.M. of Numbers

COMMON FACTOR

A *common factor* of two or more numbers is a number which divides each of them exactly.
For example, 4 is a common factor of 8 and 12.

HIGHEST COMMON FACTOR

Highest common factor of two or more numbers is the greatest number that divides each one of them exactly. For example, 6 is the highest common factor of 12, 18 and 24. Highest Common Factor is also called *Greatest Common Divisor* or *Greatest Common Measure*.

Symbolically, these can be written as H.C.F. or G.C.D. or G.C.M., respectively.

METHODS OF FINDING H.C.F.

I. Method of Prime Factors

Step 1 Express each one of the given numbers as the product of prime factors.
[A number is said to be a *prime number* if it is exactly divisible by 1 and itself, but not by any other number, e.g., 2, 3, 5, 7, etc. are prime numbers]

Step 2 Choose common factors.

Step 3 Find the product of these common factors. This is the required H.C.F. of given numbers.

Illustration 1: Find the H.C.F. of 70 and 90.

Solution: $70 = 2 \times 5 \times 7$

$90 = 2 \times 5 \times 9$

Common factors are 2 and 5.

$\therefore$ H.C.F. $= 2 \times 5 = 10$.

Illustration 2: Find the H.C.F. of 3332, 3724 and 4508.

Solution: $3332 = 2 \times 2 \times 7 \times 7 \times 17$

$3724 = 2 \times 2 \times 7 \times 7 \times 19$

$4508 = 2 \times 2 \times 7 \times 7 \times 23$

$\therefore$ H.C.F. $= 2 \times 2 \times 7 \times 7 = 196$.

Illustration 3: Find the H.C.F. of 360 and 132.

Solution: $360 = 2^3 \times 3^2 \times 5$

$132 = 2^2 \times 3^1 \times 11$

$\therefore$ H.C.F. $= 2^2 \times 3^1 = 12$.

Illustration 4: If $x = 2^3 \times 3^5 \times 5^9$ and $y = 2^5 \times 3^7 \times 5^{11}$, find H.C.F. of x and y.

Solution: The factors common to both x and y are 2^3, 3^5 and 5^9.

$\therefore$ H.C.F. $= 2^3 \times 3^5 \times 5^9$.

II. Method of Division

***A. For two numbers*:**

Step 1 Greater number is divided by the smaller one.

Step 2 Divisor of (1) is divided by its remainder.

Step 3 Divisor of (2) is divided by its remainder. This is continued until no remainder is left.
H.C.F. is the divisor of last step.

Illustration 5: Find the H.C.F. of 3556 and 3444.

$$
\begin{array}{r}
3444\overline{)3556}(\ 1 \\
3444 \\
\hline
112\overline{)3444}(\ 30 \\
3360 \\
\hline
84\overline{)112}(\ 1 \\
84 \\
\hline
28\overline{)\ 84}(\ 3 \\
84 \\
\hline
\times
\end{array}
$$

$\therefore$ H.C.F. $= 28$.

***B. For more than two numbers*:**

Step 1 Any two numbers are chosen and their H.C.F. is obtained.

Step 2 H.C.F. of H.C.F. (of (1)) and any other number is obtained.

Step 3 H.C.F. of H.C.F. (of (2)) and any other number (not chosen earlier) is obtained.
This process is continued until all numbers have been chosen. H.C.F. of last step is the required H.C.F.

Illustration 6: Find the H.C.F. of 13915, 9499 and 2553 by division method.

Solution:

$$
\begin{array}{r|l}
9499 & 13915 \\
 & 9499 \\
\hline
4416 & 9499 \\
 & 8832 \\
\hline
667 & 4416 \\
 & 4002 \\
\hline
414 & 667 \\
 & 414 \\
\hline
253 & 414 \\
 & 253 \\
\hline
161 & 253 \\
 & 161 \\
\hline
92 & 161 \\
 & 92 \\
\hline
69 & 92 \\
 & 69 \\
\hline
69 & 69 \\
 & 69 \\
\hline
 & \times
\end{array}
\quad
\begin{array}{l}
1 \\
 \\
2 \\
 \\
6 \\
 \\
1 \\
 \\
1 \\
 \\
1 \\
 \\
1 \\
 \\
3 \\
 \\
 \\

Illustration 9: Find the L.C.M. of 12, 15, 20 and 54.

Solution:

2	12,	15,	20,	54
2	6,	15,	10,	27
3	3,	15,	5,	27
5	1,	5,	5,	9
	1,	1,	1,	9

L.C.M. = 2 × 2 × 3 × 5 × 1 × 1 × 1 × 9
= 540.

Notes

Before finding the L.C.M. or H.C.F., we must ensure that all quantities are expressed in the same unit.

SHORT-CUT METHODS

01 H.C.F. and L.C.M. of Decimals

Step 1 Make the same number of decimal places in all the given numbers by suffixing zero(s) if necessary.

Step 2 Find the H.C.F./L.C.M. of these numbers without decimal.

Step 3 Put the decimal point (in the H.C.F./L.C.M. of Step 2) leaving as many digits on its right as there are in each of the numbers.

Illustration 10: Find the L.C.M. of 1.2, 0.24 and 6.

Solution: The given numbers can be written as 1.20, 0.24 and 6.00.

Now, ignoring the decimal we find the L.C.M. of 120, 24 and 600.

2	120,	24,	600
2	60,	12,	300
2	30,	6,	150
3	15,	3,	75
5	5,	1,	25
	1,	1,	5

∴ L.C.M. = 2 × 2 × 2 × 3 × 5 × 1 × 5 = 600

Thus, the required L.C.M. = 6.00, i.e., 6.

Illustration 11: Find the H.C.F. of 6.16 and 13.

Solution: The given numbers can be written as 6.16 and 13.00.

Now, ignoring the decimals we find the H.C.F. of 616 and 1300.

$$\begin{array}{r|l} 616 & 1300 \\ & 1232 \\ \hline 68 & 616 \\ & 612 \\ \hline 4 & 68 \\ & 68 \\ \hline & \times \end{array} \quad \begin{array}{l} \big) \\ \big(2 \\ \\ \big) \\ \big(9 \\ \\ \big) \\ \big(17 \\ \\ \end{array}$$

∴ H.C.F. of 616 and 1300 is 4.

Thus, the required H.C.F. = 0.04.

02 L.C.M. and H.C.F. of Fractions

$$\text{L.C.M} = \frac{\text{L.C.M. of the numbers in numerators}}{\text{H.C.F. of the numbers in denominators}}$$

$$\text{H.C.F.} = \frac{\text{H.C.F. of the numbers in numerators}}{\text{L.C.M. of the numbers in denominators}}$$

Illustration 12: Find the L.C.M. of $\frac{2}{5}$, $\frac{3}{10}$ and $\frac{6}{25}$.

Solution: L.C.M. of numerators 2, 3 and 6 is 6.

H.C.F. of denominators 5, 10 and 25 is 5.

∴ Required L.C.M. $= \frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}}$

$= \frac{6}{5}$.

Illustration 13: Find the H.C.F. of $\frac{4}{9}$, $\frac{10}{21}$ and $\frac{20}{63}$.

Solution: H.C.F. of numerators 4, 10 and 20 is 2.

L.C.M. of denominators 9, 21 and 63 is 63.

∴ Required H.C.F $= \frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}} = \frac{2}{63}$.

Notes

1. If the given set of numbers includes fractions as well as whole numbers, treat whole number too as fraction with 1 in its denominator.
2. The H.C.F. of a number of fractions is always a fraction, but the L.C.M. may be a fraction or an integer.

03 Product of two numbers
= L.C.M. of the numbers × H.C.F. of the numbers

Illustration 14: The H.C.F. and the L.C.M. of any two numbers are 63 and 1260, respectively. If one of the two numbers is 315, find the other number.

Solution: The required number

$$= \frac{\text{L.C.M.} \times \text{H.C.F.}}{\text{First Number}} = \frac{1260 \times 63}{315} = 252.$$

04 To find the greatest number that will exactly divide x, y and z.
Required number = H.C.F. of x, y and z.

Illustration 15: Find the greatest number that will exactly divide 200 and 320.

Solution: The required greatest number
= H.C.F. of 200 and 320 = 40.

05 To find the greatest number that will divide x, y and z leaving remainders a, b and c, respectively.
Required number = H.C.F. of $(x - a)$, $(y - b)$ and $(z - c)$.

Illustration 16: Find the greatest number that will divide 148, 246 and 623 leaving remainders 4, 6 and 11, respectively.

Solution: The required greatest number
= H.C.F. of (148 – 4), (246 – 6) and (623 – 11),
i.e., H.C.F. of 144, 240 and 612 = 12.

06 To find the least number which is exactly divisible by x, y and z.
Required number = L.C.M. of x, y and z.

Illustration 17: What is the smallest number which is exactly divisible by 36, 45, 63 and 80?

Solution: The required smallest number
= L.C.M. of 36, 45, 63 and 80
= 5040.

07 To find the least number which when divided by x, y and z leaves the remainders a, b and c, respectively. It is always observed that $(x - a) = (y - b) = (z - c) = k$(say)
$\therefore$ Required number = (L.C.M. of x, y and z) $- k$.

Illustration 18: Find the least number which when divided by 36, 48 and 64 leaves the remainders 25, 37 and 53, respectively.

Solution: Since, (36 – 25) = (48 – 37) = (64 – 53) = 11, therefore, the required smallest number
= (L.C.M. of 36, 48 and 64) – 11
= 576 – 11 = 565.

08 To find the least number which when divided by x, y and z leaves the same remainder r in each case.
Required number = (L.C.M. of x, y and z) + r.

Illustration 19: Find the least number which when divided by 12, 16 and 18, will leave in each case a remainder 5.

Solution: The required smallest number
= (L.C.M. of 12, 16 and 18) + 5
= 144 + 5 = 149.

09 To find the greatest number that will divide x, y and z leaving the same remainder in each case.

(a) When the value of remainder r is given:
Required number = H.C.F. of $(x - r)$, $(y - r)$ and $(z - r)$.

(b) When the value of remainder is not given:
Required number = H.C.F. of $|(x - y)|$, $|(y - z)|$ and $|(z - x)|$

Illustration 20: Find the greatest number which will divide 772 and 2778 so as to leave the remainder 5 in each case.

Solution: The required greatest number
= H.C.F. of (772 – 5) and (2778 – 5)
= H.C.F. of 767 and 2773
= 59.

Illustration 21: Find the greatest number which on dividing 152, 277 and 427 leaves equal remainder.

Solution: The required greatest number
= H.C.F. of $|(x - y)|$, $|(y - z)|$ and $|(z - x)|$
= H.C.F. of $|(152 - 277)|$, $|(277 - 427)|$ and $|(427 - 152)|$
= H.C.F. of 125, 150 and 275
= 25.

10 To find the n-digit greatest number which, when divided by x, y and z,

(a) leaves no remainder (i.e., exactly divisible)

Step 1 L.C.M. of x, y and $z = L$

Step 2 $L\overline{)n\text{-digit greatest number}}$
Remainder = R

Step 3 Required number
= n-digit greatest number $- R$

(b) leaves remainder K in each case
Required number
= (n-digit greatest number $- R$) + K.

Illustration 22: Find the greatest number of 4-digit number which, when divided by 12, 18, 21 and 28 leaves 3 as a remainder in each case.

Solution: L.C.M. of 12, 18, 21 and 28 = 252.

$$252\overline{)9999}\,(39$$
$$\underline{9828}$$
$$171$$

$\therefore$ The required number = (9999 – 171) + 3 = 9931.

Illustration 23: Find the greatest number of four digits which, when divided by 12, 15, 20 and 35 leaves no remainder.

Solution: L.C.M. of 12, 15, 20 and 35 = 420.

$$420\overline{)9999}\,(23$$
$$\underline{9660}$$
$$339$$

$\therefore$ The required number = 9999 – 339
= 9663.

11 To find the n-digit smallest number which when divided by x, y and z

(a) leaves no remainder (i.e., exactly divisible)

Step 1 L.C.M. of x, y and $z = L$

Step 2 $L\overline{)n\text{-digit smallest number}\,(}$
Remainder = R

Step 3 Required number
= n-digit smallest number + $(L - R)$.

(b) leaves remainder K in each case.
Required number
= n-digit smallest number + $(L - R) + K$.

Illustration 24: Find the least number of four digits which is divisible by 4, 6, 8 and 10.

Solution: L.C.M. of 4, 6, 8 and 10 = 120.

$$120\overline{)1000}(8$$
$$\underline{960}$$
$$40$$

$\therefore$ The required number = 1000 + (120 – 40)
= 1080.

Illustration 25: Find the smallest 4-digit number, such that when divided by 12, 18, 21 and 28, it leaves remainder 3 in each case.

Solution: L.C.M. of 12, 18, 21 and 28 = 252.

$$252\overline{)1000}\,(3$$
$$\underline{756}$$
$$244$$

$\therefore$ The required number
= 1000 + (252 – 244) + 3
= 1011.

EXERCISE-I

1. What is the H.C.F. of 27, 18 and 36?

(a) 7 (b) 11
(c) 9 (d) None of these

2. Determine the L.C.M of $\frac{2}{5}, \frac{3}{10}$ and $\frac{6}{25}$.

(a) $\frac{6}{5}$ (b) $\frac{11}{5}$
(c) $\frac{9}{5}$ (d) None of these

3. What is the L.C.M. of 25, 30, 35 and 40?

(a) 3800
(b) 4200
(c) 4400
(d) None of these

4. What is the greatest number which divides 852, 1065 and 1491 exactly?

(a) 193 (b) 183
(c) 223 (d) 213

5. What is the H.C.F. of $\frac{4}{9}, \frac{10}{21}$ and $\frac{20}{30}$?

(a) $\frac{4}{189}$ (b) $\frac{6}{23}$
(c) $\frac{2}{63}$ (d) None of these

6. Find the least number which when divided by 16, 18, 20 and 25 leaves 4 as remainder in each case but when divided by 7 leaves no remainder.

(a) 8004 (b) 13004
(c) 18004 (d) 18014

7. Area of three fields is 165 m^2, 195 m^2 and 85 m^2, respectively. In each of the fields a flower bed of equal length has to be made. If flower bed in each

of the fields is 3 m wide then what is the maximum length of the flower bed in each of the fields?

(a) 7 m (b) 9 m
(c) 5 m (d) None of these

8. Find the greatest number which will divide 2112 and 2792 leaving the remainder 4 in each case.

(a) 78 (b) 68
(c) 65 (d) 63

9. The H.C.F. of two numbers is 12 and their difference is 12. The numbers are:

(a) 66, 78 (b) 70, 82
(c) 94, 106 (d) 84, 96

10. A merchant has 435 litres, 493 litres and 551 litres of three different kinds of milk. Find the least number of casks of equal size required to store all the milk without mixing.

(a) 51 (b) 61
(c) 47 (d) 45

11. Find the greatest number which will divide 25, 73 and 97 so as to leave the same remainder in each case.

(a) 12 (b) 18
(c) 24 (d) 32

12. The sum of two numbers is 216 and their H.C.F. is 27. The numbers are:

(a) 54, 162 (b) 108, 118
(c) 27, 189 (d) None of these

13. How often will five bells toll together in one hour if they start together and toll at intervals of 5, 6, 8, 12, 20 seconds, respectively?

(a) 29 (b) 30
(c) 31 (d) 120

14. Find the greatest number that will divide 964, 1238 and 1400 leaving remainders 41, 31 and 51, respectively.

(a) 71 (b) 81
(c) 61 (d) 73

15. Find the side of the largest square slabs which can be paved on the floor of a room 5 m 44 cm long and 3 m 74 cm broad.

(a) 56 (b) 42
(c) 38 (d) 34

16. The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds, respectively. If they all change simultaneously at 8:20:00 hours; then they will again change simultaneously at:

(a) 8:27:12 hours (b) 8:27:24 hours
(c) 8:27:36 hours (d) 8:27:48 hours

17. The product of two numbers is 6760 and their H.C.F. is 13. How many such pairs can be formed?

(a) 2 (b) 3
(c) 4 (d) only one

18. Find the greatest number of four digits which when divided by 10, 15, 21 and 28 leaves 4, 9, 15 and 22 as remainders, respectively.

(a) 9654 (b) 9666
(c) 9664 (d) 9864

19. The number of prime factors in the expression $(6)^{10} \times (7)^{17} \times (11)^{27}$ is:

(a) 54 (b) 64
(c) 71 (d) 81

20. Find the greatest number which will divide 3962, 4085 and 4167 leaving the same remainder in each case.

(a) 37 (b) 39
(c) 41 (d) 43

21. A wholesale tea dealer has 408 kilograms, 468 kilograms and 516 kilograms of three different qualities of tea. He wants it all to be packed into boxes of equal size without mixing. Find the capacity of the largest possible box.

(a) 50 (b) 36
(c) 24 (d) 12

22. A room is 4 m 37 cm long and 3 m 23 cm broad. It is required to pave the floor with minimum square slabs. Find the number of slabs required for this purpose.

(a) 485 (b) 431
(c) 391 (d) 381

23. The least perfect square number which is divisible by 3, 4, 5, 6 and 8:

(a) 900 (b) 1200
(c) 2500 (d) 3600

24. Find the least number of five digits which when divided by 12, 16, 21, 36 and 40 leaves remainder 8 in each case.

(a) 10088 (b) 10072
(c) 10080 (d) None of these

25. Three pieces of timber 42 m, 49 m and 63 m long have to be divided into planks of the same length. What is the greatest possible length of each plank?

(a) 7 m (b) 14 m
(c) 42 m (d) 63 m

26. Three men start together to travel the same way around a circular track of 11 kilometres in circumference. Their speeds are 4, $5\frac{1}{2}$ and 8 Km/h, respectively. When will they meet at the starting point?

(a) 11 hours (b) 12 hours
(c) 23 hours (d) 22 hours

27. Five bells begin to toll together and toll at intervals of 36, 45, 72, 81 and 108 seconds. After what interval of time will they keep on tolling together?

(a) 3240 seconds (b) 3080 seconds
(c) 3140 seconds (d) 3200 seconds

28. Three different containers contain different quantities of mixture of milk and water, whose measurements are 403 Kg, 434 Kg and 465 Kg. What biggest measure must be there to measure all the different quantities exactly?

(a) 1 Kg (b) 7 Kg
(c) 31 Kg (d) 41 Kg

29. The L.C.M. and G.C.D. of two numbers are 1530 and 51, respectively. Find how many such pairs are possible?

(a) 2 (b) 3
(c) 4 (d) Only one

30. Find the least number of five digits which when divided by 63, 56 and 42 leaves remainder 1 in each case.

(a) 10082 (b) 10081
(c) 10001 (d) 10071

31. The H.C.F. and L.C.M. of two numbers are 44 and 264, respectively. If the first number is divided by 2, the quotient is 44. The other number is:

(a) 33 (b) 66
(c) 132 (d) 264

32. The largest natural number which exactly divides the product of any four consecutive natural numbers, is:

(a) 6 (b) 12
(c) 24 (d) 120

33. Find the least number of six digits which is exactly divisible by 15, 21 and 28:

(a) 100480 (b) 100270
(c) 100380 (d) 100340

34. Find the greatest number of five digits which when divided by 12, 15, 21, 25 and 28 leaves 5, 8, 14, 18 and 21 as remainders, respectively.

(a) 98696 (b) 98700
(c) 97693 (d) 98693

35. What is the smallest number which when increased by 3 is divisible by 16, 24, 30 and 32?

(a) 480 (b) 475
(c) 472 (d) 477

36. The least number of square tiles required to cover the ceiling of a room 15 m 17 cm long and 9 m 2 cm broad, is:

(a) 656 (b) 738
(c) 814 (d) 902

37. Find the least number which when divided by 2, 3, 4, 5 and 6 leaves 1, 2, 3, 4 and 5 as remainders, respectively, but when divided by 7 leaves no remainder.

(a) 210 (b) 119
(c) 126 (d) 154

38. Find the greatest number of five digits which when divided by 4, 6, 10 and 15 leaves the same remainder 3 in each case.

(a) 99993 (b) 99063
(c) 90093 (d) 99963

39. Find the least number which is a multiple of 31 and when divided by 15, 24 and 32 leaves the remainders 2, 11 and 19, respectively.

(a) 2418 (b) 2387
(c) 2356 (d) 2325

40. Find the two largest numbers of four digits having 531 as their H.C.F.

(a) 9231, 9762
(b) 9027, 9558
(c) 9037, 9568
(d) 9127, 9658

41. Find the greatest number of five digits which becomes exactly divisible by 10, 12, 15 and 18 when 3769 is added to it.

(a) 99811 (b) 99911
(c) 98911 (d) 99011

42. Find the least number which when decreased by 11 is divisible by 14, 15, 21, 32 and 60.

(a) 4371 (b) 3271
(c) 3371 (d) 3360

43. Find the least number of five digits which when divided by 8, 12, 16 and 20 leaves remainders 1, 5, 9 and 13, respectively.

(a) 10003 (b) 10093
(c) 10073 (d) 10013

44. The H.C.F. of two numbers is 11 and their L.C.M. is 693. If one of the numbers is 77, find the other.

(a) 909 (b) 119
(c) 66 (d) 99

45. Find the greatest number of four digits which is exactly divisible by 24, 28, 30 and 35.

(a) 9225 (b) 9240
(c) 9250 (d) 9260

46. Find the greatest number of four digits which must be added to 5231 so that the final number becomes exactly divisible by 12, 15, 27, 32 and 40.

(a) 7929 (b) 7829
(c) 9729 (d) 7729

47. A heap of stones can be made up into groups of 21. When made up into groups of 16, 20, 25 and 45, there are 3 stones left in each case. How many stones at least can there be in the heap?

(a) 7203 (b) 2403
(c) 3603 (d) 4803

48. Find the greatest number of five digits which when divided by 8, 9 and 10 leaves 3 as remainder in each case.

(a) 99996
(b) 99723
(c) 99983
(d) None of these

49. What is the least number of cut pieces of equal length that can be cut out of two lengths 10 m 857 mm and 15 m 87 mm?

(a) 174 (b) 172
(c) 164 (d) 184

EXERCISE-2
(BASED ON MEMORY)

1. *M* is the largest 4 digit number, which when divided by 4, 5, 6 and 7 leaves remainder as 2, 3, 4 and 5 respectively. When M is divided by 9?

(a) 2 (b) 1
(c) 3 (d) 6

[SSC CGL Tier-II CBE, 2018]

2. Three electronics devices make a beep after every 48 seconds, 72 seconds and 108 seconds respectively. They beeped together at 10 a.m. The time when they will next make a beep together at the earliest is

(a) 10 : 07 : 12 hours
(b) 10 : 07 : 24 hours
(c) 10 : 07 : 36 hours
(d) 10 : 07 : 48 hours

[SSC CGL Tier-II (CBE), 2017]

3. Two pipes of length 1.5 m and 1.2 m are to be cut into equal pieces without leaving any extra length of pipes. The greatest length of the pipe pieces of same size which can be cut from these two lengths will be

(a) 0.13 metre (b) 0.4 metre
(c) 0.3 metre (d) 0.41 metre

[SSC CGL Tier-II (CBE), 2017]

4. What least value which should be added to 1812 to make it divisible by 7, 11 and 14?

(a) 12 (b) 36
(c) 72 (d) 154

[SSC CHSL Tier-I CBE, 2017]

5. How many numbers are there from 300 to 700 which are divisible by 2, 3 and 7?

(a) 7 (b) 8
(c) 9 (d) 10

[SSC CGL Tier-I CBE, 2017]

6. The LCM of two prime numbers x and y. ($x > y$) is 161. The value of ($3y - x$)

(a) –2 (b) –1
(c) 1 (d) 2

[SSC CGL Tier-I (CBE), 2016]

7. The product of two 2 digit numbers is 2160 and their H.C.F. is 12. The numbers are

(a) (12, 60) (b) (72, 30)
(c) (36, 60) (d) (60, 72)

[SSC CGL Tier-I (CBE), 2016]

8. Find the greatest number that will divide 390, 495 and 300 without leaving a remainder.

(a) 5 (b) 15
(c) 25 (d) 35

[SSC CGL Tier-I (CBE), 2016]

9. In a school 391 boys and 323 girls have been divided into the largest possible equal classes, so that each class of boys numbers the same as each class of girls. What is the number of classes?

(a) 23 (b) 19
(c) 44 (d) 17

[SSC CGL Tier-I (CBE), 2016]

10. The HCF of $x^8 - 1$ and $x^4 + 2x^3 - 2x - 1$ is:

(a) $x^2 + 1$ (b) $x + 1$
(c) $x^2 - 1$ (d) $x - 1$

[SSC, 2015]

11. Three number s are in the ratio 1 : 2 : 3 and their HCF is 12. The numbers are

(a) 12, 24, 36 (b) 5, 10, 15
(c) 4, 8, 12 (d) 10, 20, 30

[SSC, 2014]

12. Find the least number which when divided separately by 15, 20, 36 and 48 leaves 3 as remainder in each case.

(a) 183 (b) 243
(c) 483 (d) 723

[SSC, 2014]

13. If the L.C.M. and H.C.F. of two expressions are $(x^2+6x+8)(x+1)$ and $(x+1)$, respectively and one of the expressions is x^2+3x+2, find the other.

(a) x^2+5x+4 (b) x^2-5x+4
(c) x^2+4x+5 (d) x^2-4x+5

[SSC, 2014]

14. What is the smallest number by which 625 must be divided so that the quotient is a perfect cube?

(a) 125 (b) 5
(c) 2 (d) 3

[SSC, 2014]

15. Find the greatest number which exactly divides 200 and 320.

(a) 10 (b) 20
(c) 16 (d) 40

[SSC, 2014]

16. The greatest 4-digit number exactly divisible by 10, 15 and 20 is:

(a) 9990 (b) 9960
(c) 9980 (d) 9995

[SSC, 2013]

17. If the students of 9th class are arranged in rows of 6, 8, 12 or 16, no student is left behind. The possible number of students in the class is:

(a) 60 (b) 72
(c) 80 (d) 96

[SSC, 2013]

18. If A and B are the H.C.F. and L.C.M., respectively of two algebraic expressions x and y, and $A + B = x + y$, then the value of $A^3 + B^3$ is:

(a) $x^3 - y^3$
(b) x^3
(c) y^3
(d) $x^3 + y^3$

[SSC Assistant Grade III, 2013]

19. The greatest number that divides 411, 684, 821 and leaves 3, 4 and 5 as remainders, respectively is:

(a) 254 (b) 146
(c) 136 (d) 204

[SSC Assistant Grade III, 2013]

20. L.C.M. and H.C.F. of two numbers x and y are 3 and 105, respectively. If $x + y = 36$, the value of $\frac{1}{x}+\frac{1}{y}$ is:

(a) 1 (b) $\frac{1}{6}$
(c) $\frac{12}{315}$ (d) $\frac{4}{35}$

[UPPCS, 2012]

21. Given: $\sqrt[3]{4}$, $\sqrt{3}$, $\sqrt[6]{25}$ and $\sqrt[12]{289}$, the greatest and least of them are respectively:

(a) $\sqrt[12]{289}$ and $\sqrt[3]{4}$
(b) $\sqrt{3}$ and $\sqrt[3]{4}$
(c) $\sqrt[6]{25}$ and $\sqrt{3}$
(d) $\sqrt[3]{4}$ and $\sqrt[6]{25}$

[SSC Assistant Grade III, 2012]

22. In four consecutive prime numbers that are in ascending order, the product of the first three is 385 and that of the last three is 1001. The largest given prime number is:

CHAPTER 5

Percentage

INTRODUCTION

The term *percent* means per hundred or for every hundred. It is the abbreviation of the Latin phrase *percentum*.

Scoring 60 per cent marks means out of every 100 marks the candidate scored 60 marks.

The term percent is sometimes abbreviated as p.c. The symbol % is often used for the term percent.

Thus, 40 percent will be written as 40%.

A fraction whose denominator is 100 is called a *percentage* and the numerator of the fraction is called *rate percent*, e.g., $\frac{5}{100}$ and 5 percent means the same thing, i.e., 5 parts out of every hundred parts.

BASIC FORMULAE

01 To convert a fraction into a percent:

To convert any fraction $\frac{l}{m}$ to rate percent, multiply it by 100 and put % sign, i.e., $\frac{l}{m} \times 100\%$

Illustration 1: What percentage is equivalent to $\frac{3}{5}$?

Solution: $\frac{3}{5} \times 100 = 60\%$.

02 To convert a per cent into a fraction:

To convert a percent into a fraction, drop the percent sign and divide the number by 100.

Illustration 2: What fraction is $16\frac{2}{3}\%$?

Solution: $16\frac{2}{3}\% = \frac{\left(\frac{50}{3}\right)}{100} = \left(\frac{50}{3} \times \frac{1}{100}\right) = \frac{1}{6}$.

03 To find a percentage of a given number:

$x\%$ of given number (N) $= \frac{x}{100} \times$ N.

Illustration 3: 75% of 400 = ?

Solution: 75% of 400 $= \frac{75}{100} \times 400 = 300$.

Illustration 4: Find a number whose 4% is 72.

Solution: Let the required number be x.

Then, 4% of $x = 72$

$\Rightarrow \quad \frac{4}{100} \times x = 72 \Rightarrow x = \frac{100}{4} \times 72 = 1800$.

Illustration 5: What per cent of 25 Kg is 3.5 Kg?

Solution: Let x% of 25 Kg be 3.5 Kg.

Then, x% of 25 Kg = 3.5 Kg

$\Rightarrow \quad \frac{x}{100} \times 25 = 3{\cdot}5 \Rightarrow x = \frac{3{\cdot}5 \times 100}{25} = 14$.

Hence, 3.5 Kg is 14% of 25 Kg.

SHORT-CUT METHODS

01 (a) If A is x% more than that of B, then B is less than that of A by

$$\left[\frac{x}{100+x} \times 100\right]\%$$

(b) If A is x% less than that of B, then B is more than that of A by

$$\left[\frac{x}{100-x} \times 100\right]\%$$

Explanation

Given: $A = B + \frac{x}{100}B = \frac{100+x}{100}B$

$\therefore \quad A - B = \frac{100+x}{100}B - B$

$= \left(\frac{100+x}{100} - 1\right)B = \frac{x}{100}B.$

So, $\frac{A-B}{A} = \frac{\frac{x}{100}B}{\frac{100+x}{100}B} = \frac{x}{100+x}$

$\Rightarrow \quad A - B = \left(\frac{x}{100+x} \times 100\right)\%$ of A.

Therefore, B is less than that of A by

$$\left(\frac{x}{100+x} \times 100\right)\%$$

Similarly, (b) can be proved.

Illustration 6: If Mohan's salary is 10% more than that of Sohan, then how much per cent is Sohan's salary less than that of Mohan?

Solution: Here, $x = 10$.

$\therefore$ Required answer $= \left(\frac{x}{100+x} \times 100\right)\%$

$= \left(\frac{x}{100+x} \times 100\right)\%$

$= 9\frac{1}{11}\%$

Illustration 7: If A's income is 40% less than B's income, then how much per cent is B's income more than A's income?

Solution: Here, $x = 40$.

$\therefore$ Required answer $= \left(\frac{x}{100-x} \times 100\right)\%$

$= \left(\frac{40}{100-40} \times 100\right)\%$

$= 66\frac{2}{3}\%$

02 If A is x% of C and B is y% of C, then

$$A = \frac{x}{y} \times 100\% \text{ of } B.$$

Explanation

Given: $A = \frac{x}{100}C \Rightarrow C = 100\frac{A}{x}$

and, $B = \frac{y}{100}C \Rightarrow C = 100\frac{B}{y}$

$\therefore \quad C = 100\frac{A}{x} = 100\frac{B}{y} \Rightarrow A = \frac{x}{y}B$

or, $\frac{x}{y} \times 100\%$ of B.

Illustration 8: If A is 20% of C and B is 25% of C, then what percentage is A of B?

Solution: Here, $x = 20$ and $y = 25$.

$A = \frac{x}{y} \times 100\%$ of B

$= \frac{20}{25} \times 100\%$ of B, i.e., 80% of B.

03 (a) If two numbers are respectively x% and y% more than a third number, then the first number is $\left(\frac{100+x}{100+y} \times 100\right)\%$ of the second and the second number is $\left(\frac{100+y}{100+x} \times 100\right)$ % of the first.

(b) If two numbers are, respectively, x% and y% less than a third number, then the first number is $\left(\frac{100-x}{100-y} \times 100\right)\%$ of the second and the second number is $\left(\frac{100-y}{100-x} \times 100\right)\%$ of the first number.

Explanation

Let A, B and C be the three numbers.

Given:

$$A = C + \frac{x}{100}C = \left(\frac{100+x}{100}\right)C \Rightarrow C = A\left(\frac{100}{100+x}\right)$$

and, $B = C + \frac{y}{100}C = \left(\frac{100+y}{100}\right)C \Rightarrow C = B\left(\frac{100}{100+y}\right)$

$\therefore \quad A\left(\frac{100}{100+x}\right) = B\left(\frac{100}{100+y}\right)$

$\Rightarrow \quad A = \left(\frac{100+x}{100+y}\right)B$ or $\left(\frac{100+x}{100+y}\right) \times 100\%$ of B

and, $B = \left(\frac{100+y}{100+x}\right)A$ or $\left(\frac{100+y}{100+x}\right)$ × 100% of A.

Similarly, (b) can be proved.

Illustration 9: Two numbers are respectively 20% and 50% more than a third number. What per cent is the first of the second?

Solution: Here, $x = 20$ and $y = 50$.

$$\therefore \quad \text{First number} = \left(\frac{100+x}{100+y}\right) \times 100\% \text{ of the second}$$

$$= \left(\frac{100+20}{100+50}\right) \times 100\% \text{ of the second}$$

i.e., 80% of the second.

Illustration 10: Two numbers are, respectively, 32% and 20% less than a third number. What per cent is the first of the second?

Solution: Here, $x = 32$ and $y = 20$.

$$\therefore \quad \text{First number} = \left(\frac{100-x}{100-y}\times 100\right)\% \text{ of the second}$$

$$= \left(\frac{100-32}{100-20}\times 100\right)\% \text{ of the second}$$

i.e., 85% of the second.

04 (a) If the price of a commodity increases by P%, then the reduction in consumption so as not to increase the expenditure is

$$\left(\frac{P}{100+P}\times 100\right)\%$$

(b) If the price of a commodity decreases by P%, then the increase in consumption so as not to decrease the expenditure is

$$\left(\frac{P}{100-P}\times 100\right)\%$$

Explanation

Let the original price of the commodity be ₹100.

Then, the increased price = $100 + \frac{P}{100}\times 100$

= ₹$(100 + P)$.

Therefore, to keep the price unchanged, there should be a reduction in the consumption of the commodity by ₹P.

$\because$ Decrease in ₹$(100 + P)$ = ₹P

$\therefore$ Decrease in ₹100 = $\frac{P}{100+P}\times 100$

$\therefore$ Required reduction in consumption is

$$\left(\frac{P}{100+P}\times 100\right)\%$$

Similarly, (b) part can be proved.

Illustration 11: If the price of sugar increases by 25%, find how much per cent its consumption be reduced so as not to increase the expenditure.

Solution: Reduction in consumption

$$= \left(\frac{P}{100+P}\times 100\right)\%$$

$$= \left(\frac{25}{100+25}\times 100\right)\% \text{ or } 20\%$$

Illustration 12: If the price of a commodity decreases by 25%, find how much per cent its consumption be increased so as not to decrease the expenditure.

Solution: Increase in consumption

$$= \left(\frac{P}{100-P}\times 100\right)\%$$

$$= \left(\frac{25}{100-25}\times 100\right)\% \text{ or } 33\frac{1}{3}\%$$

05 If a number is changed (increased/decreased) successively by x% and y%, then net % change is given by $\left(x+y+\frac{xy}{100}\right)$% which represents increase or decrease in value according as the sign is +ve or –ve.

If x or y indicates decrease in percentage, then put –ve sign before x or y, otherwise +ve sign.

Explanation

Let the given number be N.

If it is increased by x%, then it becomes

$$N + x\% \text{ of } N = N + \frac{Nx}{100} = \frac{N(x+100)}{100}.$$

If it is further increased by y%, then it becomes

$$\frac{N(x+100)}{100} + \frac{y}{100}\times\frac{N(x+100)}{100}$$

$$= \frac{N(x+100)(y+100)}{(100)^2}$$

$$\therefore \quad \text{Net change} = \frac{N(x+100)(y+100)}{(100)^2} - N$$

$$= \frac{N(100x+100y+xy)}{(100)^2}$$

$$\therefore \quad \%\text{ change} = N\left(x+y+\frac{xy}{100}\right)\times\frac{1}{100}\times\frac{100}{N}$$

$$= \left(x + y + \frac{xy}{100}\right)\%$$

Illustration 13: If salary of a person is first increased by 15% and thereafter decreased by 12%, what is the net change in his salary?

Solution: Here, $x = 15$ and $y = -12$.

$\therefore$ The net% change in the salary

$$= \left(x + y + \frac{xy}{100}\right)\% = \left(15 - 12 - \frac{15 \times 12}{100}\right)\% \text{ or } 1.2\%$$

Since the sign is +ve, the salary of the person increases by 1.2%

Illustration 14: The population of a town is decreased by 25% and 40% in two successive years. What per cent population is decreased after two years?

Solution: Here, $x = -25$ and $y = -40$.

$\therefore$ The net% change in population

$$= \left(x + y + \frac{xy}{100}\right)\%$$

$$= \left(-25 - 40 + \frac{25 \times 40}{100}\right)\% \text{ or } -55\%$$

Since the sign is –ve, there is decrease in population after two years by 55%

06 If two parameters A and B are multiplied to get a product and if A is changed (increased/decreased) by x% and another parameter B is changed (increased/decreased) by y%, then the net % change in the product ($A \times B$) is given $\left(x + y + \frac{xy}{100}\right)\%$ which represents increase or decrease in value according as the sign in +ve or –ve.

If x or y indicates decrease in percentage, then put –ve sign before x or y, otherwise +ve sign.

Illustration 15: If the side of a square is increased by 20%, its area is increased by k%. Find the value of k.

Solution: Since side × side = area

$\therefore$ Net% change in area

$$= \left(x + y + \frac{xy}{100}\right)\% = \left(20 + 20 + \frac{20 \times 20}{100}\right)\%$$

[Here, $x = 20$ and $y = 20$]

$= 44\%$

Therefore, the area is increased by 44%.

Here, $k = 44$.

Illustration 16: The radius of a circle is increased by 2%. Find the percentage increase in its area.

Solution: Since $\pi \times$ radius × radius = area

$\therefore$ Net% change in area

$$= \left(x + y + \frac{xy}{100}\right)\% = \left(2 + 2 + \frac{2 \times 2}{100}\right)\%$$

[Here, $x = 2$ and $y = 2$]

$$= 4\frac{1}{25}\%$$

Therefore, the percentage increase in area is $4\frac{1}{25}\%$

Illustration 17: The tax on a commodity is diminished by 15% and its consumption increases by 10%. Find the effect on revenue.

Solution: Since tax × consumption = revenue

$\therefore$ Net% change in revenue

$$= \left(x + y + \frac{xy}{100}\right)\% = \left(-15 + 10 - \frac{15 \times 10}{100}\right)\%$$

[Here, $x = -15$ and $y = 10$]

$= -6.5\%$

$\therefore$ The revenue decreases by 6.5%

07 If the present population of a town (or value of an item) be P and the population (or value of item) changes at r% per annum, then

(a) Population (or value of item) after n years

$$= P\left(1 + \frac{r}{100}\right)^n$$

(b) Population (or value of item) n years ago

$$= \frac{P}{\left(1 + \frac{r}{100}\right)^n}.$$

where r is +ve or –ve according as the population (or value of item) increases or decreases.

Explanation

Population at the end of first year

$$= P + \frac{r}{100}P = P\left(1 + \frac{r}{100}\right).$$

Now, the population at the beginning of second year

$$= P\left(1 + \frac{r}{100}\right).$$

$\therefore$ Population at the end of second year

$$= P\left(1+\frac{r}{100}\right)+\frac{r}{100}P\left(1+\frac{r}{100}\right) = P\left(1+\frac{r}{100}\right)^2$$

$\vdots \qquad \vdots \qquad \vdots$

Population at the end of n years $= P\left(1+\frac{r}{100}\right)^n$.

Illustration 18: The population of a town increases 5% annually. If its present population is 84000, what will it be in 2 years time?

Solution: Here, $P = 84000$, $r = 5$ and $n = 2$.

$\therefore$ Population of the town after 2 years

$\because$

$$= P\left(1+\frac{r}{100}\right)^n = 84000\left(1+\frac{5}{100}\right)^2$$

$$= 84000 \times \frac{105}{100}\times\frac{105}{100} = 92610.$$

Illustration 19: The population of a town increases at the rate of 5% annually. If the present population is 4410, what it was 2 years ago?

Solution: Here, $P = 4410$, $r = 5$ and $n = 2$.

$\therefore$ Population of the town 2 years ago

$$= \frac{P}{\left(1+\frac{r}{100}\right)^n} = \frac{4410}{\left(1+\frac{5}{100}\right)^2} = \frac{4410}{\frac{105}{100}\times\frac{105}{100}} = 4000.$$

08 If a number A is increased successively by x% followed by y% and then by z%, then the final value of A will be

$$A\left(1+\frac{x}{100}\right)\left(1+\frac{y}{100}\right)\left(1+\frac{z}{100}\right)$$

In case a given value decreases by any percentage, we will use a –ve sign before that.

Illustration 20: The population of a town is 144000. It increases by 5% during the first year. During the second year, it decreases by 10% and increases by 15% during the third year. What is the population after 3 years?

Solution: Here, $P = 144000$, $x = 5$, $y = -10$ and $z = 15$.

$\therefore$ Population of the town after 3 years

$$= A\left(1+\frac{x}{100}\right)\left(1+\frac{y}{100}\right)\left(1+\frac{z}{100}\right)$$

$$= 144000\left(1+\frac{5}{100}\right)\left(1-\frac{10}{100}\right)\left(1+\frac{15}{100}\right)$$

$$= \frac{144000\times105\times90\times115}{100\times100\times100} = 156492.$$

09 In an examination, the minimum pass percentage is x% If a student secures y marks and fails by z marks, then the maximum marks in the examination is $\frac{100(y+z)}{x}$.

Explanation

Let the maximum marks be m.

Given: x% of $m = y + z$

$$\Rightarrow \quad \frac{x}{100} \times m = y + z \text{ or } m = \frac{100(y+z)}{x}.$$

Illustration 21: In an examination, a student must get 60% marks to pass. If a student who gets 120 marks, fails by 60 marks, find the maximum marks.

Solution: Here, $x = 60$, $y = 120$ and $z = 60$.

$\therefore$ Maximum marks

$$= \frac{100(y+z)}{x} = \frac{100(120+60)}{60} = \frac{100\times180}{60} = 300.$$

10 In an examination x% and y% students respectively fail in two different subjects while z% students fail in both the subjects, then the percentage of students who pass in both the subjects will be $(100 - (x + y - z))$%

Explanation

Percentage of students who failed in one subject = $(x - z)$%

Percentage of students who failed in other subject = $(y - z)$%

Percentage of students who failed in both the subjects = z%

$\therefore$ Percentage of students who passed in both the subjects

$$= [100 - [(x - z) + (y - z) + z]]\%$$
$$= (100 - (x + y - z))\%$$

Illustration 22: In an examination, 42% students failed in Mathematics and 52% failed in Science. If 17% failed in both the subjects, find the percentage of those who passed in both the subjects.

Solution: Here, $x = 42$, $y = 52$ and $z = 17$.

$\therefore$ Percentage of students passing both the subjects

$$= (100 - (x + y - z))\%$$
$$= (100 - (42 + 52 - 17))\% \text{ or } 23\%$$

EXERCISE-I

1. What percentage is equivalent to $5\frac{1}{4}$?

(a) 525% (b) 425%
(c) 625% (d) None of these

2. 0.005 = (... ? ...)%

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$%
(c) $\frac{1}{3}$% (d) None of these

3. $6\frac{2}{3}$% expressed as a fraction in its lowest term is:

(a) $\frac{2}{15}$ (b) $\frac{1}{15}$
(c) $\frac{3}{20}$ (d) None of these

4. What fraction is 0.6%

(a) $\frac{7}{500}$ (b) $\frac{9}{500}$
(c) $\frac{3}{500}$ (d) None of these

5. 0.025 in terms of rate per cent is:

(a) 3.5% (b) 2.5%
(c) 1.5% (d) None of these

6. What per cent of 12 is 84?

(a) 800% (b) 600%
(c) 700% (d) None of these

7. Express $\frac{7}{8}$ as percentage.

(a) $67\frac{1}{2}$ (b) $87\frac{1}{2}$%
(c) $97\frac{1}{4}$% (d) None of these

8. Express $8\frac{1}{3}$% as a fraction.

(a) $\frac{1}{12}$ (b) $\frac{1}{16}$
(c) $\frac{1}{18}$ (d) None of these

9. $37\frac{1}{2}$% of ₹48 is:

(a) ₹20 (b) ₹16
(c) ₹18 (d) None of these

10. What per cent of $\frac{2}{7}$ is $\frac{1}{35}$?

(a) 15% (b) 18%
(c) 10% (d) None of these

11. 75% of 480 = (?) × 15

(a) 12 (b) 36
(c) 24 (d) None of these

12. If 200% of a number is 90, then what is the 80% of that number?

(a) 48
(b) 36
(c) 24
(d) None of these

13. If $37\frac{1}{2}$% of a number is 45, then $87\frac{1}{2}$% of the number will be:

(a) 115 (b) 135
(c) 105 (d) None of these

14. ? × 15 = 37.5% of 220.

(a) 5.5 (b) 7.5
(c) 6.5 (d) None of these

15. What per cent of 4 Km is 8 metres?

(a) 0.4 (b) 0.2
(c) 0.8 (d) None of these

16. x% of y + y% of x = ?

(a) 3% of xy
(b) 2% of xy
(c) 5% of xy
(d) None of these

17. 0.35% of a number is equivalent to multiplying it by the number:

(a) 0.0025 (b) 0.0045
(c) 0.0035 (d) None of these

18. If 8% of x = 4% of y, then 20% of x is:

(a) 15% of y (b) 10% of y
(c) 20% of y (d) None of these

19. x% of y + ?% of x = x% of (x + y).

(a) $x + y$ (b) x
(c) y (d) None of these

20. A number x is 125% of y. To compute y, the number x has to be multiplied by:

(a) 0.08 (b) 0.4
(c) 0.8 (d) None of these

21. 25% of 25% = ?

(a) 6.25 (b) 0.0625
(c) 0.625 (d) None of these

22. Which number is 60% less than 80?

(a) 24 (b) 36
(c) 32 (d) None of these

23. 20% of 30% of 20% of ₹850 is:

(a) ₹9.50 (b) ₹10.20
(c) ₹10.50 (d) None of these

24. The greatest of $16\frac{2}{3}$%, $6\frac{2}{3}$%, 0·3 is:

(a) $16\frac{2}{3}$% (b) $6\frac{2}{3}$%
(c) 0.3 (d) Cannot be compared

25. 40% of 20% + 30% of 25% + 50% of 28% is equivalent to:

(a) 29.5% (b) 28.5%
(c) 30.5% (d) None of these

26. If 90% of A = 30% of B and B = x% of A, then the value of x is:

(a) 800 (b) 300
(c) 700 (d) None of these

27. 1 quintal 25 Kg is what per cent of 1 metric tonne?

(a) $16\frac{1}{2}$% (b) $8\frac{1}{2}$%
(c) $12\frac{1}{2}$% (d) None of these

28. If 12% of x is equal to 6% of y, then 18% of x will be equal to how much per cent of y?

(a) 7% (b) 9%
(c) 11% (d) None of these

29. If a number is 20% more than the other, how much per cent is the second number less than the first?

(a) $12\frac{1}{3}$% (b) $16\frac{2}{3}$%
(c) $16\frac{1}{3}$% (d) None of these

30. If A's income is 25% less than that of B, then how much per cent is B's income more than that of A?

(a) $33\frac{1}{3}$% (b) $66\frac{2}{3}$%
(c) $11\frac{2}{3}$% (d) None of these

31. If the given two numbers are respectively 7% and 28% of a third number, then what percentage is the first of the second?

(a) 20% (b) 25%
(c) 18% (d) None of these

32. Two numbers are respectively 60% and 20% more than a third number. Second number expressed as a percentage of first is:

(a) 75% (b) 90%
(c) 80% (d) None of these

33. Two numbers are less than a third number by 30% and 37%, respectively. How much per cent is the second number less than the first?

(a) 15%
(b) 10%
(c) 20%
(d) None of these

34. Two numbers are respectively 20% and 10% more than a third number. How much per cent is the first number more than the second?

(a) $9\frac{1}{11}$% (b) $7\frac{1}{11}$%
(c) $11\frac{1}{11}$% (d) None of these

35. The price of cooking oil has increased by 15%. The percentage of reduction that a family should effect in the use of cooking oil so as not to increase the expenditure on this account is:

(a) $15\frac{2}{23}$% (b) $13\frac{1}{23}$%
(c) $17\frac{1}{23}$% (d) None of these

36. If the price of apples goes down by 10%, find the percentage of increase that a family should effect in its consumption so as not to increase expenditure on this account is:

(a) $13\frac{1}{11}$% (b) $15\frac{1}{11}$
(c) $11\frac{1}{11}$% (d) None of these

37. A number is increased by 20% and then decreased by 20%, the final value of the number:

(a) Does not change
(b) Decreases by 2%
(c) Increases by 4%
(d) Decreases by 4%.

38. A man's wages were decreased by 50%. Again, the reduced wages were increased by 50%. He has a loss of:

(a) 35% (b) 25%
(c) 20% (d) None of these

39. The population of a town is decreased by 20% and 25% in two successive years. What per cent population is decreased after two years?

(a) 50% (b) 40%
(c) 60% None of these

40. The difference between a discount of 35% and two successive discounts of 20% and 20% on a certain bill was ₹22. Find the amount of the bill.

(a) ₹3200 (b) ₹2200
(c) ₹1800 (d) None of these

41. A shopkeeper marks the prices of his goods at 25% higher than the original price. After that, he allows a discount of 12%. What profit or loss did he make?

(a) 10% profit (b) 15% profit
(c) 10% loss (d) 15% loss

42. Two shopkeepers sell a ratio of similar brand and type at the same list price of ₹1000. The first allows two successive discounts of 20% and 10% and the second allows the successive discounts of 15% and 15%. Find the difference in discounts offered by the two shopkeepers.

(a) ₹3.50 (b) ₹1.50
(c) ₹2.50 (d) None of these

43. The tax on a commodity is diminished by 10% and its consumption increases by 10%. Find the effects on revenue.

(a) 1% (b) 2%
(c) 3% (d) None of these

44. The radius of a sphere is increased by 10%. The surface area increases by

(a) 21% (b) 31%
(c) 41% (d) None of these

45. When the price of an article is reduced by 15%, the sales increases by 35%. The percentage change in the total amount of receipts is:

(a) $14\frac{3}{4}$% decrease (b) $14\frac{3}{4}$% increase
(c) $13\frac{3}{4}$% decrease (d) None of these

46. If the side of a square is increased by 30%, its area is increased by:

(a) 49% (b) 69%
(c) 79% (d) None of these

47. The length and breadth of a square are increased by 30% and 20%, respectively. The area of the rectangle so formed exceeds the area of the square by:

(a) 56% (b) 46%
(c) 66% (d) None of these

48. In measuring the sides of a rectangle, one side is taken 10% in excess and the other 20% in deficit. Find the error per cent in area calculated from the measurement.

(a) 12% deficit (b) 10% deficit
(c) 12% excess (d) None of these

49. For a rectangle, the length and breadth are increased by 10% and 20%, respectively. The percentage increase in area is:

(a) 24% (b) 48%
(c) 32% (d) None of these

50. Water tax is increased by 20% but its consumption is decreased by 20%. The increase or decrease in the expenditure is:

(a) 4% decrease
(b) 4% increase
(c) 8% decrease
(d) 8% increase

51. On decreasing the price of a colour TV by 30%, its sale is increased by 20%. The effect on the revenue is:

(a) 16% decrease (b) 16% increase
(c) 20% increase (d) None of these

52. The population of a city increases at the rate of 10% annually. Its present population is 90.51 lacs. The population 3 years ago was nearly:

(a) 72 Lakhs (b) 68 Lakhs
(c) 80 Lakhs (d) None of these

53. The value of a machine depreciates at the rate of 10% every year. It was purchased 3 years ago. If its present value is ₹8748, its purchase price was:

(a) ₹16000 (b) ₹18000
(c) ₹12000 (d) None of these

54. The income of a company increases 20% per annum. If its income is ₹2664000 in the year 1999 what was its income in the year 1997?

(a) ₹1750000 (b) ₹1650000
(c) ₹1850000 (d) None of these

55. The population of a town is 32000. It increases 15% annually. What will it be in 2 years?

(a) 52340 (b) 42320
(c) 62430 (d) None of these

56. The value of a machine is ₹6250. It decreases by 10% during the first year, 20% during the second year and 30% during the third year. What will be the value of the machine after 3 years?

(a) ₹2650 (b) ₹3050
(c) ₹3150 (d) None of these

57. The population of a town increases by 12% during first year and decreases by 10% during second year. If the present population is 50400, what it was 2 years ago?

(a) 40000 (b) 35000
(c) 50000 (d) None of these

58. Ramesh loses 20% of his pocket money. After spending 25% of the remainder he has ₹480 left. What was his pocket money?

(a) ₹600 (b) ₹800
(c) ₹900 (d) None of these

59. An army lost 10% its men in war, 10% of the remaining due to diseases and 10% of the rest were hurt. Thus, the strength was reduced to 729000 active men. Find the original strength.

(a) 1000000 (b) 1200000
(c) 1500000 (d) None of these

60. The daily wage is increased by 25% and a person now gets ₹25 per day. What was his daily wage before the increase?

(a) ₹25 (b) ₹20
(c) ₹30 (d) None of these

61. A student has to secure 15% marks to get through. If he gets 80 marks and fails by 70 marks, find the maximum marks set for the examination.

(a) 900 (b) 1000
(c) 1200 (d) None of these

62. In an examination, 30% and 35% students respectively failed in History and Geography while 27% students failed in both the subjects. If the number of students passing the examination is 248, find the total number of students who appeared in the examination.

(a) 425 (b) 380
(c) 400 (d) None of these

63. Mr Katial buys a house for ₹100000 and rents it. He puts 12.5% of each month's rent aside for upkeep and repairs, pays ₹325 per year as taxes and realizes 5.5% annually on his investment. Find the monthly rent.

(a) ₹634.76 (b) ₹554.76
(c) ₹654.76 (d) None of these

64. In an examination, there were 2000 candidates, out of which 900 candidates were boys and rest were girls. If 32% of the boys and 38% of the girls passed, then the total percentage of failed candidates is:

(a) 35.3% (b) 64.7%
(c) 68.5% (d) 70%

65. From the salary of an officer, 10% is deducted as house rent, 15% of the rest he spends on children's education and 10% of the balance, he spends on clothes. After this expenditure he is left with ₹1377. His salary is:

(a) ₹2000 (b) ₹2040
(c) ₹2100 (d) ₹2200

66. If the price of gold increases by 30%, find by how much the quantity of ornaments must be reduced so that the expenditure may remain the same as before?

(a) $27\frac{2}{13}\%$ (b) $23\frac{1}{13}\%$
(c) 30% (d) 19%

67. The price of sugar has fallen by 10%. How many quintals can be bought for the same money which was sufficient to buy 18 quintals at the higher price?

(a) 20 (b) 22
(c) 25 (d) 30

68. In an examination, there are 1000 boys and 800 girls, 60% of boys and 40% girls passed. The percentage of candidates that failed is:

(a) 48.88 (b) 45.88
(c) 50.00 (d) 49.88

69. The price of an article is cut by 20%. To restore it to its original price, the new price must be increased by:

(a) 20% (b) $22\frac{1}{2}\%$
(c) 25% (d) 40%

70. In a fraction, numerator is increased by 25% and the denominator is diminished by 10%. The new fraction obtained is $\frac{5}{9}$. The original fraction is:

(a) $\frac{2}{5}$ (b) $\frac{5}{9}$

(c) $\frac{3}{5}$ (d) None of these

71. One side of a square is increased by 30%. To maintain the same area the other side will have to be decreased by:

(a) $23\frac{1}{13}\%$ (b) $76\frac{12}{13}\%$

(c) 30% (d) 15%

EXERCISE-2
(BASED ON MEMORY)

1. 0.06% of 250% of 1600 is

(a) 24 (b) 0.24

(c) 0.024 (d) 2.4

[SSC CGL Tier-II CBE, 2018]

2. Two numbers are 90% and 75% lesser than a third number. By what % should the first number be increased so that it becomes equal to the second number?

(a) 250 (b) 200

(c) 150 (d) 100

[SSC CGL Tier-II CBE, 2018]

3. When a number is increased by 216, it becomes 140% of itself. What is the number?

(a) 540 (b) 756

(c) 450 (d) 875

[SSC CGL Tier-II CBE, 2018]

4. If $X = 600$ and $Y = 800$, then X is how much per cent less than Y?

(a) 33.33 (b) 25

(c) 75 (d) 35

[SSC CHSL (10 + 2) Tier-I CBE, 2018]

5. A man donates 30% of his wealth to charity. 30% and 25% of the remaining wealth to his wife and son respectively. The rest he divides equally between his three daughters. One of his daughter gets ₹ 42 lakh as her share. What was the man's wealth the (₹ lakhs)?

(a) 280 (b) 400

(c) 500 (d) 350

[SSC CGL Tier-II CBE, 2018]

6. A man's annual income has increased by ₹ 5 lakhs but the tax on income that he has to pay has reduced from 12% to 10%. He now pays ₹ 10,000 more income tax. What is his increased income (in ₹ lakhs)

(a) 20 (b) 25

(c) 15 (d) 10

[SSC CGL Tier-II CBE, 2018]

7. 40% are the passing marks. A students gets 250 marks yet fails by 38 marks. What is the maximum marks?

(a) 720 (b) 750

(c) 800 (d) 840

[SSC CGL Tier-II CBE, 2018]

8. Ravi is 12 years younger than Surya. Ravi's age 40% of the sum of his and Surya's age. What will be Surya's age 9 years hence?

(a) 36 (b) 24

(c) 33 (d) 45

[SSC CGL Tier-II CBE, 2018]

9. A boy found the answer for the question "subtract the sum of $\frac{1}{4}$ and $\frac{1}{5}$ from unity and express the answer in decimals" as 0.45. The percentage of error in his answer was

(a) $\left(\frac{100}{11}\right)\%$ (b) 50%

(c) 10% (d) $\left(\frac{200}{11}\right)\%$

[SSC CGL Tier-II CBE, 2017]

10. $a\%$ of $b + b\%$ of $a =$ _____

(a) $2a\%$ of b (b) $2a\%$ of $2b$

(c) $2a\%$ of $2a$ (d) $2b\%$ of $2b$

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

11. A number is increased by 84, it becomes 107% of itself. What is the number?

(a) 600 (b) 900

(c) 1500 (d) 1200

[SSC CGL Tier-I (CBE), 2017]

12. 20% of $a = b$, then b% of 20 is the same as _____ of a

(a) 8% (b) 40%
(c) 4% (d) 80%

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

13. A man wills 40% of his wealth to his wife and rest to his children. What per cent of the wealth willed to the wife do the children get?

(a) 150 (b) 66.6
(c) 50 (d) 20

[SSC CGL Tier-I CBE, 2017]

14. When 40 is subtracted from 30% of a number, the result is 50. What is the value of the numberr?

(a) 150 (b) 180
(c) 300 (d) 450

[SSC Multi-Tasking Staff, 2017]

15. If 18% of a number is 720, then what is 81% of the same number?

(a) 2280 (b) 1620
(c) 3240 (d) 3160

[SSC Multi-Tasking Staff, 2017]

16. Two numbers are 10% and 25% lesser than a third number. By how much percent is the second number to be enhanced to make it equal to the first number?

(a) 16.67 per cent
(b) 20 per cent
(c) 15 per cent
(d) 60 per cent

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

17. P is 20% more than Q and 40% less than R. If value of Q is ₹150, then what is the value of R (in ₹)?

(a) 300 (b) 320
(c) 220 (d) 250

[SSC CAPFs ASI & Delhi Police SI, 2017]

18. If 25% of a number is 6, what is the number is 50% more than the initial number?

(a) 36 (b) 24
(c) 30 (d) 16

[SSC CAPFs ASI & Delhi Police SI, 2017]

19. A, B, C are three students. A got 18% more marks than B and 12% less than C. If B got 220 marks, how much marks C has got?

(a) 230 (b) 295
(c) 240 (d) 290

[SSC CGL Tier-I CBE, 2017]

20. If A is 20% more than 360, then what is the value of A?

(a) 396 (b) 432
(c) 480 (d) 512

[SSC Multi-Tasking Staff, 2017]

21. x is 30% more than y and 25% less than z. If value of y is ₹ 300, then what is the value (in ₹) of z?

(a) 390 (b) 400
(c) 470 (d) 520

[SSC CAPFs ASI & Delhi Police SI, 2017]

22. Two numbers are more than the third number by 20% and 80% respectively. First number is what per cent of the second number?

(a) 75 (b) 50
(c) 66.66 (d) 150

[SSC Delhi Police Constable, 2017]

23. If income of R is 30% more than the income of Q and the income of Q is 20% more than the income of P the income of R is how much percent more than the income of P?

(a) 50 (b) 10
(c) 56 (d) 55

[SSC Multi-Tasking Staff, 2017]

24. The average monthly salary of all the employees in a factory is ₹ 8840. If the average salary of all the officers is ₹ 15000 and that of the remaining employees is ₹ 8000, then what is the percentage of the officers among the employees?

(a) 10% (b) 12%
(c) $8\frac{1}{3}\%$ (d) 11%

[SSC CGL Tier-II CBE, 2017]

25. The monthly salary of Mr. Sachdev gets increased by 5% there by his salary becomes ₹ 15,120 per annum. His earlier monthly salary (before the increase) was

(a) ₹ 1,320 (b) ₹ 1,200
(c) ₹ 1,240 (d) ₹ 1,440

[SSC Multi-Tasking Staff, 2017]

26. A person spends 25% of his annual income on house rent, 15% on education of children and 45% on other items. If he saves ₹ 14,400 annually, then the person's total income is:

(a) ₹ 96,000 (b) ₹ 98,000
(c) ₹ 1,00,000 (d) ₹ 1,20,000

[SSC Matric Level MTS, 2017]

27. Amit donated 205 of his income to a school and deposited 20% of the remainder in his bank. If he has ₹ 12800 now, what is the income (in ₹) of Amit?

(a) 18000 (b) 20000
(c) 24000 (d) 32000

[SSC CGL Tier-I CBE, 2017]

28. An engineering student has to secure 25% marks to pass. He gets 47 and fails by 43 marks. What are the examination?

(a) 385 marks (b) 410 marks
(c) 360 marks (d) 435 marks

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

29. Two students appeared for an examination. One of them secured 19 marks more than the other and his marks were 60% of the sum of their marks. The marks obtained by them are

(a) 78 and 59 (b) 57 and 38
(c) 45 and 26 (d) 99 and 80

[SSC CGL Tier-I CBE, 2017]

30. A person scores 45% of the total marks in the exam and still fails by 40 marks. The passing percentage of the exam is 55%. What is the maximum marks of the exam?

(a) 300 (b) 350
(c) 400 (d) 500

[SSC CGL Tier-I CBE, 2017]

31. The price of electricity has been increased by 25%. If a person wants to keep the expenditure same, then the percentage reduction in use of electricity should be:

(a) 19 (b) 18
(c) 21 (d) 20

[SSC Matric Level, 2017]

32. If the price of apple increase from ₹ 80 per kg to ₹ 100 per kg, by what per cent a person should decrease the consumption of apple so that his expenditure remains same?

(a) 25 (b) 22
(c) 20 (d) 18

[SSC CAPFs ASI & Delhi Police SI, 2017]

33. The price of a house increase every year by 20%. If the present price is ₹ 720000, then what was its price (in ₹) 2 years ago

[SSC Multi-Tasking Staff, 2017]

34. The population of a city increases at the rate of 5% per annual. If the present population of the city is 3,70,440, its population years ago was:

(a) 30,000 (b) 2,80,000
(c) 3,60,000 (d) 3,20,000

[SSC Matric Level MTS, 2017]

35. the price of table depreciates every year by 20%. If the value of the table after 2 years will ₹ 32000, then what is present price (in ₹) of the table?

(a) 48000 (b) 44000
(c) 50000 (d) 51000

[SSC CAPFs ASI & Delhi Police SI, 2017]

36. The present population of a is 26010. It increase annual at the rate of 2%. What was the population of town two years ago?

(a) 25000 (b) 25100
(c) 25200 (d) 25500

[SSC CGL Tier-I CBE, 2017]

37. The price of motor cycle depreciates every year by 10%. If the value of the motor cycle after 3 years will be ₹ 36450, then what is the present value (in ₹) of the motor cycle?

(a) 45000 (b) 50000
(c) 48000 (d) 51000

[SSC CAPFs ASI & Delhi Police SI, 2017]

38. On a certain date, Pakistan has a success rate of 60% against India in all the ODIs played between the two countries. They lost the next 30 ODIs in a row to India and their success rate comes down to 30%. The total number of ODIs played between the two countries is

(a) 50 (b) 45
(c) 60 (d) 30

[SSC CGL Tier-II CBE, 2017]

39. A businessman imported Laptops, worth ₹ 210000, Mobile phones worth ₹ 100000 and Television sets worth ₹ 150000. He had to pay 10% duty on laptops, 8% on phones and 5% on Television sets as a special case. How much total duty (in Rupees) he had to pay on all items as per above details?

(a) 36500 (b) 37000
(c) 37250 (d) 37500

[SSC CGL Tier-II CBE, 2017]

40. The weight of two iron balls are 3.5 kg and 7.5 kg. What is the percentage weight of the first ball with respect to second ball?

(a) $46\frac{2}{3}\%$ (b) 35%

(c) $46\frac{1}{3}\%$ (d) 45%

[SSC CGL Tier-I CBE, 2016]

41. One number is 25% of another number. The larger number is 12 more than the smaller. The larger number is

(a) 48 (b) 16
(c) 4 (d) 12

[SSC CGL Tier-I (CBE), 2016]

42. The number of students in a class is increased by 20% and the number now becomes 66. Initially the number was

(a) 45 (b) 50
(c) 55 (d) 60

[SSC CGL Tier-I (CBE), 2016]

43. A village lost 12% of its goals in flood and 5% of remainder died from diseases. If the number left now is 8360, what was the original number before the flood?

(a) 1000 (b) 10000
(c) 1,00,000 (d) 8360

[SSC CGL Tier-II CBE, 2016]

44. If A is equal to 20% of B and B is equal to 25% of C; then what percent of C is equal to A?

(a) 10 (b) 15
(c) 5 (d) 20

[SSC CGL Tier-I (CBE), 2016]

45. In a school there are 1500 students, 44% of them are girls. Monthly fee of each boy is ₹ 540 and the fee of each girl is 25% less than that of a boy. The sum of fees of boys and girls both is

(a) ₹ 720600 (b) ₹ 720800
(c) ₹ 720900 (d) ₹ 721000

[SSC CGL Tier-I (CBE), 2016]

46. In a marriage party 32% are women 54% are men and there are 196 children. How many men are there in the marriage party?

(a) 756 (b) 448
(c) 332 (d) 324

[SSC CGL Tier-I (CBE), 2016]

47. $6\frac{1}{4}\%$ of 1600 + $12\frac{1}{2\%}$ of 800 equals

(a) 100 (b) 200
(c) 300 (d) 400

[SSC CGL Tier-I (CBE), 2016]

48. If the salary of Manoj is 40% less than that of subhash, then by how much percentage is the salary of Subhash more than that of Manoj?

(a) 60% (b) $66\frac{1}{4}\%$
(c) $66\frac{2}{3}\%$ (d) 65%

[SSC CGL Tier-I CBE, 2016]

49. Two numbers are 50% and 80% less than a third number. By how much per cent is the second number to be enhanced to make it equal to the first number?

(a) 150 per cent
(b) 60 per cent
(c) 30 per cent
(d) 37.5 per cent

[SSC CGL Tier-I CBE, 2016]

50. There is a ratio of 5 : 4 between two numbers. If 40% of the first number is 12, then what would be 50% of the second number?

(a) 12 (b) 24
(c) 18 (d) Data Inadequate

[SSC CPO SI, ASI, 2016]

51. If 10% of x is 3 times 15% of y, then find $x : y$.

(a) 7 : 2 (b) 9 : 2
(c) 8 : 3 (d) 11 : 4

[SSC CGL Tier-I CBE, 2016]

52. If 35% of A's income is equal to 25% of B's income, then the ratio of A's income to B's income is

(a) 7 : 5 (b) 5 : 7
(c) 4 : 7 (d) 4 : 3

[SSC CGL Tier-I CBE, 2016]

53. Mukesh has twice as much money as Soham. Soham has 50% more money than Pankaj. If the average money with them is ₹ 110, then Mukesh has

(a) ₹ 155 (b) ₹ 160
(c) ₹ 180 (d) ₹ 175

[SSC CGL Tier-II CBE, 2016]

54. Christy donated 10% of his income to an orphanage and deposited 20% of the remainder in his bank. If he has now ₹ 7200 left, what is his income.

(a) ₹ 10000 (b) ₹ 8000
(c) ₹ 9000 (d) ₹ 8500

[SSC CPO, 2016]

55. The average salary of employees in a firm was ₹ 5200 and that of females was ₹ 4200. The mean salary of all the employees was ₹ 5000. What is the percentage of female employees?

(a) 80% (b) 20%
(c) 40% (d) 30%

[SSC CGL Tier-I (CBE), 2016]

56. If A's salary is 40% less than that of B, then how much percent is B's salary more than that of A?

(a) $33\frac{1}{3}$ (b) $66\frac{2}{3}$

(c) $33\frac{2}{3}$ (d) $66\frac{1}{3}$

[SSC CGL Tier-I (CBE), 2016]

57. A man spends 15% of his income. If his expenditure is ₹ 75, his income (in rupees) is:

(a) 400 (b) 300

(c) 750 (d) 500

[SSC CGL Tier-I (CBE), 2016]

58. If A's salary is 30% more than that of B then boy how much percent is B's salary less than that of A?

(a) 13.01% (b) 13.07%

(c) 23.07% (d) 23.01%

[SSC CGL Tier-I (CBE), 2016]

59. There are 1400 students in the school 25% of them wear spectacles and $\frac{2}{7}$ of them wear the spectacles are boys. How many girls in the school do wear the spectacles?

(a) 250 (b) 100

(c) 200 (d) 300

[SSC CGL Tier-I (CBE), 2016]

60. If 60% of the students in a school are boys and the number of girls is 812, how many boys are there in the school?

(a) 1128 (b) 1218

(c) 1821 (d) 1281

[SSC CGL Tier-I (CBE), 2016]

61. The average marks obtained in a class of 50 students is 70%. The average of first 25 is 60%. and that of 24 is 80%. What is the marks obtained by the last student?

(a) 90% (b) 60%

(c) 80% (d) 70%

[SSC CGL Tier-I (CBE), 2016]

62. In an examination 65% of students pass in History and 55% students pass in Hindi. If 5% students fall in both the subjects, then what is the percentage of students who have passed in both the subjects?

(a) 15 (b) 20

(c) 25 (d) 30

[SSC CGL Tier-I CBE, 2016]

63. Due to a price hike of 20% 4 kg less sugar is available for ₹ 120. What is the initial price ₹ 120. What is the initial price per kg of sugar?

(a) ₹ 5 per kg
(b) ₹ 4 per kg
(c) ₹ 6 per kg
(d) ₹ 5.5 per kg

[SSC CGL Tier-I (CBE), 2016]

64. The population of a town is 9000. It the number of females increase by 5% and the males by 7.5% what will be the total population after. The number of females currently is 3000.

(a) 9600 (b) 9200

(c) 10500 (d) 9540

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2016]

65. The population of a city is 2000. It increased by 20% during the first year and 30% during the second year. The population after two years will be:

(a) 32000 (b) 40000

(c) 31200 (d) 30000

[SSC CGL Tier-I (CBE), 2016]

66. A set A consists of integers 27, 28, 30, 32 and 33. If integers k is included in the average of set A will increase by 30%. What is the value of integer K?

(a) 68 (b) 79

(c) 84 (d) 92

[SSC CPO SI & ASI, 2016]

67. An army lost 10% of its men in war, 10% of the remaining died due to disease and 10% of the rest were declared disabled and 10% of the rest were declared disabled. Thus the strength of the army was reduced to 7,29,000 active men. The original strength of the army was

(a) 1500000 (b) 1000000

(c) 1200000 (d) 1100000

[SSC CGL Tier-II CBE, 2016]

68. Starting with 8000 workers, the company Increase the number of workers by 5%, 10% and 20% at the end of first, second and third year respectively. The number of workers in the fourth year was

(a) 10188 (b) 11088

(c) 11008 (d) 11808

[SSC CGL Tier-I (CBE), 2016]

69. If "basis points" are defined so that 1 per cent is equal to 100 basis points, then by how many basis points is 82.5 per cent greater than 62.5 per cent?

(a) 0.2 (b) 20
(c) 200 (d) 2000

[SSC CGL Tier-I (CBE), 2016]

70. In the last financial year, a car company sold 41,800 cars. In this year, the target is to sell 51,300 cars, By what per cent must the sale be increased?

(a) $11\frac{9}{22}\%$ (b) $8\frac{9}{22}\%$
(c) $8\frac{11}{23}$ (d) $22\frac{8}{11}$

[SSC CGL Tier-I (CBE), 2016]

71. In a motor of 120 machines parts, 5% parts were defective. In another motor of 80 machine parts, 10% parts were defective. For the two motors considered together, the percentage of defective machine parts was

(a) 7 (b) 6.5
(c) 7.5 (d) 8

[SSC CGL Tier-I (CBE), 2016]

72. Number of appeared candidates from State Q increased by 100% from 2006 to 2007. If the total number of qualified candidates from State Q in 2006 and 2007 together is 408, what is the number of appeared candidates from State Q in 2006?

(a) 380 (b) 360
(c) 340 (d) 320
(e) 300

[IBPS, 2015]

73. If 60% of A = 30% of B, B = 40% of C and C = x% of A, then value of x is

(a) 800 (b) 500
(c) 300 (d) 200

[SSC, 2015]

74. Water tax is increased by 20% but its consumption is decreased by 20%. Then the increase of decrease in the expenditure of the money is

(a) 4% decrease (b) 4% increase
(c) 5% decrease (d) No change

[SSC, 2015]

75. In an office, 40% of the staff is female, 70% of the female staff and 50% of the male staff are married. The percentage of the unmarried staff in the office is

(a) 42 (b) 64
(c) 60 (d) 54

[SSC, 2015]

76. A sum of ₹7,930 is divided into 3 parts and given on loan at 5% simple interest to A, B and C for 2, 3 and 4 years respectively. If the amounts of all three are equal after their respective periods of loan, then the A received a loan of

(a) ₹2,760 (b) ₹2,750
(c) ₹2,800 (d) ₹3,050

[SSC, 2015]

77. The percentage increase in the surface area of a cube when each side is doubled is:

(a) 300% (b) 150%
(c) 50% (d) 200%

[SSC, 2015]

78. A man spends 75% of his income. His income is increased by 20% and he increased his expenditure by 10%. His savings are increased by:

(a) 25% (b) $37\frac{1}{2}$
(c) 50% (d) 10%

[SSC, 2015]

79. The percentage of metals in a mine of lead ore is 60%. Now the percentage of silver is 3/4 % of metals and the rest is lead. If the mass of ore extracted from this mine is 8000kg, the mass (in kg.) of lead of:

(a) 4762 (b) 4763
(c) 4764 (d) 4761

[SSC, 2015]

80. In an examination, a student must get 36% marks to pass. A student who gets 190 margks failed by 35 marks. The total marks in that examination is:

(a) 500 (b) 625
(c) 810 (d) 450

[SSC, 2015]

81. The number that is to be added to 10% of 320 to have the sum as 30% of 230 is:

(a) 37 (b) 32
(c) 23 (d) 73

[SSC, 2014]

82. The strength of a school increases and decreases in every alternate year by 10%. It started with increase in 2000. Then the strength of the school in 2003 as compared to that in 2000 was:

(a) Increased by 8.9%
(b) Decreased by 8.9%
(c) Increased by 9.8%
(d) Decreased by 9.8%

[SSC, 2014]

83. Two years ago, the value of a motorbike was ₹62,500. If the value depreciates by 4% every year, now its value is:

(a) ₹56,700 (b) ₹57,600
(c) ₹57,500 (d) ₹55,700

[SSC, 2014]

84. to attract more visitors, Zoo authority announces 20% discount on every ticket which cost 25 p. for this reason, sale of ticket increases by 28%. Find the percentage of increase in the number of visitors.

(a) 40% (b) 50%
(c) 60% (d) No change

[SSC, 2014]

85. In two successive years, 80 and 60 students of a school appeared at the final examination of which 60% and 80% passed respectively. The average rate of students passed (in percent) is

(a) 68% (b) $68\frac{4}{7}\%$

(c) 70% (d) $72\frac{3}{7}\%$

[SSC, 2014]

86. If A's salary is 50% more than that of B, then B's salary is less than A's by

(a) 33 % (b) $40\frac{1}{3}\%$

(c) $45\frac{1}{3}\%$ (d) $33\frac{1}{3}\%$

[SSC, 2014]

87. Two numbers A and B are such that the sum of 5% of A and 4% of B is $\frac{2}{3}$rd of the sum of 6 % of A and 8 % of B. The ratio A : B is

(a) 4 : 3 (b) 3 : 4
(c) 1 : 1 (d) 2 : 3

[SSC, 2014]

88. A man spends 75% of his income. His income increases by 20% and his expenditure also increases by 10%. The percentage of increase in his savings is

(a) 40 % (b) 30 %
(c) 50 % (d) 25 %

[SSC, 2014]

89. 20% raise of price followed by a discount of 25% of the raised portion will

(a) increase the price by 15%
(b) decrease the price by 10%
(c) increase the price by 10%
(d) decrease the price by 5%

[SSC, 2014]

90. In an election, a candidate who gets 84% of the votes is elected by a majority of 476 votes. What is total number of votes polled?

(a) 600 (b) 700
(c) 900 (d) 810

[SSC, 2014]

91. A number increased by $22\frac{1}{2}\%$ gives 98. The number is:

(a) 45 (b) 18
(c) 80 (d) 81

[SSC, 2013]

92. In an examination A got 25% marks more than B, B got 10% less than C and C got 25% more than D. If D got 320 marks out of 500, the marks obtained by A were:

(a) 405 (b) 450
(c) 360 (d) 400

[SSC, 2013]

93. Three sets of 40, 50 and 60 students appeared for an examination and the pass percentage was 100, 90 and 80, respectively. The pass percentage of the whole set is:

(a) $88\frac{2}{3}$ (b) $84\frac{2}{3}$

(c) $88\frac{1}{3}$ (d) $84\frac{1}{3}$

[SSC, 2013]

94. A clerk received annual salary of ₹3,660 in the year 1975. This was 20% more than his salary in 1974. What was his salary in 1974?

(a) ₹3,005 (b) ₹3,000
(c) ₹3,500 (d) ₹3,050

[SSC Assistant Grade III, 2013]

95. Out of his total income, Mr. Kapur spends 20% on house rent and 70% of the rest on household expenses. If he saves ₹1,800, what is his total income?

(a) 7,800 (b) 7,000
(c) 8,000 (d) 7,500

[SSC Assistant Grade III, 2013]

96. 134% of 3894 + 38.94 of 134 = ?

(a) 11452 (b) 10000
(c) 10452 (d) 1100
(e) None of these

[IBPS PO/MT, 2013]

97. Rama's expenditure and savings are in the ratio3:2. His income increases by 10 per cent. His expenditure also increases by 12%. His savingsincreases by:

(a) 7% (b) 10%
(c) 9% (d) 13%

[SSC Assistant Grade III, 2012]

98. Two numbers are 30% and 40% more than the third number respectively. The first number is x% of the second. Then x = ?

(a) $105\frac{2}{13}$ (b) 140

(c) $105\frac{5}{7}$ (d) $92\frac{6}{7}$

[SSC Assistant Grade III, 2012]

99. The price of cooking oil has increased by 25%. The percentage of reduction that a family should effect in the use of cooking oil, so as not to increase the expenditure on this account, is:

(a) 15% (b) 20%
(c) 25% (d) 30%

[SSC Assistant Grade III, 2012]

100. In an examination, 52% of the candidates failed in English and 42% failed in Mathematics. If 17% failed in both the subjects, then the percentage of candidates, who passed in both the subjects, was:

(a) 23 (b) 21
(c) 25 (d) 22

[SSC, 2012]

101. In an election there were only two candidates. One of the candidates secured 40% of votes and is defeated by the other candidate by 298 votes. The total number of votes polled is:

(a) 745 (b) 1460
(c) 1490 (d) 1500

[SSC, 2012]

102. 23% of 6783 + 57% of 8431 = ?

(a) 6460 (b) 6420
(c) 6320 (d) 6630
(e) 6360

[IBPS PO/MT, 2012]

103. The sum of three consecutive numbers is 2262. What is 41% of the highest number?

(a) 301.51 (b) 303.14
(c) 308.73 (d) 306.35
(e) 309.55

[IBPS PO/MT, 2012]

104. Akash scored 73 marks in subject A. He scored 56% marks in subject B and X marks in subject C. Maximum marks in each subject were 150. The overall percentage marks obtained by Akash in all three subjects together was 54%. How many marks did he score in subject C?

(a) 84 (b) 86
(c) 79 (d) 73
(e) None of these

[IBPS PO/MT, 2012]

105. When the price of sugar decreases by 10%, a man could buy 1 Kg more for ₹270. Then the original price of sugar per Kg is:

(a) ₹25 (b) ₹30
(c) ₹27 (d) ₹32

[SSC (GL), 2011]

106. First and second numbers are less than a third number by 30% and 37%, respectively. The second number is less than the first by:

(a) 7% (b) 4%
(c) 3% (d) 10%

[SSC (GL), 2011]

107. The price of a commodity rises from ₹6 per Kg to ₹7.50 per Kg. If the expenditure cannot increase, the percentage of reduction in consumption:

(a) 15 (b) 20
(c) 25 (d) 30

[SSC (GL), 2011]

108. Raman scored 456 marks in an examination and Seeta got 54% marks in the same examination which is 24 marks less than Raman. If the minimum passing marks in the examination is 34%, then how much more marks did Raman score than the minimum passing makrs?

(a) 184 (b) 196
(c) 190 (d) 180

[Bank of Baroda PO Examination, 2011]

109. In an Entrance Examination Ritu scored 56% marks, Smita scored 92% marks and Rina scored 634 marks. The maximum marks of the examination are 875. What are the average marks scored by all the three girls together?

(a) 929 (b) 815
(c) 690 (d) 643

[IBPS Bank PO, 2011]

110. In a test, a candidate secured 468 marks out of maximum marks '*A*'. If the maximum marks '*A*' were converted to 700 marks, he would have secured 336 marks. What were the maximum marks of the test?

(a) 775 (b) 875
(c) 975 (d) 1075

[IBPS Bank PO, 2011]

111. An HR Company employs 4800 people, out of which 45% are males and 60% of the males are either 25 years or older. How many males are employed in the company who are younger than 25 years?

(a) 2640 (b) 2160
(c) 1296 (d) 864

[IBPS Bank PO, 2011]

112. Six-elevenths of a number is equal to 22% of second number. Second number is equal to the one-fourth of third number. The value of the third number is 2400. What is the 45% of first number?

(a) 109.8 (b) 111.7
(c) 117.6 (d) None of these

[IBPS Bank PO, 2011]

113. A jar contains 10 red marbles and 30 green ones. How many red marbles must be added to the jar so that 60% of the marbles will be red?

(a) 25 (b) 30
(c) 35 (d) 40

[SSC, 2011]

114. If a number multiplied by 25% of itself gives a number which is 200% more than number, then the number is:

(a) 12 (b) 16
(c) 20 (d) 24

[SSC, 2011]

115. The price of onions has been increased by 50%. In order to keep the expenditure on onions the same the percentage of reduction in consumption has to be:

(a) 50% (b) $33\frac{1}{3}\%$
(c) 33% (d) 30%

[SSC, 2011]

Directions (Q. 116): What approximate value should come in place of question mark (?) in the following question? (Note: You are not expected to calculate the exact value.)

116. 39.897% of 4331 + 58.779% of 5003 =?

(a) 4300 (b) 4500
(c) 4700 (d) 4900
(e) 5100

[IBPS PO/MT, 2011]

117. Ramola's monthly income is three times Ravina's monthly income. Ravina's monthly income is fifteen percent more than Ruchira's monthly income. Ruchira's monthly income is ₹32,000. What is Ramola's annual income?

(a) ₹1,10.400 (b) ₹13,24,800
(c) ₹36,800 (d) ₹52,200
(e) None of these

[IBPS PO/MT, 2011]

118. In a test, a candidates secured 468 marks out of maximum marks 'A'. Had the maximum marks 'A' converted to 700, he would have secured 336 marks. What was the maximum marks of the test?

(a) 775 (b) 875
(c) 975 (d) 1075
(e) None of these

[IBPS PO/MT, 2011]

119. Six-elevenths of a number is equal to 22 per cent of the second number. The second number is equal to one-fourth of the third number. The value of the third number is 2400. What is 45% of the first number?

(a) 109.8 (b) 111.7
(c) 117.6 (d) 123.4
(e) None of these

[IBPS PO/MT, 2011]

Directions (Q. 120): What will come in place of question mark (?) in the following question?

120. 32.05% of 259.99 =?

(a) 92 (b) 88
(c) 78 (d) 90
(e) 83

[SBI Associates Banks PO, 2011]

121. Mr X invested a certain amount in Debt and Equity Funds in the ratio of 4:5. At the end of one year, he earned a total dividend of 30% on his investment. After one year, he reinvested the amount including the dividend in the ratio of 6:7 in Debt and Equity Funds. If the amount reinvested in Equity Funds was ₹94,500,

what was the original amount invested in Equity Funds?

(a) ₹75,000 (b) ₹81,000
(c) ₹60,000 (d) ₹65,000
(e) None of these

[SBI Associates Banks PO, 2011]

122. The product of one-third of a number and 150% of another number is what per cent of the product of the original numbers?

(a) 80% (b) 50%
(c) 75% (d) 120%
(e) None of these

[SBI Associates Banks PO, 2011]

123. Mr Shamin's salary increases every year by 10% in June. If there is no other increase or reduction in the salary and his salary in June 2011 was ₹22,385, what was his salary in June 2009?

(a) ₹18,650 (b) ₹18.000
(c) ₹19,250 (d) ₹18,500
(e) None of these

[SBI Associates Banks PO, 2011]

124. How many students passed in first class?

Statements:

I. 85% of the students who appeared in examination have passed either in first class or in second class or in pass class.

II. 750 students have passed in second class.

III. The number of students who passed in pass class is 28% of those passed in second class.

(a) All I, II and III
(b) Only I and III
(c) Only II and III
(d) Question cannot be answered even with information in all three statements.
(e) None of these

[SBI Associates Banks PO, 2011]

125. 34.5% of 1800 + 12.4% of 1500 = $(?)^3$ + 78:

(a) 27 (b) 9
(c) 81 (d) 162
(e) None of these

[Indian Overseas Bank PO, 2011]

126. 67% of 801 – 231.17 = ? –23% of 789:

(a) 490 (b) 440
(c) 540 (d) 520
(e) 590

[Indian Overseas Bank PO, 2011]

127. Five-ninths of a number is equal to twenty five per cent of the second number. The second number is equal to one-fourth of the third number. The value of the third number is 2960. What is 30 per cent of the first number?

(a) 88.8 (b) 99.9
(c) 66.6 (d) Cannot be determined
(e) None of these

[Indian Overseas Bank PO, 2011]

128. Dinesh's monthly income is four times Suresh's monthly income. Suresh's monthly income is twenty per cent more than Jyoti's monthly income. Jyoti's monthly income is ₹22,000. What is Dinesh's monthly income?

(a) ₹1,06,500 (b) ₹1,05,600
(c) ₹1,04,500 (d) ₹1,05,400
(e) None of these

[Indian Overseas Bank PO, 2011]

129. In a school there are 250 students, out of whom 12 per cent are girls. Each girl's monthly fee is ₹450 and each boy's monthly fee is 24 per cent more than that of a girl. What is the total monthly fee of girls and boys together?

(a) ₹1,36,620 (b) ₹1,36,260
(c) ₹1,32,660 (d) ₹1,32,460
(e) None of these

[Indian Overseas Bank PO, 2011]

130. A sum of ₹731 is distributed among A, B and C, suchthat A receives 25% more than B and B receives 25% less than C. What is C's share in the amount?

(a) ₹172 (b) ₹200
(c) ₹262 (d) ₹258
(e) None of these

[Andhra Bank PO, 2011]

131. Pradeep invested 20% more than Mohit. Mohit invested 10% less than Raghu. If the total sum of their investment is ₹17,880, how much amount did Raghu invest?

(a) ₹6,000 (b) ₹8,000
(c) ₹7,000 (d) ₹5,000
(e) None of these

[Corporation Bank PO, 2011]

132. If the numerator of a fraction is increased by 150% and the denominator of the fraction is increase by 300%, the resultant fraction is $\frac{5}{18}$. What is the original fraction?

(a) $\frac{4}{9}$ (b) $\frac{4}{5}$

(c) $\frac{8}{9}$ (d) $\frac{8}{11}$

(e) None of these

[Punjab and Sind Bank PO, 2011]

133. The price of an article was first increased by 10% and then again by 20%. If the last increased price be ₹33, the original price was:

(a) ₹30 (b) ₹27.50

(c) ₹26.50 (d) ₹25

[SSC (GL), 2010]

134. If an electricity bill is paid before due date, one gets a reduction of 4% on the amount of the bill. By paying the bill before due date a person got a reduction of ₹13. The amount of his electricity bill was:

(a) ₹125 (b) ₹225

(c) ₹325 (d) ₹425

[SSC (GL), 2010]

135. In a test, minimum passing percentage for girls and boys is 35% and 40% respectively. A boy scored 483 marks and failed by 117 marks. What are the minimum passing marks for girls?

(a) 425 (b) 520

(c) 500 (d) None of these

[CBI (PO), 2010]

136. If the numerator of a fraction is increased by 150% and the denominator of the fraction is increased by 350%, the resultant fraction is $\frac{25}{51}$, what is the original fraction?

(a) $\frac{11}{17}$ (b) $\frac{11}{15}$

(c) $\frac{15}{17}$ (d) $\frac{13}{15}$

(e) None of these

[Corporation Bank PO, 2010]

137. When the price of a toy was increased by 20%, the number of toys sold was decreased by 15%. What was its effect on the total sales of the shop?

(a) 2% increase (b) 2% decrease

(c) 4% increase (d) 4% decrease

[SSC, 2010]

138. Krishnamurthy earns ₹15000 per month and spends 80% of it. Due to pay revision, his monthly income has increased by 20% but due to price rise, he has to spend 20% more. His new savings are:

(a) ₹3,400 (b) ₹3,000

(c) ₹3,600 (d) ₹4,000

[SSC, 2010]

139. Two numbers are respectively $12\frac{1}{2}$ % and 25%more than a third number. The first number is how much per cent of the second number?

(a) 90 (b) 87.5

(c) 25 (d) 12.5

[SSC, 2010]

140. Population of a town increases 2.5% annually but is decreased by 0.5% every year due to migration. What will be the percentage of increase in 2 years?

(a) 5 (b) 4.04

(c) 4 (d) 3.96

[SSC, 2010]

141. A merchant has announced 25% rebate on prices of ready-made garments at the time of sale. If a purchaser needs to have a rebate of ₹400, then how many shirts, each costing ₹320 should he purchase?

(a) 10 (b) 7

(c) 6 (d) 5

[SSC, 2010]

142. A reduction of 10% in the price of tea enables a dealer to purchase 25 Kg more tea for ₹22500. What is the reduced price per Kg of tea?

(a) ₹70 (b) ₹80

(c) ₹90 (d) ₹100

[SSC, 2010]

143. Ram donated 4% of his income to a charity and deposited 10% of the rest in a Bank. If now he has ₹8640 left with him, then his income is:

(a) ₹12,500 (b) ₹12,000

(c) ₹10,500 (d) ₹10,000

[SSC, 2010]

144. Twelve per cent of Kaushal's monthly salary is equal to sixteen per cent of Nandini's monthly salary. Suresh's monthly salary is half that of Nandini's monthly salary. If Suresh's annual salary is ₹1.08 Lakhs, what is Kaushal's monthly salary?

(a) ₹20,000 (b) ₹18,000

(c) ₹26,000 (d) ₹24,000

(e) None of these

[Central Bank of India PO, 2010]

145. Rita invested 25% more than Sunil. Sunil invested 30% less than Abhinav, who invested ₹6,000. What is the ratio of the amount that Rita invested to the total amount invested by all of them together?

(a) 35:104 (b) 13:29
(c) 101:36 (d) 35:103
(e) None of these

[Punjab and Sind Bank PO, 2010]

146. 15% of 578 + 22.5% of 644 = ?
(a) 213.4 (b) 233.6
(c) 231.8 (d) 231.6
(e) None of these

[Indian Bank PO, 2010]

147. Sonu invested 10% more than Mona. Mona invested 10% less than Raghu. If the total sum of their investments is ₹5,780, how much amount did Raghu invest?
(a) ₹2,010 (b) ₹2,000
(c) ₹2,100 (d) ₹2,210
(e) None of these

[Indian Bank PO, 2010]

ANSWER KEYS

EXERCISE-1

1. (a)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (c)	**11.** (c)	**12.** (b)	**13.** (c)
14. (a)	**15.** (b)	**16.** (b)	**17.** (c)	**18.** (b)	**19.** (b)	**20.** (c)	**21.** (b)	**22.** (c)	**23.** (b)	**24.** (c)	**25.** (a)	**26.** (b)
27. (c)	**28.** (b)	**29.** (b)	**30.** (a)	**31.** (b)	**32.** (a)	**33.** (b)	**34.** (a)	**35.** (b)	**36.** (c)	**37.** (d)	**38.** (b)	**39.** (b)
40. (b)	**41.** (a)	**42.** (c)	**43.** (a)	**44.** (a)	**45.** (b)	**46.** (b)	**47.** (a)	**48.** (a)	**49.** (c)	**50.** (a)	**51.** (a)	**52.** (b)
53. (c)	**54.** (c)	**55.** (b)	**56.** (c)	**57.** (c)	**58.** (b)	**59.** (a)	**60.** (b)	**61.** (b)	**62.** (c)	**63.** (b)	**64.** (b)	**65.** (a)
66. (b)	**67.** (a)	**68.** (a)	**69.** (c)	**70.** (a)	**71.** (a)							

EXERCISE-2

1. (d)	**2.** (c)	**3.** (a)	**4.** (b)	**5.** (b)	**6.** (b)	**7.** (a)	**8.** (d)	**9.** (d)	**10.** (a)	**11.** (d)	**12.** (c)	**13.** (a)
14. (c)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (a)	**19.** (b)	**20.** (b)	**21.** (d)	**22.** (c)	**23.** (c)	**24.** (b)	**25.** (b)	**26.** (a)
27. (b)	**28.** (c)	**29.** (b)	**30.** (c)	**31.** (d)	**32.** (c)	**33.** (a)	**34.** (d)	**35.** (c)	**36.** (a)	**37.** (b)	**38.** (c)	**39.** (a)
40. (a)	**41.** (b)	**42.** (c)	**43.** (b)	**44.** (c)	**45.** (c)	**46.** (a)	**47.** (b)	**48.** (c)	**49.** (a)	**50.** (a)	**51.** (b)	**52.** (b)
53. (c)	**54.** (a)	**55.** (b)	**56.** (b)	**57.** (d)	**58.** (c)	**59.** (a)	**60.** (b)	**61.** (c)	**62.** (c)	**63.** (a)	**64.** (a)	**65.** (c)
66. (c)	**67.** (b)	**68.** (b)	**69.** (d)	**70.** (d)	**71.** (a)	**72.** (c)	**73.** (b)	**74.** (a)	**75.** (a)	**76.** (a)	**77.** (d)	**78.** (b)
79. (c)	**80.** (b)	**81.** (a)	**82.** (a)	**83.** (b)	**84.** (c)	**85.** (b)	**86.** (d)	**87.** (a)	**88.** (c)	**89.** (b)	**90.** (b)	**91.** (c)
92. (b)	**93.** (a)	**94.** (d)	**95.** (d)	**96.** (c)	**97.** (a)	**98.** (d)	**99.** (b)	**100.** (a)	**101.** (c)	**102.** (e)	**103.** (e)	**104.** (b)
105. (b)	**106.** (d)	**107.** (b)	**108.** (a)	**109.** (d)	**110.** (c)	**111.** (d)	**112.** (d)	**113.** (c)	**114.** (a)	**115.** (b)	**116.** (c)	**117.** (b)
118. (c)	**119.** (e)	**120.** (e)	**121.** (a)	**122.** (b)	**123.** (d)	**124.** (d)	**125.** (b)	**126.** (a)	**127.** (b)	**128.** (b)	**129.** (b)	**130.** (e)
131. (a)	**132.** (a)	**133.** (d)	**134.** (c)	**135.** (d)	**136.** (c)	**137.** (a)	**138.** (c)	**139.** (a)	**140.** (c)	**141.** (d)	**142.** (c)	**143.** (d)
144. (d)	**145.** (d)	**146.** (d)	**147.** (b)									

EXPLANATORY ANSWERS

EXERCISE-I

1. (a) $5\frac{1}{4} = \frac{21}{4} = \frac{21}{4} \times 100 = 525\%$

2. (b) $0.005 = \frac{5}{1000} = \frac{5}{1000} \times 100 = \frac{1}{2}\%$

3. (b) $6\frac{2}{3}\% = \frac{\left(\frac{20}{3}\right)}{100} = \left(\frac{20}{3} \times \frac{1}{100}\right) = \frac{1}{15}$.

4. (c) $0{\cdot}6\% = \frac{0{\cdot}6}{100} = \frac{6}{1000} = \frac{3}{500}$.

5. (b) $.025 = \left(\frac{25}{1000} \times 100\right)\% = 2.5\%$

6. (c) Let x% of 12 = 84

$\Rightarrow \frac{x}{100} \times 12 = 84$

$\Rightarrow x = \frac{84 \times 100}{12} = 700.$

$\therefore$ 700% of 12 is 84.

7. (b) $\frac{7}{8} = \left(\frac{7}{8} \times 100\right)\% = \frac{175}{2}\% = 87\frac{1}{2}\%$

8. (a) $8\frac{1}{3}\% = \frac{25}{3}\% = \frac{25}{3} \times \frac{1}{100} = \frac{1}{12}$.

9. (c) $37\frac{1}{2}\%$ of ₹48 $= 48 \times \frac{75}{2 \times 100} =$ ₹18.

Alternative Solution

$37\frac{1}{2}\%$ of ₹48

$37\frac{1}{2}\% = \frac{3}{8}$

$\therefore \frac{3}{8} \times 48 =$ ₹18

10. (c) Let x% of $\frac{2}{7} = \frac{1}{35}$

$\Rightarrow x = \frac{100 \times 7}{2 \times 35} = 10.$

$\therefore$ 10% of $\frac{2}{7}$ is $\frac{1}{35}$.

11. (c) Let 75% of 480 = $x \times 15$.

Then, $\frac{75}{100} \times 480 = 15x$

or, $x = \frac{75 \times 480}{100 \times 15} = 24.$

12. (b) Let 200% of $x = 90 \Rightarrow \frac{200}{100} \times x = 90$

$\Rightarrow x = \frac{100 \times 90}{200} = 45.$

$\therefore$ 80% of 45 $= \frac{80}{100} = 45 = 36.$

13. (c) Let the number be x, then

$37\frac{1}{2}\%$ of $x = 45 \Rightarrow \frac{75}{2} \times \frac{1}{100} \times x = 45$

or, $\frac{3}{8}x = 45 \Rightarrow x = \frac{45 \times 8}{3} = 120.$

$\therefore 87\frac{1}{2}\%$ of 120 $= \frac{75}{2} \times \frac{1}{100} \times 120 = 105.$

14. (a) Let $x \times 15 = 37.5\%$ of 220

$\Rightarrow 15x = \frac{37.5}{100} \times 220$

$\Rightarrow x = \frac{37.5 \times 220}{15 \times 100} = 5.5.$

15. (b) Let x% of 4 Km = 8 metre

$\Rightarrow \frac{x}{100} \times 4000 = 8$ ($\because$ 1 Km = 1000 metre)

$\Rightarrow x = \frac{8 \times 100}{4000} = \frac{1}{5} = 0.2$

$\therefore$ 0.2% of 4 Km = 8 metre.

16. (b) x% of $y + y$% of $x = \left(\frac{x}{100} \times y\right) + \left(\frac{y}{100} \times x\right)$

$= \frac{2}{100}xy = 2\%$ of xy.

17. (c) $0.35\% = 0{\cdot}35 \times \frac{1}{100} = 0.0035.$

18. (b) We have, 8% of x = 4% of y

$\Rightarrow \frac{8}{100} \times x = \frac{4}{100} \times y \Rightarrow x = \left(\frac{4}{100} \times \frac{100}{8}\right)y = \frac{y}{2}$.

$\therefore$ 20% of $x = \frac{20}{100} \times x = \frac{20}{100} \times \frac{y}{2}$

$= \frac{10}{100} \times y$

$= 10\%$ of y.

19. (b) Let the missing figure be a.

We have, $\frac{x}{100} \times y + \frac{a}{100} \times x = \frac{x}{100} \times (x + y)$

$\Rightarrow xy + ax = x(x + y)$

$\Rightarrow ax = x^2$. $\therefore \quad a = x$.

20. (c) We have, $x = 125\%$ of $y \Rightarrow x = \frac{125}{100}y = \frac{5}{4}y$

or, $y = \frac{4}{5}x = 0.8x$.

21. (b) 25% of 25% $= \frac{25}{100} \times \frac{25}{100} = \frac{625}{10000} = 0.0625$.

22. (c) Required number = 80 – 60% of 80

$= 80 - \frac{60}{100} \times 80 = 32$.

23. (b) 20% of 30% of 20% of 850

$= \frac{20}{100} \times \frac{30}{100} \times \frac{20}{100} \times 850 = \frac{1020}{100} = ₹10.20$.

24. (c) $16\frac{2}{3}\% = \frac{50}{3} \times \frac{1}{100} \times \frac{1}{6} = 0.167 < 0.3$.

$6\frac{2}{3}\% = \frac{20}{3} \times \frac{1}{100} = \frac{1}{15} = 0.067 < 0.3$.

$\therefore$ 0.3 is greatest.

25. (a) 40% of 20% $= \frac{40}{100} \times \frac{20}{100} = \frac{8}{100} = 8\%$

30% of 25% $= \frac{30}{100} \times \frac{25}{100} = \frac{75}{100} = 7.5\%$

and, 50% of 28% $= \frac{50}{100} \times \frac{28}{100} = \frac{14}{100} = 14\%$

$\therefore$ (40% of 20% + 30% of 25% + 50% of 28%)

$= 8\% + 7.5\% + 14\% = 29.5\%$.

26. (b) We have, $\frac{90}{100} \times A = \frac{30}{100} \times B$

$= \frac{30}{100} \times \frac{x}{100} \times A$

$\therefore \quad x = \left(100 \times \frac{100}{30} \times \frac{90}{100}\right) = 300$.

Alternative Solution

$90\%\ A = 30\%\ B$

$\frac{A}{B} = \frac{1}{3}$

$B = 3A$

$\therefore x = 300$

27. (c) Let 1 quintal 25 Kg = x% of 1 metric tonne

$\Rightarrow x\% = \frac{125}{1000} = \frac{1}{8} = \frac{1}{8} \times 100\%$

$= 12\frac{1}{2}\%$

28. (b) We have, 12% of x = 6% of y

$\Rightarrow$ 2% of x = 1% of y

$\Rightarrow (2 \times 9)\%$ of $x = (1 \times 9)\%$ of y

$\Rightarrow$ 18% of x = 9% of y.

29. (b) Here, $x = 20$.

$\therefore$ Required answer $= \left(\frac{x}{100 + x} \times 100\right)\%$

$= \left(\frac{20}{100 + 20} \times 100\right)\% = 16\frac{2}{3}\%$

30. (a) Here, $x = 25$.

$\therefore$ Required answer $= \left(\frac{x}{100 - x} \times 100\right)\%$

$= \left(\frac{25}{100 - 25} \times 100\right)\%$

$= 33\frac{1}{3}\%$

Alternative Solution

(a) Let B's income = ₹100

$\therefore$ A's income = 75% (100) = ₹75

Req. % $= \frac{25}{75} \times 100 = 33\frac{1}{3}\%$

31. (b) Here, $l = 7$ and $m = 28$.

$\therefore$ First number $= \frac{l}{m} \times 100\%$ of second number

$= \frac{7}{28} \times 100\%$ of second number

or, 25% of second number.

32. (a) Here, $x = 60$ and $y = 20$.

$\therefore$ Second number $= \left(\frac{100 + y}{100 + x} \times 100\right)\%$ of the first

$= \left(\frac{100 + 20}{100 + 60} \times 100\right)\%$ of the first

i.e., 75% of the first.

33. (b) Here, $x = 30$ and $y = 37$.

$\therefore$ Second number $= \left(\frac{100 - y}{100 - x} \times 100\right)\%$ of the first

$= \left(\frac{100 - 37}{100 - 30} \times 100\right)\%$ of the first

i.e., 90% of the first.

34. (a) Here $x = 20$ and $y = 10$.

$\therefore$ First number $= \left(\frac{100+x}{100+y}\times 100\right)\%$ of the second

$= \left(\frac{100+20}{100+10}\times 100\right)\%$ of the second

i.e., $109\frac{1}{11}\%$ of the second.

$\therefore$ The first number is $9\frac{1}{11}\%$ more than the second.

35. (b) Reduction in consumption

$= \left(\frac{P}{100+P}\times 100\right)\%$

$= \left(\frac{15}{100+15}\times 100\right)\%$ or $13\frac{1}{23}\%$

Alternative Solution

(b) Price $\uparrow = 15\% \Rightarrow \frac{15}{100} = \frac{3}{20}$

$\frac{a}{b} = \frac{3}{20}$

Consumption $\downarrow = \left(\frac{a}{a+b}\times 100\right)\%$

$= \left(\frac{3}{23}\times 100\right)\%$

$= 13\frac{1}{23}$

36. (c) Increase in consumption

$= \left(\frac{P}{100-P}\times 100\right)\%$

$= \left(\frac{10}{100-10}\times 100\right)\%$ or $11\frac{1}{9}\%$

Alternative Solution

(c) Price $\downarrow = 10\% = \frac{1}{10}$

$\frac{a}{b} = \frac{1}{10}$

Consumption $\uparrow = \left(\frac{a}{a-b}\times 100\right)\%$

$= \left(\frac{1}{9}\times 100\right)\% = 11\frac{1}{9}$

37. (d) Here, $x = 20$ and $y = -20$.

$\therefore$ The net % change in value

$= \left(x+y+\frac{xy}{100}\right)\%$

$= \left(20-20-\frac{20\times 20}{100}\right)\%$ or -4%.

Since the sign is –ve, there is decrease in value by 4%

38. (b) Here, $x = -50$ and $y = 50$.

$\therefore$ The net % change in wages

$= \left(x+y+\frac{xy}{100}\right)\%$

$= \left(-50+50-\frac{50\times 50}{100}\right)\%$

or, -25%.

Since the sign is –ve, he has a loss of 25%

Alternative Solution

(b) If there is $x\%$ $\uparrow$ the $x\%$ $\downarrow$ is made, then there will be always $\left(\frac{x}{10}\right)^2\%$ loss occur

$x = 50$

$\therefore$ loss $= \left(\frac{50}{10}\right)^2\% = 25\% \downarrow$

39. (b) Here, $x = -20$ and $y = -25$.

$\therefore$ The net % change in population

$= \left(x+y+\frac{xy}{100}\right)\%$

$= \left(-20-25+\frac{20\times 25}{100}\right)$ or -40%

Since the sign is –ve, the population is decreased by 40% after two years.

40. (b) The equivalent discount of two successive discounts of 20% and 20%

$= \left(x+y+\frac{xy}{100}\right)\%$

$= \left(-20-20+\frac{20\times 20}{100}\right)\%$ or -36%

Given: 36% – 35% = ₹22.

$\therefore$ Amount of the bill = 22 × 100 = ₹2200.

41. (a) Here, $x = 25$ and $y = -12$.

$\therefore$ The net% change in original price

$= \left(x+y+\frac{xy}{100}\right)$

$= \left(25-12-\frac{25\times 12}{100}\right)\%$ or 10%

Since the sign is +ve, there is a profit of 10%

42. (c) The equivalent discount of two successive discounts of 20% and 10%

$= \left(x+y+\frac{xy}{100}\right)\% = \left(-20-10+\frac{20\times 10}{100}\right)\%$ or 28%

CHAPTER 9

Profit and Loss

INTRODUCTION

Nowadays, transactions have become a common feature of life. When a person deals in the purchase and sale of any item, he either gains or loses some amount generally. The aim of any business is to earn profit. The commonly used terms in dealing with questions involving sale and purchase are:

Cost Price: The cost price of an article is the price at which an article has been purchased. It is abbreviated as C.P.

Selling Price: The selling price of an article is the price at which an article has been sold. It is abbreviated as S.P.

Profit or Gain: If the selling price of an article is more than the cost price, there is a gain or profit.

Thus, Profit or Gain = S.P. − C.P.

Loss: If the cost price of an article is greater than the selling price, the seller suffers a loss.

Thus, Loss = C.P. − S.P.

Note that profit and loss are always calculated with respect to the cost price of the item.

Illustration 1: (i) If C.P. = ₹235, S.P. = ₹240, then profit = ?

(ii) If C.P. = ₹116, S.P. = ₹107, then loss = ?

Solution: (i) Profit = S.P. − C.P. = 240 − 235 = ₹5.

(ii) Loss = C.P. − S.P. = 116 − 107 = ₹9.

BASIC FORMULAE

01 Gain on ₹100 is *Gain per cent*

$$\text{Gain \%} = \frac{\text{Gain} \times 100}{\text{C.P.}}$$

Loss on ₹100 is *Loss per cent*

$$\text{Loss \%} = \frac{\text{Loss} \times 100}{\text{C.P.}}$$

Illustration 2: The cost price of a shirt is ₹200 and selling price is ₹250. Calculate the % of profit.

Solution: We have, C.P. = ₹200, S.P. = ₹250.

Profit = S.P. − C.P. = 250 − 200 = ₹50.

$$\therefore \quad \text{Profit \%} = \frac{\text{Profit} \times 100}{\text{C.P.}} = \frac{50 \times 100}{200} = 25\%$$

Illustration 3: Anu bought a necklace for ₹750 and sold it for ₹675. Find her percentage of loss.

Solution: Here, C.P. = ₹750, S.P. = ₹675.

Loss = C.P. − S.P. = 750 − 675 = ₹75.

$$\therefore \quad \text{Loss \%} = \frac{\text{Loss} \times 100}{\text{C.P.}} = \frac{75 \times 100}{750} = 10\%$$

02 When the selling price and gain % are given:

$$\text{C.P.} = \left(\frac{100}{100 + \text{Gain\%}}\right) \times \text{S.P.}$$

03 When the cost and gain per cent are given:

$$\text{S.P.} = \left(\frac{100 + \text{Gain\%}}{100}\right) \times \text{C.P.}$$

Explanation

$$\text{Since Profit \%} = \frac{\text{Profit} \times 100}{\text{C.P.}}$$

$$= \left[\frac{(\text{S.P.} - \text{C.P.}) \times 100}{\text{C.P.}}\right]$$

$$\therefore \quad \frac{\text{Profit \%}}{100} = \frac{\text{S.P.}}{\text{C.P.}} - 1$$

$$\text{or,} \quad \frac{\text{S.P.}}{\text{C.P.}} = 1 + \frac{\text{Profit \%}}{100}$$

$$\therefore \quad \text{S.P.} = \left(\frac{100 + \text{Profit \%}}{100}\right) \times \text{C.P.}$$

$$\text{and, C.P.} = \left(\frac{100}{100 + \text{Profit \%}}\right) \times \text{S.P.}$$

04 When the cost and loss per cent are given:

$$\text{S.P.} = \left(\frac{100 - \text{Loss \%}}{100}\right) \times \text{C.P.}$$

05 When the selling price and loss per cent are given:

$$\text{C.P.} = \left(\frac{100}{100 - \text{Loss \%}}\right) \times \text{S.P.}$$

Explanation

$$\text{Since Loss \%} = \frac{\text{Loss} \times 100}{\text{C.P.}}$$

$$= \left[\frac{(\text{C.P.} - \text{S.P.}) \times 100}{\text{C.P.}}\right]$$

$$\therefore \quad \frac{\text{Loss\%}}{100} = 1 - \frac{\text{S.P.}}{\text{C.P.}}$$

$$\text{or} \quad \frac{\text{S.P.}}{\text{C.P.}} = 1 - \frac{\text{Loss\%}}{100}$$

$$\therefore \quad \text{S.P.} = \left(\frac{100 - \text{Loss\%}}{100}\right) \times \text{C.P.}$$

$$\text{and,} \quad \text{C.P.} = \left(\frac{100}{100 - \text{Loss\%}}\right) \times \text{S.P.}$$

Illustration 4: Mr Sharma buys a cooler for ₹4500. For how much should he sell it to gain 8%?

Solution: We have, C.P. = ₹4500, gain % = 8%

$$\therefore \quad \text{S.P.} = \left(\frac{100 + \text{Gain \%}}{100}\right) \times \text{C.P.}$$

$$= \left(\frac{100 + 8}{100}\right) \times 4500$$

$$= \frac{108}{100} \times 4500 = ₹4860.$$

Illustration 5: By selling a fridge for ₹7200, Pankaj loses 10%. Find the cost price of the fridge.

Solution: We have, S.P. = ₹7200, loss % = 10%

$$\therefore \quad \text{C.P.} = \left(\frac{100}{100 - \text{Loss\%}}\right) \times \text{S.P.}$$

$$= \left(\frac{100}{100 - 10}\right) \times 7200$$

$$= \frac{100}{90} \times 7200 = ₹8000.$$

Illustration 6: By selling a pen for ₹99, Mohan gains $12\frac{1}{2}$%. Find out cost price of the pen.

Solution: Here, S.P. = ₹99, gain % = $12\frac{1}{2}$% or $\frac{25}{2}$%

$$\therefore \quad \text{C.P.} = \left(\frac{100}{100 + \text{Gain\%}}\right) \times \text{S.P.}$$

$$= \left(\frac{100}{100 + \frac{25}{2}}\right) \times 99$$

$$= \left(\frac{100 \times 2}{225}\right) \times 99 = ₹88.$$

SOME USEFUL SHORTCUT METHODS

01 If a man buys x items for ₹y and sells z items for ₹w, then the gain or loss per cent made by him is

$$\left(\frac{xw}{zy} - 1\right) \times 100\%$$

Explanation

S.P. of z iterms = ₹w

S.P. of x items = ₹$\frac{w}{z}x$

Net profit = $\frac{w}{z}x - y$.

$$\therefore \quad \% \text{ Profit} = \frac{\frac{w}{z}x - y}{y} \times 100\%$$

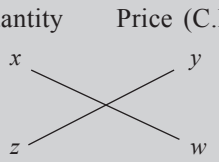
i.e., $\left(\frac{xw}{zy} - 1\right) \times 100\%$

which represents loss, if the result is negative.

Notes

In the case of gain per cent, the result obtained bears positive sign whereas in the case of loss per cent the result obtained bears sign negative.

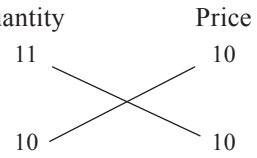
How to Remember?



1. Cross-multiply the numbers connected by the arrows (xw and zy).
2. Mark the directon of the arrows for cross-multiplicaton. The arrow going down forms the numerator while the arrow going up forms the denominator $\left(\frac{xw}{zy}\right)$.

Illustration 7: If 11 oranges are bought for ₹10 and sold at 10 for ₹11, what is the gain or loss %?

Solution:



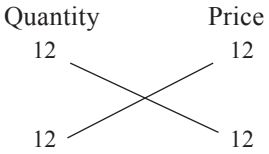
$$\% \text{ profit} = \left(\frac{xw}{zy} - 1\right) \times 100\%$$

$$= \left(\frac{11 \text{ x } 11}{10 \text{ x } 10} - 1\right) \times 100\%$$

$$= \frac{21}{100} \times 100\% = 21\%$$

Illustration 8: A fruit seller buys apples at the rate of ₹12 per dozen and sells them at the rate of 15 for ₹12. Find out his percentage gain or loss.

Solution:



$$\% \text{ gain or loss} = \left(\frac{xw}{zy} - 1\right) \times 100\%$$

$$= \left(\frac{12 \times 12}{15 \times 12} - 1\right) \times 100\%$$

$$\left(\frac{144}{180} - 1\right) \times 100\%$$

$$\left(\frac{-36}{180} \times 100\%\right)$$

Since the sign is –ve, there is a loss of 25%

02 If the cost price of m articles is equal to the selling price of n articles, then

$$\% \text{ gain or loss} = \left(\frac{m-n}{n}\right) \times 100$$

[If $m > n$, it is % gain and, if $m < n$, it is % loss]

Explanation

Let, the C.P. of an article be ₹1.

$\therefore$ C.P. of m articles = ₹m × 1 = ₹m

$\because$ S.P. of n articles = ₹m

$\therefore$ S.P. of an article = ₹$\frac{m}{n}$

$\therefore$ Profit on 1 article = ₹$\left(\frac{m}{n} - 1\right)$

i.e., ₹$\left(\frac{m-n}{n}\right)$

$\therefore$ % profit = $\left(\frac{m-n}{n}\right) \times 100$ i.e., $\left(\frac{m-n}{n}\right) \times 100$

Illustration 9: A shopkeeper professes to sell his goods on cost price, but uses 800 gm, instead of 1 Kg. What is his gain %?

Solution: Here, cost price of 1000 gm is equal to selling price of 800 gm,

$$\therefore \quad \% \text{ gain} = \left(\frac{m-n}{n}\right) \times 100$$

$$= \left(\frac{1000 - 800}{800}\right) \times 100$$

$$= \frac{200}{800} \times 100 = 25\%$$

Illustration 10: If the selling price of 12 articles is equal to the cost price of 18 articles, what is the profit %?

Solution: Here, $m = 18$, $n = 12$

$$\therefore \quad \text{Profit \%} = \left(\frac{m-n}{n}\right)\times 100$$

$$= \left(\frac{18-12}{12}\right)\times 100$$

$$= \frac{6}{12}\times 100 = 50\%$$

03 If an article is sold at a price $S.P._1$., then % gain or % loss is x and if it is sold at a price $S.P_2$., then % gain or % loss is y. If the cost price of the article is C.P., then

$$\frac{S.P._1}{100+x} = \frac{S.P._2}{100+y} = \frac{C.P.}{100} = \frac{S.P._1 - S.P._2}{x-y},$$

where x or y is –ve, if it indicates a loss, otherwise it is +ve.

Illustration 11: By selling a radio for ₹1536, Suresh lost 20%. What per cent shall he gain or lose by selling it for ₹2000?

Solution: Here, $S.P._1 = 1536$, $x = -20$

(–ve sign indicates loss)

$S.P._2 = ₹2000$, $y = ?$

Using the formula:

$$\frac{S.P._1}{100+x} = \frac{S.P._2}{100+y},$$

we get, $\frac{1536}{100-20} = \frac{2000}{100+y}$

$$\Rightarrow \quad 100 + y = \frac{2000\times 80}{1536} = 104\frac{1}{6}$$

$$\Rightarrow \quad y = 4\frac{1}{6}\%$$

Thus, Suresh has a gain of $4\frac{1}{6}\%$ by selling it for ₹2000.

04 If 'A' sells an article to 'B' at a gain/loss of $m\%$ and 'B' sells it to 'C' at a gain/loss of $n\%$. If 'C' pays ₹z for it to 'B', then the cost price for 'A' is

$$\left[\frac{100^2 z}{(100+m)(100+n)}\right]$$

where m or n is –ve, of it indicates a loss, otherwise it is +ve.

Illustration 12: Mohit sells a bicycle to Rohit at a gain of 10% and Rohit again sells it to Jyoti at a profit of 5%. If Jyoti pays ₹462 to Rohit, what is the cost price of the bicycle for Mohit?

Solution: Here, $m = 10$, $n = 5$, $z = ₹462$.

Using the formula,

$$\text{C.P.} = \left[\frac{100^2 z}{(100+m)(100+n)}\right]$$

we get, C.P. for Mohit $= \left[\frac{100^2\times 462}{(100+10)(100+5)}\right]$

$$= \frac{462\times 10000}{110\times 105} = ₹400.$$

Illustration 13: 'A' sells a DVD to 'B' at a gain of 17% and 'B' again sells it to 'C' at a loss of 25%. If 'C' pays ₹1053 to 'B', what is the cost price of the DVD to 'A'?

Solution: We have, $m = 17$, $n = -25$, $z = ₹1053$.

$\therefore$ Cost price of DVD to

$$= \left[\frac{100^2 z}{(100+m)(100+n)}\right]$$

$$= \frac{100\times 100\times 1053}{(100+17)(100-25)}$$

$$= \frac{100\times 100\times 1053}{117\times 75} = ₹1200.$$

05 If 'A' sells an article to 'B' at a gain/loss of $m\%$, and 'B' sells it to 'C' at a gain/loss of $n\%$, then the resultant profit/loss per cent is given by

$$\left(m+n+\frac{mn}{100}\right) \qquad \cdots(1)$$

Where m or n is –ve, if it indicates a loss, otherwise it is +ve.

Notes

The expression given by eq. (1) represents resultant profit or loss according as it is +ve or –ve.

Illustration 14: 'A' sells a horse to 'B' at a profit of 5% and 'B' sells it: to 'C' at a profit of 10%. Find out the resultant profit per cent.

Solution: We have, $m = 5$ and $n = 10$.

$$\therefore \quad \text{Resultant profit \%} = \left(m+n+\frac{mn}{100}\right)$$

$$= \left(5+10+\frac{5\times 10}{100}\right)$$

$$= \frac{31}{2}\% \quad \text{or,} \quad 15\frac{1}{2}\%$$

Illustration 15: Manoj sells a shirt to Yogesh at a profit of 15%, and Yogesh sells it to Suresh at a loss of 10%. Find the resultant profit or loss.

Solution: Here, $m = 15$, $n = -10$

$$\therefore \quad \text{Resultant profit/loss \%} = \left(m + n + \frac{mn}{100}\right)$$

$$= \left(15 - 10 + \frac{15 \times -10}{100}\right) = \left(15 - 10 - \frac{150}{100}\right)$$

$$= 7/2\% \quad \text{or} \quad 3\frac{1}{2}\%,$$

which represents profit as the sign is +ve.

06 When two different articles are sold at the same selling price, getting gain/loss of $x\%$ on the first and gain/loss of $y\%$ on the second, then the overall % gain or % loss in the transaction is given by

$$\left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)}\right]\%$$

The above expression represent overall gain or loss according as its sign is +ve or –ve.

07 When two different articles are sold at the same selling price getting a gain of $x\%$ on the first and loss of $x\%$ on the second, then the overall % loss in the transaction is given by

$$\left(\frac{x}{10}\right)^2 \%$$

Note that in such questions, there is always a loss.

Explanation

Let, each article be sold at ₹z.

Since gain/loss of $x\%$ is made on the first, cost price of the first article

$$= ₹z\left(\frac{100}{100 + x}\right)$$

Also, gain/loss of $y\%$ is made on the second. Therefore, cost price of the second article

$$= ₹z\left(\frac{100}{100 + y}\right)$$

$$\therefore \quad \text{Total C.P.} = z\left(\frac{100}{100 + x}\right) + z\left(\frac{100}{100 + y}\right)$$

$$= z\left[\frac{100(100 + y) + 100(100 + x)}{(100 + x)(100 + y)}\right]$$

Total S.P. $= 2z$.

$$\therefore \quad \text{Overall \% gain or loss} = \frac{\text{S.P.} - \text{C.P.}}{\text{C.P.}} \times 100$$

$$= \frac{2z - \dfrac{100z[100 + x + 100 + y]}{(100 + x)(100 + y)}}{\dfrac{100z[100 + x + 100 + y]}{(100 + x)(100 + y)}} \times 100$$

$$= \frac{2(100 + x)(100 + y) - 100(200 + x + y)}{100(200 + x + y)} \times 100$$

$$= \frac{100x + 100y + 2xy}{(100 + x) + (100 + y)}\%$$

$$= \left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)}\right]\%.$$

Notes

In case $y = -x$, we have

Overall % gain or loss $= -\dfrac{x^2}{100}\%$,

Since the sign is –ve, there is always a loss.

Illustration 16: Mahesh sold two scooters, each for ₹24000. If he makes 20% profit on the first and 15% loss on the second, what is his gain or loss per cent in the transactions?

Solution: Here, $x = 20$ and $y = -15$.

$$\therefore \quad \text{Over all gain/loss \%}$$

$$= \left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)}\right]\%$$

$$= \left[\frac{100(20 - 15) + 2 \times 20 \times -15}{(100 + 20) + (100 - 15)}\right]\%$$

$$= -\frac{100}{205}\% = -\frac{20}{41}\%$$

which represents loss, being a –ve expression.

Illustration 17: Rajesh sold two horses for ₹990 each; gaining 10% on the one and losing 10% on the other. Find out his total gain or loss per cent.

Solution: Here, $x = 10$.

$$\therefore \quad \text{Overall loss \%} = \left(\frac{x}{10}\right)^2 \% = \left(\frac{10}{10}\right)^2 \% = 1\%$$

08 A merchant uses faulty measure and sells his goods at gain/loss of $x\%$. The overall % gain/loss(g) is given by

$$\frac{100 + g}{100 + x} = \frac{\text{True measure}}{\text{Faulty measure}}$$

Notes

If the merchant sells his goods at cost price, then $x = 0$.

09 A merchant uses $y\%$ less weight/length and sells his goods at gain/loss of $x\%$. The overall % gain/loss is given by

$$\left[\left(\frac{y+x}{100-y}\right)\times 100\right]\%$$

Illustration 18: A dishonest shopkeeper professes to sell cloth at the cost price, but he uses faulty meter rod. His meter rod measures 95 cm only. Find his gain per cent.

Solution: Here, true measure = 100 cm.

False measure = 95 cm.

Since the shopkeeper sells the cloth at cost price, $\therefore x = 0$.

$\therefore$ Overall gain % is given by

$$\frac{100+g}{100+x} = \frac{\text{True measure}}{\text{Faulty measure}}$$

$$\Rightarrow \quad \frac{100+g}{100} = \frac{100}{95} \Rightarrow 100 + g = \frac{100\times 100}{95}$$

$$\Rightarrow \quad g = \frac{10000}{95} - 100 = 5\frac{5}{19}\%$$

Illustration 19: A dishonest shopkeeper professes to sell his goods at cost price, but he uses a weight of 800 g for the Kg weight. Find out his gain per cent.

Solution: True measure = 1000 g.

False measure = 800 g. Also, $x = 0$.

$\therefore$ Overall gain % is given by

$$\frac{100+g}{100+x} = \frac{\text{True measure}}{\text{False measure}}$$

$$\Rightarrow \quad \frac{100+g}{100} = \frac{1000}{800} \Rightarrow 100 + g = \frac{1000\times 100}{800}$$

$$\Rightarrow \quad g = \frac{1000}{8} - 100 = 25\%$$

Illustration 20: A shopkeeper sells goods at 44% loss on cost price, but uses 30% less weight. What is his percentage profit or loss?

Solution: Here, $x = -44$ and $y = 30$.

$$\therefore \text{ Overall gain/loss\%} = \left(\frac{y+x}{100-y}\right)\times 100\%$$

$$= \left(\frac{30-44}{100-30}\times 100\right)\%$$

$$= \left(\frac{-14}{70}\times 100\right)\% = -20\%,$$

which represents loss being a negative expression.

10 A person buys two items for ₹A and sells one at a loss of $l\%$ and the other at a gain of $g\%$. If each item was sold at the same price, then

(a) The cost price of the item sold at loss

$$= \frac{A(100 + \%\text{gain})}{(100 - \%\text{loss}) + (100 + \%\,\text{gain})}$$

(b) The cost price of the item sold at gain

$$= \frac{A(100 - \%\,\text{loss})}{(100 - \%\text{loss}) + (100 + \%\,\text{gain})}$$

Illustration 21: Ramesh buys two books for ₹410. He sells one at a loss of 20% and the other at a gain of 25%. If both the books are sold at the same price, find out the cost price of two books.

Solution: Cost price of the book sold at a loss of 20%

$$= \frac{410\,(100+25)}{(100-20)+(100+25)}$$

$$= \frac{410\times 125}{80+125} = ₹250.$$

Cost price of the book sold at a profit of 25%

$$= \frac{410\,(100-20)}{(100-20)+(100+25)} = \frac{410\times 80}{80+125}$$

$$= ₹160.$$

11 If two successive discounts on an article are $m\%$ and $n\%$ respectively, then a single discount equivalent to the two successive discounts will be:

$$\left(m + n - \frac{mn}{100}\right)\%$$

Explanation

Let, the marked price of the article be ₹100.

$\therefore$ S.P. after the first discount = ₹$(100 - m)$ and discount at $n\%$ on ₹$(100 - m)$ = ₹$\dfrac{(100-m)\times n}{100}$.

$\therefore$ Single equivalent discount

$$= \left[m + \frac{(100-m)\times n}{100}\right]\%$$

$$= \left(\frac{100m + 100n - mn}{100}\right)\%$$

$$= \left(m + n - \frac{mn}{100}\right)\%$$

12 If three successive discounts on an article are *l%*, *m%* and *n%* respectively, then a single discount equivalent to the three successive discounts will be

$$\left[l + m + n - \frac{(lm + ln + mn)}{100} + \frac{lmn}{100^2}\right]\%$$

Explanation

Let, the marked price of the article be ₹100.

∴ S.P. after the first discount = ₹(100 − *l*).

Second discount at *m*% on ₹(100 − *l*)

$$= ₹\frac{(100 - l) \times m}{100}$$

∴ S.P. after second the discount

$$= ₹(100 - l) - \frac{(100 - l)m}{100}$$

$$= ₹\frac{100\,(100 - l) - (100 - l)\,m}{100}$$

$$= ₹\frac{(100 - l) \cdot (100 - m)}{100}$$

Third discount at *n%* on $₹\frac{(100 - l)\,(100 - m)}{100}$

$$= ₹\frac{(100 - l)\,(100 - m)n}{100 \times 100}$$

∴ S.P. after the third discount

$$= ₹\frac{(100 - l)\,(100 - m)}{100} - \frac{(100 - l)\,(100 - m)n}{100 \times 100}$$

$$= ₹\frac{(100 - l)\,(100 - m)\,(100 - n)}{100 \times 100}$$

$$= \left(l + m + n - \frac{(lm + ln + mn)}{100} + \frac{lmn}{(100)^2}\right)$$

∴ Single equivalent discount

$$= \left(l + m + n - \frac{(lm + ln + mn)}{100} + \frac{lmn}{(100)^2}\right)\%$$

Illustration 22: Find a single discount equivalent to two successive discounts of 10% and 20%.

Solution: The equivalent single discount is given by

$$\left(10 + 20 - \frac{10 \times 20}{100}\right)\%, \text{ i.e., } 28\%$$

Illustration 23: Find out a single discount equivalent to three successive discounts of 10%, 20% and 30%.

Solution: The equivalent single discount is given by

$$\left(10 + 20 + 30 - \frac{(10 \times 20 + 10 \times 30 + 20 \times 30)}{100} + \frac{10 \times 20 \times 30}{100^2}\right)\%$$

i.e., $\left(60 - 11 + \frac{6}{10}\right)\% = \frac{496}{10}\%$ or 49.6%

Illustration 24: Two shopkeepers sell machines at the same list price. The first allows two successive discounts of 30% and 16% and the second 20% and 26%. Which discount series is more advantageous to the buyers?

Solution: A single discount equivalent to the two successive discounts of 30% and 16% is

$$\left(30 + 16 - \frac{30 \times 16}{100}\right)\%$$

or, $\left(46 - \frac{24}{5}\right)\%$ or, $41\frac{1}{5}\%$

Also, a single discount equivalent to the two successive discounts of 20% and 26% is

$$\left(20 + 26 - \frac{20 \times 26}{100}\right)\%$$

or, $\left(46 - \frac{26}{5}\right)\%$ or $40\frac{4}{5}\%$

Clearly, the discount series being offered by the first shopkeeper is more advantageous to the buyers.

13 A shopkeeper sells an item at ₹*z* after offering a discount of *d%* on labelled price. Had he not offered the discount, he would have earned a profit of *p%* on the cost price.

The cost price of each item is given by

$$\text{C.P.} = \left[\frac{100^2 z}{(100 - d)\,(100 + p)}\right].$$

Illustration 25: A shopkeeper sold sarees at ₹266 each after giving 5% discount on labelled price. Had he not given the discount, he would have earned a profit of 12% on the cost price. What was the cost price of each saree?

Solution: We have, labelled price *z* = ₹266, discount *d* = 5% and profit *p* = 12%

Using the formula

$$\text{C.P.} = \left[\frac{100^2 z}{(100 - d)\,(100 + p)}\right]$$

we get the cost price of each saree

$$= \left[\frac{100 \times 100 \times 266}{(100-5)(100+12)}\right]$$

$$= \frac{100 \times 100 \times 266}{95 \times 112} = ₹250.$$

EXERCISE-I

1. Mohan buys a watch for ₹350 and sells it for ₹392. Find out his percentage of profit.

(a) 9% (b) 12%
(c) 14% (d) None of these

2. Ramesh purchased a bicycle for ₹5200 and spent ₹800 on its repairs. He had to sell it for ₹5500. Find out his profit or loss per cent.

(a) $8\frac{1}{3}\%$ loss (b) $7\frac{1}{2}\%$ gain
(c) 9% (d) None of these

3. A man buys 10 articles for ₹8 and sells them at ₹1.25 per article. His gain per cent is:

(a) 55% (b) $56\frac{1}{4}\%$
(c) 40% (d) None of these

4. A toothpaste labeled at ₹80 is sold for ₹68. The rate of discount is:

(a) 12% (b) 14%
(c) 15% (d) None of these

5. Sardar Singh bought 200 dozen oranges at ₹10 a dozen. He spent ₹500 on transportation. He sold them at ₹1 each. What was his profit or loss per cent?

(a) 4% (b) 6%
(c) 5% (d) None of these

6. Mr Verma sold his scooter for ₹10500 at a gain of 5%. Find out the cost price of the scooter.

(a) ₹10300 (b) ₹10700
(c) ₹10000 (d) None of these

7. Suresh buys a camera for ₹1800 and sells it at 10% loss. Find out its selling price.

(a) ₹1620 (b) ₹1730
(c) ₹1650 (d) None of these

8. Hemant purchased 120 rims of paper at ₹80 per rim. He spent ₹280 on transportation, paid octroi at the rate of 40 paise per rim and paid ₹72 to the coolie. If he wants to have a gain of 8%, the selling price per rim must be:

(a) ₹89 (b) ₹90
(c) ₹95 (d) None of these

9. A shopkeeper loses 7% by selling a cricket ball for ₹31. For how much should he sell the ball so as to gain 5%?

(a) ₹50 (b) ₹65
(c) ₹35 (d) None of these

10. A shopkeeper sold some articles at ₹35 per piece and gained 40%. What would be the selling price of each article to earn 60% profit?

(a) ₹40 (b) ₹45
(c) ₹50 (d) None of these

11. A man bought apples at the rate of 6 for ₹20 and sold them at 4 for ₹16. His estimated profit % is:

(a) 23% (b) 18%
(c) 20% (d) None of these

12. A fruit vendor buys 10 bananas for ₹14 and sells them at 12 for ₹15. Find his percentage gain or loss.

(a) $10\frac{5}{7}\%$ loss (b) $10\frac{5}{9}\%$ gain
(c) 9% gain (d) None of these

13. If eggs are bought 12 for ₹10 and sold at 10 for ₹12. What is the gain or loss%.

(a) 40% loss (b) 44% gain
(c) 44% loss (d) None of these

14. If the cost price of 21 watches is equal to the selling price of 18 pieces, then what would be the gain per cent in this transaction?

(a) $6\frac{1}{2}\%$ (b) 7%
(c) $6\frac{2}{3}\%$ (d) None of these

15. A shopkeeper gains the cost of 8 metres thread by selling 40 metres thread. Find his gain per cent.

(a) 19% (b) 20%
(c) 22% (d) None of these

16. If the selling price of 2/3 of a certain quantity of milk be equal to the cost price of whole milk, then what will be the gain per cent in this transaction?

(a) 50% (b) 48%
(c) 53% (d) None of these

17. A shopkeeper sells 20 pencils for the same amount of money as he paid for 25 pencils. What is his gain per cent?

(a) 20% (b) 25%
(c) 24% (d) None of these

18. Mohit lost 18% by selling a bicycle for ₹1230. What per cent shall he gain or loss by selling it for ₹1600?

(a) $6\frac{2}{3}$% loss (b) 4% gain
(c) $6\frac{2}{3}$% gain (d) None of these

19. A shopkeeper sells an article at a gain of 10%. Had he sold it at a loss of 20%, its selling price would have been ₹180 less. What is the cost price of the article?

(a) ₹630 (b) ₹600
(c) ₹580 (d) None of these

20. A person sells 36 oranges per ₹1 and makes a loss of 4%. Find how many oranges per ₹ to be sold to have a gain of 8%?

(a) 1/32 (b) 5
(c) 1/16 (d) None of these

21. A person sells a colour TV at 10% below the cost price. Had he received ₹1494 more, he would have made a profit of $12\frac{1}{2}$%. What was the cost price of the colour TV?

(a) ₹6400 (b) ₹7200
(c) ₹6640 (d) None of these

22. Vijay sold a watch at a gain of 5%. Had he sold it for ₹72 more, he would have gained 13%. Find out the cost price of the watch.

(a) ₹900 (b) ₹910
(c) ₹870 (d) None of these

23. Sita sells a calculator to Gita at a gain of 17% and Gita sells it to Anu at a loss of 25%. If Anu pays ₹1842.75 for it, then what did Sita pay for it?

(a) ₹2080 (b) ₹2100
(c) ₹2110 (d) None of these

24. 'A' buys an article and sells it to 'B' at a profit of 10%, 'B' sells it to 'C' gaining 20%. If 'C' gives ₹924, what amount did 'A' give?

(a) ₹700 (b) ₹724
(c) ₹780 (d) None of these

25. 'A' sells an article to 'B' at a gain of 20% and 'B' sells it to 'C' at a gain of 10% and 'C' sells it to 'D' at a gain of $12\frac{1}{2}$%. If 'D' pays ₹29.70 what did it cost to 'A'?

(a) ₹20 (b) ₹24
(c) ₹18 (d) None of these

26. Rajesh sells taperecorder to Mihir at a loss of 10% and Mihir sells it to Shiv at a loss of 20%. If Shiv pays ₹1440 for it, at what price did Rajesh buy?

(a) ₹1920 (b) ₹2000
(c) ₹1800 (d) None of these

27. A man sells a scooter to his friend at 10% loss. If the friend sells it for ₹54000 and gains 20%, find out the original cost price of the scooter.

(a) ₹50000 (b) ₹45000
(c) ₹40000 (d) None of these

28. 'A' sells a good to 'B' at a profit of 10% and B sells it to 'C' at a profit of 20%. Find out the resultant profit.

(a) ₹35% (b) 20%
(c) ₹32% (d) None of these

29. A manufacturer sells an article to a wholesale dealer at a profit of 20%. The wholesale dealer sells it to a retail merchant at a loss of 5%. Find out the resultant profit or loss.

(a) 14% loss (b) 14% gain
(c) 12% gain (d) None of these

30. A man sold two watches for ₹3750 each. On one he gained 5%, and on the other, he lost 5%. What was his total gain or loss percentage?

(a) $1\frac{1}{4}$% (b) $\frac{1}{2}$%
(c) $\frac{1}{4}$% (d) None of these

31. A men sells two houses at the rate of ₹1.995 Lakhs each. On one house he gains 20% and on the other he loses 20%. His gain or loss per cent in the whole transaction is:

(a) 5% loss (b) 4%
(c) 4% loss (d) None of these

32. A shopkeeper sold two bicycles for ₹1500 each. On one, he gains 25% and on the other he loses 20%. His gain or loss per cent in the whole transaction is:

(a) $2\frac{18}{41}$% loss (b) $2\frac{18}{41}$% gain
(c) 2% gain (d) None of these

33. A man sells two articles, each for ₹640. He earns 20% profit on the first, and 40% profit on the second. Find his overall per cent profit.

(a) $29\frac{1}{2}$% (b) $28\frac{1}{2}$%
(c) $29\frac{3}{13}$% (d) None of these

34. A person sells two articles, each for ₹1040. He incurs 20% loss on the first and 10% loss on the second. Find out overall per cent loss.

(a) $12\frac{5}{17}$% (b) $15\frac{5}{17}$
(c) $13\frac{3}{4}$% (d) None of these

35. A grocer sells rice at a profit of 20% and uses a weight which is 25% less. Find out overall gain percentage.

(a) 60% (b) 65%
(c) 58% (d) None of these

36. A shopkeeper sells goods at 10% loss on cost price, but uses 20% less weight. What is his profit or loss percentage?

(a) 2% gain (b) $2\frac{1}{2}$% loss
(c) $2\frac{1}{2}$% gain (d) None of these

37. A cloth merchant says that due to slump in the market, he sells cloth at 10% loss, but he uses a inaccurate metre scale and actually gains 15%. Find out the actual length of the scale.

(a) 72.4 cm (b) 71.34 cm
(c) 78.25 cm (d) None of these

38. A cloth dealer professes to sell cotton at cost price, but uses a meter having a length of 80 cm only and charges for the meter. Find his gain per cent.

(a) 25% (b) 30%
(c) 40% (d) None of these

39. Sudeep buys two CDs for ₹380 and sells one at a loss of 22% and the other at a gain of 12%. If both the CDs are sold at the same price, then the cost price of two CDs is:

(a) ₹196, ₹225 (b) ₹230, ₹140
(c) ₹224, ₹156 (d) None of these

40. An article is listed at ₹65. A customer bought this article for ₹56.16 and received two successive discounts of which one is 10%. Find out the other discount in this discount scheme offered by the shopkeeper.

(a) 4% (b) 3%
(c) 6% (d) None of these

41. A cash payment that will settle a bill for 250 chairs at ₹50 per chair less 20% and 15% with a further discount of 5% on cash payment is:

(a) ₹7025 (b) ₹8075
(c) ₹8500 (d) None of these

42. A person sells taperecorders at ₹1134 each after giving a discount of 19% on the marked price. Had he not given the discount, he would have earned a profit of 40% on the cost price. The cost price of each taperecorder is:

(a) ₹1000 (b) ₹1200
(c) ₹1400 (d) None of these

EXERCISE-2
(BASED ON MEMORY)

1. Arun buys one kilogram of apples for ₹ 120 and sells it to Swati gaining 25%. Swati sells it to Divya who again sells it for ₹ 198, making a profit of 10%. What is the profit percentage made by Swati?

(a) 25% (b) 20%
(c) 16.67% (d) 15%

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2018]

2. A vendor buys bananas at 9 for ₹ 8 and sells at 8 for ₹ 9. What will be the profit or loss (in %)?

(a) 13.28% profit
(b) 26.56% loss
(c) 26.56% profit
(d) 13.28% loss

[SSC CGL Tier-II CBE, 2018]

3. If a stall sell a pizza at ₹ 200 he makes 20% loss if he wants to make 10% profit then at what price (in ₹) should he sell?

(a) 250 (b) 300
(c) 275 (d) 325

[SSC CGL Tier-II CBE, 2018]

4. If the selling price is tripled and cost price doubled the profit would become 65%. What is the present profit (in %)?

(a) 20 (b) 15
(c) 25 (d) 10

[SSC CGL Tier-II CBE, 2018]

5. If a vendor sells a coconut at ₹ 14.4 he makes 10% loss. If he wants to make 25% profit, there at what price (in ₹) should he sell?

(a) 18 (b) 20
(c) 16 (d) 22

[SSC CGL Tier-II CBE, 2018]

6. Selling price of a chair is ₹ 1386. If loss per cent is 23% then what is the cost price (in ₹) of chair?

(a) 1600 (b) 1800
(c) 1900 (d) 1067

[SSC CHSL (10 + 2) Tier-I CBE, 2018]

7. Selling price of a glass is ₹ 1965 and loss per cent is 25%. If selling price is ₹ 3013, then what will be the profit per cent?

(a) 13 (b) 10.4
(c) 15 (d) 20

[SSC CHSL (10 + 2) Tier-I CBE, 2018]

8. On a certain item profit is 150%. If the cost price increase by 25% what will be the new profit margin (in %)

(a) 25 (b) 50
(c) 100 (d) 75

[SSC CGL Tier-II CBE, 2018]

9. A wholesaler had 200 dozens of mangoes. He sold some of these mangoes at 20% profit and the rest at 10% profit, so that he made 13% profit on selling all the mangoes. How many mangoes (in dozens) did he sells at 20% profit?

(a) 140 (b) 60
(c) 80 (d) 120

[SSC CGL Tier-II CBE, 2018]

10. At a village trade fair a man buys a horse and a camel together for ₹ 51,250. He sold the horse at a profit of 25% and the camel at a loss of 20%. If he sold both the animals at the same price, then the cost price of the cheaper animal was ₹ _____.

(a) 6600 (b) 7500
(c) 25000 (d) 20000

[SSC CGL Tier-II CBE, 2018]

11. Oil equal to 20% of the weight of ground nut is extracted in a mill. The matter left after extraction is sold as cattle feed at the rate of ₹ 12.5 per kg. The ground nuts are bought at ₹ 20 per kg. the processing cost is ₹ 5 per kg. At what price (₹ per kg) should the oil be sold to earn 20% profit on total costs (Total cost = cost of groundnuts and processing costs)?

(a) 250 (b) 150
(c) 200 (d) 100

[SSC CGL Tier-II CBE, 2018]

12. A man buys 10 oranges for a rupee and sells 8 oranges for a rupee. What is the profit per cent?

(a) 10 (b) 12.5
(c) 20 (d) 25

[SSC Multi-Tasking Staff, 2017]

13. A dishonest dealer defrauds to the extent of x% in buying as well as selling his goods by using faulty weight. What will be the gain per cent on his outlay?

(a) $2x$% (b) $\left(\frac{10}{x}+x^2\right)$%

(c) None of these (d) $\left(x+\frac{x^2}{100}\right)$%

[SSC CGL Tier-II (CBE), 2017]

14. A fan costing ₹ 1200 is being sold for ₹ 1500. What is the profit per cent?

(a) 10 (b) 12.5
(c) 25 (d) 20

[SSC Multi-Tasking Staff, 2017]

15. Raman sells a machine for ₹ 39 lakh at a loss. Had he sold it for ₹ 49 lakh, his gain would nhave been three times the loss. What is the cost price of the machine?

(a) ₹ 46.5 lakh (b) ₹ 62 lakh
(c) ₹ 36.5 lakh (d) ₹ 41.5 lakh

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

16. A rice trader buys 8 quintals of rice for ₹ 3,600, 10% rice is lost in transportation. At what rate should he sell to earn 15 % profit?

(a) ₹ 352.1 per quintal
(b) ₹ 517.5 per quintal
(c) ₹ 575 per quintal
(d) ₹ 582.3 per quintal

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

17. A rice trader buys 16 quintals of rice for ₹ 5632. 20% rice is lost in transportation. At what rate should he sell to earn 25 % profit?

(a) ₹ 225.2 per quintal
(b) ₹ 550 per quintal
(c) ₹ 440 per quintal
(d) ₹ 563.2 per quintal

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

18. Raheem sells a machine for ₹48 lakh at a loss. Had he sold it for ₹ 60 lakh, his gain would have been 5 times the former loss. Find the cost price of the machine.

(a) ₹ 58 lakh (b) ₹ 69.6 lakh
(c) ₹ 42 lakh (d) ₹ 50 lakh

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

19. A man purchased 120 reams of paper at ₹ 80 per ream. He spend ₹ 280 on transportation, paid octrol at the rate of 40 palse per ream and paid ₹ 72 to a porter. In order to gain 8% he must sell each ream of paper for:

(a) ₹ 90 (b) ₹ 87.5
(c) ₹ 89 (d) ₹ 85

[SSC Matric Level, 2017]

20. A man bought 15 mangoes for a rupee. How many mangoes were sold for a rupee so that there is a loss of 25%?

(a) 10 (b) 12
(c) 18 (d) 20

[SSC CAPFs ASI & Delhi Police SI, 2017]

21. By selling a fan for ₹ 1900 a man has a loss of 5%, then at what price (in ₹) should he sell the fan to gain 20%?

(a) 2000 (b) 2400
(c) 2600 (d) 2800

[SSC CAPFs ASI & Delhi Police SI, 2017]

22. Profit obtained on selling an article for ₹ 310 is equal to the loss incurred on selling that article for ₹ 230. What will be the loss percentage when selling price is ₹ 180?

(a) $16\frac{1}{3}$ (b) $16\frac{2}{3}$
(c) $33\frac{1}{3}$ (d) $33\frac{2}{3}$

[SSC CAPFs ASI & Delhi Police SI, 2017]

23. A man bought nine pens for a rupee. How many pens should be sell for a rupee to gain 50%?

(a) 3 (b) 6
(c) 7 (d) 8

[SSC CAPFs ASI & Delhi Police SI, 2017]

24. A shopkeeper professes to sell his goods at cost price but uses a 930 gm weight instead of 1 kilogram weight. What will be the profit per cent of the shopkeeper?

(a) 3.76 (b) 7.52
(c) 9.85 (d) 10.36

[SSC CAPFs ASI & Delhi Police SI, 2017]

25. While selling an article for ₹ 18450, a person suffered a loss of 50%. At what price he should have sold the article (in ₹) to earn a profit of 50%?

(a) 13837 (b) 52000
(c) 55350 (d) 56775

[SSC CGL Tier-I (CBE), 2017]

26. A man gains 15% by selling a calculator for a certain price. If he sells it at the triple the price, then what will be the profit percentage?

(a) 125 (b) 175
(c) 225 (d) 245

[SSC CGL Tier-I CBE, 2017]

27. By selling 90 pens for ₹ 80 a man loses 20%. What should be the selling price (in ₹) of 90 pens for 20% profit?

(a) 90 (b) 100
(c) 110 (d) 120

[SSC CGL Tier-I CBE, 2017]

28. A trader buys 500kg of cotton for ₹ 9,000. 10% of this cotton is spoiled due to rain. At what rate (₹ / kg) should he sell the rest to earn 10% profit?

(a) 25 (b) 30
(c) 22 (d) 20

[SSC CGL Tier-I CBE, 2017]

29. A trader had 1200 kg of rice. He sold a part of it at 5% profit and the rest at 11% profit, so that he made a total profit of 7%. How much (in kg) rice did he sell at 5% profit?

(a) 900 (b) 600
(c) 400 (d) 800

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

30. If cost price is ₹ 150 less than the selling price and profit is 20%, then what is the selling price (in ₹)?

(a) 750 (b) 800
(c) 850 (d) 900

[SSC Multi-Tasking Staff, 2017]

31. If profit is $\left(\frac{1}{11}\right)$th of selling price, what is the profit per cent?

(a) $9\frac{1}{11}$ (b) 10

(c) $8\frac{1}{3}$ (d) $11\frac{1}{9}$

[SSC Multi-Tasking Staff, 2017]

32. The ratio of cost price and selling price of an article is 25 : 26. The per cent of profit will be

(a) 26% (b) 25%

(c) 1% (d) 4$

[SSC CGL Tier-II (CBE), 2017]

33. Profit of ₹ 144000 has to be divided among three partners Akram, Bipin and Chintan in the ratio 3 : 2 : 7. How much ₹ does Chinatn get?

(a) ₹ 84000 (b) ₹ 24000

(c) ₹ 36000 (d) ₹ 42000

[SSC CGL Tier-I (CBE), 2017]

34. *P* invests ₹ 9100 for 3 months. *Q* invests ₹ 6825 for 2 months and *R* ₹ 8190 for 5 months in a business. If the total profit amounts to ₹ 4158, how much profit should *Q* get?

(a) ₹ 682.50 (b) ₹ 693

(c) ₹ 1386 (d) ₹ 346.50

[SSC Matric Level, 2017]

35. Raman, Manan and Kamal are partners and invest in a business such that Raman invests $\frac{2}{5}$ th of total and Manan invests $\frac{3}{8}$th of the total. What is the ratio of profits of Raman, Manan and Kamal respectively?

(a) 16 : 15 : 9 (b) 16 : 15 : 31

(c) 2 : 3 : 5 (d) 15 : 16 :9

[SSC CAPFs ASI & Delhi Police SI, 2017]

36. If the ratio of the cost price and selling price of an article is 20 : 21, then what will be the profit per cent?

(a) 5% (b) 4.5%

(c) 6% (d) 7.5%

[SSC Multi-Tasking Staff, 2017]

37. On an article the profit is 210% of the cost price. If the cost price increase by 40% but the selling price remains constant, approximately what per cent of selling price will be the profit?

(a) 55 (b) 62

(c) 74 (d) 85

[SSC CAPFs ASI & Delhi Police SI, 2017]

38. The profit earned by a shopkeeper by selling a bucket at a gain of 8% is ₹ 28 more than when he sells it at a loss of 8%. The cost price (in Rupees) of the bucket is

(a) 170 (b) 190

(c) 175 (d) 165

[SSC CGL Tier-II CBE, 2017]

39. A trader sold an article at a gain of 20%. Had he purchased it for 40% more and sold for ₹ 24 less, he would have incurred a loss of 20%. What is the cost price (in ₹) of the article?

(a) 150 (b) 300

(c) 450 (d) 600

[SSC CAPFs ASI & Delhi Police SI, 2017]

40. Mr. Kapur purchased two toy cycles for ₹ 750 each. He sold these cycles, gaining 6% on one and losing 4% on the other. The gain or loss per cent in the whole transaction is

(a) 1% loss (b) 1% gain

(c) 1.5% loss (d) 1.5% gain

[SSC CGL Tier-II (CBE), 2017]

41. A man bought 500 metres of electronic wire at 50 paise per metre. He sold 50% of it at a profit of 5%. At what per cent should he sell the remainder so as to gain 10% on the whole transaction?

(a) 13% (b) 12.5%

(c) 15% (d) 20%

[SSC CGL Tier-II (CBE), 2017]

42. A shopkeeper sold one-third of his goods at a loss of 15%. To get a profit of 10% on the whole transaction, he should sell the remaining articles at a profit of

(a) $22\frac{1}{2}\%$ (b) $16\frac{2}{3}\%$

(c) 15% (d) 25%

[SSC Multi-Tasking Staff, 2017]

43. A trader buys two articles for ₹ 4000 each. While selling if he gains 12.5% on one and loses 20% on the other, then what will be the overall loss per cent?

(a) 2.5 (b) 3.75

(c) 5 (d) 5.25

[SSC CAPFs ASI & Delhi Police SI, 2017]

44. On selling an article for ₹170, a shopkeeper loses 15%. In order to gain 20%, he must sell that article at rupees:

(a) 215.50 (b) 212.50
(c) 240 (d) 210

[SSC CGL 2017]

45. By Selling 80 ball pens for ₹140, a retailer loses 30%. How many ball pens should he sell for ₹104 so as to make a profit of 30%?

(a) 32 (b) 52
(c) 48 (d) 42

[SSC CGL 2017]

46. If the selling price of 40 articles is equal to the cost price of 50 articles, the loss or gain per cent is

(a) 25% gain (b) 20% gain
(c) 25% loss (d) 20% loss

[SSC CGL Tier-I (CBE), 2016]

47. By what fraction selling price (S.P.) must be multiplied to get the cost price (C.P.) if the loss is 20%?

(a) $\frac{4}{5}$ (b) $\frac{8}{5}$
(c) $\frac{5}{4}$ (d) $\frac{6}{5}$

[SSC CGL Tier-II (CBE), 2016]

48. A shopkeeper sells rice at 10% profit and uses weight 30% less than the actual measure. His gain per cent is

(a) $57\frac{2}{3}\%$ (b) $57\frac{1}{7}\%$
(c) $57\frac{2}{5}\%$ (d) $57\frac{3}{7}\%$

[SSC CGL Tier-II (CBE), 2016]

49. A man bought 4 dozen eggs at ₹ 24 per dozen and 2 dozen eggs at ₹ 32 per dozen. To gain 20% on the whole, he should sell the eggs at

(a) ₹ 16 per dozen (b) ₹ 21 per dozen
(c) ₹ 32 per dozen (d) ₹ 35 per dozen

[SSC CGL Tier-I (CBE), 2016]

50. By selling cloth at ₹ 9 per meter, a shopkeeper loses 10%. Find the rate at which it should be sold so as to earn profit of 15%.

(a) ₹ 11.20 (b) ₹ 11.30
(c) ₹ 11.40 (d) ₹ 11.50

[SSC CGL Tier-I (CBE), 2016]

51. Kamal has some apples. He sold 40% more than he ate. If he sold 70 apples, how many dis he eat?

(a) 18 (b) 42
(c) 50 (d) 90

[SSC CGL Tier-I (CBE), 2016]

52. *A* and *B* jointly made a profit of ₹ 1650 and they decided to share it such that $\frac{1}{3}$ of *A*'s profit is equal to $\frac{2}{5}$ of *B*'s profit. Then profit of *B* is

(a) ₹ 700 (b) ₹ 750
(c) ₹ 850 (d) ₹ 800

[SSC CGL Tier-II CBE, 2016]

53. Anil started a business with an investment of ₹ 25,000. After 3 months, Vishal joined his business with a capital of ₹ 30, 000. At the end of the year, they have made a profit of ₹ 19,000 what will be Anil's share in the profit?

(a) ₹ 10,000 (b) ₹ 12,500
(c) ₹ 10,250 (d) ₹ 14,000

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2016]

54. Instead of dividing 391 cookies among 3 children *A*, *B*, *C* in the ratio $\frac{1}{5}:\frac{1}{4}:\frac{1}{8}$, it was divided in to the ratio 5 : 4 : 8. Who gains the most and how many?

(a) *A*, 21 cookies
(b) *B*, 78 cookies
(c) *C*, 98 cookies
(d) *C*, 78 cookies

[SSC CPO SI & ASI, 2016]

55. After selling 5% of a quantity of sugar, 5 kg of sugar remains. Find the total quantity of sugar.

(a) 19 kg. (b) $5\frac{5}{19}$
(c) 100 kg (d) 95 kg

[SSC CGL Tier-I (CBE), 2016]

56. A shopkeeper purchases two items for ₹ 520. One of them is sold gaining 16% and the other at a loss of 10%, thus making no profit or loss. What is the selling price of the item sold at loss?

(a) ₹ 288 (b) ₹ 232
(c) ₹ 320 (d) ₹ 200

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2016]

57. The marked price of an article is ₹ 5000. But due to a special festive offer a certain per cent of discount is declared. Mr. X availed this opportunity and bought the article at reduced price. He then sold it at ₹ 5000

and thereby made a profit of $11\frac{1}{9}\%$. The percentage of discount allowed was

(a) 10 (b) $3\frac{1}{3}$

(c) $7\frac{1}{2}$ (d) $11\frac{1}{9}$

[SSC CGL Tier-II CBE, 2016]

58. A man buys a table and a chair for ₹ 500. He sells the table at a loss of 10% and the chair at a gain of 10%. He still gains ₹ 10 on the whole. The cost price of chair in rupees is:

(a) ₹ 200 (b) ₹ 250
(c) ₹ 300 (d) ₹ 350

[SSC CGL Tier-I (CBE), 2016]

59. If a commission at the rate of 10% is given to a bookseller on the marked price of a book by the publisher, the publisher gains 20%. If the commission is increased to 15% then the gain percent would be:

(a) $16\frac{2}{3}\%$ (b) $13\frac{1}{3}\%$

(c) $15\frac{5}{6}\%$ (d) $12\frac{1}{2}\%$

[SSC CPO SI, ASI, 2016]

60. 'A' bought a certain quantity of oranges at total cost of ₹1200/-. He sold 1/3rd of those oranges at 20% loss. If A earns an overall profit of 10%, at what percent profit did A sell the rest of the oranges?

(a) 16% (b) 15%
(c) 22% (d) 25%
(e) 20%

[IBPS, 2015]

61. Present age of Bob is equal to Abby's age 8 years ago. Four years hence, the respective ratio between Bob's age and Abby's age will be 4 : 5 at that time. What is Bob's present age?

(a) 24 years (b) 32 years
(c) 40 years (d) 20 years
(e) 28 years

[IBPS, 2015]

Directions (62-65): *In these questions, two equations numbered I and II are given. You have to solve both the equations and mark the appropriate option.*

Give Answer:

(1) If $x > y$

(2) If $x \geq y$

(3) If $x < y$

(4) If relationship between x and y cannot be determined

(5) $x \leq y$

62. I. $2x^2 + 19x + 45 = 0$

II. $2y^2 + 11y + 12 = 0$

[IBPS, 2015]

63. I. $3x^2 - 13x + 12 = 0$

II. $2y^2 - 15y + 28 = 0$

[IBPS, 2015]

64. I. $x^2 = 16$

II. $2y^2 - 17y + 36 = 0$

[IBPS, 2015]

65. I. $6x^2 + 19x + 15 = 0$

II. $3y^2 + 11y + 10 = 0$

[IBPS, 2015]

66. I. $2x^2 - 11x + 15 = 0$

II. $2y^2 - 11y + 14 = 0$

[IBPS, 2015]

67. 10% discount and then 20% discount in succession is equivalent to total discount of

(a) 15% (b) 30%
(c) 24% (d) 28%

[SSC, 2015]

68. The marked price of a watch was ₹720. A man bought the same for ₹550.80 after getting two successive discounts, the first being 10%. The second discount rate is

(a) 12% (b) 14%
(c) 15% (d) 18%

[SSC, 2015]

69. Allowing 20% and 15% successive discounts, the selling price of an article becomes ₹3,060; then the marked price will be

(a) ₹4,400 (b) ₹5,000
(c) ₹4,500 (d) ₹4,000

[SSC, 2015]

70. If a shopkeeper wants to give 20% discount on a toy, he has to sell it for ₹300. If he sells it at ₹405, then his gain percent is

(a) 5% (b) 8%
(c) 4% (d) 6%

[SSC, 2015]

71. Ram sold two horses at the same price. In one he gets a profit of 10% and in the other he gets a loss of 10%. Then Ram gets

(a) no loss or profit (b) 1% profit
(c) 1% loss (d) 2% loss

[SSC, 2015]

72. A manufacturer fixes his selling price at 33% over the cost of production. If cost of production goes up by 12% and manufacturer raises his selling price by 10%, his percentage profit is

(a) $36\frac{5}{9}\%$ (b) $28\frac{3}{8}\%$

(c) $30\frac{5}{8}\%$ (d) 35%

[SSC, 2015]

73. The marked price of a tape recorder is ₹12,600. A festival discount of 5% is allowed on it. Further for cash payment, a second discount of 2% is given. The cash payment, in rupees, is to be made for buying it is

(a) 11,370.60 (b) 11,073.60
(c) 11,703.60 (d) 11,730.60

[SSC, 2015]

74. A man purchases some oranges at the rate of 3 for ₹40 and the same quantity at 5 for ₹60. If he sells all the oranges at the rate of 3 for ₹50, find his gain or loss percent (to the nearest integer).

(a) 31% loss (b) 31% profit
(c) 34% loss (d) 32% profit

[SSC, 2015]

75. A dealer fixed the price of an article 40% above the cost of production. While selling it he allows a discount of 20% and makes a profit of ₹48. The cost of production (in ₹) of the article is

(a) 360 (b) 420
(c) 400 (d) 320

[SSC, 2015]

76. There would be a 10% loss, if rice is sold at ₹54 per kg. To earn a profit of 20%, the price of rice per kg will be

(a) ₹70 (b) ₹72
(c) ₹65 (d) ₹63

[SSC, 2015]

77. A man sells an article at 5% above its cost price. If he had bought it at 5% less than what he had paid for it and sold it at ₹2 less, he would have gained 10%. The cost price of the article is

(a) ₹400 (b) ₹100
(c) ₹200 (d) ₹300

[SSC, 2015]

78. Articles are marked at a price which gives a profit of 25%. After allowing a certain discount the profit reduces to 12½%. The discount percent is

(a) 12½% (b) 11.1%
(c) 10% (d) 12%

[SSC, 2015]

79. A shopkeeper gains 17% after allowing a discount of 10% on the marked price of an article. Find his profit percent if the article is sold at marked price allowing no discount.

(a) 27% (b) 30%
(c) 37% (d) 23%

[SSC, 2015]

80. List price of a book is ₹100. A dealer sells three such books for ₹274.50 after allowing discount at a certain rate. Find the rate of discount.

(a) 8.16% (b) 8.5%
(c) 8.34% (d) 8.33%

[SSC, 2015]

81. The printed price of an article is 40% higher than its cost price. Then the rate discount so that he gains 12% profit is:

(a) 15% (b) 21%
(c) 18% (d) 20%

[SSC, 2015]

82. Cost price of 100 books is equal to the selling price of 60 books. The gain percentage or less percentage is:

(a) $66\frac{2}{3}\%$ (b) $66\frac{3}{2}\%$
(c) 67% (d) 66%

[SSC, 2015]

83. If the discount of 10% is given on the marked price of a radio, the gain is 20%. If the discount is increased to 20%, the gain is:-

(a) $6\frac{1}{4}\%$ (b) 5%

(c) $7\frac{5}{8}\%$ (d) $6\frac{2}{3}\%$

[SSC, 2015]

84. A dealer buys an article listed at ₹100 and gets successive discounts of 10% and 20%. He spends 10% of the Cost Price on transportation. At what price should he sell the article to earn a profit of 15%?

(a) ₹92.00 (b) ₹91.08
(c) ₹90.80 (d) ₹91.20

[SSC, 2015]

85. A shopkeeper allows a discount of 10% on the marked price of a camera. Marked price of the camera, which costs him ₹600, to make a profit of 20% should be:-

(a) ₹700 (b) ₹800
(c) ₹750 (d) ₹650

[SSC, 2015]

86. A dealer sold a bicycle at a profit of 10%. Had he brought the bicycle at 10% less price and sold it at a price ₹60 more, he would have gained 25%. The cost price of the bicycle was-

(a) ₹2600 (b) ₹2000
(c) ₹2200 (d) ₹2400

[SSC, 2015]

87. The difference between successive discounts of 40% followed by 30% and 45% followed by 20% on the marked price of an article is ₹12. The marked price of the article is:

(a) ₹400 (b) ₹200
(c) ₹800 (d) ₹600

[SSC, 2015]

88. Find a simple discount equivalent to a discount series of 10%, 20% and 25%

(a) 45% (b) 55%
(c) 52% (d) 46%

[SSC, 2015]

89. Cost price of 100 books is equal to the selling price of 60 books. The gain or loss percentage will be:

(a) $66\frac{2}{3}\%$ (b) $66\frac{1}{4}\%$

(c) 66% (d) $66\frac{3}{4}\%$

[SSC, 2015]

90. An article which is marked at ₹975 is sold for ₹897. The % discount is?

(a) 6% (b) 10%
(c) 12% (d) 8%

[SSC, 2015]

91. A shopkeeper sold an item at 10% loss after giving a discount equal to half the marked price. The cost price of the item is:

(a) $\frac{1}{9}$ of marked price

(b) $\frac{4}{9}$ of marked price

(c) $\frac{5}{9}$ of marked price

(d) $\frac{7}{9}$ of marked price

[SSC, 2014]

92. A person purchased a saree for ₹7710 after availing a net discount of ₹1285. The percentage of discount, the saree-shop offers, is:

(a) $14\frac{1}{7}\%$ (b) $14\frac{2}{7}\%$

(c) $14\frac{3}{7}\%$ (d) $14\frac{4}{7}\%$

[SSC, 2014]

93. A cycle dealer offers a discount of 10% and still makes a profit of 26%. What does he pay for a cycle whose marked price is ₹840?

(a) ₹600 (b) ₹650
(c) ₹700 (d) ₹750

[SSC, 2014]

94. If the cost price of an item is $\frac{2}{5}$ of its marked price and if it is sold at a discount of 10%, then there will be:

(a) 25% profit (b) 40% profit
(c) 50% profit (d) 125% profit

[SSC, 2014]

95. An item costing ₹200 is being sold at 10% loss. If the price is further reduced by 5%, the selling price will be:

(a) ₹170 (b) ₹171
(c) ₹175 (d) ₹179

[SSC, 2014]

96. A shopkeeper buys 144 items at 90 paisa each. On the way 20 items are broken. He sells the remainder

at ₹1.20 each. His gain per cent correct to one place of decimal is:

(a) 13.8% (b) 14.6%
(c) 14.8% (d) 15.8%

[SSC, 2014]

97. There is a profit of 20% on the cost price of an article. The per cent of profit, when calculated on selling price is:

(a) $16\frac{2}{3}\%$ (b) 20%

(c) $33\frac{1}{3}\%$ (d) None of these

[SSC, 2014]

98. By selling an article for ₹102, there is a loss of 15%. When the article is sold for ₹134.40, the net result in the transaction is:

(a) 12% gain (b) 12% loss
(c) 10% loss (d) 15% gain

[SSC, 2014]

99. Two toys are sold at ₹504 each. One toy brings the dealer a gain of 12% and the other a loss of 4%. The gain or loss per cent by selling both the toys is:

(a) $3\frac{5}{13}\%$ Profit (b) $4\frac{5}{13}\%$ Profit

(c) $5\frac{1}{13}\%$ Profit (d) $2\frac{3}{13}\%$ Loss

[SSC, 2014]

100. A sold a horse to B for ₹4800 by losing 20%. B sells it to C at a price which would have given A a profit of 15%. B's gain is:

(a) ₹1800 (b) ₹1900
(c) ₹2000 (d) ₹2100

[SSC, 2014]

101. A reduction of 21% in the price of an item enables a person to buy 3 Kg more for ₹100. The reduced price of the item per Kg is:

(a) ₹5.50 (b) ₹7.50
(c) ₹10.50 (d) ₹7.00

[SSC, 2014]

102. A bakery bakes cake with the expectation that it will earna profit of 40% by selling each cake at marked price. But during the delivery to showroom 16% of the cakes were completely damaged and hence could not be sold. 24% of the cakes were slightly damaged and hence could be sold at 80% of the cost price. The remaining 60% of the cakes were sold at marked price. What is the percentage profit in the whole consignment?

(a) 3.2 (b) 2.4
(c) 2.8 (d) 4.2
(e) 3.6

[IBPS PO/MT, 2014]

103. The list price of a shirt is ₹440 and a customer pays ₹396 for it. The discount rate is

(a) 10 % (b) $10^{1/2}$ %
(c) 20 % (d) 12 %

[SSC, 2014]

104. A plate was sold for ₹6,300 after giving two successive discounts of $12^{1/2}$ % and 10 %. Find the marked price.

(a) ₹7,300 (b) ₹7,700
(c) ₹8,000 (d) ₹7,250

[SSC, 2014]

105. Nisha bought a number of oranges at 2 for a rupee and an equal number at 3 for a rupee. To make a profit of 20% she should sell a dozen for

(a) ₹6 (b) ₹8
(c) ₹10 (d) ₹12

[SSC, 2014]

106. A shopkeeper listed the price of goods at 30% above the cost price. He sells half the stock at this price, one fourth of the stock at a discount of 15% and the remaining at 30% discount. His overall profit is

(a) $15\frac{3}{8}\%$ (b) 15 %

(c) $-15\frac{3}{5}\%$ (d) $15\frac{2}{3}\%$

[SSC, 2014]

107. A trader buys good at 20% discount on marked price. If he wants to make a profit of 25% after allowing a discount of 20%, by what percent should his marked price be greater than the original marked price?

(a) 15 % (b) 65 %
(c) 25 % (d) 20 %

[SSC, 2014]

108. A shopkeeper allows a discount of 10% on the marked price of an item but charges a sales tax of 8% on the discounted price. If the customer pays ₹3,402 as the price including the sales tax, then the marked price is

(a) ₹3,400 (b) ₹3,500
(c) ₹3,600 (d) ₹3,800

[SSC, 2014]

109. A fruit-seller buys some oranges and by selling 40% of them he fealises the cost price of all the oranges. As the oranges being to grow over-ripe, he reduces the price and sells 80% of the remaining oranges at half the previous rate of profit. The rest of the oranges being rotten are thrown away. The overall percentage of profit is

(a) 80 (b) 84
(c) 94 (d) 96

[SSC, 2014]

110. A merchant advertises 10% off on the items bought from his store. The total discount got by a customer who bought a cooker worth \$650, a heater worth \$500 and a bag worth \$65 is

(a) \$128.50 (b) \$121.50
(c) \$120.50 (d) \$123.50

[SSC, 2014]

111. By selling an article for \$450, a man loses 10%. The gain or loss percent if he sells it for \$540 is

(a) loss 9% (b) gain 8%
(c) loss 8% (d) gain 9%

[SSC, 2014]

112. A man loses $20\frac{1}{2}$ % of his money and after spending 80% of the remainder, he is left with \$159. How much did he have at first?

(a) \$1,000 (b) \$1,200
(c) \$500 (d) \$800

[SSC, 2014]

113. The cost price of a table is ₹3,200. A merchant wants to make 25% profit by selling it. At the time of sale he declares a discount of 20% on the marked price. The marked price (in ₹) is

(a) 4,000 (b) 4,500
(c) 5,000 (d) 6,000

[SSC, 2014]

114. A shopkeeper allows a discount of 12.5% on the marked price of a certain article and makes a profit of 20%. If the article costs the shopkeeper ₹210, then the marked price of the article will be

(a) ₹386 (b) ₹288
(c) ₹387 (d) ₹350

[SSC, 2014]

115. A businessman allows a discount of 10% on the marked price. What percent above the cost price must he mark his goods to make a profit of 17%?

(a) 30% (b) 20%
(c) 27% (d) 18%

[SSC, 2014]

116. Charging 30% above its production cost a radio maker puts a label of ₹286 on a ratio as its price. But at the time of selling it, he allows 10% discount on the labeled price. What will his gain be?

(a) ₹198 (b) ₹37.40
(c) ₹257.40 (d) ₹254.40

[SSC, 2014]

117. A tradesman marks his goods 30% more than the cost price. If he allows a discount of $6\frac{1}{4}$% then his gain per cent is:

(a) $23\frac{3}{4}$% (b) 22%

(c) $21\frac{7}{8}$% (d) 30%

[SSC, 2013]

118. A shopkeeper purchased a chair marked at ₹600 at two successive discounts of 15% and 20%. He spent ₹28 on transportation and sold the chair for ₹545. His gain per cent was:

(a) 25% (b) 30%
(c) 35% (d) 20%

[SSC, 2013]

119. The marked price of a piano was ₹15,000. At the time of sale, there were successive discounts of 20% 10% and 10% on it. The sale price was:

(a) ₹9,720 (b) ₹9,750
(c) ₹9,760 (d) ₹9,780

[SSC, 2013]

120. By selling 25 metres of cloth a trader gains the selling price of 5 metres of cloth. The gain of the trader in % is:

(a) 25 (b) 20
(c) 28 (d) 29

[SSC, 2013]

121. A sells a suitcase to B at 10% profit. B cells it to C at 30% profit. If C pays ₹2,860 for it, then the price at which A bought it is:

(a) ₹1,000 (b) ₹1,600
(c) ₹2,000 (d) ₹2,500

[SSC, 2013]

122. Gita buys a plot of land for ₹96,000. She sells $\frac{2}{5}$ of it at a loss of 6%. She wants to make a profit of 10% on the whole transaction by selling the remaining

land. The gain percentage on the remaining land is:

(a) 20 (b) $20\frac{2}{3}$

(c) 14 (d) 7

[SSC, 2013]

123. An article is sold at a gain of 15%. Had it been sold for ₹27 more, the profit would have been 20%. The cost price of the article is:

(a) ₹500 (b) ₹700

(c) ₹540 (d) ₹545

[SSC, 2013]

124. On selling 17 balls at₹720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is:

(a) 45 (b) 50

(c) 55 (d) 60

[SSC, 2013]

125. Two items A and B are sold at a profit of 10% and 15%, respectively. If the amount of profit received is the same, then the cost price of A and B may be:

(a) ₹1,000, ₹1,500 (b) ₹5,000, ₹2,000

(c) ₹3,000, ₹2,000 (d) ₹3,000, ₹5,000

[SSC, 2013]

126. Arun marks up the computer he is selling by 20% profit and sells them at a discount of 15%. Arun's net gain per cent is:

(a) 4 (b) 2

(c) 3.5 (d) 2.5

[SSC Assistant Grade III, 2013]

127. A dealer buys a table listed at ₹1,500 and gets successive discounts of 20% and 10%. He spends ₹20 on transportation and sells at a profit of 20%. Find the selling price of the table.

(a) ₹1,320 (b) ₹1,080

(c) ₹1,200 (d) ₹1,230

[SSC Assistant Grade III, 2013]

128. A sells an article to B at a gain of 20% and B sells it to C at a gain of 10% and C sells it to D at a gain of $12\frac{1}{2}\%$. If D pays ₹29.70, then A purchased the article for:

(a) ₹40 (b) ₹10

(c) ₹20 (d) ₹30

[SSC Assistant Grade III, 2013]

129. By selling 80 ball pens for ₹140 a retailer loses 30%. How many ball pens should he sell for ₹104 so as to make a profit of 30%?

(a) 32 (b) 52

(c) 48 (d) 42

[SSC Assistant Grade III, 2013]

130. A shopkeeper sells two watches for ₹308 each. On one watch he earns 12% profit and on the others he suffers 12% loss. His profit or loss in the entire transaction was:

(a) $1\frac{11}{25}\%$ loss (b) $1\frac{11}{25}\%$ gain

(c) $3\frac{2}{25}\%$ loss (d) $3\frac{2}{25}\%$ gain

(e) None of these

[IBPS PO/MT, 2013]

131. A discountof 40% on the marked price of atrouser enables Ajit to purchase a shirt also which costs him ₹320. How much did Ajit payfor the trouser?

(a) ₹ 480 (b) ₹ 540

(c) ₹ 800 (d) ₹ 400

[SSC Assistant Grade III, 2012]

132. Rahim bought a gift item for ₹510 after getting a discount of 15%. He then sells it 5% above the marked price. The profit earned in this deal is:

(a) ₹150 (b) ₹120

(c) ₹100 (d) ₹90

[SSC Assistant Grade III, 2012]

133. A shopkeeper marks his goods at 40% above their cost price. He is able to sell $\frac{3}{4}$th of his goods at this price, and the remaining at 40% discount. Assuming that the shopkeeper is able to sell the goods he buys, find his loss or gain as % of the whole transaction.

(a) 20% loss (b) 23% loss

(c) 26% gain (d) 30% gain

[SSC Assistant Grade III, 2012]

134. A fruit seller bought 240 bananas at the rate of ₹48 per dozen. He sells half of them at the rate of ₹5 per banana, $\frac{1}{6}$th of the remaining are found to be rotten. The price per banana at which he has to sell the remaining bananas to get a profit of 25% on his entire investment is:

(a) ₹5.5 (b) ₹6.0

(c) ₹5.0 (d) ₹6.5

[SSC Assistant Grade III, 2012]

135. A and B started a business by investing ₹3,50,000 and ₹1,40,000 respectively. A gets 20% of the yearly profit for managing the business. Thereafter the profit is divided in the ratio of the capital. If A receives

totally ₹38,000 more than B at the end of a year, then the profit is:
(a) ₹28,000 (b) ₹2,80,000
(c) ₹1,05,000 (d) ₹70,000

[SSC, 2012]

136. A fan in a shop is offered at a discount of 10%. It is sold during clearance sale at 6% discount over the already discounted price at ₹846. The original marked price of the fan is:
(a) ₹1000 (b) ₹900
(c) ₹850 (d) ₹896

[SSC, 2012]

137. A trader allows a trade discount of 20% and a cash discount of $6\frac{1}{4}\%$ on the marked price of the goods and gets a net gain of 20% of the cost. By how much above the cost should the goods be marked for the sale?
(a) 40% (b) 50%
(c) 60% (d) 70%

[SSC, 2012]

138. A discount series of 10%, 20% and 40% is equal to a single discount of:
(a) 56.80% (b) 50%
(c) 70% (d) 43.20%

[SSC, 2012]

139. Tarun bought a TV with 20% discount on the labelled price. Had he bought it with 25% discount, he would have saved ₹500. At what price did he buy the TV?
(a) ₹7,500 (b) ₹8,500
(c) ₹8,000 (d) ₹7,400

[SSC, 2012]

140. A manufacturer sells an article to a wholesale dealer at a profit of 10%. The wholesale dealer sells it to a shopkeeper at 20% profit. The shopkeeper sells it to a customer for ₹56,100 at a loss of 15%. Then the cost price of the article to the manufacturer is:
(a) ₹25,000 (b) ₹10,000
(c) ₹50,000 (d) ₹55,000

[SSC, 2012]

141. A loss of 19% gets converted into a profit of 17% when the selling price is increased by ₹162. The cost price of the article is:
(a) ₹450 (b) ₹600
(c) ₹360 (d) ₹540

[SSC, 2012]

142. A man purchased 150 pens at the rate of ₹12 per pen. He sold 50 pens at gain of 10%. The percentage gain at which he must sell the remaining pens so as to gain 15% on the whole outlay is:
(a) $21\frac{1}{2}\%$ (b) 20%
(c) 17% (d) $17\frac{1}{2}\%$

[SSC, 2012]

143. A dealer sold two types of goods for ₹10,000 each. On one of them, he lost 20% and on the other he gained 20%. His gain or loss per cent in the entire transaction was:
(a) 2% loss (b) 2% gain
(c) 4% gain (d) 4% loss

[SSC, 2012]

144. The cost price of 40 articles is the same as the selling price of 25 articles. Find the gain per cent.
(a) 65% (b) 60%
(c) 15% (d) 75%

[SSC, 2012]

145. A sells an article to B making a profit of $\frac{1}{5}th$ his outlay. B sells it to C, gaining 20%. If C sells it for ₹600 and incurs a loss of $\frac{1}{6}th$ his outlay, the cost price of A is:
(a) ₹600 (b) ₹500
(c) ₹720 (d) ₹800

[SSC, 2012]

146. A man had a certain amount with him. He spent 20% of that to buy an article and 5% of the remaining on transport. Then he gifted ₹120. If he is left with ₹1,400, the amount he spent on transport is:
(a) ₹76 (b) ₹61
(c) ₹95 (d) ₹80

[SSC, 2012]

147. An article was purchased for ₹78,350. Its price was marked up by 30%. It was sold at a discount of 20% on the marked-up price. What was the profit per cent on the cost price?
(a) 4% (b) 7%
(c) 5% (d) 3%
(e) 6%

[IBPS PO/MT, 2012]

148. 20% loss on selling price is what per cent loss on the cost price?
(a) 25% (b) $16\frac{2}{3}\%$
(c) 15% (d) $16\frac{1}{3}\%$

[SSC (GL), 2011]

149. X sells two articles for ₹4,000 each with no loss and no gain in the interaction. If one was sold at a gain of 25% the other is sold at a loss of:

(a) 25% (b) $18\frac{2}{9}\%$

(c) $16\frac{2}{3}\%$ (d) 20%

[SSC (GL), 2011]

150. A man purchased some eggs at 3 for ₹5 and sold them at 5 for ₹12. Thus, he gained ₹143 in all. The number of eggs he bought is:

(a) 210 (b) 200

(c) 195 (d) 190

[SSC (GL), 2011]

151. The cost price of an article is 64% of the marked price. The gain percentage after allowing a discount of 12% on the marked price is:

(a) 37.5% (b) 48%

(c) 50.5% (d) 52%

[SSC (GL), 2011]

152. By selling an article for ₹144, a person gained such that the percentage gain equals the cost price of the article. The cost price of the article is:

(a) ₹90 (b) ₹80

(c) ₹75 (d) ₹60

[SSC (GL), 2011]

153. A man sells two article for ₹5000 each neither losing nor gaining in the deal. If he sold one of them at a gain of 25%, the other article is sold at a loss of:

(a) $15\frac{2}{3}\%$ (b) $16\frac{2}{3}\%$

(c) $17\frac{1}{3}\%$ (d) $18\frac{1}{3}\%$

[SSC (GL), 2011]

154. A man bought orange at the rate of 8 for ₹34 and sold them at the rate of 12 for ₹57. How many oranges should be sold to earn a net profit of ₹45?

(a) 90 (b) 100

(c) 135 (d) 150

[SSC (GL), 2011]

155. A shopkeeper allows 23% commission on his advertised price and still makes a profit of 10%. If he gains ₹56 on one item. His advertised price of the item, in ₹, is:

(a) 820 (b) 780

(c) 790 (d) 800

[SSC (GL), 2011]

156. Seema purchased an item for ₹9600 and sold it for loss of 5 per cent. From that money she purchased another item and sold it for gain of 5 per cent. What is her overall gain/loss?

(a) Loss of ₹36 (b) Profit of ₹24

(c) Loss of ₹54 (d) None of these

[Bank of Baroda PO Examination, 2011]

157. By selling an article at $\frac{3}{4}th$ of the marked price, there is gain of 25%. The ratio of the marked price and the cost price is:

(a) 5:3 (b) 3:5

(c) 3:4 (d) 4:3

[SSC, 2011]

158. Successive discounts of 10%, 20% and 50% will be equivalent to a single discount of:

(a) 36% (b) 64%

(c) 80% (d) 56%

[SSC, 2011]

159. A retailer offers the following discount schemes for buyers on an article:

I. Two successive discounts of 10%.

II. A discount of 12% followed by a discount of 8%.

III. Successive discounts of 15% and 5%.

IV. A discount of 20%.

The selling price will be minimum under the scheme

(a) I (b) II

(c) III (d) IV

[SSC, 2011]

160. The value of an article depreciates every year at the rate of 10% of its value. If the present value of the article is ₹729, then its worth 3 years ago was:

(a) ₹1250 (b) ₹1000

(c) ₹1125 (d) ₹1200

[SSC, 2011]

161. Nitin bought some oranges at ₹40 a dozen and an equal number at ₹30 a dozen. He sold them at ₹45 a dozen and made a profit of ₹480. The number of oranges, he bought, was:

(a) 48 (b) 60

(c) 72 (d) 84

[SSC, 2011]

162. A man buys two chairs for a total cost of ₹900. By selling one for $\frac{4}{5}$ of its cost and the other for $\frac{5}{4}$

of its cost, he makes a profit of ₹90 on the whole transaction. The cost of the lower priced chair is:

(a) ₹360 (b) ₹400
(c) ₹420 (d) ₹300

[SSC, 2011]

163. By selling 100 oranges, a vendor gains the selling price of 20 oranges. He gain per cent is:

(a) 20 (b) 25
(c) 30 (d) 32

[SSC, 2011]

164. 60% of the cost price of an article is equal to 50% of its selling price. Then the percentage of profit or loss on the cost price is:

(a) 20% loss (b) $16\frac{2}{3}$% profit
(c) 20% profit (d) 10% loss

[SSC, 2011]

165. Maninder bought two horses at ₹40,000 each. He sold one horse at 15% gain, but had to sell the second horse at a loss. If he had suffered a loss of ₹3,600 on the whole transaction, then the selling price of the second horse is:

(a) ₹30,000 (b) ₹30,200
(c) ₹30,300 (d) ₹30,400

[SSC, 2011]

166. A fruit-seller buys x guavas for ₹y and sells y guavas for ₹x. If $x>y$, then he made:

(a) $\frac{x^2-y^2}{xy}$% loss (b) $\frac{x^2-y^2}{xy}$% gain
(c) $\frac{x^2-y^2}{y^2}$% loss (d) $\frac{x^2-y^2}{y^2}\times 100$% gain

[SSC, 2011]

167. Profit earned by an organization is distributed among officers and clerks in the ratio of 5:3. If the number of officers is 45 and the number of clerks is 80 and the amount received by each officer is ₹25,000, what was the total amount of profit earned?

(a) ₹22 Lakhs (b) ₹18.25 Lakhs
(c) ₹18 Lakhs (d) ₹23.25 Lakhs
(e) None of these

[SBI Associates Banks PO, 2011]

168. A shopkeeper labeled the price of his articles in order to earn a profit of 30% on the cost price. He, then, sold the articles by offering a discount of 10% on the labelled price. What is the actual rate of profit he earned in the deal?

(a) 18% (b) 8%
(c) 20% (d) Cannot be determined
(e) None of these

[SBI Associates Banks PO, 2011]

169. Kamya purchased an item for ₹46,000 and sold it at aloss of 12 per cent. With that amount she purchased another item which he sold at a gain of 12%. What was her overall gain/loss?

(a) Loss of ₹662.40 (b) Profit of ₹662.40
(c) Loss of ₹642.80 (d) Profit of ₹642.80
(e) None of these

[Allahabad Bank PO, 2011]

170. Rehaan purchased a bike for ₹54,000. He sold it at a loss of 8%. With that money, he again purchased another bike and sold that at a profit of 10 per cent. What is his overall loss/profit?

(a) loss ₹657 (b) profit ₹567
(c) loss ₹648 (d) profit₹648
(e) None of these

[Corporation Bank PO, 2011]

171. A shopkeeper earns a profit of 12% on selling a book at 10% discount on the printed price. The ratio of the cost price and the printed price of the book is:

(a) 45:56 (b) 45:51
(c) 47:56 (d) 47:51

[SSC (GL), 2010]

172. What profit/loss per cent did Ravi earn if he purchased an item of ₹5600 and sold it at 3/4 of its cost price?

(a) Loss of 20 per cent
(b) Gain of 25 per cent
(c) Neither gain nor loss
(d) None of these

[OBC PO Examination, 2010]

173. A shopkeeper sells notebooks at the rate of ₹45 each and earns a commission of 4%. He also sells pencil box at the rate of ₹80 each and earns a commission of 20%. How much amount of commission will he earn in two weeks if he sells 10 notebooks and 6 pencil boxes a day?

(a) ₹1956 (b) ₹1586
(c) ₹1496 (d) ₹1596
(e) None of these

[CBI PO Examination, 2010)]

174. A shopkeeper bought 30 Kg of wheat at the rate of ₹45 per Kg. He sold 40% of the total quantity at the rate of ₹50 per Kg. Approximately, at what price per Kg should he sell the remaining quantity to make 25 per cent overall profit ?

(a) ₹54 (b) ₹52
(c) ₹50 (d) ₹60

[Allahabad Bank PO Examination, 2010]

175. A person sold a horse at a gain of 15%. Had he bought it for 25% less and sold it for ₹60 less, he would have made a profit of 32%. The cost price of the horse was:

(a) ₹370 (b) ₹372
(c) ₹375 (d) ₹378

[SSC, 2010]

176. A sells an article to B at a gain of 25%, B sells it to C at a gain 20% and C sells it to D at a gain of 10%. If D pays ₹330 for it, how much did it cost A?

(a) ₹200 (b) ₹250
(c) ₹275 (d) ₹290

[SSC, 2010]

177. By selling an article for ₹21, a man lost such that the percentage loss was equal to the cost price. The cost price of the article was:

(a) ₹30 or ₹70 (b) ₹35 or ₹60
(c) ₹45 (d) ₹50

[SSC, 2010]

178. Half of 100 articles were sold at a profit of 20% and the rest at a profit of 40%. If all of the articles had been sold at a profit of 25%, the total profit would have been ₹100 less than earlier profit. The cost price of each article was:

(a) ₹10 (b) ₹15
(c) ₹20 (d) ₹30

[SSC, 2010]

179. The marked price of a clock is ₹3200. It is to be sold at ₹2448 at two successive discounts. If the first discount is 10%, then the second discount is:

(a) 5% (b) 10%
(c) 15% (d) 20%

[SSC, 2010]

180. A dealer marks his goods 30% above his cost price and then allows 15% discount on it. What is the cost price of an article on which he gains ₹84?

(a) ₹800 (b) ₹560
(c) ₹373.33 (d) ₹280

[SSC, 2010]

181. A shopkeeper wishes to give 5% commission on the marked price of an article but also wants to earn a profit of 10%. If his cost price is ₹95, then the marked price is:

(a) ₹100 (b) ₹110
(c) ₹120 (d) ₹130

[SSC, 2010]

182. A shopkeeper sells sugar in such a way that the selling price of 950 g of sugar is the same as the cost price of 1 Kg of sugar. What is his gain per cent?

(a) $5\frac{5}{19}$ (b) $5\frac{1}{5}$
(c) 5 (d) $4\frac{1}{19}$

[SSC, 2010]

183. A person bought a horse and a carriage for ₹20000. Later, he sold the horse at 20% profit and the carriage at 10% loss. Thus, he gained 2% in the whole transaction. The cost price of the horse was:

(a) ₹7200 (b) ₹7500
(c) ₹8000 (d) ₹9000

[SSC, 2010]

184. A sells an article to B at 15% profit. B sells it to C at 10% loss. If C pays ₹517.50 for it then A purchased it at:

(a) ₹500 (b) ₹750
(c) ₹1000 (d) ₹1250

[SSC, 2010]

185. An article is sold at a certain fixed price. By selling it at $\frac{2}{3}$ of that price, one loses 10%. The gain per cent on selling it at the original price is:

(a) 20 (b) $33\frac{1}{3}$
(c) 35 (d) 40

[SSC, 2010]

186. A sells an article to B for ₹45,000 losing 10% in the transaction. B sells it to C at a price which would have given a profit of 10% to A. By what per cent does B gain?

(a) $\frac{75}{2}$ (b) $\frac{100}{3}$
(c) $\frac{200}{9}$ (d) $\frac{150}{7}$

[SSC, 2010]

187. The cost price of an article is 80% of its marked price for sale. How much per cent does the tradesman gain after allowing a discount of 12%?

(a) 20 (b) 12
(c) 10 (d) 8

[SSC, 2010]

188. A merchant purchases a wrist watch for ₹450 and fixes its list price in such a way that after allowing a discount of 10%, he earns a profit of 20%. Then the list price (in rupees) of the wrist watch is:

(a) ₹500 (b) ₹600
(c) ₹750 (d) ₹800

[SSC, 2010]

189. A, B, C are partners in a business. During a particular year, A received one third of the profit, B received $\frac{1}{4}$ of the profit and C received the remaining ₹5000. How much amount of money did A receive?

(a) ₹1000 (b) ₹3000
(c) ₹4000 (d) ₹5000

[SSC, 2010]

190. A shopkeeper sells notebooks at the rate of ₹457 each and earns a commission of 4%. He also sells pencil boxes at the rate of ₹80 each and earns a commission of 20%. How much amount of commission will he earn in two weeks if he sells 10 notebooks and 6 pencil boxes a day?

(a) ₹1,956 (b) ₹1,586
(c) ₹1,496 (d) ₹1,596
(e) None of these

[CBI PO, 2010]

191. Meenal purchased a car for ₹2,50,000 and sold it for ₹3,48,000. What is the per cent profit she made on the car?

(a) 40 (b) 39.2
(c) 38.4 (d) 38
(e) None of these

[Corporation Bank PO, 2010]

192. A shopkeeper bought 30 Kg of wheat at the rate of ₹45 per Kg. He sold 40% of the total quantity at the rate of ₹50 per Kg. Approximately, at what price per Kg should he sell the remaining quantity to make 25 per cent overall profit?

(a) ₹54 (b) ₹52
(c) ₹50 (d) ₹60
(e) ₹56

[Allahabad Bank PO, 2010]

ANSWER KEYS

EXERCISE-1

1. (b)	**2.** (a)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (c)	**7.** (a)	**8.** (b)	**9.** (c)	**10.** (a)	**11.** (c)	**12.** (a)	**13.** (b)
14. (c)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (c)	**19.** (b)	**20.** (a)	**21.** (c)	**22.** (a)	**23.** (b)	**24.** (a)	**25.** (a)	**26.** (b)
27. (a)	**28.** (c)	**29.** (b)	**30.** (c)	**31.** (c)	**32.** (a)	**33.** (c)	**34.** (b)	**35.** (a)	**36.** (c)	**37.** (c)	**38.** (a)	**39.** (c)
40. (a)	**41.** (b)	**42.** (a)										

EXERCISE-2

1. (b)	**2.** (c)	**3.** (c)	**4.** (d)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (c)	**9.** (b)	**10.** (d)	**11.** (d)	**12.** (d)	**13.** (c)
14. (c)	**15.** (d)	**16.** (c)	**17.** (b)	**18.** (d)	**19.** (a)	**20.** (d)	**21.** (b)	**22.** (c)	**23.** (b)	**24.** (b)	**25.** (c)	**26.** (d)
27. (d)	**28.** (c)	**29.** (d)	**30.** (d)	**31.** (b)	**32.** (d)	**33.** (a)	**34.** (b)	**35.** (a)	**36.** (a)	**37.** (a)	**38.** (c)	**39.** (b)
40. (b)	**41.** (c)	**42.** (a)	**43.** (b)	**44.** (c)	**45.** (a)	**46.** (a)	**47.** (c)	**48.** (b)	**49.** (c)	**50.** (d)	**51.** (c)	**52.** (b)
53. (a)	**54.** (c)	**55.** (b)	**56.** (a)	**57.** (a)	**58.** (c)	**59.** (b)	**60.** (d)	**61.** (b)	**62.** (c)	**63. (4)**	**64.** (e)	**65.** (b)
66. (d)	**67.** (d)	**68.** (c)	**69.** (c)	**70.** (b)	**71.** (a)	**72.** (c)	**73.** (d)	**74.** (b)	**75.** (c)	**76.** (b)	**77.** (a)	**78.** (b)
79. (b)	**80.** (b)	**81.** (d)	**82.** (a)	**83.** (d)	**84.** (b)	**85.** (b)	**86.** (d)	**87.** (d)	**88.** (d)	**89.** (a)	**90.** (d)	**91.** (c)
92. (b)	**93.** (a)	**94.** (d)	**95.** (b)	**96.** (c)	**97.** (a)	**98.** (a)	**99.** (a)	**100.** (d)	**101.** (d)	**102.** (a)	**103.** (a)	**104.** (c)
105. (a)	**106.** (a)	**107.** (c)	**108.** (b)	**109.** (b)	**110.** (b)	**111.** (b)	**112.** (a)	**113.** (c)	**114.** (b)	**115.** (a)	**116.** (b)	**117.** (c)
118. (a)	**119.** (a)	**120.** (a)	**121.** (c)	**122.** (b)	**123.** (c)	**124.** (d)	**125.** (c)	**126.** (b)	**127.** (a)	**128.** (c)	**129.** (a)	**130.** (a)
131. (a)	**132.** (b)	**133.** (c)	**134.** (b)	**135.** (d)	**136.** (a)	**137.** (c)	**138.** (a)	**139.** (c)	**140.** (c)	**141.** (a)	**142.** (d)	**143.** (d)
144. (b)	**145.** (b)	**146.** (d)	**147.** (a)	**148.** (b)	**149.** (d)	**150.** (c)	**151.** (a)	**152.** (b)	**153.** (b)	**154.** (a)	**155.** (d)	**156.** (d)
157. (a)	**158.** (b)	**159.** (d)	**160.** (b)	**161.** (a)	**162.** (d)	**163.** (b)	**164.** (c)	**165.** (d)	**166.** (d)	**167.** (d)	**168.** (e)	**169.** (a)
170. (d)	**171.** (a)	**172.** (d)	**173.** (d)	**174.** (d)	**175.** (c)	**176.** (a)	**177.** (a)	**178.** (c)	**179.** (c)	**180.** (a)	**181.** (b)	**182.** (a)
183. (c)	**184.** (a)	**185.** (c)	**186.** (c)	**187.** (c)	**188.** (b)	**189.** (c)	**190.** (e)	**191.** (b)	**192.** (d)			

EXPLANATORY ANSWERS

EXERCISE-1

1. (b) Here, C.P. = ₹350, S.P. = ₹392

Profit = S.P. − C.P. = 392 − 350 = ₹42.

$\therefore$ Profit % $= \frac{\text{Profit} \times 100}{\text{C.P.}} = \frac{42 \times 100}{350} = 12\%$

2. (a) C.P. of the bicycle = 5200 + 800 = ₹6000

S.P. = ₹5500. Since S.P. < C.P.,

$\therefore$ Loss = C.P. − S.P. = 6000 − 5500 = ₹500.

$\therefore$ Loss % $= \frac{\text{Loss} \times 100}{\text{C.P.}} = \frac{500 \times 100}{6000}$

$= \frac{25}{3}\%$ or $8\frac{1}{3}\%$

3. (b) Cost price of 10 articles (C.P.) = ₹8.

Selling price of 10 articles (S.P.) = 1.25 × 10

= ₹12.50

Profit = S.P. − C.P. = 12.50 − 8 = ₹4.50.

$\therefore$ Gain % = $\frac{\text{Gain} \times 100}{\text{C.P}} = \frac{4.50 \times 100}{8} = 56\frac{1}{4}\%$

4. (c) Marked price (M.P.) = ₹80.

Selling price (S.P.) = ₹68.

Discount = M.P. − S.P. = 80 − 68 = ₹12.

$\therefore$ Rate of discount = $\frac{\text{Discount} \times 100}{\text{Marked Price}} = \frac{12 \times 100}{80} = 15\%$

5. (a) Cost price of 200 dozen oranges = 200 × 10 = ₹2000

Transportation cost = ₹500.

$\therefore$ Cost Price (C.P.) = 2000 + 500 = ₹2500.

Selling Price (S.P.) = 200 × 12 × 1 = ₹2400.

Loss = C.P. − S.P. = 2500 − 2400 = ₹100.

$\therefore$ Loss % = $\frac{\text{Loss} \times 100}{\text{C.P.}} = \frac{100 \times 100}{2500} = 4\%$

6. (c) We have, S.P. = ₹10500, gain% = 5%

$\therefore$ C.P. = $\left(\frac{100}{100 + \text{Gain}\%}\right) \times \text{S.P.}$

$= \left(\frac{100}{100 + 5}\right) \times 10500$

$= \frac{100}{105} \times 10500 = ₹10000.$

7. (a) Here, C.P. = ₹1800, loss % = 10%

$\therefore$ S.P. = $\left(\frac{100 - \text{Loss}\%}{100}\right) \times \text{C.P.}$

$= \left(\frac{100 - 10}{100}\right) \times 1800$

$= \frac{90}{100} \times 1800 = ₹1620.$

8. (b) We have,

Cost price (C.P.) = (120 × 80 + 280 + 72 + 120 × 40)

= ₹10000.

Gain % = 8%

$\therefore$ Selling Price (S.P.) of 120 rims

$= \left(\frac{100 + \text{Gain}\%}{100}\right) \times \text{C.P.}$

$= \left(\frac{100 + 8}{100}\right) \times 10000$

$= \frac{108}{100} \times 10000 = ₹10800.$

Thus, selling price per rim = $\frac{10800}{120} = ₹90.$

9. (c) In the first case, S.P. = ₹31 and loss % = 7%

$\therefore$ C.P. = $\left(\frac{100}{100 - \text{Loss}\%}\right) \times \text{S.P.} = \left(\frac{100}{100 - 7}\right) \times 31$

$= \frac{100}{93} \times 31 = ₹33\frac{1}{3}.$

In the second case, C.P. = $₹33\frac{1}{3}$ and gain % = 5%

$\therefore$ S.P. = $\left(\frac{100 + \text{Gain}\%}{100}\right) \times \text{C.P.}$

$= \left(\frac{100 + 5}{100}\right) \times \frac{100}{3} = \frac{105}{100} \times \frac{100}{3} = ₹35.$

Alternative Solution

93% C.P of Cricket ball = ₹31

To gain 5%

$\therefore$ 105% of C.P of cricket ball = $\frac{31}{93} \times 105$

= ₹35

10. (a) In the first case, we have, S.P. = ₹35 and gain % = 40%

$\therefore$ C.P. = $\left(\frac{100}{100 + \text{Gain}\%}\right) \times \text{S.P.}$

$= \left(\frac{100}{100 + 40}\right) \times 35$

$= \frac{100}{140} \times 35 = ₹25.$

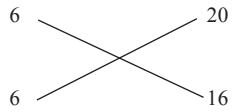
In the second case, C.P. = ₹25 and gain% = 60%

$\therefore$ S.P. = $\left(\frac{100 + \text{Gain}\%}{100}\right) \times \text{C.P.}$

$= \left(\frac{100 + 60}{100}\right) \times 25$

$= \frac{160}{100} \times 25 = ₹40.$

11. (c) Quantity Price



% profit = $\left(\frac{xw}{zy} - 1\right) \times 100\% = \left(\frac{6 \times 16}{4 \times 20} - 1\right) \times 100\%$

$= \frac{16}{80} \times 100\%$

= 20%

CHAPTER 17

Simple Interest

INTRODUCTION

When a person A borrows some money from another person B, then A has to pay certain amount to B in exchange of using the money. This amount paid by A is called *Interest*. The total amount of money borrowed by A from B is called the *Principal*. The money paid back to B, which comprises the principal and the interest is called the *Amount.*

In other words,

Amount = Principal + Interest

The interest is usually charged according to a specified term, which is expressed as some per cent of the principal and is called the *rate of interest* for a fixed period of time. This fixed period may be a year, six months, three months or a month, and correspondingly the rate of interest is charged annually, semi-annually, quarterly or monthly basis. For Examinationple, the rate of interest is 5% per annum means, the interest payable on ₹100 for one year is ₹5.

Interest is of two types:

1. Simple Interest
2. Compound Interest

SIMPLE INTEREST

When interest is payable on the principal amount only, it is called *Simple interest*. For example, simple interest on ₹100 at 5% per annum will be ₹5 each year, that is, at the end of the first year, total amount will be ₹105. At the end of the second year, it will be ₹110, and so on.

Thus, simple interest is the interest computed on the principal amount for the entire period it is borrowed.

In this chapter, we shall limit ourselves to simple interest. Compound interest will be discussed in the next chapter.

BASIC FORMULAE

If P stands for principal, R is the rate per cent per annum, T is the number of years, I is the simple interest and A is the amount, then

01 $$\text{Simple Interest} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

or, $$I = \frac{P \times R \times T}{100}$$

Illustration 1: Find the simple interest on ₹5200 for 2 years at 6% per annum.

Solution: Here, P = ₹5200, T = 2 years and R = 6%

$$\therefore \quad \text{Simple interest} = \frac{P \times R \times T}{100} = \frac{5200 \times 6 \times 2}{100}$$

= ₹624.

02 $$\text{Principal} = \frac{100 \times \text{Simple Interest}}{\text{Rate} \times \text{Time}}$$

or, $$P = \frac{100 \times I}{R \times T}$$

Illustration 2: A man earns ₹450 as interest in 2 years on a certain sum of money invested in a company at the rate of 12 per cent per annum. Find the sum invested by the man in the company.

Solution: We have, I = ₹450, T = 2 years,

R = 12% per annum

$$\therefore \quad P = \frac{I \times 100}{R \times T} = \frac{450 \times 100}{12 \times 2} = ₹1875.$$

Thus, the money invested by the man was ₹1875.

03 $\text{Rate} = \dfrac{100 \times \text{Simple Interest}}{\text{Principal} \times \text{Time}}$

or $R = \dfrac{100 \times I}{P \times T}$

Illustration 3: At what interest rate per annum, in 4 years, a sum of ₹5000 will become ₹6000?

Solution: Here, P = ₹5000, A = ₹6000, T = 4 years

So, $I = A - P$ = ₹(6000 − 5000) = ₹1000

$\therefore \quad R = \dfrac{100 \times I}{P \times T} = \dfrac{100 \times 1000}{5000 \times 4} = 5\%$

04 $\text{Time} = \dfrac{100 \times \text{Simple Interest}}{\text{Rate} \times \text{Principal}}$

or, $T = \dfrac{100 \times I}{R \times P}$

Illustration 4: In what time ₹1200 will earn an interest of ₹240 at 5% per annum?

Solution: Here, P = ₹1200, I = ₹240, R = 5%

$\therefore \quad T = \dfrac{100 \times I}{R \times P} = \dfrac{100 \times 240}{1200 \times 5} = 4$ years.

05 Amount = Principal + Simple Interest

$= \text{Principal} + \dfrac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$

$= \text{Principal}\left(1 + \dfrac{\text{Rate} \times \text{Time}}{100}\right)$

or, $A = P\left(1 + \dfrac{R \times T}{100}\right)$

Illustration 5: Mahesh borrowed ₹3000 from his friend Suresh at 15 per cent per annum for 3 years. Find the interest and money returned by Mahesh to Suresh.

Solution: Here, P = ₹3000, R = 15% per annum, T = 3 years

$\therefore \quad I = \dfrac{P \times R \times T}{100} = \dfrac{3000 \times 15 \times 3}{100} = ₹1350.$

$\therefore \quad A = P + I$ = ₹3000 + ₹1350 = ₹4350.

Thus, Mahesh paid ₹1350 as interest to Suresh, the total amount Mahesh returned to Suresh = ₹4350.

SHORT-CUT METHODS

01 If a certain sum in T years at R% per annum amounts to ₹A, then the sum will be

$$P = \frac{100 \times A}{100 + R \times T}$$

Explanation

Let the principal be ₹x

$\therefore$ Simple interest = ₹$(A - x)$

$\therefore \quad A - x = \dfrac{x \times R \times T}{100}$

$\Rightarrow \quad 100A - 100x = xRT$

$\Rightarrow \quad (100 + RT)x = 100A$

$\therefore \quad x = \dfrac{100 \times A}{100 + R \times T}.$

Illustration 6: What principal will amount to ₹570 at 4% per annum in $3\frac{1}{2}$ years?

Solution: We have, A = ₹570, R = 4% per annum, $T = \dfrac{7}{2}$ years.

$\therefore \quad P = \dfrac{100 \times A}{100 + R \times T} = \dfrac{100 \times 570}{100 + 4 \times 7/2}$

$= \dfrac{100 \times 570}{114} = ₹500$

Thus, ₹500 will be ₹570 at 4% per annum in $3\frac{1}{2}$ years.

02 The annual payment that will discharge a debt of ₹A due in T years at R% per annum is

$$\text{Annual payment} = ₹\left(\frac{100A}{100T + \dfrac{RT(T-1)}{2}}\right)$$

Explanation

Let the annual payment be ₹x.

Since the first instalment is paid at the end of the first year,

$\therefore$ Amount of the first instalment at the end of t years

$$= x + \frac{(T-1) \times R \times x}{100}$$

Similarly, amount of the second instalment at the end of t years

$$= x + \frac{(T-2)\times R\times x}{100}, \text{ and so on.}$$

Thus, total amount of T instalments

$$A = \left[x + \frac{(T-1)\times R\times x}{100}\right] + \left[x + \frac{(T-2)\times R\times x}{100}\right] + \cdots + x$$

$$= Tx + \frac{Rx}{100}[(T-1)+(T-2)+\cdots+1]$$

$$= Tx + \frac{Rx}{100}\left[\frac{(T-1)\times T}{2}\right]$$

or, $100\ Tx + Rx\left[\frac{T(T-1)}{2}\right] = 100A$

or, $x\left[100T + \frac{RT(T-1)}{2}\right] = 100A$

$$\therefore\quad x = \frac{100A}{100T + \frac{RT(T-1)}{2}}.$$

Illustration 7: Find out the annual instalment that will discharge a debt of ₹12,900 due in 4 years at 5% per annum simple interest.

Solution: Here, A = ₹12900, T = 4 years, R = 5% per annum.

$$\therefore\quad \text{Annual instalment} = \frac{100\times A}{100\times T + \frac{RT(T-1)}{2}}$$

$$= \frac{100\times 12900}{(100\times 4) + \frac{5(4-1)\times 4}{2}}$$

$$= \frac{100\times 12900}{400+30} = \frac{100\times 12900}{430}$$

$$= ₹3000.$$

03 If a certain sum is invested in n types of investments in such a manner that equal amount is obtained on each investment where interest rates are R_1, R_2, R_3, ..., R_n respectively, and time periods are T_1, T_2, T_3, ...,T_n respectively, then the ratio in which the amounts are invested is:

$$\frac{1}{100+R_1T_1} : \frac{1}{100+R_2T_2} :$$

$$\frac{1}{100+R_3T_3} : \ldots \frac{1}{100+R_nT_n}.$$

Explanation

Let P_1, P_2, ..., P_n be invested in n types of investments whose interest rates are R_1, R_2, ..., R_n and time periods are T_1, T_2, ..., T_n.

Then, $P_1 = \frac{100\times A}{100+R_1T_1}$

$P_2 = \frac{100\times A}{100+R_2T_2}$

$\vdots \qquad \vdots$

$P_n = \left(\frac{100\times A}{100+R_nT_n}\right).$

$\therefore\quad P_1:P_2:\ldots:P_n$

$$= \frac{100\times A}{100+R_1T_1} : \frac{100\times A}{100+R_2T_2} : \ldots \frac{100\times A}{100+R_nT_n}$$

$$= \frac{1}{100+R_1T_1} : \frac{1}{100+R_2T_2} : \ldots \frac{1}{100+R_nT_n}$$

[$\because$ the amount A remains same for all]

Illustration 8: A sum of ₹1586 is divided among three such parts that amount obtained on these three parts of money after 2, 3 and 4 years, respectively, at the rate of 5% per annum remains equal. Find out such three parts of the sum.

Solution: Since the amount accured from each of the three parts of ₹1586 at the rate of 5% p.a. in 2, 3 and 4 years, respectively, remains equal, such three parts of ₹1586 will be in the ratio of

$$\frac{1}{100+R_1T_1} : \frac{1}{100+R_2T_2} : \frac{1}{100+R_3T_3}$$

Hence, the ratio is

$$= \frac{1}{100+5\times 2} : \frac{1}{100+5\times 3} : \frac{1}{100+5\times 4}$$

$$= \frac{1}{110} : \frac{1}{115} : \frac{1}{120}$$

$$= \frac{1\times 30360}{110} : \frac{1\times 30360}{115} : \frac{1\times 30360}{120}$$

($\because$ L.C.M. of 110, 115 and 120 is 30360)

$\therefore$ Ratio = 276:264:253

Sum of proportionals = 276 + 264 + 253 = 793

$\therefore$ 1st part = $\frac{276}{793}\times 1586$ = ₹552,

2nd part = $\frac{264}{793}\times 1586$ = ₹528

and, 3rd part = $\frac{253}{793}\times 1586$ = ₹506.

04 If a certain sum of money becomes n times itself in T years at simple interest, then the rate of interest per annum is

$$R = \frac{100(n-1)}{T}\%$$

Explanation

Let, ₹P become ₹nP in t years.

$\therefore$ Simple interest I is given by

$$I = P - P = (n-1)P$$

$\therefore$ Rate of interest R is given by

$$R = \frac{100\times I}{P\times T} = \frac{100\times(n-1)P}{P\times T} = \frac{100(n-1)}{T}.$$

Illustration 9: A certain sum of money trebles itself in 5 years simple interest. Find the rate per cent per annum.

Solution: Here, $n = 3$, $T = 5$ years

$$\therefore \quad R = \frac{100(n-1)}{T}\% = \frac{100(3-1)}{5}\% = 40\%$$

05 If a certain sum of money becomes n times itself at R% per annum simple interest in T years, then

$$T = \left(\frac{n-1}{R}\right) \times 100 \text{ years.}$$

Illustration 10: In what time a sum of money will double itself at a rate of simple interest of 8% p.a.?

Solution: Required time $(T) = \dfrac{(n-1)\times 100}{R}$ years

$$= \frac{(2-1)\times 100}{8} \text{ years}$$

$$= 12\frac{1}{2} \text{ years.}$$

06 If a certain sum of money becomes n times itself in T years at a simple interest, then the time T in which it will become m times itself is given by

$$T' = \left(\frac{m-1}{n-1}\right)\times T \text{ years.}$$

Explanation

Let the principal be ₹P.

Let it become m times in T' years.

Then, the amount in T years = ₹nP

and the amount in T' years = ₹mP.

$$\therefore \quad nP - P = \frac{P\times R\times T}{100}$$

$$\text{or,} \quad (n-1)P = \frac{P\times R\times T}{100} \qquad \cdots(1)$$

$$\text{and,} \quad (m-1)P = \frac{P\times R\times T}{100} \qquad \cdots(2)$$

$$\therefore \quad \frac{(m-1)P}{(n-1)P} = \frac{P\times R\times T'}{100}\times\frac{100}{P\times R\times T}$$

$$\text{or,} \quad \frac{m-1}{n-1} = \frac{T'}{T}$$

$$\therefore \quad T' = \left(\frac{m-1}{n-1}\right) T \text{ years.}$$

Illustration 11: A sum of money put out on simple interest doubles itself in $12\frac{1}{2}$ years. In how many years would it treble itself?

Solution: Here, $n = 2$, $m = 3$, $T = \dfrac{25}{2}$ years.

$$\therefore \quad \text{Required time } (T') = \left(\frac{m-1}{n-1}\right) \times T \text{ years}$$

$$= \left(\frac{3-1}{2-1}\right)\times\frac{25}{2} \text{ years}$$

$$= 25 \text{ years.}$$

07 Effect of change of P, R and T on simple interest is given by the following formula:

Change in Simple Interest

$$= \frac{\text{Product of fixed parameter}}{100}$$

$\times$ [difference of product of variable parameters]

For example, if rate (R) changes from R_1 to R_2 and P, T are fixed, then

$$\text{Change in SI} = \frac{PT}{100}\times(R_1 - R_2)$$

Similarly, if principal (P) changes from P_1 to P_2 and R, T are fixed, then change in SI $= \dfrac{RT}{100}\times(P_1 - P_2)$

Also, if rate (R) changes from R_1 to R_2 and time (T) changes from T_1 to T_2, but principal (P) is fixed, then change in SI $= \dfrac{P}{100}\times(R_1T_1 - R_2T_2)$.

Illustration 12: If simple interest on ₹600 increases by ₹30, when the rate % increases by 4% per annum, find out the time.

Solution: Here, $P = 600$, change in SI $= 30$, $R_1 - R_2 = 4$, $T = ?$

Using, change in SI $= \dfrac{PT}{100}\times(R_1 - R_2)$

we have, $30 = \frac{600T}{100} \times 4$

$\Rightarrow \quad T = \frac{5}{4}$, i.e., $1\frac{1}{4}$ years.

Illustration 13: If the simple interest on ₹1400 be more than the interest on ₹1000 by ₹60 in 5 years, find the out the rate per cent per annum.

Solution: Here, change in SI = 60, $P_1 - P_2 = 400$, $T = 5$, $R = ?$

Using change in SI $= \frac{RT}{100} \times (P_1 - P_2)$

We have, $60 = \frac{5R}{100} \times 400 \quad \Rightarrow \quad R = 3\%$

Illustration 14: If the simple interest on a certain sum at 4% per annum for 4 years is ₹80 more than the interest on the same sum for 3 years at 5% per annum, find out the sum.

Solution: Here, change in SI = 80, $R_1 = 4$, $R_2 = 5$, $T_1 = 4$, $T_2 = 3$, $P = ?$

Using change in SI $= \frac{P}{100} \times (R_1T_1 - R_2T_2)$

We have, $80 = \frac{P}{100} \times (4 \times 4 - 5 \times 3)$

$\Rightarrow \quad P =$ ₹8000.

08 If a debt of ₹Z is paid in 'n' number of instalments and if the value of each instalment is ₹a, then the borrowed (debt) amount is given by

$$Z = na + \frac{Ra}{100 \times b} \times \frac{n(n-1)}{2}$$

where, R = rate of interest per annum
b = number of instalments/year
$b = 1$, when each instalment is paid yearly
$b = 2$, when each instalment is paid half-yearly
$b = 4$, when each instalment is paid quarterly
$b = 12$, when each instalment is paid monthly.

Illustration 15: A sum of ₹2 is lent to be paid back in 3 equal monthly instalments of ₹1 each. Find the rate per cent.

Solution: Here, $Z =$ ₹2, $a =$ ₹1, $n = 3$, $b = 12$, $R = ?$

Using the formula

$$Z = na + \frac{Ra}{100 \times b} \times \frac{n(n-1)}{2},$$

we have, $2 = 3 \times 1 + \frac{R \times 1}{100 \times 12} \times \frac{3 \times 2}{2} \quad \Rightarrow \quad R = 400\%$

$\therefore$ The rate % p.a. is 400%

09 If a certain sum of money P lent out at SI amounts to A_1 in T_1 years and to A_2 in T_2 years, then

$$P = \frac{A_1T_2 - A_2T_1}{T_2 - T_1}$$

and, $$R = \frac{A_1 - A_2}{A_1T_2 - A_2T_1} \times 100\%$$

Illustration 16: If a certain sum of money at simple interest amounts to ₹5184 in 2 years and to ₹5832 in 3 years, what is the sum and the rate of interest?

Solution: Principal $= \frac{A_1T_2 - A_2T_1}{T_2 - T_1}$

$$\left[\begin{array}{l}\text{Here, } A_1 = 5184,\ A_2 = 5832\\ T_1 = 2,\ T_2 = 3\end{array}\right]$$

$$= \frac{5184 \times 3 - 5832 \times 2}{3 - 2} = ₹3888$$

and, Rate $= \frac{(A_2 - A_1) \times 100}{T_1A_2 - T_2A_1} = \frac{(5832 - 5184) \times 100}{2 \times 5832 - 3 \times 5184}$

$$= \frac{64800}{3888} = 16\frac{2}{3}\%$$

10 If a certain sum of money P lent out for a certain time T amounts to A_1 at R_1% per annum and to A_2 at R_2% per annum, then

$$P = \frac{A_2R_1 - A_1R_2}{R_1 - R_2}$$

and, $T = \frac{A_1 - A_2}{A_2R_1 - A_1R_2} \times 100$ years.

Illustration 17: A certain sum is invested for certain time. It amounts to ₹450 at 7% per annum. But, when invested at 5% per annum, it amounts to ₹350. Find out the sum and time.

Solution: Here, $A_1 = 450$, $R_1 = 7$, $A_2 = 350$, $R_2 = 5$.

Using the formula,

$$P = \frac{A_2R_1 - A_1R_2}{R_1 - R_2}$$

We get, $P = \frac{350 \times 7 - 450 \times 5}{7 - 5} =$ ₹100

Also, using the formula,

$$T = \left(\frac{A_1 - A_2}{A_2R_1 - A_1R_2}\right) \times 100$$

we get, $T = \left(\frac{450-350}{350\times 7-450\times 5}\right)\times 100 = 5$ years.

11 If an amount P_1 lent at simple interest rate of R_1% per annum, and another amount P_2 at simple interest rate of R_2% per annum, then the rate of interest for the whole sum is

$$R = \left(\frac{P_1R_1 + P_2R_2}{P_1 + P_2}\right).$$

Illustration 18: Mohan deposits ₹5000 in NSC at 2% per annum and ₹2000 in mutual funds at 4% per annum. Find out the rate of interest for the whole sum.

Solution: Here, $P_1 = 5000$, $R_1 = 2$, $P_2 = 2000$, $R_2 = 4$. Using the formula

$$R = \left(\frac{P_1R_1 + P_2R_2}{P_1 + P_2}\right)$$

We get, $R = \frac{5000\times 2 + 2000\times 4}{5000 + 2000} = 2\frac{4}{7}\%$

12 If a certain sum of money is lent out in n parts in such a manner that equal sum of money is obtained as simple interest on each part where interest rates are $R_1, R_2, ..., R_n$ and time periods are $T_1, T_2, ..., T_n$, respectively, then the ratio in which the sum will be divided in n parts is given by

$$\frac{1}{R_1T_1} : \frac{1}{R_2T_2} : ... \frac{1}{R_nT_n}.$$

Explanation

Let the n equal parts be $P_1, P_2, ..., P_n$ and, let I be the equal interest earned on each part.

Then, $P_1 = \frac{I\times 100}{R_1T_1}$

$P_2 = \frac{I\times 100}{R_2T_2}$

$\vdots \qquad\qquad \vdots$

$P_n = \frac{I\times 100}{R_nT_n}$.

$\therefore \quad P_1:P_2: ... :P_n = \frac{I\times 100}{R_1T_1} : \frac{I\times 100}{R_2T_2} : ... \frac{I\times 100}{R_nT_n}$

$= \frac{1}{R_1T_1} : \frac{1}{R_2T_2} : ... \frac{1}{R_nT_n}$.

Illustration 19: If a sum of ₹1600 is divided into two such parts that the simple interest on the first part for 2 $\frac{1}{2}$ years at the rate of 4% p.a. equals the simple interest on the second part for 5 years at the rate of 3% p.a., then find two such divisions of the sum.

Solution: Ratio of one part to other part of ₹1600

$$= \frac{1}{R_1T_1} : \frac{1}{R_2T_2}$$

$\therefore$ 1st part:2nd part $= \frac{1}{4\times\frac{5}{2}} : \frac{1}{3\times 5}$

[Here, R_1 = 4% p.a., $T_1 = \frac{5}{2}$ years, R_2 = 3% p.a., T_2 = 5 years]

or, 1st part:2nd part $= \frac{1}{10} : \frac{1}{15} = 3:2$

Sum of proportionals = 3 + 2 = 5

$\therefore$ 1st part $= \frac{3}{5}\times 1600 = ₹96$

and, 2nd part $= \frac{2}{5}\times 1600 = ₹640$.

13 When there is a change in principal (P), Rate (R) and Time (T), then the value of simple interest I also changes and is given by

$$\frac{I_1}{I_2} = \frac{P_1\times R_1\times T_1}{P_2\times R_2\times T_2}$$

$$\Rightarrow \quad \frac{A_1 - P_1}{A_2 - P_2} = \frac{P_1\times R_1\times T_1}{P_2\times R_2\times T_2}$$

as $I_1 = A_1 - P_1$ and $I_2 = A_2 - P_2$.

Illustration 20: If ₹85 amounts to ₹95 in 3 years, what ₹102 will amount to in 5 years at the same rate per cent?

Solution: Here, P_1 = ₹85, A_1 = ₹95, T_1 = 3 years, P_2 = ₹102, T_2 = 5 years, $R_1 = R_2 = R$ (say).

Then, using the formula

$$\frac{A_1 - P_1}{A_2 - P_2} = \frac{P_1\times R_1\times T_1}{P_2\times R_2\times T_2}$$

We have, $\frac{95-85}{A_2 - 102} = \frac{85\times R\times 3}{102\times R\times 5}$

$\Rightarrow \quad A_2 - 102 = 20$

$\Rightarrow \quad A_2 = 122$

$\therefore$ The amount is ₹122.

14 Out of a certain sum P, $\frac{1}{a}$ part is invested at R_1%, $\frac{1}{b}$ part at R_2% and the remainder $\left(1-\frac{1}{a}-\frac{1}{b}\right)$ say

$\frac{1}{c}$ part at R_3%. If the annual income from all these investments is ₹A, then the original sum is given by

$$P = \left(\frac{A \times 100}{\frac{R_1}{a} + \frac{R_2}{b} + \frac{R_3}{c}} \right).$$

Illustration 21: Out of a certain sum, $\frac{1}{3}$ is invested at 3%, $\frac{1}{6}$ at 6% and the rest at 8%. If the annual income is ₹300, then the original sum is

Solution: Here,

$$\frac{1}{a} = \frac{1}{3}, \frac{1}{b} = \frac{1}{6},$$

$$\frac{1}{c} = 1 - \left(\frac{1}{3} + \frac{1}{6} \right) = \frac{1}{2},$$

R_1 = 3%, R_2 = 6%, R_3 = 8%,

A = ₹300.

$$\therefore \text{ The original sum } = \frac{A \times 100}{\frac{R_1}{a} + \frac{R_2}{b} + \frac{R_3}{c}}$$

$$= \frac{300 \times 100}{\frac{3}{3} + \frac{6}{6} + \frac{8}{2}} = \frac{30000}{6}$$

= ₹5000.

EXERCISE-I

1. The simple interest on ₹500 at 6% per annum from May 3rd to July 15th in the same year is:
(a) ₹9 (b) ₹6
(c) ₹4 (d) None of these

2. Mr Irani borrowed a sum of ₹10000 from a finance company for 6 years at 8% per annum. The amount returned by Mr Irani to the finance company is:
(a) ₹14800 (b) ₹12600
(c) ₹13300 (d) None of these

3. The principal that will yield ₹60 as simple interest at 6% per annum in 5 years is:
(a) ₹175 (b) ₹350
(c) ₹200 (d) None of these

4. The sum of money that will produce ₹1770 interest in $7\frac{1}{2}$ years at 8% simple interest per annum is:
(a) ₹2950 (b) ₹3120
(c) ₹2800 (d) None of these

5. If the simple interest on a certain sum of money after $6\frac{1}{4}$ years is 3/8 of the principal, then the rate of interest per annum is:
(a) 5% (b) 6%
(c) 4% (d) None of these

6. Rakesh borrowed ₹5000 from Ganesh at simple interest. If Ganesh received ₹500 more than his capital after 5 years, then the rate of interest per annum is:
(a) 2% (b) 3%
(c) 4% (d) None of these

7. The rate per cent per annum at which ₹1200 amount to ₹1440 in 4 years, is:
(a) 5% (b) 4%
(c) 6% (d) None of these

8. If simple interest on a certain sum of money is ₹256 and the rate of interest per annum equals the number of years, then the rate of interest is:
(a) 13% (b) 14%
(c) 16% (d) None of these

9. If the simple interest on a certain sum of money for 2 years is one-fifth of the sum, then the rate of interest per annum is:
(a) 9% (b) 10%
(c) 8% (d) None of these

10. If the simple interest on a certain sum of money is $\frac{4}{25}$ of the sum and the rate per cent equals the number of years, then the rate of interest per annum is:
(a) 2% (b) 3%
(c) 4% (d) None of these

11. If a certain sum of money borrowed at 5% per annum simple interest amounts to ₹1020 in 4 years, then the sum of money borrowed is:
(a) ₹850 (b) ₹925
(c) ₹750 (d) None of these

12. In what time ₹1200 will become ₹1344 at 6% per annum?

(a) $2\frac{1}{2}$ years (b) 3 years

(c) 2 years (d) None of these

13. In what time ₹8100 will produce the same income at 3% as ₹225 in 4 years at 3%.

(a) $\frac{1}{7}$ years (b) $\frac{1}{9}$ years

(c) $\frac{1}{6}$ years (d) None of these

14. If ₹1000 be invested at interest rate of 5% and the interest be added to the principal every 10 years, then the number of years in which it will amount to ₹2000 is:

(a) $16\frac{2}{3}$ years (b) $16\frac{1}{4}$ years

(c) 16 years (d) None of these

15. If ₹500 amounts to ₹725 at 9% simple interest in some time, what will ₹600 amount to at 11% in the same time?

(a) ₹870 (b) ₹930

(c) ₹910 (d) None of these

16. Sumit lends ₹10000 for 2 years at 20% per annum simple interest. After 1 year, he receives ₹6000. How much will he receive next year?

(a) ₹5900 (b) ₹6400

(c) ₹7200 (d) None of these

17. What principal will amount to ₹15000 at 10% per annum in 5 years?

(a) ₹10000 (b) ₹8700

(c) ₹10500 (d) None of these

18. The annual payment that will discharge a debt of ₹47250 due 3 years, hence at the rate of 5% simple interest is:

(a) ₹8000 (b) ₹10000

(c) ₹15000 (d) None of these

19. The annual instalment that will discharge a debt of ₹4200 due in 5 years at 10% simple interest is:

(a) ₹700 (b) ₹750

(c) ₹800 (d) None of these

20. If the amount obtained by Mahesh by investing ₹1500 for $2\frac{1}{2}$ years at the rate of 8% p.a. is equal to the amount obtained by Suresh by investing a certain sum for 2 years at 5% p.a. simple interest, then the sum invested by Suresh is:

(a) ₹$1636\frac{4}{11}$ (b) ₹1636

(c) ₹$1636\frac{1}{2}$ (d) None of these

21. A man invests ₹3965 in the names of his three daughters, Neeta, Sita and Gita in such a way that they would receive the same amount after 2, 3 and 4 years. If the rate of interest is 5% p.a., then the amount invested for Neeta, Sita and Gita is:

(a) ₹1380, ₹1320, ₹1265
(b) ₹1330, ₹1360, ₹1380
(c) ₹1265, ₹1320, ₹1340
(d) None of these

22. A sum of money at simple interest becomes four times in 24 years. The rate per cent of interest per annum is:

(a) $13\frac{3}{4}\%$ (b) $12\frac{1}{2}\%$

(c) $11\frac{3}{4}\%$ (d) None of these

23. In how many years will a sum of money treble itself at 10% per annum simple interest?

(a) 15 years (b) 19 years

(c) 20 years (d) None of these

24. A sum of money doubles itself in 8 years. In how many years will it treble?

(a) 16 years (b) 15 years

(c) 14 years (d) None of these

25. A sum was put at simple interest at a certain rate for 4 years. Had it been put at 2% higher rate, it would have fetched ₹56 more. Find the sum.

(a) ₹680 (b) ₹700

(c) ₹720 (d) None of these

26. If the interest on ₹800 be more than the interest on ₹400 by ₹40 in 2 years, then the rate of interest per annum is:

(a) 5% (b) $5\frac{1}{2}\%$

(c) 6% (d) None of these

27. If the difference between the simple interest on a certain sum for 4 years at $2\frac{1}{2}\%$ per annum and the simple interest on the same sum for the same period at 3% per annum is ₹60, then the sum is:

(a) ₹3000 (b) ₹2900

(c) ₹3100 (d) None of these

28. If a certain sum of money at simple interest amounts to ₹2800 in 2 years and ₹3250 in 5 years, then the rate of interest per annum is:

(a) 4% (b) 6%
(c) 5% (d) None of these

29. If a certain sum of money amounts to ₹1760 in two years and ₹2000 in 5 years at simple interest, then the sum is:

(a) ₹1960 (b) ₹1590
(c) ₹1600 (d) None of these

30. A certain sum is invested for certain time. It amounts to ₹450 at 7% per annum. But when invested at 5% per annum, it amounts to ₹350. Find the sum.

(a) ₹60 (b) ₹100
(c) ₹120 (d) None of these

31. A certain sum is invested for T years. It amounts to ₹400 at 10% per annum. But, when invested at 4% per annum, it amounts to ₹200. Find the time (T).

(a) 41 years (b) 39 years
(c) 50 years (d) None of these

32. If a sum of ₹9 is lent to be paid back in 10 equal monthly instalments of ₹1 each, then the rate of interest is:

(a) $266\frac{2}{3}\%$ (b) $265\frac{3}{4}\%$
(c) 266% (d) None of these

33. A sum of ₹7700 is to be divided among three brothers Vikas, Vijay and Viraj in such a way that simple interest on each part at 5% per annum after 1, 2 and 3 years, remains equal. The share of Vikas is more than that of Viraj by:

(a) ₹2800 (b) ₹2500
(c) ₹3000 (d) None of these

34. If simple interest on a certain sum of money for 4 years at 5% p.a. is same as the simple interest on ₹560 for 10 years at the rate of 4% p.a., then the sum of money is:

(a) ₹1190 (b) ₹1120
(c) ₹1210 (d) None of these

35. Mr Mani invested an amount of ₹12000 at a simple interest rate of 10% per annum and another amount at a simple interest rate of 20% per annum. The total interest earned at the end of one year on the total amount invested became 14% per annum. Find the total amount invested.

(a) ₹20000 (b) ₹20800
(c) ₹21000 (d) None of these

36. Mr Gupta deposits ₹3000 in a bank at 10% per annum and ₹5000 in another bank at 8% per annum. The rate of interest for the whole sum is:

(a) $8\frac{1}{2}\%$ (b) $8\frac{3}{4}\%$
(c) 8% (d) None of these

37. A person invested $\frac{2}{3}$ of his capital at 3%, $\frac{1}{6}$ at 6% and the remainder at 12%. If his annual income is ₹25, then the capital is:

(a) ₹490 (b) ₹510
(c) ₹500 (d) None of these

38. The simple interest on a sum of money will be ₹600 after 10 years. If the principal is trebled after 5 years, then what will be the total interest at the end of the tenth year?

(a) ₹1200 (b) ₹1190
(c) ₹1210 (d) None of these

39. ₹1500 is invested at a rate of 10% simple interest and interest is added to the principal after every 5 years. In how many years will it amount to ₹2500.

(a) $6\frac{1}{9}$ years (b) $6\frac{1}{4}$ years
(c) 7 years (d) None of these

40. Sumit lent some money to Mohit at 5% per annum simple interest. Mohit lent the entire amount to Birju on the same day at $8\frac{1}{2}\%$ per annum. In this transaction, after a year Mohit earned a profit of ₹350. Find out the sum of money lent by Sumit to Mohit.

(a) ₹9000 (b) ₹10000
(c) ₹10200 (d) None of these

41. Brinda borrowed ₹1000 to build a hut. She pays 5% simple interest. She lets the hut to Ramu and receives a rent of $₹12\frac{1}{2}$ per month from Ramu. In how many years Brinda would clear off the debt?

(a) 10 years (b) $10\frac{1}{4}$ years
(c) $10\frac{1}{2}$ years (d) None of these

42. The rate of interest on a sum of money is 4% per annum for the first 2 years, 6% per annum for the next 4 years, and 8% per annum for the period beyond 6 years. If the simple interest accrued by the sum for a total period of 9 years is ₹1120, then the sum is:

(a) ₹2400 (b) ₹2200
(c) ₹2000 (d) None of these

EXERCISE-2
(BASED ON MEMORY)

1. The simple interest on a sum for 8 years is ₹47500. The rate of 10% per annum and for the next 3 years is 15% per annum. What is the value (in ₹) of sum?

(a) 50000 (b) 60000
(c) 45000 (d) 62500

[SSC CHSL (10+2) Tier-I CBE, 2018]

2. A sum of ₹8000 is divided into two parts. The simple interest on first part at the rate of 21% per annum is equal to the simple interest on second part at the rate of 35% per annum. What is the interest (in ₹) of each part?

(a) 1050 (b) 840
(c) 1400 (d) 1220

[SSC CHSL (10+2) Tier-I CBE, 2018]

3. An investor invested his saving in the stock market. The value of his investments increased by 12% and 9% in the first year and the second year respectively. If the value of his investments after two years became ₹97,664 then how much had he invested (in ₹)?

(a) 81000 (b) 75000
(c) 80000 (d) 72000

[SSC CGL Tier-II CBE, 2018]

4. In how much time (in years) will a sum of ₹9500 amounts to ₹17622.5 at the rate of 4.5% per annum at simple interest?

(a) 21 (b) 23
(c) 19 (d) 20

[SSC Delhi Police Constable, 2017]

5. Dalajit lent ₹10800 to Jaabir for 3 years and ₹7500 to kabir for 2 years on simple interest and received ₹1422 in all from both of them as interest. The rate of interest per annum is

(a) 3.5 per cent (b) 4 per cent
(c) 3 per cent (d) 4.5 per cent

[SSC CGL Tier-I CBE, 2017]

6. Deepak lent ₹8800 to Jaichand for 13 years on simple interest at the same rate of interest and received ₹14432 in all from both of them as interest. The rate of interest per annum is

(a) 8.5 per cent (b) 9 per cent
(c) 9.5 per cent (d) 8 per cent

[SSC CGL Tier-I CBE, 2017]

7. A sum was doubled with $12\frac{1}{2}\%$ rate of simple interest, per annum. Then taken for that sum is:

(a) $8\frac{1}{2}$ years (b) 8 years
(c) 10 years (d) $12\frac{1}{2}$ years

[SSC Matric Level MTS, 2017]

8. What is the simple interest on ₹5400 in 5 years at the rate of 12% per annum?

(a) ₹2700 (b) ₹2950
(c) ₹3120 (d) ₹3240

[SSC CAPFs ASI and Delhi Police SI, 2017]

9. A certain sum becomes ₹1020 in 5 years and ₹1200 in B years at simple interest. What is the value of Principal?

(a) ₹820 (b) ₹780
(c) ₹700 (d) ₹720

[SSC CAPFs ASI and Delhi Police SI, 2017]

10. A certain sum becomes 5 times in 3 years, at simple interest, then in how many years it will become 13 times?

(a) 6 (b) 15
(c) 9 (d) 12

[SSC CAPFs ASI and Delhi Police SI, 2017]

11. A certain sum of money amounts to ₹918 in 2 years and ₹969 in 3.5 years at simple interest, What is the rat of intrest (in % per annum)?

(a) 4 (b) 5
(c) 6 (d) 8

[SSC CGL Tier-I CBE, 2017]

12. Simple interest received by a person in 10 years on a principal of ₹9500 is 130% of the principal. What is the rate of interest (in %) per annum?

(a) 12 (b) 13
(c) 15 (d) 19

[SSC CGL Tier-I CBE, 2017]

13. An amount fetched a total simple interest of ₹3200 at the rate of 6.25% per year in 4 years. What is the amount (in ₹)?

(a) 13800 (b) 11800
(c) 12800 (d) 14800

[SSC CGL Tier-I CBE, 2017]

14. The effective annual rate of interest corresponding to a nominal rate of 13% per annum payable half-yearly is

(a) 26% (b) 26.85%
(c) 13% (d) 13.42%

[SSC CHSL (10+2) Tier-I CBE, 2017]

15. A person lent ₹10000 to B for 3 years and ₹6000 to C for 4 years on simple interest at same rate of interest and received ₹5400 in all from both of them as interest. What is the rate of the interest (in%)?

(a) 10 (b) 12.5
(c) 15 (d) 20

[SSC CHSL (10+2) Tier-I CBE, 2017]

16. Albert invested and amount of x rupees in a fixed deposit scheme offering 10% per annum for 1st Year and 15% per annum for 2nd year and received an amount of ₹20,240 after two years. What is x (in ₹)?

(a) 15000 (b) 16000
(c) 14000 (d) 18000

[SSC CGL Tier-I CBE, 2017]

17. Alipta got some amount of money from her father. In how many years will the ratio of the money and the interest obtained from it be 10:3 at the rate of 6% simple interest per annum?

(a) 7 years (b) 3 years
(c) 5 years (d) 4 years

[SSC CGL Tier-I (CBE), 2016]

18. The simple interest on a certain sum of money at the rate of 5% per annum for 8 years is ₹840. Rate of interest for which the same amount of interest can be received on the same sum, after 5 years is:

(a) 7% per annum (b) 8% per annum
(c) 9% per annum (d) 10% per annum

[SSC CHSL (10+2) Tier-I (CBE), 2016]

19. A sum of ₹2800 is divided into two parts in such a way that the interest on both the parts is equal. If the first part is lent at 9% p.a. for 5 years and second part is for 6 years at 10% 9.a., find the two sums.

(a) ₹1800, ₹1000 (b) ₹1600, ₹1200
(c) ₹1400, ₹1400 (d) ₹1300, ₹1500

[SSC CAPFs (CPO) SI and ASI, Delhi Police, 2016]

20. If the simple interest of ₹1 for month is paisa, then the rate per cent per annum will be

(a) 10% (b) 8%
(c) 12% (d) 6%

[SSC CGL Tier-I (CBE), 2016]

21. In simple interest rate per annum a certain sum amounts to ₹5,182 in 2 years and ₹5,832 in 3 years. The principal rupees is

(a) ₹2882 (b) ₹5000
(c) ₹3882 (d) ₹4000

[SSC CGL Tier-I (CBE), 2016]

22. For what sum will the simple interest at R% per annum for 2 years will be R?

(a) ₹$\frac{100}{2R}$ (b) ₹50

(c) ₹$\frac{100}{R}$ (d) ₹$\frac{200}{R}$

[SSC CGL Tier-I (CBE), 2016]

23. A certain sum doubles in 7 years at simple interest. The same sum under the same interest rate will become 4 times in how many years.

(a) 14 (b) 28
(c) 21 (d) 10

[SSC CPO SI, ASI, 2016]

24. A certain sum of money amounts to ₹2200 at 5% p.a. rate of interest, ₹2320 at 8% interest in the same period of time, The period of time is:

(a) 3 years (b) 4 years
(c) 5 years (d) 2 years

[SSC CAPFs (CPO) SI and ASI, Delhi Police, 2016]

25. At what per cent of simple interest will a sum of money double itself in 15 years?

(1) $6\frac{1}{3}\%$ (2) $6\frac{2}{3}\%$

(3) $6\frac{1}{2}\%$ (4) 6%

[SSC CGL Tier-I (CBE), 2016]

26. The rate of simple interest for which a sum money becomes 5 times of itself in 8 years is:

(a) 30% (b) 40%
(c) 50% (d) 55%

[SSC CGL Tier-I (CBE), 2016]

27. If a borrowed ₹P at x% and B borrowed ₹Q (> P) at y% per annum at simple interest at the same time, then the amount of their debts will be equal after

(a) $100\left(\frac{Q-P}{Px-Qy}\right)$ years

(b) $100\left(\frac{Px-Qy}{Q-P}\right)$ years

(c) $100\left(\frac{Px-Qy}{P-Q}\right)$ years

(d) $100\left(\frac{P-Q}{Px-Qy}\right)$ years

[SSC CGL Tier-II, 2016]

28. A money lender claims to lend money at the rate of 10% per annum simple interest, However, he takes the interest in advance when he lends a sum for one year. At what interest rate does he lend the money actually?

(a) 10% (b) $10\frac{1}{9}\%$

(c) 11% (d) $11\frac{1}{9}\%$

[SSC CPO SI, ASI, 2016]

29. A man buys a watch for ₹1950 in cash and sells it for ₹2200 at a credit of 1 year. If the rate of interest be 10% per annum, then how much profit or loss will he have?

(a) ₹55 gain (b) ₹50 Profit
(b) ₹30 gain (d) ₹30 Profit

[SSC CGL Tier-I (CBE), 2016]

30. A money lender lends ₹400 for 3 years to a person and lends ₹500 for 4 years to the other person at the same rate of simple interest. If altogether he receives ₹160 as interest per annum?

(a) 5% (b) 7%
(c) 9% (d) 10%

[SSC CGL Tier-I (CBE), 2016]

31. The simple interest on a sum of money is $\frac{8}{25}$ of the sum. If the number of years is numerically half the rate percent per annum, then the rate percent per annum is

(a) 4 (b) $6\frac{1}{4}$

(c) 5 (d) 8

[SSC, 2015]

32. A sum of money lent out at simple interest amounts to ₹720 after 2 years and ₹1020 after a further period of 5 years. Find the principal.

(a) ₹600 (b) ₹1740
(c) ₹120 (d) ₹6000

[SSC, 2015]

33. Ram deposited a certain sum of money in a company at 12% per annum simple interest for 4 years and deposited equal amount in fixed deposit in a bank for 5 years at 15% per annum simple interest. If the difference in the interest from two sources is ₹1350 then the sum deposited in each case is:-

(a) ₹5000 (b) ₹4000
(c) ₹6500 (d) ₹3000

[SSC, 2015]

34. Prakash lends a part of ₹20,000 at 8% simple interest and remaining at $\frac{4}{3}\%$ simple interest. His total income after a year was ₹800. Find the sum lent at 8%.

(a) ₹8,000 (b) ₹12,000
(c) ₹6,000 (d) ₹10,000

[SSC, 2014]

35. Nitin borrowed some money at the rate of 6% p.a. for the first three years, 9% p.a. for the next five years and 13% p.a. for the period beyond eight years. If the total interest paid by him at the end of eleven years is ₹8,160, the money borrowed by him was:

(a) 12,000 (b) 6,000
(c) 8,000 (d) 10,000

[SSC Assistant Grade III, 2013]

36. A sum of ₹16800 is divided into two parts. One part is lent at a simple rate of interest 6% per annum and the other at 8% per annum. After 2 years the total sum received is ₹19000. The sum lent at the rate of 6% simple interest is:

(a) ₹12200 (b) ₹12000
(c) ₹11000 (d) ₹10000
(e) None of these

[IBPS PO/MT, 2013]

37. Arun invested a sum of money at a certain rate of simple interest for a period of four years. Had he invested the same sum for a period of six years, the total interest earned by him would have been fifty per cent more than the earlier interest amount? What was the rate of interest per cent per annum?

(a) 4 (b) 8
(c) 5 (d) Cannot be determined

[Gramin Bank U.P. (SO) Examination, 2012]

38. The simple interest on a sum of money is $\frac{1}{9}$th of the principal and the number of years is equal to the rate per cent per annum. The rate per cent per annum is equal to:

(a) 3% (b) $\frac{1}{3}\%$

(c) $\frac{1}{10}\%$ (d) $3\frac{1}{3}\%$

[SSC Assistant Grade III, 2012]

39. Arun lends ₹20,000 to two of his friends. He gives ₹12,000 to the first at 8% p.a. simple interest. Arun wants to makes a profit of 10% on the whole. The simple interest rate at which he should lend the remaining sum of money to the second friend is:

(a) 8% (b) 16%
(c) 12% (d) 13%

[SSC, 2012]

40. In what time will a sum of money double itself at the rate of 20% per annum simple interest

(a) 10 years (b) 5 years
(c) 2 years (d) 14 years

[SSC (GL), 2011]

41. 800 becomes ₹956 in 3 years at a certain rate of simple interest. If the rate of interest is increased by 4%, what amount will ₹800 become in 3 years?

(a) ₹1020.80 (b) ₹1025
(c) ₹1052 (d) ₹1050

[SSC (GL), 2011]

42. Simple interest on a certain sum is $\frac{16}{25}$ of the sum. The rate per cent if the rate per cent and time (in years) are equal, is:

(a) 6% (b) 8%
(c) 10% (d) 12%

[SSC (GL), 2011]

43. If the simple interest on ₹x at a rate of a% for m years is same as that on ₹y at a rate of a^2% for m^2 years, then x:y is equal to:

(a) m:a (b) am:1

(c) $\frac{1}{m}:\frac{1}{a}$ (d) $\frac{1}{am}:1$

[SSC, 2011]

44. A took two loans altogether of ₹1200 from B and C. B claimed 14% simple interest per annum, while C claimed 15% per annum. The total interest paid by A in one year was ₹172. Then a borrowed:

(a) ₹800 from C (b) ₹625 from C
(c) ₹400 from B (d) ₹800 from B

[SSC, 2011]

45. A person has left an amount of ₹1,20,000 to be divided between his two sons aged 14 years and 12 years such that they get equal amounts when each attains 18 years of age. If the amount gets a simple interest of 5% per annum, the younger son's share at present is:

(a) ₹48,800 (b) ₹57,600
(c) ₹62,400 (d) ₹84,400

[SSC, 2011]

46. The simple interest accrued on an amount of ₹22,500 at the end of four years is ₹10,800. What would be the compound interest accrued on the same amount at the same rate of interest at the end of two years?

(a) ₹16,908 (b) ₹5,724
(c) ₹28,224 (d) ₹8,586
(e) None of these

[IBPS PO/MT, 2011]

47. The simple interest accrued on a sum of certain principal is ₹1200 in 4 years at the rate of 8% per annum.. What would be the simple interest accrued on thrice of that principal at the rate of 6% per annum in 3 years?

(a) ₹2025 (b) ₹3025
(c) ₹2250 (d) ₹2150

[OBC PO, 2010]

48. A man invested $\frac{1}{3}$ of his capital at 7%, $\frac{1}{4}$ at 8% and the remaining at 10% rate of simple interest. If his annual income from interests is ₹561, then the capital invested was:

(a) ₹6000 (b) ₹5600
(c) ₹6600 (d) ₹7200

[SSC, 2010]

49. A person rece a simple interest of ₹1,000 on a certain principal at the rate of 5% p.a. in 4 years. What compound interest will would that person receive on twice the principal in two years at the same rate?

(a) ₹1,000 (b) ₹1,005
(c) ₹11,025 (d) ₹10,125
(e) None of these

[Punjab and Sind Bank PO, 2010]

ANSWER KEYS

EXERCISE-1

1. (b)	**2.** (a)	**3.** (c)	**4.** (a)	**5.** (b)	**6.** (a)	**7.** (a)	**8.** (c)	**9.** (b)	**10.** (c)	**11.** (a)	**12.** (c)	**13.** (b)
14. (a)	**15.** (b)	**16.** (c)	**17.** (a)	**18.** (c)	**19.** (a)	**20.** (a)	**21.** (a)	**22.** (b)	**23.** (c)	**24.** (a)	**25.** (b)	**26.** (a)
27. (a)	**28.** (b)	**29.** (c)	**30.** (b)	**31.** (c)	**32.** (a)	**33.** (a)	**34.** (b)	**35.** (a)	**36.** (b)	**37.** (c)	**38.** (a)	**39.** (a)
40. (b)	**41.** (a)	**42.** (c)										

EXERCISE-2

1. (a)	**2.** (a)	**3.** (c)	**4.** (c)	**5.** (c)	**6.** (d)	**7.** (b)	**8.** (d)	**9.** (d)	**10.** (c)	**11.** (a)	**12.** (b)	**13.** (c)
14. (d)	**15.** (a)	**16.** (b)	**17.** (c)	**18.** (b)	**19.** (b)	**20.** (c)	**21.** (c)	**22.** (b)	**23.** (c)	**24.** (d)	**25.** (b)	**26.** (c)
27. (a)	**28.** (d)	**29.** (b)	**30.** (a)	**31.** (d)	**32.** (a)	**33.** (a)	**34.** (a)	**35.** (c)	**36.** (a)	**37.** (d)	**38.** (d)	**39.** (d)
40. (b)	**41.** (c)	**42.** (b)	**43.** (b)	**44.** (d)	**45.** (b)	**46.** (b)	**47.** (a)	**48.** (c)	**49.** (e)			

EXPLANATORY ANSWERS

EXERCISE-1

1. (b) Time from May 3rd to July 15th

= 28 days of May + 30 days of June and 15 days of July

= 73 days = $\frac{73}{365}$ years, i.e., $\frac{1}{5}$ years.

$\therefore\ I = \frac{P \times R \times T}{100} = \frac{500 \times 6 \times \frac{1}{5}}{100} = ₹6.$

2. (a) We have, P = ₹10000, R = 8% per annum, T = 6 years.

$\therefore\ I = \frac{P \times R \times T}{100} = \frac{10000 \times 8 \times 6}{100} = ₹4800$

$\therefore\ A = P + I = 10000 + 4800 = ₹14800$

Thus, Mr Irani returned ₹14800 to the finance company.

3. (c) Here, I = ₹60, R = 6% per annum, T = 5 years.

$\therefore$ Principal $(P) = \frac{100 \times I}{R \times T} = \frac{100 \times 60}{6 \times 5} = ₹200.$

4. (a) Here, I = ₹1770, R = 8% per annum, $T = \frac{15}{2}$ years.

$\therefore$ Principal $(P) = \frac{100 \times I}{R \times T} = \frac{100 \times 1770}{8 \times \frac{15}{2}}$

= ₹2950.

5. (b) Let the sum of money be ₹x.

Then, simple interest = $\frac{3}{8}x$

Also, time = $6\frac{1}{4}$ years, i.e., $\frac{25}{4}$ years

$\therefore$ Rate $(R) = \frac{100 \times I}{P \times T} = \frac{100 \times \frac{3x}{8}}{x \times \frac{25}{4}}$

$= \frac{100 \times 3}{2 \times 25} = 6\%.$

6. (a) Here, P = ₹5000, I = ₹500, T = 5 years.

Therefore, using the formula

$R = \frac{100 \times I}{P \times T}$

We have, rate of interest $(R) = \frac{100 \times 500}{5000 \times 5} = 2\%$ p.a.

7. (a) We have, P = ₹1200, T = 4 years,

I = 1440 −1200 = ₹240.

$\therefore$ Rate $(R) = \frac{100 \times I}{P \times T} = \frac{100 \times 240}{1200 \times 4} = 5\%$ per annum.

CHAPTER 19

Logarithms

INTRODUCTION

Logarithm, in Mathematics, is the 'exponent' or 'power' to which a stated number called the *base*, is raised to yield a specific number. For example, in the expression $10^2 = 100$, the logarithm of 100 to the **base** 10 is 2. This is written as $\log_{10} 100 = 2$. Logarithms were originally invented to help simplify the arithmetical processes of multiplication, division, expansion to a power and extraction of a 'root', but they are nowadays used for a variety of purposes in pure and applied Mathematics.

Logarithm

If for a positive real number ($a \neq 1$), $a^m = b$, then the index m is called the logarithm of b to the base a. We write this as

$$\log_a b = m$$

'log' being the abbreviation of the word 'logarithm'. Thus,

$$a^m = b \Leftrightarrow \log_a b = m$$

where, $a^m = b$ is called the *exponential form* and $\log_a b = m$ is called the *logarithmic form*.

Illustration 1: Refer to the following Table

Exponential form	logarithmic form
$3^5 = 243$	$\log_3 243 = 5$
$2^4 = 16$	$\log_2 16 = 4$
$3^0 = 1$	$\log_3 1 = 0$
$8^{1/3} = 2$	$\log_8 2 = \frac{1}{3}$

LAWS OF LOGARITHMS

1. **Product formula**

   The logarithm of the product of two numbers is equal to the sum of their logarithms.
   i.e., $\log_a (mn) = \log_a m + \log_a n$.
   Generalisation: In general, we have
   $\log_a (mnpq...) = \log_a m + \log_a n + \log_a p + \log_a q + ...$

2. **Quotient formula**

   The logarithm of the quotient of two numbers is equal to the difference of their logarithms.
   i.e., $\log_a \left(\frac{m}{n}\right) = \log_a m - \log_a n$,
   where, a, m, n are positive and $a \neq 1$.

3. **Power formula**

   The logarithm of a number raised to a power is equal to the power multiplied by logarithm of the number.
   i.e., $\log_a (m^n) = n \log_a m$,
   where, a, m are positive and $a \neq 1$.

3. **Base changing formula**

   $\log_n m = \frac{\log_a m}{\log_a n}$. So, $\log_n m = \frac{\log m}{\log n}$.
   where, m, n, a are positived and $n \neq 1$, $a \neq 1$.

4. **Reciprocal relation**

   $\log_b a \times \log_a b = 1$,
   where, a, b are positive and not equal to 1.

5. $\log_b a = \frac{1}{\log_a b}$

6. $a^{\log_a x} = x$, where, a and x are positive, $a \neq 1$.

7. If $a > 1$ and $x > 1$, then $\log_a x > 0$.

8. If $0 < a < 1$ and $0 < x < 1$, then $\log_a x > 0$.

9. If $0 < a < 1$ and $x > 1$, then $\log_a x > 0$.

10. If $a > 1$ and $0 < x < 1$, then $\log_a x < 0$.

SHORT-CUT METHODS

01 Logarithm of 1 to any base is equal to zero.

i.e., $\log_a 1 = 0$, where $a > 0$, $a \neq 1$.

02 Logarithm of any number to the same base is 1.

i.e., $\log_a a = 1$, where $a > 0$, $a \neq 1$.

Common Logarithms

There are two bases of logarithms that are extensively used these days. One is base e (e = 2. 71828 approx.) and the other is base 10. The logarithms to base e are called natural logarithms. The logarithms to base 10 are called the common logarithms.

$\log_{10}10 = 1$, since $10^1 = 10$.

$\log_{10}100 = 2$, since $10^2 = 100$.

$\log_{10}10000 = 4$, since $10^4 = 10000$.

$\log_{10}0.01 = -2$, since $10^{-2} = 0.01$.

$\log_{10}0.001 = -3$, since $10^{-3} = 0.001$

and, $\log_{10}1 = 0$, since $10° = 1$.

EXERCISE-I

1. Find $\log_{3/2} 3.375$.
(a) 2 (b) 3
(c) 5/2 (d) 17/2

2. If $x = \log_{2a} a$, $y = \log_{3a} 2a$ and $z = \log_{4a} 3a$, find yz $(2 - x)$.
(a) 1 (b) –1
(c) 2 (d) –2

3. $\dfrac{\log x}{l+m-2m} = \dfrac{\log y}{m+n-2l} = \dfrac{\log z}{n+l-2m}$, find $x^2y^2z^2$.
(a) 2 (b) –1
(c) 4 (d) 1

4. If $\log \dfrac{x+y}{5} = \dfrac{1}{2}(\log x + \log y)$, then $\dfrac{x}{y} + \dfrac{y}{x}$ =
(a) 20 (b) 23
(c) 22 (d) 21

5. If $\log(x + y) = \log\left(\dfrac{3x-3y}{2}\right)$, then $\log x - \log y$ =
(a) $\log 2$ (b) $\log 3$
(c) $\log 5$ (d) $\log 6$

6. If $\log_2 x + \log_4 x + \log_{16} x = 21/4$, then x =
(a) 8 (b) 4
(c) 2 (d) 16

7. $7\log\dfrac{16}{15} + 5\log\dfrac{25}{24} + 3\log\dfrac{81}{80}$ =
(a) $\log 2$ (b) $\log 3$
(c) $\log 5$ (d) None of these

8. If $0 < a \leq x$, the minimum value of $\log_a x + \log_x a$ is:
(a) 1 (b) 2
(c) 3 (d) 5

9. If $\dfrac{\log x}{b-c} = \dfrac{\log y}{c-a} = \dfrac{\log z}{a-b}$, then $xyz = x^a \cdot y^b \cdot z^c$ $= x^{b+c} \cdot y^{c+a} \cdot z^{a+b}$ =
(a) 1 (b) 0
(c) 2 (d) None of these

10. $x^{\log y - \log z} \cdot y^{\log z - \log x} \cdot z^{\log x - \log y}$ =
(a) 0 (b) 2
(c) 1 (d) None of these

11. If $\log_{10}[98 + \sqrt{x^2 - 12x + 36}] = 2$, then x =
(a) 4 (b) 8
(c) 12 (d) 4,8

12. If $x = \log_a bc$, $y = \log_b ca$, $z = \log_c ab$, then
(a) $xyz = x + y + z + 2$
(b) $xyz = x + y + z + 1$
(c) $x + y + z = 1$
(d) $xyz = 1$.

13. If $a^x = b^y = c^z = d^w$, then $\log_a (bcd)$ =

(a) $\frac{1}{x}\left(\frac{1}{y}+\frac{1}{z}+\frac{1}{w}\right)$ (b) $x\left(\frac{1}{y}+\frac{1}{z}+\frac{1}{w}\right)$

(c) $\frac{y+z+w}{x}$ (d) None of these

14. If $\log_{10} 2 = 0.3010$, then $\log_{10} (1/2)$ =

(a) –0.3010 (b) 0.6990

(c) 1.6990 (d) –0.3010

15. If $\log_2 (3^{2x-2} + 7) = 2 + \log_2 (3^{x-1} + 1)$, then x =

(a) 0 (b) 1

(c) 2 (d) 1 or 2

16. If $\log_a b = \log_b c = \log_c a$, then

(a) $a > b \geq c$ (b) $a < b < c$

(c) $a = b = c$ (b) $a < b \leq c$.

17. If $\frac{1}{\log_x 10} = \frac{2}{\log_a 10} - 2$, then x =

(a) $a/2$ (b) $a/100$

(c) $a^2/10$ (d) $a^2/100$

18. If $a^2 + b^2 = c^2$, then $\frac{1}{\log_{c+a} b} + \frac{1}{\log_{c-a} b}$ =

(a) 1 (b) 2

(c) –1 (d) –2

19. If $\log_{10} 87.5 = 1.9421$, then the number of digits in $(875)^{10}$ is:

(a) 30 (b) 29

(c) 20 (d) 19

20. If $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$, then the number of zeros between the decimal point and the first significant figure in $(0.0432)^{10}$ is:

(a) 10 (b) 13

(c) 14 (d) 15

21. If $(4.2)^x = (0.42)^y = 100$, then $\frac{1}{x} - \frac{1}{y}$ =

(a) 1 (b) 2

(c) 1/2 (d) –1

22. $\frac{\log_9 11}{\log_5 13} - \frac{\log_3 11}{\log_{\sqrt{5}} 13}$ =

(a) 1 (b) –1

(c) 0 (d) None of these

23. If $\frac{\log x}{2} = \frac{\log y}{3} = \frac{\log z}{5}$, then yz in terms of x is:

(a) x (b) x^2

(c) x^3 (d) x^4

24. If $4^x + 2^{2x-1} = 3^{x+\frac{1}{2}} + 3^{x-\frac{1}{2}}$, then x =

(a) 1/2 (b) 3/2

(c) 5/2 (d) 1

25. $\frac{\log 49\sqrt{7} + \log 25\sqrt{5} - \log 4\sqrt{2}}{\log 17.5}$ =

(a) 5 (b) 2

(c) 5/2 (d) 3/2

26. $\log_{10} \tan 40° \cdot \log_{10} 41° \ldots \log_{10} \tan 50°$ =

(a) 1 (b) 0

(c) –1 (d) None of these

27. If $\log_8 p = 2.5$, $\log_2 q = 5$, then p in terms of q is

(a) $q\sqrt{q}$ (b) $2q$

(c) q (d) $q/2$

28. If $y = \frac{1}{a^{1-\log_a x}}$, $z = \frac{1}{a^{1-\log_a y}}$ and $x = a^k$, then k =

(a) $\frac{1}{a^{1-\log_a z}}$ (b) $\frac{1}{1-\log_a z}$

(c) $\frac{1}{1+\log_z a}$ (d) $\frac{1}{1-\log_z a}$

29. If $\log_e 2 \cdot \log_b 625 = \log_{10}{}^{16} \cdot \log_e 10$, then b =

(a) 4 (b) 5

(c) 1 (d) e

30. $5^{\sqrt{\log_5 7}} - 7^{\sqrt{\log_7 5}}$

(a) $\log 2$ (b) 1

(c) 0 (d) None of these

31. $2\log_3 7 - 7\log_3 2$

(a) $\log_2 7$ (b) $\log 7$

(c) $\log 2$ (d) 0

32. If $\log_{30} 3 = a$, $\log_{30} 5 = b$, then $\log_{30} 8$ =

(a) $3(1 - a - b)$ (b) $a - b + 1$

(c) $1 - a - b$ (d) $3(a - b + 1)$

33. If $0 < a < 1$, $0 < x < 1$ and $x < a$, then $\log_a x$:

(a) < 1 (b) > 1

(c) < 0 (d) ≤ 1

34. $\log_5 2$ is

(a) an integer (b) a rational number

(c) an irrational number (d) a prime number

35. $\log_5\left(1+\frac{1}{5}\right) + \log_5\left(1+\frac{1}{6}\right) + \log_5\left(1+\frac{1}{7}\right) + \cdots + \log_5\left(1+\frac{1}{624}\right)$

(a) 5 (b) 4
(c) 3 (d) 2

36. If $\log_{10} 2986 = 3.4751$, then $\log_{10} 0.02986 =$
(a) $\bar{1}.2986$ (b) $\bar{2}.4751$
(c) 0.34751 (d) None of these

37. If $\log(2a - 3b) = \log a - \log b$, then $a =$
(a) $\frac{3b^2}{2b-1}$ (b) $\frac{3b}{2b-1}$
(c) $\frac{b^2}{2b+1}$ (d) $\frac{3b^2}{2b+1}$

38. If $\log(x - y) - \log 5 - \frac{1}{2}\log x - \frac{1}{2}\log y = 0$, then $\frac{x}{y} + \frac{y}{x} =$
(a) 25 (b) 26
(c) 27 (d) 28

39. If $\log x{:}3 = \log y{:}4 = \log z{:}5$, then $zx =$
(a) $2y$ (b) y^2
(c) $8y$ (d) $4y$

40. If $3 + \log_5 x = 2\log_{25} y$, then $x =$
(a) $y/125$ (b) $y/25$
(c) $y^2/625$ (d) $3 - y^2/25$

41. If $\frac{\log_2 a}{2} = \frac{\log_3 b}{3} = \frac{\log_4 c}{4}$ and $a^{1/2} \cdot b^{1/3} \cdot c^{1/4} = 24$, then
(a) $a = 24$ (b) $b = 81$
(c) $c = 64$ (d) $c = 256$

42. If $\frac{\log_2 x}{3} = \frac{\log_2 y}{4} = \frac{\log_2 z}{5k}$ and $\frac{z}{x^3 y^4} = 1$, then $k =$
(a) 3 (b) 4
(c) 5 (d) -5

43. $\frac{3 + \log_{10} 343}{2 + \frac{1}{2}\log\left(\frac{49}{4}\right) + \frac{1}{3}\log\left(\frac{1}{125}\right)} =$
(a) 3 (b) 3/2
(c) 2 (d) 1

44. If $\frac{\log x}{a^2 + ab + b^2} = \frac{\log y}{b^2 + bc + c^2} = \frac{\log z}{c^2 + ca + a^2}$, then $x^{a-b} \cdot y^{b-c} \cdot z^{c-a} =$
(a) 0 (b) -1
(c) 1 (d) 2

45. If $3^{x-2} = 5$ and $\log_{10} 2 = 0.20103$, $\log_{10} 3 = 0.4771$, then $x =$
(a) $1\frac{22187}{47710}$ (b) $2\frac{22187}{47710}$
(c) $3\frac{22187}{47710}$ (d) None of these

46. If $\log_2 = 0.30103$ and $\log_3 = 0.4771$, then number of digits in $(648)^5$ is:
(a) 12 (b) 13
(c) 14 (d) 15

47. If $\log x = \frac{\log y}{2} = \frac{\log z}{5}$, then $x^4 \cdot y^3 \cdot z^{-2} =$
(a) 2 (b) 10
(c) 1 (d) 0

48. $\frac{\log\sqrt{27} + \log\sqrt{1000} + \log 8}{\log 120}$
(a) 1/2 (b) 1
(c) 3/2 (d) 2

49. For $x > 0$, if $y = \frac{{}_{10}\log_{10}^{x}}{x^2}$ and $x = y^a$, then $a =$
(a) 1 (b) -1
(c) 0 (d) 2

50. If $x = 100_{4/3}(1/2)$, $y = \log_{1/2}(1/3)$, then
(a) $x > y$ (b) $x < y$
(c) $x = y$ (d) $x \geq y$

ANSWER KEYS

EXERCISE-I

1. (b)	**2.** (a)	**3.** (d)	**4.** (b)	**5.** (c)	**6.** (a)	**7.** (a)	**8.** (b)	**9.** (a)	**10.** (c)	**11.** (d)	**12.** (a)	**13.** (b)
14. (c)	**15.** (d)	**16.** (c)	**17.** (d)	**18.** (b)	**19.** (a)	**20.** (b)	**21.** (c)	**22.** (c)	**23.** (d)	**24.** (b)	**25.** (c)	**26.** (b)
27. (a)	**28.** (b)	**29.** (b)	**30.** (c)	**31.** (d)	**32.** (a)	**33.** (b)	**34.** (c)	**35.** (b)	**36.** (b)	**37.** (a)	**38.** (c)	**39.** (b)
40. (a)	**41.** (d)	**42.** (c)	**43.** (a)	**44.** (c)	**45.** (c)	**46.** (d)	**47.** (c)	**48.** (c)	**49.** (b)	**50.** (b)		

EXPLANATORY ANSWERS

EXERCISE-I

1. (b) $\log_{3/2} 3.375 = x \Rightarrow \left(\frac{3}{2}\right)^x = 3.375$

$\Rightarrow (1.5)^x = (1.5)^3 \Rightarrow x = 3.$

2. (a) $yz(2 - x) = 2yz - xyz = 2\log_{4a} 2a - \log_{4a} a$

$= \log_{4a}\left(\frac{4a^2}{a}\right) = 1.$

3. (d) Each is equal to k

$\Rightarrow \log x = k\,(l + m - 2n)$,

$\log y = k\,(m + n - 2l)$, $\log z = k\,(n + l - 2m)$.

$\Rightarrow \log xyz = k\,(0) \Rightarrow xyz = e^0 = 1 \Rightarrow x^2y^2z^2 = 1.$

4. (b) $\log\left(\frac{x+y}{5}\right) = \frac{1}{2}[\log x + \log y]$

$\Rightarrow x + y = 5\sqrt{xy} \Rightarrow x^2 + y^2 = 23xy$

$\Rightarrow \frac{x}{y} + \frac{y}{x} = 23.$

5. (c) $x + y = \frac{3x - 3y}{2} \Rightarrow x = 5y \Rightarrow \frac{x}{y} = 5$

$\Rightarrow \log x - \log y = \log 5.$

6. (a) $\log_2 x + \frac{1}{2}\log_2 x + \frac{1}{4}\log_2 x = \frac{21}{4}$

$\Rightarrow \log_2^x\left(1 + \frac{1}{2} + \frac{1}{4}\right) = \frac{21}{4} \Rightarrow \log_2 x = 3 \Rightarrow x = 8.$

7. (a) $7\log\left(\frac{2^4}{5\times 3}\right) + 5\log\left(\frac{5^2}{2^3\times 3}\right) + 3\log\left(\frac{3^4}{2^4\times 5}\right)$

$= 28\log 2 - 7\log 5 - 7\log 3 + 10\log 5 - 15\log 2$

$-5\log 3 + 12\log 3 - 12\log 2 - 3\log 5 = \log 2.$

8. (b) $0 < a \le x$; Min. value of $\log_a x + \log_x a$ is 2 when we put $x = a$.

9. (a) $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b} = k$ (say)

$\Rightarrow \log x = k\,(b - c)$, $\log y = k\,(c - a)$, $\log z = k\,(a - b)$

$\Rightarrow \log x + \log y + \log z = 0 \Rightarrow xy\,z = 1.$

Also, $a\log x + b\log y + c\log z = 0 \Rightarrow x^a \cdot y^b \cdot z^c = 1.$

Again $(b + c)\log x + (c + a)\log y + (a + b)\log z = 0.$

$\Rightarrow x^{b+c} \cdot y^{c+a} \cdot z^{a+b} = 1.$

$\therefore xyz = x^a \cdot y^b \cdot z^c = x^{b+c} \cdot y^{c+a} \cdot z^{a+b} = 1.$

10. (c) $x^{\log y - \log z} \cdot y^{\log z - \log x} \cdot z^{\log x - \log y} = k$ (say)

$\Rightarrow (\log y - \log z)\log x + (\log z - \log x)\log y$

$+ (\log x - \log y)\log z = \log k = 0$

$\Rightarrow k = 1.$

11. (d) $98 + \sqrt{x^2 - 12x + 36} = 100$

$\Rightarrow \sqrt{x^2 - 12x + 36} = 2$

$\Rightarrow x^2 - 12x + 32 = 0$

$\Rightarrow x = 8, 4.$

12. (a) $x = \log_a bc \Rightarrow a^x = bc \Rightarrow a^{x+1} = abc$

$\Rightarrow a = (abc)^{1/x+1}.$

Similarly, $b = (abc)^{1/y+1}$ and $c = (abc)^{1/z+1}$

$\therefore abc = (abc)^{\frac{1}{x+1}+\frac{1}{y+1}+\frac{1}{z+1}}$

$\Rightarrow 1 = \frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}$

$\Rightarrow (x + 1)(y + 1)(z + 1) = (y + 1)(z + 1)$

$+ (x + 1)(z + 1) + (x + 1)(y + 1)$

$\Rightarrow xyz = x + y + z + 2.$

13. (b) $b^y = a^x \Rightarrow b = a^{\frac{x}{y}}, c = a^{\frac{x}{z}}, d = a^{\frac{x}{w}}$

$\log_a(bcd) = \log_a\left(a^{\frac{x}{y}}.a^{\frac{x}{z}}.a^{\frac{x}{w}}\right) = \frac{x}{y} + \frac{x}{z} + \frac{x}{w} = x\left(\frac{1}{y} + \frac{1}{z} + \frac{1}{w}\right).$

14. (c) $\log_{10}\left(\frac{1}{2}\right) = -\log_{10} 2 = -0.3010$

$= 1 - 0.3010 - 1 = 1.6990.$

15. (d) $\log_2(3^{2x-2} + 7) = \log_2 4 + \log_2(3^{x-1} + 1)$

$[\because 2 = 2\log_2^2 = \log_2 2^2]$

$\Rightarrow 3^{2x-2} + 7 = 4(3^{x-1} + 1)$

$\Rightarrow t^2 + 7 = 4(t + 1)$, where, $3^{x-1} = t$

$\Rightarrow t^2 - 4t + 3 = 0 \Rightarrow t = 1, 3$

When $t = 1 \Rightarrow 3^{x-1} = 1 \Rightarrow x = 1$

When $t = 3 \Rightarrow 3^{x-1} = 3^1 \Rightarrow x = 2.$

16. (c) $\log_a b = \log_b c = \log_c a = k$ (say)

$b = a^k$, $c = b^k$, $a = c^k$

$\Rightarrow c = (a^k)^k = a^{k^2} = (c^{k^2})^k = c^{k^3}$

$\Rightarrow k^3 = 1 \Rightarrow k = 1. \therefore a = b = c.$

17. (d) $\log_{10} x = 2\log_{10} a - 2$

$\Rightarrow \log_{10} x = 2(\log_{10} a - 1)$

$\Rightarrow \log_{10} x = 2\log_{10}\left(\frac{a}{10}\right) \Rightarrow x = \frac{a^2}{100}.$

18. **(b)** $\log_b(c + a) + \log_b(c - a)$
$= \log_b(c^2 - a^2) = \log_b b^2 = 2.$

19. **(a)** $x = (875)^{10} = (87.5 \times 10)^{10}$
$\therefore \quad \log_{10} x = 10(\log_{10} 87.5 + 1)$
$= 10(1.9421 + 1)$
$= 10(2.9421) = 29.421.$
$\therefore \quad x = \text{Antilog } (29.421).$
$\therefore \quad$ Number of digits in $x = 30.$

21. **(c)** $(4.2)^x = 100 \quad \Rightarrow \quad (42)^x = 10^{2+x}$

$$\Rightarrow \quad 42 = \left(\frac{42}{100}\right)^y \qquad \cdots(1)$$

$$\frac{2}{x} - \frac{2}{y} = 100 \quad \Rightarrow \quad (42)^y = 10^{2+2y}$$

$$\Rightarrow \quad 42 = 10^{\frac{2}{y}+2} \qquad \cdots(2)$$

From (1) and (2), $\dfrac{2}{x} - \dfrac{2}{y} = 1 \Rightarrow \dfrac{1}{x} - \dfrac{1}{y} = \dfrac{1}{2}.$

22. **(c)** $\dfrac{\log_9 11}{\log_5 13} - \dfrac{\log_3 11}{\log_{\sqrt{5}} 13} = \dfrac{\log_3 11}{2.\log_5 13} - \dfrac{\log_3 11}{2.\log_5 13} = 0.$

23. **(d)** $\dfrac{\log x}{2} = \dfrac{\log y}{3} = \dfrac{\log z}{5} = k$ (say)
$\Rightarrow \quad \log x = 2k,\ \log y = 3k,\ \log z = 5k$
$\Rightarrow \quad \log yz = 3k + 5k = 8k;\ \log x^4 = 8k$
$\therefore \quad \log yz = \log x^4 \quad \Rightarrow \quad yz = x^4.$

24. **(b)** $4^x + \dfrac{4^x}{2} = \dfrac{3^x}{\sqrt{3}} + 3^x.\sqrt{3}$

$$\Rightarrow \quad 4^x \cdot \frac{3}{2} = 3^x \cdot \frac{4}{\sqrt{3}} \Rightarrow \left(\frac{4}{3}\right)^x = \frac{8}{3\sqrt{3}}$$

$$\Rightarrow \quad \left(\frac{4}{3}\right)^x = \left(\frac{4}{3}\right)^{3/2} \Rightarrow x = \frac{3}{2}.$$

25. **(c)** $\dfrac{\log 7^{5/2} + \log 5^{5/2} - \log 2^{5/2}}{\log 17.5}$

$$= \frac{5(\log 7 + \log 5 - \log 2)}{2\log\left(\frac{35}{2}\right)} = \frac{5}{2}$$

26. **(b)** $\log_{10} \tan 40° \cdot \log_{10} \tan 41° \cdots \log_{10} \tan 50°$
$= 0,$ since $\log_{10} \tan 45 = 0.$

27. **(a)** $\log_8 p = \dfrac{5}{2} \quad \Rightarrow \quad p = (8)^{5/2} = 2^{\frac{15}{2}} = (2^5)^{3/2}$
$\log_2 q = 5 \quad \Rightarrow \quad q = 2^5.$
$\therefore \quad p = q^{3/2}.$

28. **(b)** $\log_a y = \dfrac{1}{1 - \log_a x}$, $\log_a z = \dfrac{1}{1 - \log_a y}$

$$\therefore \quad \log_a z = \frac{1}{1 - \left(\frac{1}{1 - \log_a x}\right)} = \frac{1 - \log_a x}{-\log_a x}$$

$$\Rightarrow \quad -\log_a z = -1 + \frac{1}{\log_a x}$$

$$\Rightarrow \quad \frac{1}{\log_a x} = 1 - \log_a z$$

$$\therefore \quad \log_a x = \frac{1}{1 - \log_a z} \Rightarrow x = \frac{1}{a^{(1-(\log_a(z)))}} = a^k \text{ (given)}$$

$$\therefore \quad k = \frac{1}{1 - \log_a z}.$$

29. **(b)** $\log_e 2 \cdot 4\log_b 5 = 4 \cdot \log_{10} 2 \cdot \log_e 10 = 4\log_e 2$
$\Rightarrow \quad \log_b 5 = 1 \quad \Rightarrow \quad b = 5.$

30. **(c)** $5^{\sqrt{\log_5 7}} - (7^{\log_7 5})^{\frac{1}{\sqrt{\log_7 5}}}$

$$= 5^{\sqrt{\log_5 7}} - \frac{1}{5^{\sqrt{\log_7 5}}}$$

$$\Rightarrow \quad 5^{\sqrt{\log_5 7}} - 5^{\sqrt{\log_5 7}} = 0.$$

31. **(d)** $2\log_3 7 - 7\log_3 2 = 2\log_{27} \cdot \log_3 2 - 7\log_3 2$
$= 7\log_3 2 - 7\log_3 2 = 0.$

32. **(a)** $a + b = \log_{30} 15 = \log_{30}\left(\dfrac{30}{2}\right) = 1 - \log_{30} 2$
$\Rightarrow \quad \log_{30} 2 = 1 - a - b.$
$\therefore \quad \log_{30} 8 = 3(1 - a - b).$

33. **(b)** $0 < a < 1,\ 0 < x < 1$ and $x < a$
$\Rightarrow \quad \log_a x > \log_a a \quad \Rightarrow \quad \log_a x > 1.$

34. **(c)** $\log_5 2 = \dfrac{p}{q} \quad \Rightarrow \quad 2 = 5^{p/q} = 2^q = 5^p$
$\Rightarrow$ even number = odd number,
which is a contradiction.
$\therefore \quad \log_5 2$ is an irrational number.

35. **(b)** $\log_5 \dfrac{6}{5} + \log_5 \dfrac{7}{6} + \log_5 \dfrac{8}{7} + \cdots + \log_5 \dfrac{625}{624}$

$$= \log_5\left(\frac{6}{5} \cdot \frac{7}{6} \cdot \frac{8}{7} \cdots \frac{625}{624}\right) = \log_5\left(\frac{625}{5}\right) = 4.$$

36. **(b)** $\log_{10}(0.02986) = \log_{10}\left(\dfrac{2986}{100000}\right)$
$= 3.4751 - 5 = -1.5249$
$= 2.4751.$

37. **(a)** $2a - 3b = \frac{a}{b} \Rightarrow 2ab - 3b^2 = a$

$\Rightarrow 3b^2 = a(2b - 1)$

$\Rightarrow a = \frac{3b^2}{2b-1}$.

38. **(c)** $(x - y)^2 = 25xy \Rightarrow x^2 + y^2 = 27xy \Rightarrow \frac{x}{y} + \frac{y}{x} = 27$.

39. **(b)** $\frac{\log x}{3} = \frac{\log y}{4} = \frac{\log z}{5} = k$

$\Rightarrow \log x = 3k;\ \log y = 4k;\ \log z = 5k.$

$\Rightarrow \log(zx) = \log z + \log x = 8k = 2\log y$

$\therefore\ zx = y^2.$

40. **(a)** $3 + \log_5 x = \log_5 y \Rightarrow \log_5(125x) = \log_5 y \Rightarrow x = \frac{y}{125}$.

41. **(d)** $\frac{\log_2 a}{2} = \frac{\log_3 b}{3} = \frac{\log_4 c}{4} = k$

$\Rightarrow a = 2^{2k},\ b = 3^{3k},\ c = 4^{4k}$ and

$a^{1/2} \cdot b^{1/3} \cdot c^{1/4} = 2^k \cdot 3^k \cdot 4^k = 24$

$\Rightarrow 24^k = 24^1 \Rightarrow k = 1.$

$\therefore\ a = 4,\ b = 27,\ c = 256.$

42. **(c)** $\frac{z}{x^3y^4} = 1 \Rightarrow \log_2 z - 3\log_2 x - 4\log_2 y = 0$

$\Rightarrow \log_2 z - 3 \cdot \frac{3}{5k} \cdot \log_2 z - 4 \cdot \frac{4}{5k} \cdot \log_2 z = 0$

$\Rightarrow 1 - \frac{9}{5k} - \frac{16}{5k} = 0$

$\Rightarrow 5k - 25 = 0 \Rightarrow k = 5.$

43. **(a)** $\frac{3(1+\log_{10} 7)}{2 + \log\frac{7}{2} + \log\frac{1}{5}} = \frac{3(1+\log_{10} 7)}{2 + \log\left(\frac{7}{10}\right)}$

$= \frac{3(1+\log_{10} 7)}{1+\log_{10} 7} = 3.$

44. **(c)** Each ratio $= k \Rightarrow \log x = k(a^2 + ab + b^2)$

$\Rightarrow (a - b)\log x = k(a^3 - b^3)$

$\Rightarrow \log x^{a-b} = k(a^3 - b^3) \Rightarrow x^{a-b} = e^{k(a^3-b^3)}$

Similarly, $y^{b-c} = e^{k(b^3-c^3)},\ z^{c-a} = e^{k(c^3-a^3)}.$

$\therefore\ x^{a-b} \cdot y^{b-c} \cdot z^{c-a} = e^0 = 1.$

45. **(c)** $3^{x-2} = 5 \Rightarrow 3^x = 45 = \left(\frac{90}{2}\right)$

$\Rightarrow x\log_{10} 3 = \log_{10} 90 - \log_{10} 2$

$= 2\log_{10} 3 + 1 - \log_{10} 2$

$\Rightarrow x(0.4771) = 1.65317$

$\Rightarrow x = \frac{165317}{47710} = 3\frac{22187}{47710}$.

46. **(d)** $\log(648)^5 = 5\log(81 \times 8) = 20\log 3 + 15\log 2$

$= 20(0.4771) + 15(0.30103)$

$= 14.05745.$

$\therefore$ Number of digits in $(648)^5$ is 15.

47. **(c)** $\frac{\log x}{1} = \frac{\log y}{2} = \frac{\log z}{5} = k$

$\Rightarrow \log x = k,\ \log y = 2k,\ \log z = 5k.$

$\therefore\ \log(x^4 \cdot y^3 \cdot z^{-2}) = 4\log x + 3\log y - 2\log z = 0$

$\Rightarrow x^4 \cdot y^3 \cdot z^{-2} = 1.$

48. **(c)** $\frac{\log\sqrt{27} + \log\sqrt{1000} + \log 8}{\log 120}$

$= \frac{\frac{3}{2}(\log 3 + \log 10 + \log 4)}{\log 3 + \log 10 + \log 4} = \frac{3}{2}$

49. **(b)** $y = \frac{10^{\log_{10} x}}{x^2} = \frac{1}{x} = \frac{1}{y^a} = y^{-a} \Rightarrow a = -1.$

50. **(b)** $x = \log_{4/3}(1/2) = -\log_{4/3} 2 < 0$

and, $y = \log_{1/2}(1/3) = \log_2 3 > 0 \Rightarrow y > x.$

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CHAPTER 28

Quadratic Equations

INTRODUCTION

An equation of degree two is called a *quadratic equation*. The general form of a quadratic equation is $ax^2 + bx + c = 0$, where a, b, c are real numbers, $a \neq 0$ and x is a real variable. Some examples of quadratic equations are $x^2 + 4x + 3 = 0$, $3x^2 - 4x + 5 = 0$ and $3x^2 + 2x - 3 = 0$.

Roots of a Quadratic Equation

A *root* of the equation $f(x) = 0$ is that value of x which makes $f(x) = 0$. In other words, $x = a$ is said to be a root of $f(x) = 0$, where $f(a)$ is the value of the polynomial $f(x)$ at $x = a$ and is obtained by replacing x by a in $f(x)$.

For example, -1 is a root of the quadratic equation $x^2 + 6x + 5 = 0$ because $(-1)^2 + 6(-1) + 5 = 0$.

Solution of a Quadratic Equation

If there is a quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$, the roots of this equation are

$$\frac{-b+\sqrt{b^2-4ac}}{2a} \text{ and } \frac{-b-\sqrt{b^2-4ac}}{2a}.$$

Illustration 1: Solve the following quadratic equations:

(i) $6x^2 + x - 2 = 0$

(ii) $2x^2 + x - 1 = 0$

Solution: (i) Using formula:

$$\text{The roots are } x = \frac{-1\pm\sqrt{(1)^2-4(6)(-2)}}{2\times 6}$$

$$= \frac{-1\pm\sqrt{49}}{12} = \frac{6}{12}, \frac{-8}{12}.$$

i.e., $\frac{1}{2}, \frac{-2}{3}$.

Using factorization:

$$6x^2 + x - 2 = 0 \Leftrightarrow 6x^2 + 4x - 3x - 2 = 0$$

$$\Leftrightarrow 2x(3x + 2) - 1(3x + 2) = 0$$

$$\Leftrightarrow (2x + 1)(3x + 2) = 0$$

$$\Leftrightarrow x = \frac{1}{2} \text{ or } x = -\frac{2}{3}.$$

(ii) Using formula:

$$\text{The roots are } x = \frac{-1\pm\sqrt{(1)^2-4(2)(-1)}}{2\times 2}$$

$$= \frac{-1\pm\sqrt{9}}{4} = \frac{-1\pm 3}{4}$$

$$= \frac{2}{4}, \frac{-4}{4} \text{ i.e., } \frac{1}{2}, -1.$$

Using factorization:

$$2x^2 + x - 1 = 0 \Leftrightarrow 2x^2 + 2x - x - 1 = 0$$

$$\Leftrightarrow 2x(x + 1) - 1(x + 1) = 0$$

$$\Leftrightarrow (2x - 1)(x - 1) = 0.$$

$$\Leftrightarrow x = \frac{1}{2} \text{ or } x = -1.$$

Nature of Roots

A quadratic equation has exactly two roots may be real or imaginary or coincident.

If $ax^2 + bx + c$, $a \neq 0$, then $D = b^2 - 4ac$ is called *discriminant*.

1. If $D > 0$, then there are two distinct and real roots given by

$$\alpha = \frac{-b+\sqrt{b^2-4ac}}{2a}, \ \beta = \frac{-b-\sqrt{b^2-4ac}}{2a}.$$

2. If $D = 0$, then there is a repeated real root given by

$$\alpha = -\frac{b}{2a} \text{ i.e., roots are real and equal.}$$

3. If $D < 0$, then there are no real roots.

Notes

The roots are rational if $D > 0$ and D is a perfect square whereas the roots are irrational if $D > 0$ but D is not a perfect square.

Illustration 2: Find the nature of the roots of the equations:

(i) $2x^2 + x - 1 = 0$

(ii) $x^2 + x + 1 = 0$

(iii) $x^2 + 5x + 5 = 0$

(iv) $\frac{4}{3}x^2 - 2x + \frac{3}{4} = 0$

Solution: (i) $D = (1)^2 - 4 \times 2 \times (-1) = 9 > 0$.

Also, D is a perfect square.

So, the roots are real, distinct and rational.

(ii) $D = (1)^2 - 4 \times 1 \times 1 = -3 < 0$

So, the roots are imaginary.

(iii) $D = (5)^2 - 4 \times 1 \times 5 = 5 > 0$.

Also, D is not a perfect square.

So, the roots are real, distinct and irrational.

(iv) $D = (-2)^2 - 4 \times \frac{4}{3} \times \frac{3}{4} = 0$.

So, the roots are real and equal.

Illustration 3: For what value of k will the quadratic equation $kx^2 - 2\sqrt{5}\,x + 4 = 0$ have real and equal roots.

Solution: $D = (-2\sqrt{5})^2 - 4 \times k \times 4 = 20 - 16k$.

The given equation will have real and equal roots if $D = 0$.

i.e., $20 - 16k = 0$ or $k = \frac{20}{16} = \frac{5}{4}$.

Notes

1. If $p + \sqrt{q}$ is a root of a quadratic equation, then its other root is $p - \sqrt{q}$.

Illustration 4: If $2 + \sqrt{3}$ is one root of a quadratic equation, find the other root.

Solution: The other root is $2 - \sqrt{3}$.

2. $ax^2 + bx + c$ can be expressed as a product of two linear factors only when $D \geq 0$.

Illustration 5: For what value of k, the quadratic polynomial $kx^2 + 4x + 1$ can be factorized into two real linear factors.

Solution: $D = (4)^2 - 4 \times k \times 1 = 16 - 4k$.

The given quadratic polynomial can be factorized into real linear factors if $D \geq 0$.

i.e., $16 - 4k \geq 0$ or $-4k \geq -6$ or $k \leq 4$.

Relation Between Roots and Coefficients

Let, α, β be the roots of the equation,

$$ax^2 + bx + c = 0$$

Then, sum of the roots

$$= \alpha + \beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$$

and product of the roots

$$= \alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

Illustration 6: Find the sum and the product of the roots of the quadratic equation $2x^2 + 5\sqrt{3}\,x + 6 = 0$.

Solution: Here $a = 2$, $b = 5\sqrt{3}$, $c = 6$.

$$\therefore \quad \text{Sum of the roots} = -\frac{b}{a} = -\frac{5\sqrt{3}}{2}.$$

$$\text{Product of the roots} = \frac{c}{a} = \frac{6}{2} = 3.$$

Formation of a Quadratic Equation with Given Roots

If α, β are the roots of a quadratic equation the equation can he written as

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

i.e., $x^2 - (\text{sum of roots})x + \text{product of roots} = 0$.

Illustration 7: Find the quadratic equation whose roots are 5 and -6.

Solution: Sum of roots $= 5 + (-6) = -1$,

Product of roots $= 5 \times (-6) = -30$.

$\therefore$ The required quadratic equation is

$x^2 - (-1)x + (-30) = 0$ i.e., $x^2 + x - 30 = 0$.

EXERCISE–I

1. In the following determine the set of value of P for which the given quadratic equation has real roots.

$$Px^2 + 4x + 1 = 0$$

(a) $P \neq 4$ (b) $P > 4$
(c) $P \leq 4$ (d) $P \geq 4$

2. If one root of the quadratic equation $2x^2 + Px + 4 = 0$ is 2, find the second root and value of P.

(a) 1, −6 (b) 1, 6
(c) −1, 6 (d) −1, −6

3. One root of the quadratic equation $x^2 - 5x + 6 = 0$ is 3. Find the other root.

(a) 2 (b) −2
(c) 1 (d) −1

4. The roots of the equation

$\sqrt{7}x^2 - 6x - 13\sqrt{7} = 0$ are

(a) $-\sqrt{7}, \frac{-13\sqrt{7}}{7}$ (b) $\sqrt{7}, \frac{-13\sqrt{7}}{7}$

(c) $-\sqrt{7}, \frac{13\sqrt{7}}{7}$ (d) None of these

5. The roots of the equation

$3a^2x^2 - abx - 2b^2 = 0$ are

(a) $\frac{b}{a}, \frac{-2b}{3a}$ (b) $\frac{b}{a}, \frac{2b}{3a}$

(c) $\frac{-b}{a}, \frac{-2b}{3a}$ (d) None of these

6. The roots of the equation

$a^2x^2 - 3abx - 2b^2 = 0$ are

(a) $\frac{2b}{a}, \frac{-b}{a}$ (b) $\frac{2b}{a}, \frac{b}{a}$

(c) $\frac{-2b}{a}, \frac{b}{a}$ (d) None of these

7. Construct a quadratic equation whose roots are $\sqrt{2}$ and $2\sqrt{2}$

(a) $x^2 - 3\sqrt{2}x - 4 = 0$ (b) $x^2 - 3\sqrt{2}x + 4 = 0$

(c) $x^2 + 3\sqrt{2}x - 4 = 0$ (d) $x^2 + 3\sqrt{2}x + 4 = 0$

8. The roots of the equation

$ax^2 + (4a^2 - 3b)x - 12ab = 0$ are

(a) $4a, \frac{3b}{a}$ (b) $-4a, \frac{3b}{a}$

(c) $-4a, \frac{3b}{a}$ (d) $-4a, \frac{-3b}{a}$

9. Construct a quadratic equation whose roots have the sum = 6 and product = −16.

(a) $x^2 - 6x - 16 = 0$ (b) $x^2 + 6x - 16 = 0$

(c) $x^2 - \sqrt{3}x - 6 = 0$ (d) None of these

10. In the following, find the value (s) of P so that the given equation has equal roots

$3x^2 - 5x + P = 0$

(a) −25/12 (b) 25/6
(c) 25/12 (d) − 25/6

11. If α and β are the roots of the equation $ax^2 + bx + c = 0$, find the value of $\alpha^2 + \beta^2$

(a) $\frac{b^2 - 2ac}{2a^2}$ (b) $\frac{b^2 + 2ac}{a^2}$

(c) $\frac{b^2 + 2ac}{2a^2}$ (d) $\frac{b^2 - 2ac}{a^2}$

12. If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, the value of $\alpha^3 + \beta^3$ is

(a) $\frac{b(b^2 - 3ac)}{a^3}$ (b) $\frac{b(3ac - b^2)}{a^3}$

(c) $\frac{b(3ac + b^2)}{a^3}$ (d) None of these

13. If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is

(a) $\frac{b^2 - 2ac}{ac}$ (b) $\frac{b^2 - 2ac}{2ac}$

(c) $\frac{b^2 - ac}{2ac}$ (d) $\frac{b^2 + 2ac}{ac}$

14. The quadratic equation with rational coefficients, whose one root is $\sqrt{5}$, is:

(a) $x^2 + 5 = 0$ (b) $x^2 - 10 = 0$
(c) $x^2 - 5 = 0$ (d) None of these

15. The equation $x^2 - px + q = 0$, $p, q \in R$ has on real root if:

(a) $p^2 \le 4q$ (b) $p^2 < 4q$
(c) $p^2 > 4q$ (d) None of these

16. Determine p so that the equation $x^2 + 5px + 16 = 0$ has on real root.

(a) $\frac{-4}{5} < p < \frac{4}{5}$ (b) $\frac{-8}{5} < p < \frac{8}{5}$

(c) $p < -\frac{4}{5}$ or $p > \frac{4}{5}$ (d) None of these

17. For what value of k the quadratic polynomial $3z^2 + 5z + k$ can be factored into product of real linear factors?

(a) $k \le \frac{25}{6}$ (b) $k \le \frac{25}{12}$

(c) $k \ge \frac{25}{12}$ (d) $k \ge \frac{25}{6}$

18. $x = 3$ i a solution of the equation $3x^2 + (k - 1)x + 9 = 0$ if k has value

(a) 13 (b) –13
(c) 11 (d) –11

19. One root of the equation $3x^2 - 10x + 3 = 0$ is $\frac{1}{3}$. Find the other root.

(a) 3 (b) 1/3
(c) –3 (d) None of these

20. The expression $x^4 + 7x^2 + 16$ can be factored as:

(a) $(x^2 + x + 1)(x^2 + x + 16)$

(b) $(x^2 + x + 1)(x^2 - x + 16)$

(c) $(x^2 + x + 4)(x^2 - x + 4)$

(d) $(x^2 + x - 4)(x^2 - x - 4)$

21. The common root of the equations $x^2 - 7x + 10 = 0$ and $x^2 - 10x + 16 = 0$ is:

(a) –2 (b) 3
(c) 5 (d) 2

22. The roots of the equation $x^2 + px + q = 0$ are equal if:

(a) $p^2 = 2q$ (b) $p^2 = 4q$
(c) $p^2 = -4q$ (d) $p^2 = -2q$

23. An equation equivalent to the quadratic equation $x^2 - 6x + 5 = 0$ is:

(a) $6x^2 - 5x + 1 = 0$ (b) $x^2 - 5x + 6 = 0$
(c) $5x^2 - 6x + 1 = 0$ (d) $|x - 3| = 2$

24. Divide 16 into 2 parts such the twice the square of the larger part exceeds the square of the smaller part by 164

(a) 10, 6 (b) 8, 8
(c) 12, 4 (d) None of these

25. With respect to the roots of $x^2 - x - 2 - 0$, we can say that:

(a) both of them are natural numbers
(b) both of them are integers
(c) the latter of the two is negative
(d) None of these

26. The solution of $2 - x = \frac{x - 2}{x}$ would include:

(a) –2, –1 (b) 2, –1
(c) –4, 2 (d) 4, –2

27. If $\log_{10}(x^2 - 6x + 45) = 2$, then the values of x are

(a) 6, 9 (b) 9, –5
(c) 10, 5 (d) 11, –5

28. If α, β are the roots of the equation $x^2 - 5x + 6 = 0$, construct a quadratic equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.

(a) $6x^2 + 5x - 1 = 0$ (b) $6x^2 - 5x - 1 = 0$
(c) $6x^2 - 5x + 1 = 0$ (d) $6x^2 + 5x + 1 = 0$

29. The roots of $\frac{x+4}{x-4} + \frac{x-4}{x+4} = \frac{10}{3}$ are:

(a) ± 4 (b) ± 6
(c) ± 8 (d) $2 \pm \sqrt{3}$

30. The roots of the equation $ax^2 + bx + c = 0$ will be reciprocal if

(a) $a = b$ (b) $b = c$
(c) $c = a$ (d) None of these

31. Form a quadratic equation whose one root is $3-\sqrt{5}$ and the sum of roots is 6

(a) $x^2 - 6x + 4 = 0$ (b) $x^2 + 6x + 4 = 0$
(c) $x^2 - 6x - 4 = 0$ (d) None of these

32. The value of k for which the roots α, β of the equation: $x^2 - 6x + k = 0$ satisfy the relation $3\alpha + 2\beta = 20$, is

(a) 8 (b) −8
(c) 16 (d) −16

33. Find two consecutive positive odd integers whose squares have the sum 290.

(a) 11, 13 (b) 13, 15
(c) 9, 11 (d) None of these

34. Consider the equation $px^2 + qx + r = 0$, where p, q, r are real. The roots are equal in magnitude but opposite in sign when:

(a) $q = 0$, $r = 0$, $p \neq 0$
(b) $p = 0$, $qr \neq 0$
(c) $r = 0$, $pr \neq 0$
(d) $q = 0$, $pr \neq 0$

35. Determine k such that the quadratic equation $x^2 - 2(1+3k)x + 7(3+2k) = 0$ has equal roots.

(a) $2, \frac{-10}{9}$ (b) $2, \frac{10}{9}$

(c) $-2, \frac{10}{9}$ (d) $-2, \frac{-10}{9}$

36. If the equations $x^2 + 2x - 3 = 0$ and $x^2 + 3x - k = 0$ have a common root, then the non–zero value of k is:

(a) 1 (b) 2
(c) 3 (d) 4

37. The roots of the equation $4x - 3.2x^2 + 32 = 0$ would include:

(a) 1, 2 and 3 (b) 1 and 2
(c) 1 and 3 (d) 2 and 3

38. The positive value of m for which the roots of the equation $12x^2 + mx + 5 = 0$ are in the ratio 3:2 is:

(a) $5\sqrt{10}$ (b) $\frac{5}{2}\sqrt{10}$

(c) $\frac{5}{12}$ (d) $\frac{12}{5}$

39. If α, β are the roots of the equation $2x^2 - 3x + 1 = 0$, form an equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

(a) $2x^2 + 5x + 2 = 0$ (b) $2x^2 - 5x - 2 = 0$
(c) $2x^2 - 5x + 2 = 0$ (d) None of these

40. Find the quadratic equation whose roots are reciprocal of the roots of the equation

$3x^2 - 20x + 17 = 0$

(a) $17x^2 - 20x + 3 = 0$
(b) $17x^2 + 20x + 3 = 0$
(c) $17x^2 - 20x - 3 = 0$
(d) None of these

41. If α and β are the roots of the equation

$x^2 - 3\lambda x + \lambda^2 = 0$, find λ if $\alpha^2 + \beta^2 = \frac{7}{4}$

(a) $\pm\frac{1}{2}$ (b) $\pm\frac{\sqrt{7}}{2}$

(c) $\pm\frac{\sqrt{3}}{2}$ (d) None of these

42. If α, β are the roots of the equation $ax^2 + bx + b = 0$, then

$\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} =$

(a) 1 (b) 0
(c) 2 (d) 3

43. The expression $x^2 - x + 1$ has:

(a) one proper linear factor
(b) two proper linear factors
(c) no proper linear factor
(d) None of these

44. The length of a rectangular plot is 8 m greater than its breadth. If the area of the plot is 308 m^2, find the length of the plot.

(a) 22 m (b) 18 m
(c) 20 m (d) None of these

45. If α, β are the roots of the equation $x^2 + kx + 12 = 0$ such that $\alpha - \beta = 1$, the value of k is

(a) 0 (b) ± 5
(c) ± 1 (d) ± 7

46. The value of x in the equation

$$\left(x+\frac{1}{x}\right)^2-\frac{3}{2}\left(x-\frac{1}{x}\right)=4 \text{ is:}$$

(a) -2 (b) $\frac{1}{2}$

(c) -1 (d) 0

47. If α, β are the roots of the quadratic equation $x^2 - 8x + k = 0$, find the value of k such the $\alpha^2 + \beta^2 = 40$

(a) 12 (b) 14

(c) 10 (d) 16

48. Find the value of k so that the sum of the roots of the equation $3x^2 + (2x + 1)x - k - 5 = 0$ is equal to the product of the roots:

(a) 4 (b) 6

(c) 2 (d) 8

EXERCISE–2
(BASED ON MEMORY)

1. Which of the following equations has equal roots?

(a) $3x^2 - 6x + 2 = 0$
(b) $3x^2 - 6x + 3 = 0$
(c) $x^2 - 8x + 8 = 0$
(d) $8x^2 - 8x + 2 = 0$

[SSC CHSL (10+2) Tier-I (CBE), 2017]

2. If sum of the roots of a quadratic equation is 1 and product of the roots is –20. Find the quadratic equation.

(a) $x^2 - x - 20 = 0$
(b) $x + x + 20 = 0$
(c) $x^2 + x - 20 = 0$
(d) $x^2 - x + 20 = 0$

[SSC CGL Tier-I CBE, 2017]

3. If $a(x + y) - b(x - y) = 2ab$ then the value of $2(x^2 + y^2)$ is:

(a) $2(a^2-b^2)$ (b) $2(a^2+b^2)$

(c) $4(a^2-b^2)$ (d) $4(a^2+b^2)$

[SSC CGL Tier-I (CBE), 2016]

4. If $(x - 2)(x - p) = x^2 - ax + 6$, then the value of $(a - p)$ is

(a) 0 (b) 1

(c) 2 (d) 3

[SSC CGL Tier-I (CBE), 2016]

5. If $a^2 = by + cz$, $b^2 = cz + zx$, $c^2 = ax + by$, then the value of $\frac{x}{a+x}+\frac{y}{b+y}+\frac{z}{c+z}$ is

(a) 1 (b) $a + b + c$

(c) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ (d) 0

[SSC CGL Tier-I (CBE), 2016]

6. If $x = 332$, $y = 333$, $z = 335$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is

(a) 7000 (b) 8000

(c) 9000 (d) 10000

[SSC, 2015]

7. If $m = -4$, $n = -2$, then the value of $m^3 - 3m^2 + 3m + 3n + 3n^2 + n^3$ is

(a) 124 (b) −124

(c) 126 (d) −126

[SSC, 2015]

8. If $\frac{m-a^2}{b^2+c^2}+\frac{m-b^2}{c^2+a^2}+\frac{m-c^2}{a^2+b^2}=3$, then the value of m is

(a) $a^2 + b^2$ (b) $a^2 + b^2 + c^2$

(c) $a^2 - b^2 - c^2$ (d) $a^2 + b^2 - c^2$

[SSC, 2015]

9. Let $x=\frac{\sqrt{13}+\sqrt{11}}{\sqrt{13}-\sqrt{11}}$ and $y=\frac{1}{x}$, then the value of $3x^2-5xy+3y^2$ is

(a) 1717 (b) 1171

(c) 1771 (d) 1177

[SSC, 2015]

10. If $x^2+y^2+z^2=xy+yz+zx$, then the value of $\frac{3x^4+7y^4+5z^4}{5x^2y^2+7y^2z^2+3z^2x^2}$ is

(a) 0 (b) −1

(c) 2 (d) 1

[SSC, 2015]

11. If $x-\sqrt{3}-\sqrt{2}=0$ and $y-\sqrt{3}+\sqrt{2}=0$, then value of $\left(x^3-20\sqrt{2}\right)-\left(y^3+2\sqrt{2}\right)$

(a) 1 (b) 0
(c) 2 (d) 3

[SSC, 2015]

12. If $3(a^2 + b^2 + c^2) = (a + b + c)^2$, then the relation between a, b and c is

(a) $a \neq b \neq c$ (b) $a = b \neq c$
(c) $a = b = c$ (d) $a \neq b = c$

[SSC, 2015]

13. If $(x^3-y^3):(x^2+xy+y^2)=5:1$ and $(x^2-y^2):(x-y)$ = 7 : 1, then the ratio $2x:3y$ equals

(a) 4 : 3 (b) 4 : 1
(c) 2 : 3 (d) 3 : 2

[SSC, 2015]

14. If $x=a^{\frac{1}{2}}+a^{-\frac{1}{2}}, y=a^{\frac{1}{2}}-a^{-\frac{1}{2}}$, then the value of $(x^4-x^2y^2-1)+(y^4-x^2y^2+1)$

(a) 12 (b) 14
(c) 16 (d) 13

[SSC, 2015]

15. If $a + b = 1$, find the value of $a^3 + b^3 - ab - (a^2 - b^2)^2$

(a) 2 (b) 1
(c) 0 (d) –1

[SSC, 2015]

16. If $a-\frac{1}{a-3}=5$, then the value of $(a-3)^3-\frac{1}{(a-3)^3}$ is

(a) 7 (b) 2
(c) 5 (d) 14

[SSC, 2015]

17. If $\left(\frac{p^{-1}q^2}{p^3q^{-2}}\right)^{\frac{1}{3}}\div\left(\frac{p^5q^{-3}}{p^{-2}q^3}\right)^{\frac{1}{3}}=p^aq^b$, then the value of $a + b$, where p and q are different positive primes, is

(a) 1 (b) –1
(c) 2 (d) 0

[SSC, 2015]

18. If $(3x-2y):(2x+3y)=5:6$, then one of the value of $\left(\frac{\sqrt[3]{x}+\sqrt[3]{y}}{\sqrt[3]{x}-\sqrt[3]{y}}\right)^2$ is

(a) 5 (b) 25
(c) $\frac{1}{5}$ (d) $\frac{1}{25}$

[SSC, 2015]

19. If m – 5n, = 2, then the value of (m^3 – $125n^3$ – 30mn) is

(a) 7 (b) 8
(c) 9 (d) 6

20. Given that $x^3 + y^3 = 72$ and $xy = 6$ with $x > y$, then the value of $x - y$ is:

(a) 2 (b) –2
(c) –4 (d) 4

[SSC, 2015]

21. If $x+\frac{1}{x}=2$, then the value of $x^{12}-\frac{1}{x^{12}}$ is:

(a) 4 (b) 2
(c) 0 (d) – 4

[SSC, 2015]

22. If $x=\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ and $y=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$ then the value of: $\frac{x^2+xy+y^2}{x^2-xy+y^2}$ = ?

(a) $\frac{65}{63}$ (b) $\frac{67}{65}$
(c) $\frac{63}{61}$ (d) $\frac{69}{67}$

[SSC, 2015]

23. If $x^2 + x = 5$ then the value of: $(x+3)^3+\frac{1}{(x+3)^3}$

(a) 120 (b) 130
(c) 140 (d) 110

[SSC, 2015]

24. If $4a-\frac{4}{a}+3=0$ then the value of: $a^3-\frac{1}{a^3}+3=?$

(a) $\frac{7}{16}$ (b) $\frac{21}{64}$

(c) $\frac{3}{16}$ (d) $\frac{21}{16}$

[SSC, 2015]

25. If $x = z = 225$ and $y = 226$ then the value of: $x^3 + y^3 + z^3 - 3xyz$

(a) 576 (b) 674
(c) 765 (d) 676

[SSC, 2015]

26. If $x^2 + y^2 + z^2 = 2(x + z - 1)$. Then the value of: $x^3 + y^3 + z^3 = ?$

(a) 1 (b) –1
(c) 0 (d) 2

[SSC, 2015]

27. If $5x + 9y = 5$ and $125x^3 + 729y^3 = 120$ then the value of the product of x and y is

(a) 135 (b) $\frac{1}{135}$
(c) $\frac{1}{9}$ (d) 45

[SSC, 2015]

28. The factors of $(a^2 + 4b^2 + 4b - 4ab - 2a - 8)$ are:

(a) $(a-2b-4)(a-2b+2)$

(b) $(a-b+2)(a+4b+4)$

(c) $(a+2b-4)(a+2b+2)$

(d) $(a+2b-1)(a-2b+1)$

[SSC, 2014]

Directions (29–33): In this question two equations numbered I and II are given. You have to solve both the equations and find out the correct option.

29. I. $6x^2 + 41x + 63 = 0$

II. $4y^2 + 8y + 3 = 0$

(a) Relationship between x and y cannot be established
(b) $x \geq y$
(c) $x < y$
(d) $x > y$
(e) $x \leq y$

[IBPS PO/MT, 2014]

30. I. $x^2 + 10x + 24 = 0$

II. $4y^2 - 17y + 18 = 0$

(a) $x \leq y$
(b) $x \geq y$
(c) Relationship between x and y cannot be established
(d) $x > y$
(e) $x < y$

[IBPS PO/MT, 2014]

31. I. $24x^2 + 38x + 15 = 0$

II. $12y^2 + 28y + 15 = 0$

(a) $x \leq y$ (b) $x > y$
(c) $x \geq y$ (d) $x < y$
(e) $x = y$, or Relationship between x and y cannot be established

[IBPS PO/MT, 2014]

32. I. $3x^2 - 20x - 32 = 0$

II. $2y^2 - 3y - 20 = 0$

(a) $x < y$
(b) $x \leq y$
(c) $x > y$
(d) Relationship between x and y cannot be established
(e) $x \geq y$

[IBPS PO/MT, 2014]

33. I. $x^2 - 20x + 91 = 0$

II. $y^2 - 32y + 247 = 0$

(a) $x > y$
(b) Relationship between x and y cannot be established
(c) $x \geq y$
(d) $x \leq y$
(e) $x < y$

[IBPS PO/MT, 2014]

34. If $x + \frac{1}{x} = 5$, then $x^6 + \frac{1}{x^6}$ is

(a) 12098 (b) 12048
(c) 14062 (d) 12092

[SSC, 2014]

35. If $x^2 - 3x + 1 = 0$, then the value of $\frac{x^6 + x^4 + x^2 + 1}{x^3}$ will be

(a) 18 (b) 15
(c) 21 (d) 30

[SSC, 2014]

36. If $x^4 + \frac{1}{x^4} = 119$ & $x > 1$, then find the positive value of $x^3 - \frac{1}{x^3}$

(a) 25 (b) 27
(c) 36 (d) 49

[SSC, 2014]

37. If x is a rational number and $\frac{(x+1)^3-(x-1)^3}{(x+1)^2-(x-1)^2}=2$, then the sum of numerator and denominator of x is

(a) 3 (b) 4
(c) 5 (d) 7

[SSC, 2014]

38. If $x=\sqrt{5}+2$, then the value $\frac{2x^2-3x-2}{3x^2-4x-3}$ is equal to

(a) 0.185 (b) 0.525
(c) 0.625 (d) 0.785

[SSC, 2014]

39. If $x^2 + y^2 + 1 = 2x$, then the value of $x^3 + y^5$ is

(a) 2 (b) 0
(c) –1 (d) 1

[SSC, 2014]

40. If $x^4 + \frac{1}{x^4} = 119$, then the value of $x^3-\frac{1}{x^3}$ is

(a) ±36 (b) ±33
(c) 36 (d) –36

[SSC, 2014]

41. If $\left\{\frac{1}{2}(a-b)\right\}^2+ab=p(a+b)^2$, then the value of p is (assume that $a \neq -b$)

(a) $\frac{1}{4}$ (b) $\frac{1}{8}$
(c) 1 (d) $\frac{1}{2}$

[SSC, 2014]

42. The reciprocal of $x+\frac{1}{x}$ is

(a) $x-\frac{1}{x}$ (b) $\frac{1}{x}+x$
(c) $\frac{x}{x^2+1}$ (d) $\frac{x}{x+1}$

[SSC, 2014]

43. If $x(x - 3) = -1$, then the value of $x^3(x^3 - 18)$ is

(a) 1 (b) 0
(c) –1 (d) 2

[SSC, 2014]

44. If $a\left(2+\sqrt{3}\right)=b\left(2-\sqrt{3}\right)=1$, then the value of $\frac{1}{a^2+1}+\frac{1}{b^2+1}$ is

(a) 4 (b) 9
(c) –5 (d) 1

[SSC, 2014]

45. Find the value of $\sqrt{30+\sqrt{30+\sqrt{30+\cdots}}}$.

(a) 5 (b) $3\sqrt{10}$
(c) 6 (d) 7

[SSC, 2013]

Directions (46–50): In each of these questions, two equations (I) and (II) are given. You have to solve both the equations and give answer

(a) If $x < y$ (b) If $x > y$
(c) If $x = y$ (d) If $x \geq y$
(e) If $x \leq y$ or no relationship can be established between x and y.

[IBPS PO/MT, 2013]

46. I. x^2–24x + 144 = 0
II. y^2– 26y + 169 = 0

47. I. 2x^2+ 3x– 20 = 0
II. 2y^2 + 19y + 44 = 0

48. I. 6x^2 + 77x + 121 = 0
II. y^2 + 9y – 22 = 0

49. I. x^2– 6x = 7
II. 2y^2 + 13y + 15 = 0

50. I. 10x^2– 7x + 1 = 0
II. 35y^2 –12y + 1 = 0

[IBPS PO/MT, 2013]

51. $\sqrt{6+\sqrt{6+\sqrt{6+\cdots}}}$ is equal to

(a) 2 (b) 5
(c) 4 (d) 3

[SSC, 2011]

52. The sum of the squares of two natural consecutive odd numbers is 394. The sum of the numbers is:

(a) 24 (b) 32
(c) 40 (d) 28

[SSC, 2011]

Directions (53–57): In the following questions, two equations numbered I and II are given. You have to solve both questions and give answer

(a) if $x > y$
(b) if $x \geq y$
(c) if $x < y$
(d) if $x \leq y$
(e) if $x = y$ or the relationship cannot be established.

53. I. $x^2 - 19x + 84 = 0$

II. $y^2 - 25y + 156 = 0$

[IOB PO, 2011]

54. I. $x^3 - 468 = 1729$

II. $y^2 - 1733 + 1564 = 0$

[IOB PO, 2011]

55. I. $\frac{9}{\sqrt{x}} + \frac{19}{\sqrt{x}} = \sqrt{x}$

II. $y^5 - \frac{(2\times14)^{11/12}}{\sqrt{y}} = 0$

[IOB PO, 2011]

56. I. $\sqrt{784}x + 1234 = 1486$

II. $\sqrt{1089}y + 2081 = 2345$

[IOB PO, 2011]

57. I. $\frac{12}{\sqrt{x}} - \frac{23}{\sqrt{x}} = 5\sqrt{x}$

II. $\frac{\sqrt{y}}{12} - \frac{5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

[IOB PO, 2011]

Directions (58–62): In each of these questions, two equations are given. You have to solve these equations and find out the values of x and y and

Give answer
(a) if $x < y$ (b) if $x > y$
(c) if $x \leq y$ (d) if $x \geq y$
(e) if $x = y$

[Andhra Bank PO, 2011]

58. I. $4x + 7y = 209$

II. $12x - 14y = -38$

[Andhra Bank PO, 2011]

59. I. $17x^2 + 48x = 9$

II. $13y^2 = 32y - 12$

[Andhra Bank PO, 2011]

60. I. $16x^2 + 20x + 6 = 0$

II. $10y^2 + 38y + 24 = 0$

[Andhra Bank PO, 2011]

61. I. $8x^2 + 6x = 5$

II. $12y^2 - 22y + 8 = 0$

[Andhra Bank PO, 2011]

62. I. $18x^2 + 18x + 4 = 0$

II. $12y^2 + 29y + 14 = 0$

[Andhra Bank PO, 2011]

Directions (63–29): In the following questions two equations numbered I and II are given. You have to solve both the equations and give answer

(a) $x > y$ (b) $x \geq y$
(c) $x < y$ (d) $x \leq y$
(e) $x = y$ or the relationship cannot be established

[Corporation Bank PO, 2011]

63. I. $x^2 - 11x + 24 = 0$

II. $2y^2 - 9y + 9 = 0$

[Corporation Bank PO, 2011]

64. I. $x^3 \times 13 = x^2 \times 247$

II. $y^{1/3} \times 14 = 294 \div y^{2/3}$

[Corporation Bank PO, 2011]

65. I. $\frac{12\times4}{x^{4/7}} - \frac{3\times4}{x^{4/7}} = x^{10/7}$

II. $y^3 + 783 = 999$

[Corporation Bank PO, 2011]

66. I. $\sqrt{500}x + \sqrt{402} = 0$

II. $\sqrt{360}y + (200)^{1/2} = 0$

[Corporation Bank PO, 2011]

67. I. $(17)^2 + 144 \div 18 = x$

II. $(26)^2 - 18\times21 = y$

[Corporation Bank PO, 2011]

Directions (68–72): In each of these questions, two equations are given. You have to solve these equations and find out the values of x and y and give answer

(a) $x < y$ (b) $x > y$
(c) $x \leq y$ (d) $x \geq y$
(e) $x = y$

68. **I.** $16x^2 + 20x + 6 = 0$

II. $10y^2 + 38y + 24 = 0$

[Punjab and Sind Bank PO, 2011]

69. **I.** $18x^2 + 18x + 4 = 0$

II. $12y^2 + 29y + 14 = 0$

[Punjab and Sind Bank PO, 2011]

70. **I.** $8x^2 + 6x = 5$

II. $12y^2 - 22y + 8 = 0$

[Punjab and Sind Bank PO, 2011]

71. **I.** $17x^2 + 48x = 9$

II. $13y^2 = 32y - 21$

[Punjab and Sind Bank PO, 2011]

72. **I.** $4x + 7y = 209$

II. $12x - 14y = -38$

[Punjab and Sind Bank PO, 2011]

Directions (73–77): In the following questions two equations numbered I and II are given. You have to solve both the equations and give answer

(a) if $x > y$ (b) if $x \geq y$

(c) if $x < y$ (d) if $x \leq y$

(e) if $x = y$ or the relationship cannot be established.

[Indian Bank PO, 2010]

73. **I.** $x^2 - 4 = 0$

II. $y^2 + 6y + 9 = 0$

[Indian Bank PO, 2010]

74. **I.** $x^2 - 7x + 12 = 0$

II. $y^2 + y - 12 = 0$

[Indian Bank PO, 2010]

75. **I.** $x^2 = 729$

II. $y = \sqrt{729}$

[Indian Bank PO, 2010]

76. **I.** $x^4 - 227 = 398$

II. $y^2 + 321 = 346$

[Indian Bank PO, 2010]

77. **I.** $2x^2 + 11x + 14 = 0$

II. $4y^2 + 12y + 9 = 0$

[Indian Bank PO, 2010]

ANSWER KEYS

EXERCISE–1

1. (c)	**2.** (a)	**3.** (a)	**4.** (c)	**5.** (a)	**6.** (b)	**7.** (b)	**8.** (c)	**9.** (a)	**10.** (c)	**11.** (d)	**12.** (b)	**13.** (a)
14. (c)	**15.** (b)	**16.** (b)	**17.** (b)	**18.** (d)	**19.** (a)	**20.** (c)	**21.** (d)	**22.** (b)	**23.** (d)	**24.** (a)	**25.** (b)	**26.** (b)
27. (d)	**28.** (c)	**29.** (c)	**30.** (c)	**31.** (a)	**32.** (a)	**33.** (a)	**34.** (d)	**35.** (a)	**36.** (d)	**37.** (d)	**38.** (a)	**39.** (c)
40. (a)	**41.** (a)	**42.** (b)	**43.** (c)	**44.** (c)	**45.** (d)	**46.** (c)	**47.** (a)	**48.** (a)				

EXERCISE–2

1. (b)	**2.** (d)	**3.** (d)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (d)	**11.** (b)	**12.** (c)	**13.** (b)
14. (c)	**15.** (c)	**16.** (d)	**17.** (b)	**18.** (b)	**19.** (b)	**20.** (a)	**21.** (b)	**22.** (c)	**23.** (d)	**24.** (b)	**25.** (d)	**26.** (d)
27. (b)	**28.** (a)	**29.** (c)	**30.** (e)	**31.** (c)	**32.** (d)	**33.** (d)	**34.** (a)	**35.** (c)	**36.** (c)	**37.** (b)	**38.** (c)	**39.** (d)
40. (c)	**41.** (a)	**42.** (c)	**43.** (a)	**44.** (d)	**45.** (c)	**46.** (a)	**47.** (d)	**48.** (e)	**49.** (b)	**50.** (d)	**51.** (d)	**52.** (d)
53. (d)	**54.** (b)	**55.** (e)	**56.** (a)	**57.** (a)	**58.** (e)	**59.** (a)	**60.** (b)	**61.** (c)	**62.** (d)	**63.** (b)	**64.** (c)	**65.** (d)
66. (c)	**67.** (c)	**68.** (b)	**69.** (d)	**70.** (c)	**71.** (a)	**72.** (e)	**73.** (a)	**74.** (b)	**75.** (d)	**76.** (e)	**77.** (c)	

EXPLANATORY ANSWERS

EXERCISE-I

1. (c) $Px^2 + 4x + 1 = 0$

Compare with $Ax^2 + Bx + C = 0$, we get

$$A = P,\ B = 4,\ C = 1$$

For real roots,

$B^2 - 4AC \geq 0 \Rightarrow 16 - 4P \geq 0$

$\Rightarrow 16 \geq 4P \Rightarrow P \leq 4.$

2. (a) The given equation is $2x^2 + Px + 4 = 0$

$\Rightarrow P(x) = 0$ where $P(x) = 2x^2 + Px + 4 = 0$

If 2 is a root of $P(x) = 0$, then $P(2) = 0$

$\Rightarrow 2(2)^2 + P(2) + 4 = 0 \Rightarrow 8 + 2P + 4 = 0$

$$\Rightarrow 2P = -12 \Rightarrow P = \frac{-12}{2} = -6$$

Hence the given equation is

$$2x^2 - 6x + 4 = 0 \Rightarrow 2x^2 - 2x - 4x + 4 = 0$$

$$\Rightarrow 2x(x-1) - 4(x-1) = 0$$

$$\Rightarrow 2(x-2)(x-1) = 0$$

$$\Rightarrow x = 1 \text{ or } x = 2$$

Hence second root is 1.

3. (a) The given equation is

$$x^2 - 5x + 6 = 0 \Rightarrow x^2 - 2x - 3x + 6 = 0$$
$$\Rightarrow x(x-2) - 3(x-2) = 0$$
$$\Rightarrow (x-2)(x-3) = 0$$
$$\Rightarrow x - 2 = 0 \text{ or } x - 3 = 0$$

Thus, the other root of the given quadratic equation is 2.

4. (c) $\sqrt{7}x^2 - 6x - 13\sqrt{7} = 0$

$$\Rightarrow \sqrt{7}x^2 - 13x + 7x - 13\sqrt{7} = 0$$

$$\Rightarrow x(\sqrt{7}x - 13) + \sqrt{7}(\sqrt{7}x - 13) = 0$$

$$\Rightarrow (x + \sqrt{7})(\sqrt{7}x - 13) = 0$$

$$\Rightarrow x + \sqrt{7} = 0 \text{ or } \sqrt{7}x - 13 = 0$$

$$\Rightarrow x = -\sqrt{7} \text{ or } x = \frac{13}{\sqrt{7}} = \frac{13\sqrt{7}}{7}$$

Thus, the two roots of given quadratic equation are

$-\sqrt{7}$ and $\frac{13\sqrt{7}}{7}$.

5. (a) The given quadratic equation is

$$3a^2x^2 - abx - 2b^2 = 0 \Rightarrow 3a^2x^2 - 3abx + 2abx - 2b^2 = 0$$
$$\Rightarrow 3ax(ax - b) + 2b(ax - b) = 0$$
$$\Rightarrow (ax - b)(3ax + 2b) = 0$$
$$\Rightarrow ax = b \text{ or } 3ax = -2b$$
$$\Rightarrow x = \frac{b}{a} \text{ or } x = -\frac{2b}{3a}.$$

6. (b) The given quadratic equation is

$$a^2x^2 - 3abx + 2b^2 = 0 \Rightarrow a^2x^2 - 2abx - abx + 2b^2 = 0$$
$$\Rightarrow ax(ax - 2b) - b(ax - 2b) = 0$$
$$\Rightarrow (ax - 2b)(ax - b) = 0$$
$$\Rightarrow ax - 2b = 0 \text{ or } ax - b = 0$$
$$\Rightarrow x = \frac{2b}{a} \text{ or } x = \frac{b}{a}$$

Thus, the two roots of the given quadratic equation are

$\frac{2b}{a}$ and $\frac{b}{a}$.

7. (b) Sum of the roots $= \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$

Product of the roots $= (\sqrt{2})(2\sqrt{2}) = 4$

Hence the required quadratic equation is x^2 = (sum of the roots) x + (Product of two roots) = 0

$$\Rightarrow x^2 - 3\sqrt{2x} + 4 = 0.$$

8. (c) The given quadratic equation is

$$ax^2 + (4a^2 - 3b)x - 12ab = 0$$

$$\Rightarrow ax^2 + 4a^2x - 3bx - 12ab = 0$$

$$\Rightarrow ax(x + 4a) - 3b(x + 4a) = 0$$

$$\Rightarrow (ax - 3b)(x + 4a) = 0$$

$$\Rightarrow ax - 3b = 0 \text{ or } x + 4a = 0$$

$$\Rightarrow x = \frac{3b}{a} \text{ or } x = -4a$$

Thus, the two roots of the given quadratic equation are

$-4a$ and $\frac{3b}{a}$.

9. (a) The required quadratic equation is x^2 – (sum of the roots) x + (product of the roots) = 0

$\Rightarrow x^2 - 6x - 16 = 0.$

10. (c) The given quadratic equation is

$3x^2 - 5x + P = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 3,\ b = -5,\ c = P$

If the given quadratic equation has equal roots then its discriminant = 0

$\Rightarrow b^2 - 4ac = 0 \Rightarrow (-5)^2 - 4(3)(P) = 0$

$\Rightarrow 25 - 12P = 0 \Rightarrow P = \frac{25}{12}$

11. (d) Since α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$,

$\therefore \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(-\frac{b}{a}\right)^2 - 2\frac{c}{a} = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}.$

12. (b) Since α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$

$\therefore \alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{a}{c}$

Now, $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$= \left(-\frac{b}{a}\right)^3 - 3\frac{c}{a}\left(\frac{-b}{a}\right)$

$= \frac{-b^3}{a} + \frac{3bc}{a^2} = \frac{-b^2 + 3abc}{a^3}$

$= \frac{b(3ac - b^2)}{a^3}$

13. (a) Since α and β are the roots of the equation $ax^2 + bx + c = 0$

$\therefore \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

Now, $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$

$= \frac{\left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right)}{\frac{c}{a}} = \frac{\frac{b^2}{a^2} - \frac{2c}{a}}{\frac{c}{a}}$

$= \frac{\frac{b^2 - 2ac}{a^2}}{\frac{c}{a}} = \frac{b^2 - 2ac}{ac}.$

14. (c) One root is $\sqrt{5}$

So the other root is $-\sqrt{5}$

$\therefore$ Sum of the roots is = 0

and product of the roots $= (\sqrt{5})(-\sqrt{5}) = -5$

$\therefore$ Required equation is x^2 – (sum of the roots)x + (product of the roots) = 0

$\Rightarrow x^2 - 5 = 0.$

15. (b) The equation $x^2 - px + q = 0$; $p, q \in R$ has no real root if

$B^2 < 4AC$

$\Rightarrow (-p)^2 < 4.1.q\ [\because A = 1, B = -p, C = q]$

$\Rightarrow p^2 < 4q.$

16. (b) The given quadratic equation is $x^2 + 5px + 16 = 0$(1)

Comparing it with $ax^2 + bx + c = 0$, we get $a = 1$, $b = 5p$, $c = 16$

If equation (1) has no real roots, then discriminant < 0

$\Rightarrow b^2 - 4ac < 0 \Rightarrow (5p)^2 - 4(1)(16) < 0$

$\Rightarrow 25p^2 - 64 < 0 \Rightarrow 25p^2 < 64$

$\Rightarrow p^2 < \frac{64}{25} \Rightarrow p^2 - \frac{64}{25} < 0$

$\Rightarrow \left(p - \frac{8}{5}\right)\left(p + \frac{8}{5}\right) < 0$

$\Rightarrow$ either $p - \frac{8}{5} > 0$ and $p + \frac{8}{5} < 0$

i.e., $p > \frac{8}{5}$ and $p < -\frac{8}{5}$,

which is not possible

or, $p - \frac{8}{5} < 0$ and $p + \frac{8}{5} > 0$

i.e., $p < \frac{8}{5}$ and $p > -\frac{8}{5}$ i.e., $-\frac{8}{5} < p < \frac{8}{5}$.

17. (b) We have $3z^2 + 5z + k$

Here $a = 3$, $b = 5$, $c = k$

$D = b^2 - 4ac = 25 - 12k$

For equal linear factors to exist, $D \geq 0$

i.e., $25 - 12k \geq 0 \Rightarrow 25 \geq 12k$

$\Rightarrow k \leq \frac{25}{12}$

Therefore, the set of real numbers $\leq \frac{25}{12}$ gives the set of value of k for which the given quadratic polynomial can be factored into the product of real linear factors.

18. (d) Putting $x = 3$, we get

$27 + 3(k - 1) + 9 = 0$

or $27 + 3k - 3 + 9 = 0$

or $3k = -33$ or $k = -11$.

19. (a) The given quadratic equation is $3x^2 - 10x + 3 = 0$

Comparing with $ax^2 + bx + c = 0$, we get

$a = 3,\ b = -10,\ c = 3$

$\therefore$ Sum of the roots $= -\dfrac{b}{a} = \dfrac{10}{3}$

$\because$ One root $= \dfrac{1}{3}$

$\therefore$ The other root $= \dfrac{10}{3} - \dfrac{1}{3} = \dfrac{9}{3} = 3.$

20 . (c) $x^4 + 7x^2 + 16 = (x^4 + 8x^2 + 16) - x^2$

$= (x^2 + 4)^2 - x^2$

$= (x^2 + 4 + x)(x^2 + 4 - x)$

$= (x^2 + x + 4)(x^2 - x + 4).$

21. (d) $x^2 - 7x + 10 = 0 \Leftrightarrow (x - 5)(x - 2) = 0$

$\Leftrightarrow x = 5, 2$

$x^2 - 10x + 16 = 0 \Leftrightarrow (x - 8)(x - 2) = 0$

$\Leftrightarrow x = 8, 2$

$\therefore$ Common root is 2.

22. (b) Here $a = 1,\ b = p,\ c = q$

The roots of the equation $x^2 + px + q = 0$ are equal if

$b^2 - 4ac = 0 \Rightarrow p^2 - 4q = 0 \Rightarrow p^2 = 4q.$

23. (d) $x^2 - 6x + 5 = 0 \Leftrightarrow (x - 5)(x - 1) = 0$

$\Leftrightarrow x = 5$ or 1

Also $|x - 3| = 2 \Leftrightarrow x - 3 = 2$ or

$-(x - 3) = 2 \Leftrightarrow x = 5$ or $x = 1$

$\therefore x^2 - 6x + 5 = 0$ and $|x - 3| = 2$ are equivalent.

24. (a) Let the smaller part be x. Then the larger part = $16 - x$. Now

$2(16 - x)^2 - x^2 = 164 \Rightarrow 2(256 + x^2 - 32x) - x^2 = 164$

$\Rightarrow x^2 - 64x + 348 = 0$

$\Rightarrow x^2 - 6x - 58x + 348 = 0$

$\Rightarrow x(x - 6) - 58(x - 6) = 0$

$\Rightarrow (x - 6)(x - 58) = 0$

$x = 6$ or $x = 58$

But $m = 58$ is not possible, since sum of the two parts is 16

$\therefore$ $x = 6$, $\therefore$ other part = 10.

25. (b) The given equation is of the form

$ax^2 + bx + c = 0$

Also, $D = \sqrt{9} = 3$

So, roots are rational

Hence, both the roots must be integers.

26. (b) Given equation is

$2x - x^2 = x - 2 \Leftrightarrow x^2 - x - 2 = 0$

$\Leftrightarrow (x + 1)(x - 2) = 0$

$\Leftrightarrow x = 2$ or $= -1.$

27. (d) $\log_{10}(x^2 - 6x + 45) = 2$

$\Leftrightarrow x^2 - 6x + 45 = 10^2 = 100$

$\Leftrightarrow x^2 - 6x - 55 = 0$

$\Leftrightarrow (x - 11)(x + 5) = 0$

$\Leftrightarrow x = 11$ or $x = -5.$

28. (c) Comparing $x^2 - 5x + 6 = 0$ with

$ax^2 + bx + c = 0$

$a = 1, b = -5, c = 6$

$\therefore\ \alpha + \beta = \dfrac{-b}{a} = \dfrac{5}{1} = 5$

$\alpha\beta = \dfrac{c}{a} = 6$

Now, we are to form an equation whose roots are $\dfrac{1}{\alpha}, \dfrac{1}{\beta}$.

So the required equation is

$x^2 -$ (sum of roots)x + (Product of roots) = 0

$x^2 - \left(\dfrac{1}{\alpha} + \dfrac{1}{\beta}\right)x + \left(\dfrac{1}{\alpha} \cdot \dfrac{1}{\beta}\right) = 0$

$x^2 - \left(\dfrac{\alpha + \beta}{\alpha\beta}\right)x + \left(\dfrac{1}{\alpha\beta}\right) = 0$

$x^2 - \dfrac{5}{6}x + \dfrac{1}{6} = 0$

$6x^2 - 5x + 1 = 0.$

29. (c) Given equation is: $y + \dfrac{1}{y} = \dfrac{10}{3}$,

where $y = \dfrac{x + 4}{x - 4}$

$\therefore\ 3y^2 - 10y + 3 = 0 \Rightarrow y = 3, \dfrac{1}{3}$

$\therefore\ y = \dfrac{x+4}{x-4} = 3$ or, $y = \dfrac{x+4}{x-4} = \dfrac{1}{3}$

$3x + 12 = x - 4$ or, $x + 4 = 3x - 12$

$\Rightarrow\ x = -8$ or, $x = 8$

30. (c) For reciprocal roots, product of roots must be 1

$\therefore \quad \frac{c}{a} = 1$ i.e., $c = a$

31. (a) Sum of the roots = 6

One root $= 3 - \sqrt{5}$

$\therefore$ the other root $= 6 - (3 - \sqrt{5}) = 3 + \sqrt{5}$

$\therefore$ Product of roots $= (3 - \sqrt{5})(3 + \sqrt{5})$

$= 9 - 5 = 4$

Hence the required equation is

$x^2 -$ (sum of the roots)x + (product of roots) = 0

$\Rightarrow \quad x^2 - 6x + 4 = 0.$

32. (a) $\alpha + \beta = 6$ and $3\alpha + 2\beta = 20$

$\Rightarrow \quad \alpha = 4, \beta = 2$

Product of the roots = k

So, $k = \alpha\beta = 4 \times 2 = 8.$

33. (a) Let, the two consecutive odd positive integers be $2x + 1$ and $2x + 3$ where x is a whole number.

Now,

$(2x + 1)^2 + (2x + 3)^2 = 290$

$\Rightarrow \quad 4x^2 + 4x + 1 + 4x^2 + 12x + 9 = 290$

$\Rightarrow \quad 8x^2 + 16x - 280 = 0$

$\Rightarrow \quad x^2 + 2x - 35 = 0$

$\Rightarrow \quad (x + 7)(x - 5) = 0 \Rightarrow x = 7, -5$

But, $x = -7$ is not possible, since -7 is not a whole number

$\therefore \quad x = 5.$

34. (d) Let, the roots be α and $-\alpha$. Then, sum of roots = 0

Also, roots being not equal, discriminant $\neq 0$

$\therefore \frac{q}{p} = 0$ and $q^2 - 4pr \neq 0$

$\Leftrightarrow q = 0$ and $pr \neq 0.$

35. (a) Comparing $x^2 - 2(1 + 3x)x + 7(3 + 2k) = 0$ with $ax_2 + bx + c = 0$, we get

$a = 1, b = -2(1 + 3k), c = 7(3 + 2k)$

For equal roots $D = b^2 - 4ac = 0$

$\therefore 4(1 + 3k)^2 - 4 \times 1 \times 7(3 + 2k) = 0$

$\Rightarrow \quad 4(1 + k)^2 + 6k) - 84 - 56k = 0$

$\Rightarrow \quad 36k^2 - 32k - 80 = 0$

$\Rightarrow \quad 9k - 8k - 20 = 0$

$k = \frac{8 \pm \sqrt{64 - 4(9)(-20)}}{2 \times 9}$

$= \frac{8 \pm \sqrt{784}}{18} = \frac{8 \pm 28}{18}$

$= \frac{36}{18}, \frac{-20}{18} = 2, \frac{-10}{9}.$

36. (d) Let, α be a common root of the given equations.

Then, $\alpha^2 + 2\alpha - 3 = 0$ and $\alpha^2 + 3\alpha - k = 0$

$\therefore \quad \frac{\alpha^2}{-2k + 9} = \frac{\alpha}{-3 + k} = \frac{1}{3 - 2}$

So, $\alpha^2 = \frac{9 - 2k}{1}$ and $\alpha = \frac{k - 3}{1}$

So, $(9 - 2k) = (k - 3)^2$ or $k^2 - 4k = 0$

or, $k(k - 4) = 0$, so $k = 4.$

37. (d) Given equation is: $2^{2x} - 3 \cdot 2^x \times 2^2 + 32 = 0$

or, $2^{2x} - 12 \times 2^x + 32 = 0$

$\Rightarrow y^2 - 12y + 32 = 0$, where $2^x = y$

$\Rightarrow (y - 8)(y - 4) = 0 \Rightarrow y = 8, y = 4$

$\therefore \quad 2^x = 8$ or, $2^x = 4$

$\Rightarrow 2^x = 2^3$ or, $2^x = 2^2$

$\Rightarrow x = 3$ or, $x = 2.$

38. (a) Let, the roots be 3α and 2α

Then, $3\alpha + 2\alpha = \frac{m}{12} \Rightarrow \alpha = \frac{m}{60}$

$\therefore \quad \frac{5}{72} = \left(\frac{m}{60}\right)^2 \Leftrightarrow \frac{5}{72} = \frac{m^2}{3600}$

$\Leftrightarrow m^2 = \frac{3600 \times 5}{72} = 250$

$\therefore \quad m = \sqrt{250} = 5\sqrt{10}.$

39. (c) $\because$ α, β are the roots of the equation $2x^2 - 3x + 1 = 0$

$\therefore \quad \alpha + \beta = \frac{3}{2}$ …(1)

and $\alpha\beta = \frac{1}{2}$ …(2)

We are to form a quadratic equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ S = sum of the roots

$= \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$

$$= \frac{\left(\frac{3}{2}\right)^2 - 2\left(\frac{1}{2}\right)}{\frac{1}{2}} \qquad \text{[using (1) and (2)]}$$

$$= \frac{\frac{9}{4} - 1}{\frac{1}{2}} = \frac{5}{4} \times \frac{2}{1} = \frac{5}{2}$$

P = Product of the roots

$$= \frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 1$$

Hence the required quadratic equation is x^2 – (sum of the roots)x + (Product of the roots) = 0

$$\Rightarrow \quad x^2 - \frac{5}{2}x + 1 = 0$$

$$\Rightarrow \quad 2x^2 - 5x + 2 = 0.$$

40. (a) The given quadratic equation is

$3x^2 - 20x + 17 = 0$ (1)

Compare with $ax^2 + bx + c = 0$, we get

$a = 3,\ b = -20,\ c = 17$

The roots of (1) are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{20 \pm \sqrt{400 - 4(3)(17)}}{2 \times 3}$$

$$= \frac{20 \pm \sqrt{196}}{6} = \frac{20 + 14}{6}, \frac{20 - 14}{6}$$

$$= \frac{34}{6}, \frac{6}{6} = \frac{17}{3}, 1$$

Hence the roots of (1) are $\frac{17}{3}$ and 1. So we have to form an equation whose are $\frac{3}{17}$ and 1

Sum of the roots $= \frac{3}{17} + 1 = \frac{20}{17}$

Product of the roots $= \frac{3}{17} \times 1 = \frac{3}{17}$

Hence, the required equation is

$$x^2 - \frac{20}{17}x + \frac{3}{17} = 0 \Rightarrow 17x^2 - 20x + 3 = 0.$$

41. (a) $\because$ α, β are the roots of the equation $x^2 - 3\lambda x + \lambda^2 = 0$

$\alpha + \beta = 3\lambda$ …(1)

and $\alpha\beta = \lambda^2$ …(2)

Now, $\alpha^2 + \beta^2 = \frac{7}{4}$ (given)

$$\Rightarrow \quad (\alpha + \beta)^2 - 2\alpha\beta = \frac{7}{4}$$

$$\Rightarrow (3\lambda)^2 - 2\lambda^2 = \frac{7}{4} \Rightarrow 7\lambda^2 = \frac{7}{4}$$

$$\Rightarrow \lambda^2 = \frac{1}{4} \Rightarrow \lambda = \pm\frac{1}{2}.$$

42. (b) $\because$ α, β are the roots of the equation

$ax^2 + bx + b = 0$

$\therefore \alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{b}{a}$

Now, $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} = \frac{\alpha + \beta}{\sqrt{\alpha\beta}} + \sqrt{\frac{b}{a}}$

$$= \frac{-\frac{b}{a}}{\sqrt{\frac{b}{a}}} + \sqrt{\frac{b}{a}}$$

$$= -\sqrt{\frac{b}{a}} + \sqrt{\frac{b}{a}} = 0.$$

43. (c) Comparing $x^2 - x + 1$ with $ax^2 + bx + c$ we have

$a = 1,\ b = -1,\ c = 1$

Here $D = b^2 - 4ac = (-1)^2 - 4(1)\ (1)$

$= 1 - 4 = -3$

Since $D < 0$, so the given expression has no proper linear factor.

44. (c) Let, the breadth of the rectangular plot be x m. Then the length of rectangular plot = $(x + 8)$ m

$\therefore$ Area = Length × Breadth = $x(x + 8) m^2$ But the area of the plot is given to be 308 m^2

$$\therefore \quad x(x+8) = 308 \Rightarrow x^2 + 8x - 308 = 0$$

$$\Rightarrow x^2 + 22x - 14x - 308 = 0$$

$$\Rightarrow x(x+22) - 14(x-22) = 0$$

$$\Rightarrow (x+22)(x-14) = 0$$

$$\Rightarrow x = 14, -22$$

But, $x = -22$ is not possible, since breath cannot be negative

$\therefore \quad x = 14$

Hence the breadth of the rectangular plot = 14 m Length of the rectangular plot = (14 + 8) m = 22 m.

45. (d) Let, α, β be the roots of the equation $x^2 + kx + 12 = 0$

$\therefore \quad \alpha + \beta = -k$ and $\alpha\beta = 12$

Now $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$(1)^2 = k^2 - 4(12) \Rightarrow k^2 = 49 \Rightarrow k = \pm 7.$

46. (c) Put $x - \frac{1}{x} = y$

$$\therefore \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = \left(x + \frac{1}{x}\right)^2 + 4 = y^2 + 4$$

So, given equation becomes

$$\Rightarrow \quad y\left(y - \frac{3}{2}\right) = 0 \Rightarrow y = 0 \text{ or } y = \frac{3}{2}$$

$$\therefore \quad x - \frac{1}{x} = 0 \text{ or } x - \frac{1}{x} = \frac{3}{2}$$

$\Rightarrow x^2 - 1 = 0$ or $2x^2 - 3x - 2 = 0$

$\Rightarrow x = \pm 1 \quad$ or $(2x + 1)(x - 2) = 0$

or $x = -1/2$ or $x = 2$.

47. (a) $\because \quad \alpha, \beta$ are the roots of the equation $x^2 - 8x + 1 = 0$

$$\therefore \quad \alpha + \beta = \frac{-b}{a} = \frac{-(-8)}{1} = 8$$

and $\alpha\beta = \frac{c}{a} = \frac{k}{1} = k$

Now, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\Rightarrow 40 = (8)^2 - 2k \Rightarrow 2k = 24 \Rightarrow k = 12.$

48. (a) The given equation is $3x^2 + (2k + 1)x - k - 5 = 0$

Compare with $ax^2 + bx + c = 0$, we get

$a = 3, b = 2k + 1, c = -k - 5$

$$\therefore \quad \text{Sum of the roots} = \frac{-b}{a} = \frac{-(2k+1)}{3}$$

$$\text{and Product of the roots} = \frac{c}{a} = \frac{-k-5}{3} = \frac{-(k+5)}{3}$$

$\because$ Sum of the roots = Product of the roots

$$\therefore \frac{-(2k+1)}{3} = -\frac{(k+5)}{3} \Rightarrow 2k + 1 = k + 5$$

$\Rightarrow 2k - k = 5 - 1$

$\Rightarrow k = 4.$

EXERCISE–2
(BASED ON MEMORY)

1. (b) For equal roots

$a = b^2 - 4ac = 0$

Going by option

(a) $30 - 4(b) \neq 0$

(b) $36 - 4(a) = 0$

$3x^2 - 6x + 3 = 0$ has equal roots.

2. (d) Sum of roots $= \frac{-b}{a}$

$$1 = \frac{-b}{a}$$

Product of root $= \frac{c}{a}$

$$20 = \frac{c}{a}$$

Equation $= x^2 - x + 20 = 0$

3. (d) $a(x + y) = b(x - y) = 2ab$

$x + y = 2b$ (i)

$x - y = 2a$ (ii)

Adding (i) and (ii)

$x = a + b$

Subtracting (i) and (ii)

$y = b - a$

Now,

$2(x^2 + y^2)$

from (i) and (ii)

$2((a + b)^2 + (b - a)^2)$

$2(a^2 + b^2 + 2ab + a^2 + b^2 - 2ab)$

$2(2a^2 + 2b^2)$

$4(a^2 + b^2)$

4. (c) $(x - 2)(x - p) = x^2 - ax + 6$

$x^2 - px - 2x + 2p = x^2 - ax + 6$

$-px - 2x + 2p = ax + b$

Equating coefficients of x

$p + 2 = a$

$a - p = 2$

5. **(a)** $a^2 = by + cz$

$b^2 = cz + ax$

$c^2 = ax + by$

$a^2 + ax = ax + by + cz$

$a(a + x) = ax + by + cz$

$$\frac{1}{a+x} = \frac{a}{ax+by+cz}$$

Similarly

$$\frac{1}{b+x} = \frac{b}{ax+by+cz},$$

$$\frac{1}{c+x} = \frac{c}{ax+by+cz}.$$

$$\frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z}$$

$$\frac{ax}{ax+by+cz} + \frac{by}{ax+by+cz} + \frac{cz}{ax+by+cz}$$

$$\frac{ax+by+cz}{ax+by+cz} = 1$$

6. **(a)** Given $x = 332$, $y = 333$, $z = 335$

$$x^3 + y^3 + z^3 - 3xyz = (x+y+z)$$

$$(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$(x+y+z)\frac{1}{2}\left[(x-y)^2 + (y-z)^2 + (z-x)^2\right]$$

$$\frac{1}{2}[332+333+335][1+4+9]$$

$= 7000.$

7. **(d)** $m = -4$ and $n = -2$

$$m^3 - 3m^2 = 3m + 3n + 3n^2 + n^3$$

$$\Rightarrow \left(m^3 - 3m^3 + 3m - 1\right) + \left(n^3 + 3n^2 + 3n + 1\right)$$

$$\Rightarrow (m-1)^3 + (n+1)^3$$

$$= (-4-1)^3 + (-2+1)^3$$

$$\Rightarrow (-5)^3 + (-1)^3 \Rightarrow -125 - 1 \Rightarrow -126.$$

9. **(a)** $x = \dfrac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}}, y = \dfrac{1}{x} = \dfrac{\sqrt{13}+\sqrt{11}}{\sqrt{13}+\sqrt{11}}$

$$x + y = \frac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}} + \frac{\sqrt{13}+\sqrt{11}}{\sqrt{13}-\sqrt{11}}$$

$$= \left(\frac{B+11}{13-11}\right) \times 2 = \frac{24}{2} \times 2 = 24$$

$$x \cdot y = \frac{\sqrt{13}-\sqrt{11}}{\sqrt{13}+\sqrt{11}} \times \frac{\sqrt{13}+\sqrt{11}}{\sqrt{13}-\sqrt{11}} = 1$$

$$3x^2 - 5xy + 3y^2 = 3(x+y)^2 - 11xy.$$

$$= 3(24)^2 - 11(1)$$

$$= 1717$$

10. **(d)** $x^2 + y^2 + z^2 - xy - yz - zx = 0$

$$\frac{1}{2}\left[(x-y)^2 + (y-z)^2 + (z-x)^2\right] = 0$$

$x = y = z = 1$

$$\therefore \frac{3x^4 + 7y^4 + 5z^4}{5x^2y^2 + 7y^2z^2 + 3z^2x^2} = \frac{3+7+5}{5+7+3} = 1.$$

11. **(b)** $x - \sqrt{3} - \sqrt{2} = 0$

$$\Rightarrow x = \sqrt{3} + \sqrt{2}$$

Again, $y = \sqrt{3} + \sqrt{2} = 0$

$$y = \sqrt{3} - \sqrt{2}$$

$$\therefore x - y = \sqrt{3} + \sqrt{2} - \sqrt{3} + \sqrt{2} = 2\sqrt{2}$$

and $xy = \left(\sqrt{3}+\sqrt{2}\right)\left(\sqrt{3}-\sqrt{2}\right) = 3 - 2 = 1$

$$\therefore \text{ Expression } \Rightarrow x^2 - 20\sqrt{2} - y^2 - 2\sqrt{2}$$

$$\Rightarrow x^3 - y^3 - 22\sqrt{2}$$

$$= (x-y)^3 + 3xy(x-y) - 22\sqrt{2}$$

$$= \left(2\sqrt{2}\right)^3 + 3\left(2\sqrt{2}\right) - 22\sqrt{2}$$

$$= 16\sqrt{2} + 6\sqrt{2} - 22\sqrt{2} = 0.$$

12. **(c)** $3\left(a^2 + b^2 + c^2\right) = (a+b+c)^2$

$$3a^2 + 3b^2 + 3c^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$$

$$\Rightarrow 3a^2 + 3b^2 + 3c^2 - a^2 - b^2 - c^2 = 2ab + 2bc + 2ac$$

$$\Rightarrow 2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ac = 0$$

$$\Rightarrow a^2 + b^2 - 2ab + b^2 + c^2 - 2bc + a^2 + c^2 - 2ac = 0$$

$$\Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 = 0$$

$\Rightarrow a - n = 0 \Rightarrow a = b$

$\Rightarrow b - c = 0 \Rightarrow b = c$

$\Rightarrow c - a = 0 \Rightarrow c = a$

$\therefore a = b = c.$

13. **(b)** $\dfrac{x^3 - y^3}{x^2 + xy + y^2} = \dfrac{5}{1}$

$$\frac{(x-y)\left(x^2 + xy + y^2\right)}{x^2 + xy + y^2} = 5$$

$$x - y = 5 \qquad (1)$$